TM Forum Standard

Information Framework (SID)

Product Domain Business Entities

Information Framework Suite

GB922 Domain Business Entities

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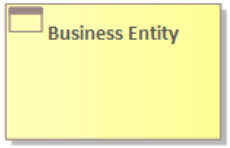
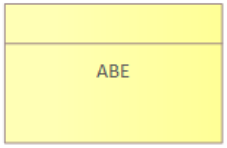
1 General Information

To find the Information framework figure, refer to “GB991 Core Frameworks Concepts and Principles” guidebook.

2 Typographic Conventions

- Relationships starting by a “/” correspond to shortcut and represent navigation relationships (aka derived relationship). A derived relationship corresponds to several relationships replaced by a unique derived relationship to give a synthetic view. It isn’t needed to implement such relationships.
- In diagrams, on Business Entities from a different ABE than the ABE being referred to in the diagram, the name after “from” represents the ABE where the Business Entity is located.
- A Business Entity is represented by a *decorator* located to the top left. 

- The Aggregated Business Entity (ABE) is represented similarly to ArchiMate. 

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3 Glossary

Name	Description
ABE	Aggregate Business Entity
BE	Business Entity
eTOM	enhanced Telecom Operation Map (TMF Framework)
ODA	Open Digital Architecture
SID	Shared Information Data model (TMF Framework) is the previous name of Information Framework
TAM	Telecom Applications Map (TMF Framework) is the previous name of Application Framework

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4 Product Domain

The Product Domain represents roles, information and activities carried out by parties (e.g. individuals / organizations) playing roles that are involved in the strategic planning, definition, development and operational aspects of Products that are offered. Activities include management of Product: strategies, capabilities, lifecycles, offerings, instances, performance, contract operations, usage statistics and support of goods and services (products) that are offered to customers by an enterprise.

Note that Products covers 2 cases: 1) Products bought by a Customer from the Service Provider, 2) Products bought by the Service Provider from Business Partners.

In looking at 'Product' for the SID model, the modelers looked at Product concepts in the eTOM and member company offerings and implementations. The key concepts that were looked at were the detailed specification of products (ProductSpecifications); the way in which they are offered into the market (ProductOfferingSpecifications); and the way in which they are maintained and perform while in use (Products). The SID model reflects Product entities from a business-oriented point-of-view and uses the best of standard modeling patterns for this area.

Figure [PrD-01] Product ABEs Level 1

The diagram below lists the Product Domain's ABEs.

Product Party Roles ABE

- Product Party Roles ABE contains all PartyRoles related to the Product Domain such as ProductManager or ProductUser.

Product Specification ABE

- The Product Specification ABE contains all that represents a product specification as perceived by the business user and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available…).

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Product Offering Specification ABE

- The Product Offering Specification ABE describes tangible and intangible goods made available for a certain price to the market in the form of product catalogs. This ABE is also responsible for targeting market segments based on the appropriate market strategy.

Product Order ABE

- The Product Order ABE contains all entities required to specify a communication used to procure or update one or many Products and ProductOfferingInstances in the context of ProductSpecifications and ProductOfferingSpecifications.
- This pattern is used to define CustomerProductOrder corresponding to Products ordered by a Customer from the Service Provider and BusinessPartnerProductOrder corresponding to Products ordered by the Service Provider from a Business Partner

Product and Product Offering Instance ABE

- Product and ProductOfferingInstance represent, in the product inventory, respectively an instance of a ProductSpecification and ProductOfferindSpecification subscribed or not yet subscribed to by a Customer from the CSP or by the CSP from a BusinessParnter.

Product Usage ABE

- Product Usage ABE represents specification of usages enabled by Products and usages of products tracked.

Loyalty ABE

- A loyalty program is one of the tools used by the loyalty process to retain customers. The Loyalty ABE contains all entities useful to specify and instantiate loyalty programs. A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.

Product Configuration ABE

- The Product Configuration ABE represents the specification of the different possible Product configurations as well as the Product Configurations instantiated for a Product.

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Product Test ABE

- Product Test ABE represents all information to specify tests that will be performed on Products as well as the results in measures being produced that reflect the functioning of the product under test.

Product Performance ABE (not fully developed)

- The Product Performance ABE handles product performance goals, the results of end-to-end product performance assessments, and the comparison of assessments against goals. The results may include the identification of potential capacity issues.

Product Capacity ABE

- Product Capacity ABE represents specific ability of Products measured in quantity and units of quantity over an extended period.

Strategic Product Portfolio Plan ABE (not fully developed)

- A strategy employed to promote a ProductOffering. Strategic Product Portfolio Plan ABE is concerned with the plans of the product portfolio, which product offerings to make available to each market segment and the plans to development and deploy product offerings, as well as retirement of products.

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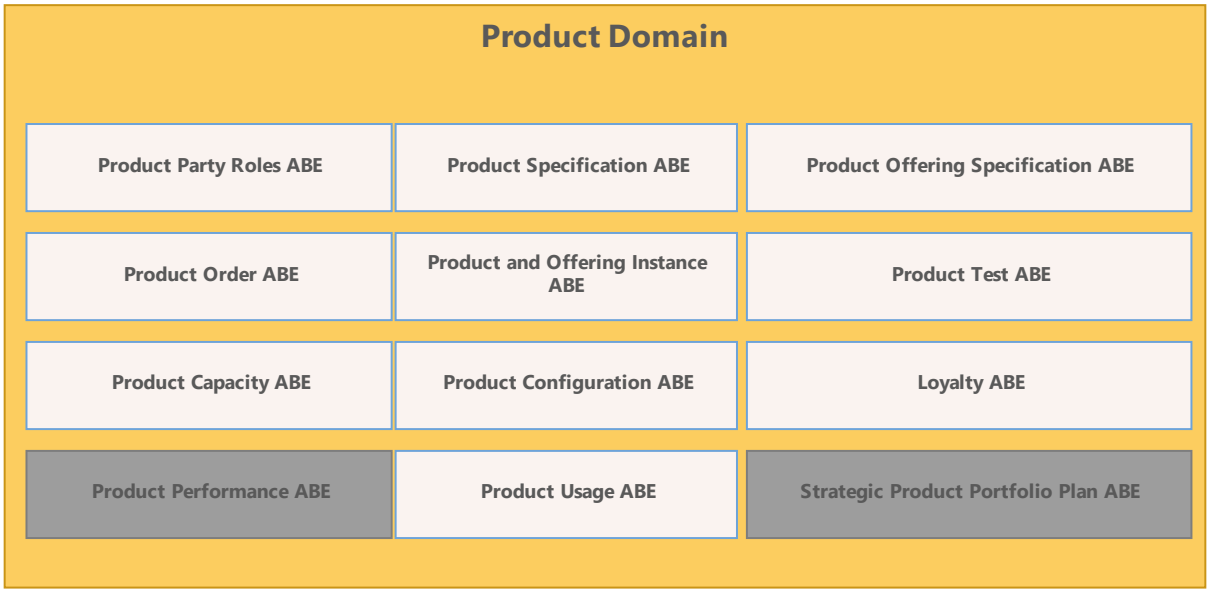

Figure [PrD-01] Product ABEs Level 1

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Figure [PrD-02] Product ABEs Level 1 & 2

The diagram below lists the Product Domain's ABEs Level 1 & 2.

Product ABE

- Represents an instance in the product inventory of a ProductSpecification subscribed or not yet subscribed by a Customer to the CSP or by the CSP to a BusinessParnter, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses.

Product Offering Instance ABE

- A ProductOfferingInstance represents the instantiation of a ProductOfferingSpecification for a specific Customer. It may carry ProductPrice and commitment choice.

Loyalty Program ABE

- The Loyalty Program ABE contains all entities needed to instantiate a loyalty program for a specific customer such as LoyaltyProgram and LoyaltyAccount.

Loyalty Program Specification ABE

- The Loyalty Program Specification ABE contains all entities specifying a loyalty program such as LoyaltyProgramProdSpec and LoyaltyRule.

Network Product Spec ABE

- The Network Product Spec ABE contains the NetworkProductSpec description and relationships.

Usage Volume Product Spec ABE

- The Usage Volume Product Spec ABE contains the UsageVolumeProductSpec and all the entities specific to this type of product.

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Goods and Shipping Product Spec ABE

- Goods and Shipping Product Spec ABE describe specificity of Goods and Shipping Product Spec and their relationships.

Product Action ABE

- The Product Action ABE contains Business Entities describing possible actions for ProductSpecifications, ProductOfferings and ProductConfigSpecs.

Product Usage Spec ABE

- The Product Usage Spec ABE specifies types of usage of ProductSpecifications.

Product Usage Entity ABE

- The Product Usage Entity ABE collects product consumption information, and generates ProductUsage according to ProductUsageSpecification for use by other business entities.

Product Catalog ABE

- Product Catalog ABE is a collection of entities that represent the specification of ProductCatalogs and catalogs described by them that contain ProductOfferings and their associated ProductOfferingPrices.

Product Offering Price ABE

- The Product Offering Price ABE describes all the prices related to ProductOfferings.

Product Offering Price Rule ABE

- The Product Offering Price Rule ABE describes all rules related to prices concerning ProductOfferings.

Pricing Logic Algorithm ABE

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- The Pricing Logic Algorithm ABE contains entities representing an instantiation of an interface specification to external rating function (without a modeled behavior in SID).

Product Offering Price Tax ABE

- The Product Offering Price Tax ABE describes how taxation is modelled in the context of ProductOfferingPrice and ProductPrice.

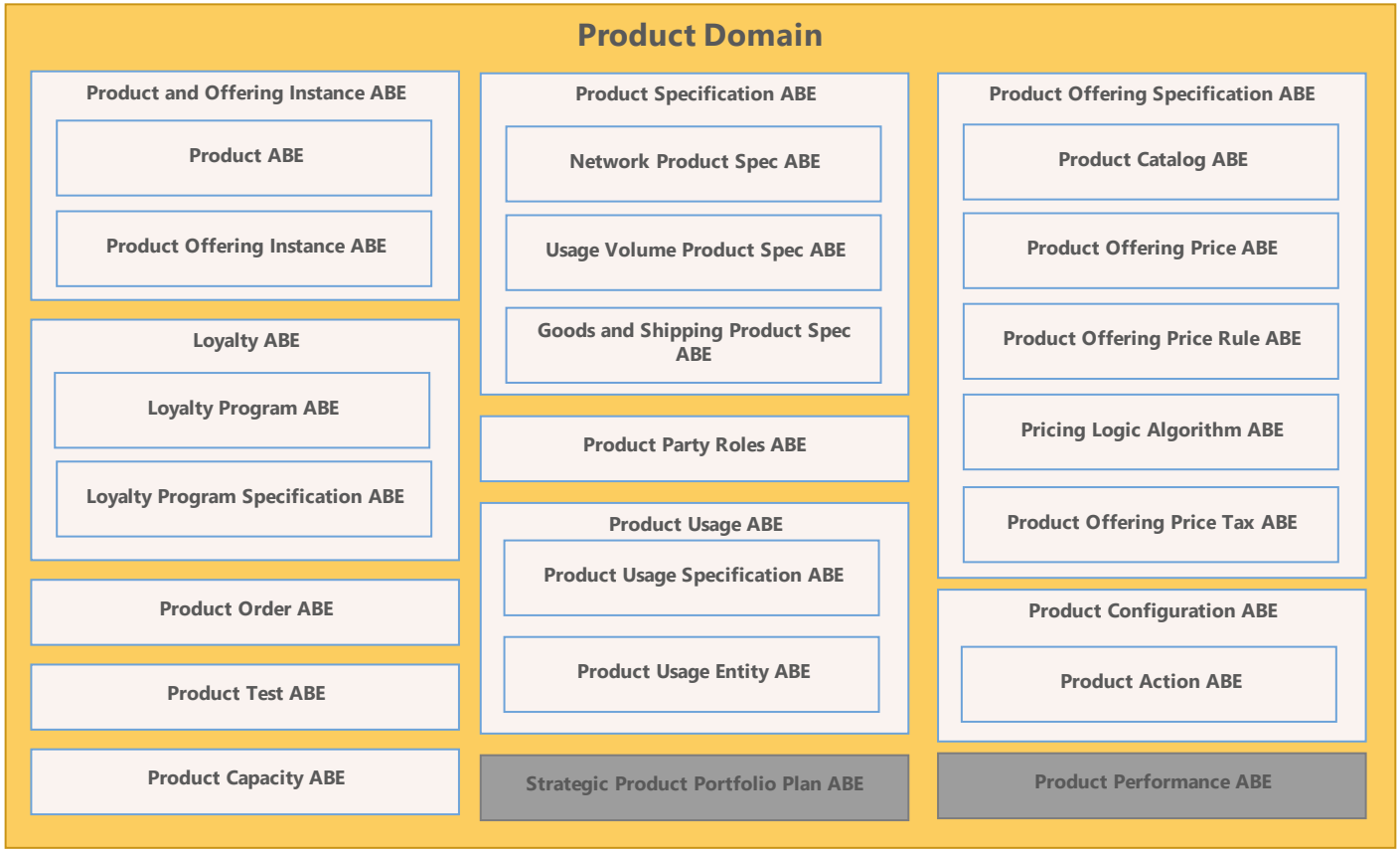

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Figure [PrD-02] Product ABEs Level 1 & 2

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4.1 Product Party Roles ABE

Product Party Roles ABE contains all PartyRoles related to the Product Domain such as ProductManager or ProductUser.

Note: *Roles related to Customer such as Payer are located in Customer Domain*

Figure Pr.00 - Product Party Roles

The Product Domain party roles presently identified are:

- **ProductUser**: A ProductUser is a type of PartyRole corresponding to the role of using a Product.
- **ProductManager**: Agreed & done: A ProductManager is a party role which is responsible for the development of products for an organization. Product managers own the business strategy behind a product, specify its functional requirements and generally manage the launch of corresponding ProductOfferings. They coordinate work done by many other functions (like software engineers, data scientists and product designers) and are ultimately responsible for the business success of the product.
- **TaxJurisdiction**: TaxJurisdiction is a specific party role played by an organization that has the authority to levy tax. The geographic area over which tax can be levied is derived from the base PartyRole class relationship to AdministrativeArea.
- **LoyaltyProgramMember**: A LoyaltyProgramMember is defined on each LoyaltyAccount. A LoyaltyProgramMember is a type of PartyRole and is granted with the rights on the LoyaltyAccount such as to execute LoyaltyBurn used to realize a CustomerPayment.

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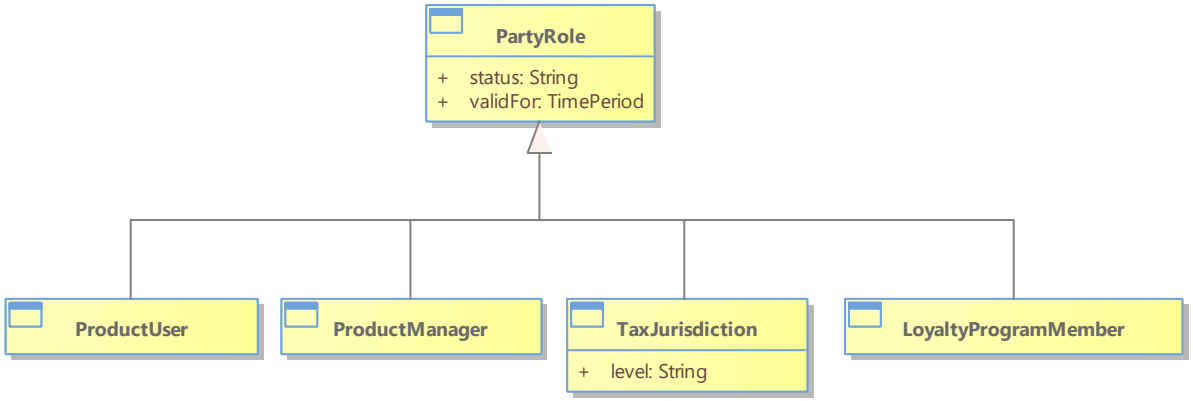

Figure Pr.00 - Product Party Roles

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4.2 Product Specification ABE

The **Product Specification ABE** contains all that represents a product specification as perceived by the business user and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available…).

Figure Pr.01 - Primary Product Domain Business Entities

An overview of how the main Product domain business entities are related is shown in the figure below. This model is developed as we work through this document.

The Product domain is essential to the overall SID model. Customers need to be able to express their functional needs in plain English in order to determine which ProductSpecifications can support their requirements and which ProductOfferingSpecifications can propose these ProductSpecifications.

The ProductSpecification represents a product specification as perceived by the Product user and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available…).

A ProductOfferingSpecification represents what is externally presented to the market for the market's use. It includes commercial constraints and price.

The ProductOfferingSpecification Business Entity is abstract. Using the atomic/composite pattern, it further specializes into two concrete Business Entities: AtomicProductOfferingSpecification and CompositeProductOfferingSpecification.

The ProductSpecification Business Entity is also abstract. Using also the atomic/composite pattern, it further specializes into two concrete Business Entities: AtomicProductSpecification and CompositeProductSpecification.

An AtomicProductOfferingSpecification is always associated with exactly one ProductSpecification, which itself can be atomic or composite.

A CompositeProductOfferingSpecification is not directly associated with any ProductSpecification. Instead, it contains ProductOfferingSpecifications, each of which may in turn be an AtomicProductOfferingSpecification that is associated with a ProductSpecification, or with a CompositeProductOfferingSpecification.

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An example of a CompositeProductOfferingSpecification is a home-based internet service representing the aggregation of three AtomicProductOfferingSpecifications: internet connection, email service, and web hosting. The home-based internet ProductOfferingSpecification is not related to any ProductSpecification, since it merely represents a grouping of other ProductOfferingSpecifications.

Vice versa, a ProductSpecification may be made available as one or more AtomicProductOfferingSpecifications (which in turn may be components of a CompositeProductOfferingSpecification).

For example, an email ProductSpecification may be used in a number of different ProductOfferingSpecifications: one such offering may be bundled as part of a Broadband Internet offering, while another offering may represent stand-alone email.

Note that ProductSpecification and ProductOfferingSpecification inherit the lifecycleStatus and validFor attributes from EntitySpecification, and ID, name and description from RootEntity.

A Product represents the instantiation of a ProductSpecification.

If the product can be used (based on a contract) over a period of time (and not for example a one-time purchase of a device), this is colloquially known as a Subscription. For example, Helen has subscribed to company ABC's internet ProductOfferingSpecification. Typically, when a Customer acquires a ProductOfferingSpecification, a ProductOfferingInstance will be instantiated for the parent ProductOfferingSpecification, for each contained ProductOfferingSpecification, and for each ProductSpecification referred to by the contained AtomicProductOfferingSpecifications in the offering tree.

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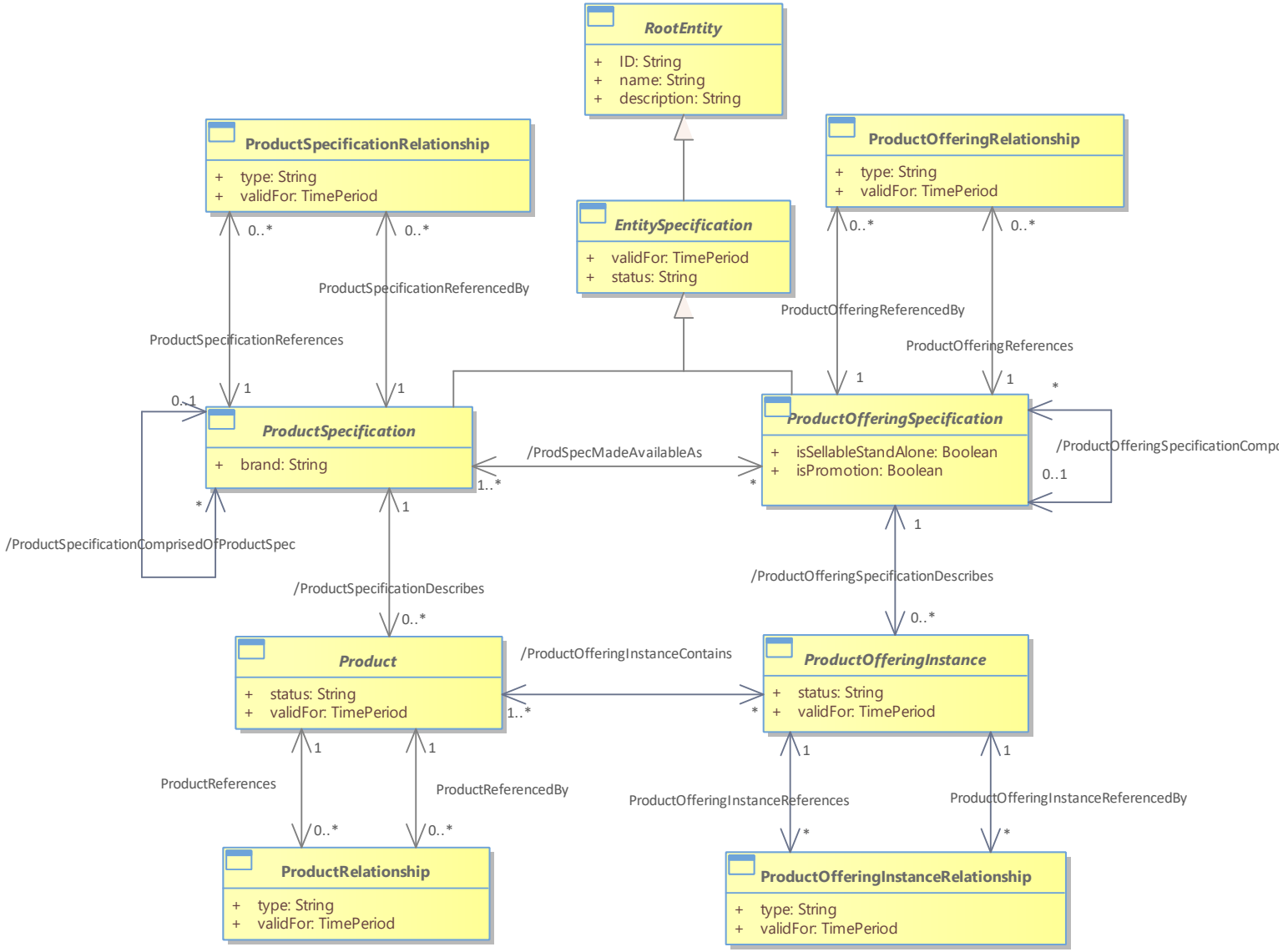

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Figure Pr.02 - Product, Service, Resource Specification Relationships

The following figure gives an overview of the relationships between Product, Service and Resource Specifications

A ProductOffering represents how the ProductSpecification is sold (i.e. packaging rules, prices, alterations, commitments…).

The ProductSpecification represents a product specification as perceived by the business user and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available…).

It can represent:

- tangible goods (Terminal, phone, mobile, fax, modem)
- or intangible goods (like an Anti-Spam service for email).

When the ProductSpecification represents a tangible goods, it is realized as ResourceSpecification. Corresponding Resources are stored in warehouses or bought to a supplier on demand.

When the ProductSpecification is an intangible good, it is either realized as a CustomerFacingService (when the Service Provider is able to realize it) or bought as a ProductSpecification proposed by a Supplier (when the Service Provider is not able to realize it). For example, a Broadband know-how might be built or bought depending on the location of it.

The CustomerFacingServiceSpec represents all the Service Provider's know-how of intangible goods at a functional level. It would be more appropriate to name it KonwHowSpec or ProductFacingServiceSpec, but the name is not changed as it is well known.

The ProductSpecification corresponds to a sub-set of the Service Provider's know-how according to what marketing people want to sell. Therefore, a ProductSpecification is a restriction of CustomerFacingServiceSpecs.

When the ProductSpecification corresponds to a tangible good, then it is realized as a ResourceSpecification.

The ResourceFacingServiceSpec represents the technical solutions that the Service Provider can implement for CustomerFacingServiceSpec.

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Each technical solution (ResourceFacingServiceSpec) requires ResourceSpecifications to be realized and might require buying part of the solution to a supplier (for example the Service Provider might have to buy the Local Loop to provide a broadband access).

Not presented in the diagram: Sometimes the company is not able to realize the ProductSpecification on its own; in these cases, they will have to buy all or part of the technical solution. This is the case for MVNOs that always have to buy what they sell from a Supplier.

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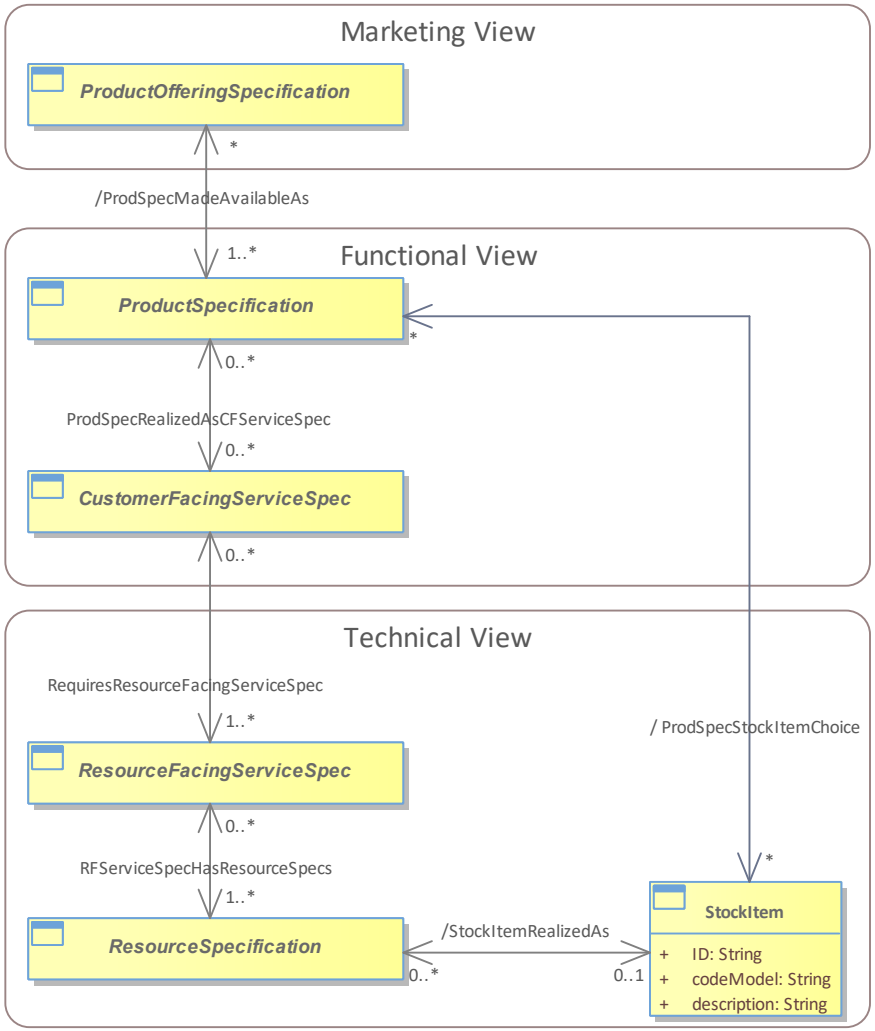

Figure Pr.02 - Product, Service, Resource Specification Relationships

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Illustration Figure Pr.02 – I01 Catalog Lifecycle Management

The Catalog Lifecycle Management process builds the specifications starting with the technical view, then the functional view and finally the marketing view.

On the contrary, the Order Handling process starts looking for what can functionally answer to the customer requirements and then find the ProductOfferings making available the ProductSpecifications.

The Delivery process has to identify the know-how (CFSSpec) corresponding to the ProductSpecification, then identify the different technical solutions (RFSSpec) corresponding to the know-how, chose the technical solution according to rules, and finally implement the ResourceSpecifications corresponding to the technical solution (RFSSpec).

Like this,

- OSS people specify functionally what can be done with OSS (Know-How i.e. CFSSpec)
- IT and Network are isolated from Marketing changes
  - Changing ProductOfferings and/or ProductSpecifications does impact only the order handling and catalog management
  - Marketing people can define new ProductSpecifications without impacting the know-how (CFSSpec)
- The delivery process is only impacted by CFSSpec / RFSSpec and Resource changes
  - Technical solutions might be changed without changing what is seen by the Customer (ProductSpecification and ProductOffering)
  - The know-how might be improved without impacting existing ProductSpecifications

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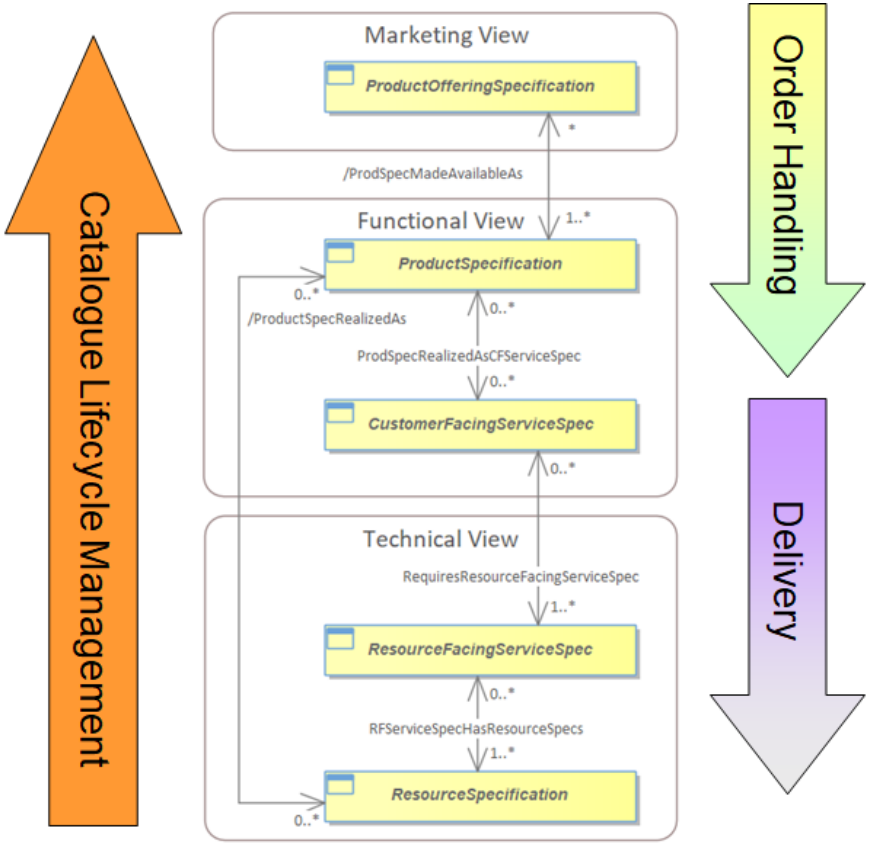

Illustration Figure Pr.02 – I01 Catalog Lifecycle Management

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Illustration Figure Pr.02 – I02: Voice Over IP

This illustration shows how the Voice Over IP can be described.

Notes:

- The H323 solution is not described here
- The SIP solution described here is close to the real situation but not complete.

The Voice over IP product is derived from the know-how of the telecommunication operator to be marketed to customers.

The know-how Voice over IP may be either realized as Software SIP solution or as a Software H323 solution.

In case of a Software SIP solution, it always requires a Service platform with type AS Base.

In addition to this, either a Huawei or Ericsson (a.k.a. E///) platform type is needed.

The Huawei platform type requires an IMS Core Huawei resource and the Ericsson platform type requires an ENUM and HSS resource.

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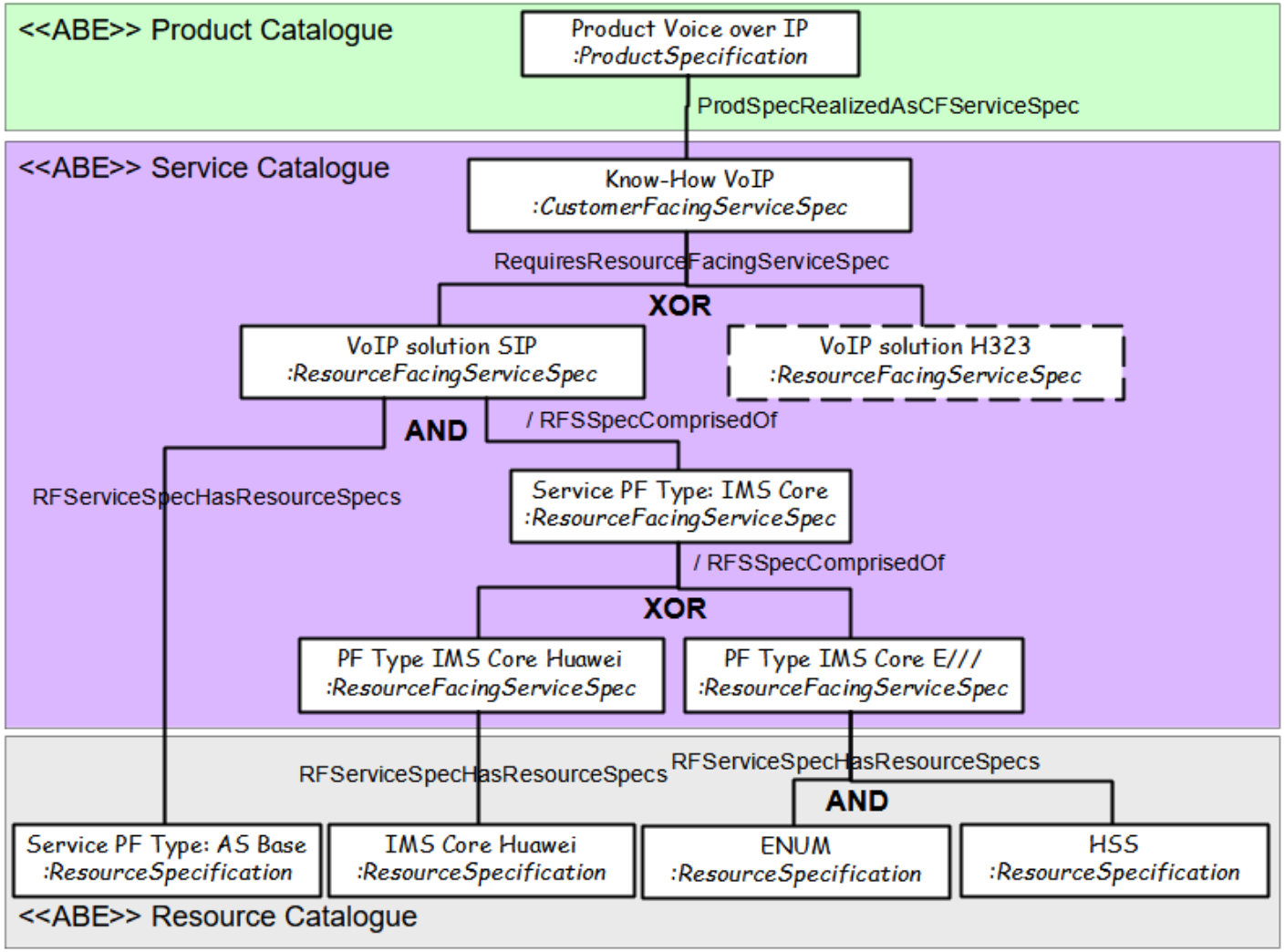

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Figure Pr.02a - Product, Service, Resource Domain Relationships

The figure below depicts the relationship among Product, Service, and Resource domain business entities.

Business entities in the Product domain have a close relationship with business entities in the Service and Resource domains.

While Product business entities represent what the market sees of a provider's offerings, Service and Resource business entities represent the realization of the offerings from a provider's perspective.

For example, a Broadband Internet ProductOfferingSpecification is realized by two Services, one is a bandwidth connectivity service to the provider's network; the second is a virtual connectivity service to the Internet Service Provider.

Each ProductSpecification may be derived from CustomerFacingServiceSpecifications, or realized as StockItems or realized as ProductSpecifications proposed by Suppliers.

The relation between ProductSpecification and StockItems only concerns tangible products such as Box (represented as a CompoundResourceSpec -Physical and Logical-) or such as a PSTN phone.

The relation between ProductSpecification and CustomerFacingServiceSpecification concerns non-tangible products such as mobile line, call forwarding…

And the relation between ProductSpecification and ProductSpecification proposed by Suppliers (through ProdSpecBPProdSpecChoice) concerns tangible or non-tangible products that we can buy to a Supplier.

If it corresponds to a tangible goods it will be shipped directly to the customer by the supplier.

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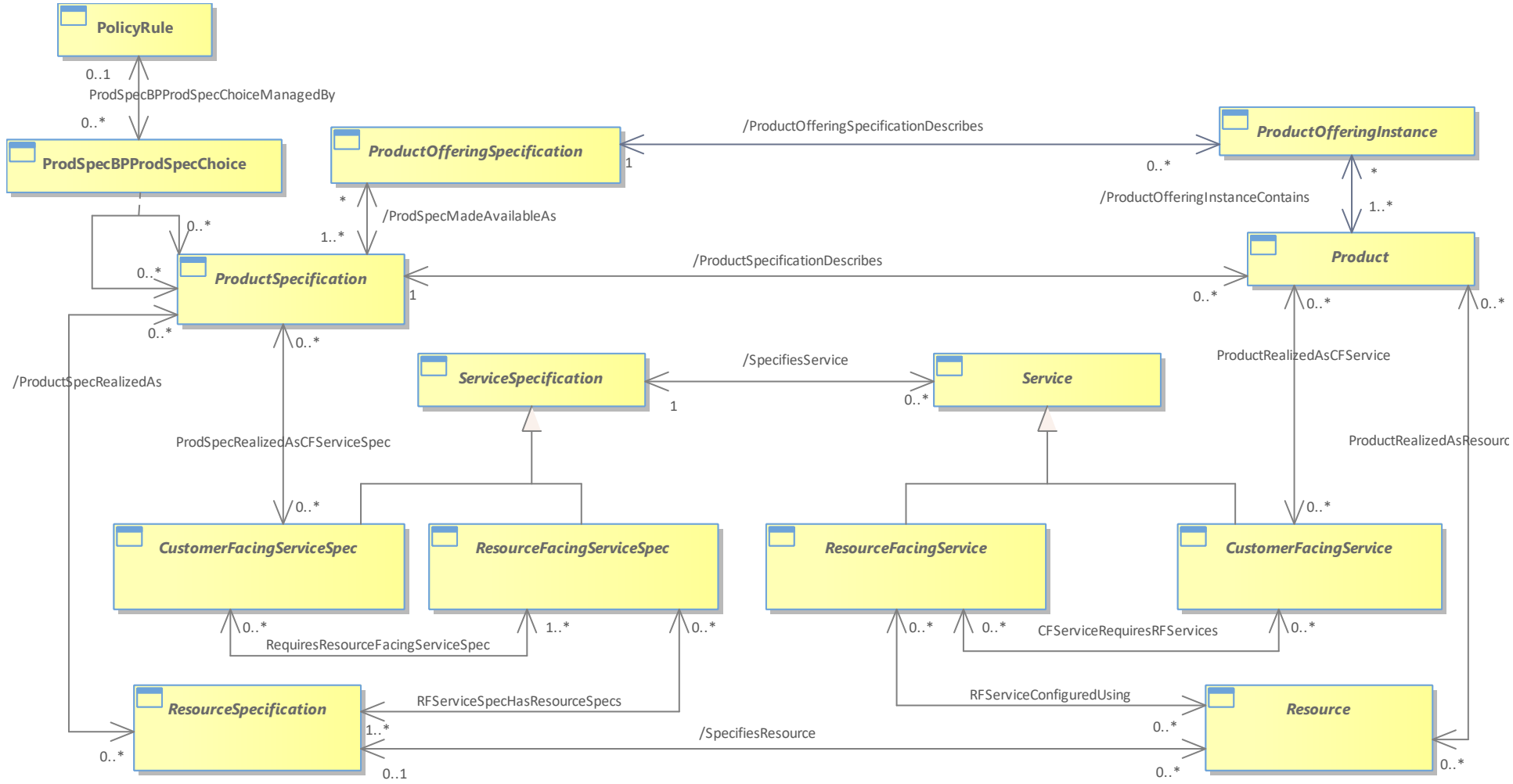

Figure Pr.02a - Product, Service, Resource Domain Relationships

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Figure Pr.02b - StockItem and other domains overview (copy from SI.01 in Resource Domain)

A StockItem represents items used by Supply Chain for Stock and Shipping Management.

The codeModel specifies the type of codification used for the stockItemCode.

Two main types exist: EAN (European Article Numbering) or GTIN (Global Trade Item Numbering). A GTIN consists of a GS1 Company Prefix, an Item Reference Number and a Check Digit.

Each of them is further defined by the length of the number such as EAN 13, GTIN-12 (USA / Canada for individual products), GTIN-13 (Rest of World for Individual products) and GTIN-14 (Worldwide for Product groupings).

A ProductConfigSpec corresponds to a configuration of the related ProductSpecification. Ex: A Red Iphone with 128 GB memory. A GoodsProductSpec is realized by a unique StockItem according to the ProductConfigSpec.

A StockItem might be realized by several ResourceSpecifications when different manufacturers produce the same StockItem (same characteristics and behaviour).

A GoodsProductSpec might be created but it can't be modified! I.e. only Create and Return operations are available for a GoodsProductSpec. For example, if the disk space of a TV decoder's record capacity can be enhanced (realized through Software update), then we create a specific ProductSpecification for the Record Capacity.

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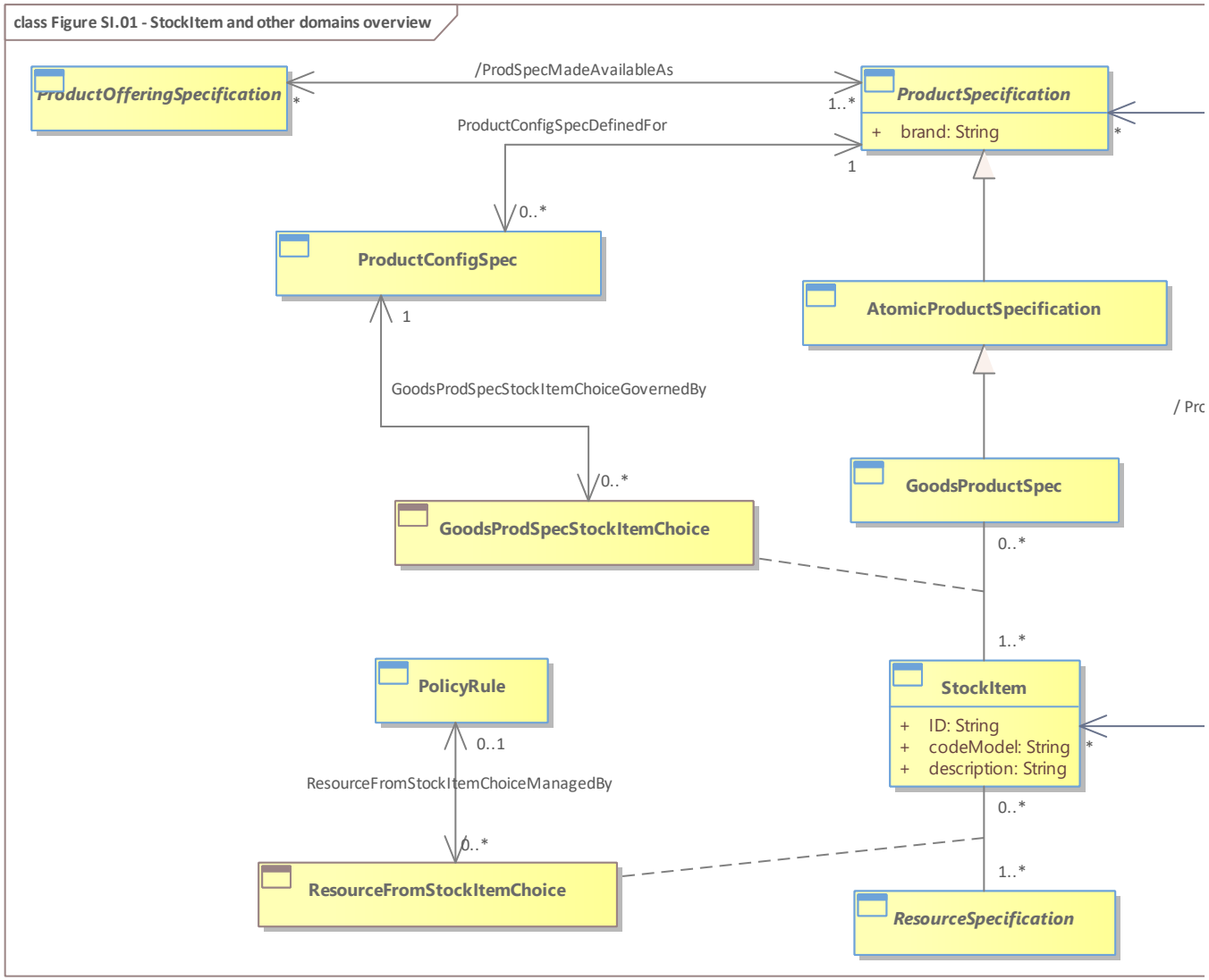

Figure Pr.02b - StockItem and other domains overview (copy from SI.01 in Resource Domain)

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Figure PR.02c - Global catalog view

This figure presents a global view on ProductSpecification and main associated BE used to represent a catalogue.

A non tangible ProductSpecification may be realized using operator's CustomerFacingServiceSpec (Example: A Mobile Connectivity ProductSpecification is generally a restriction of a CustomerFacingServiceSpec built by the operator).

In case of tangible product, a ProductSpecification can be provided as a StockItem realized as ResourceSpecifications (example: Most tangible goods are realized as StockItems bought in advance by the operator and stored in warehouses).

CustomerFacingServiceSpec may be realized using a technical solution (ResourceFacingServiceSpec) owned by the operator (example: In order to build its Mail CustomerFacingServiceSpec, an operator can use its own ResourceFacingServiceSpec).

A ResourceFacingServiceSpec may be built using ResourcesSpecification owned by the operator (example: and operator uses its own Virtual Machines and software to build its ResourceFacingServiceSpec).

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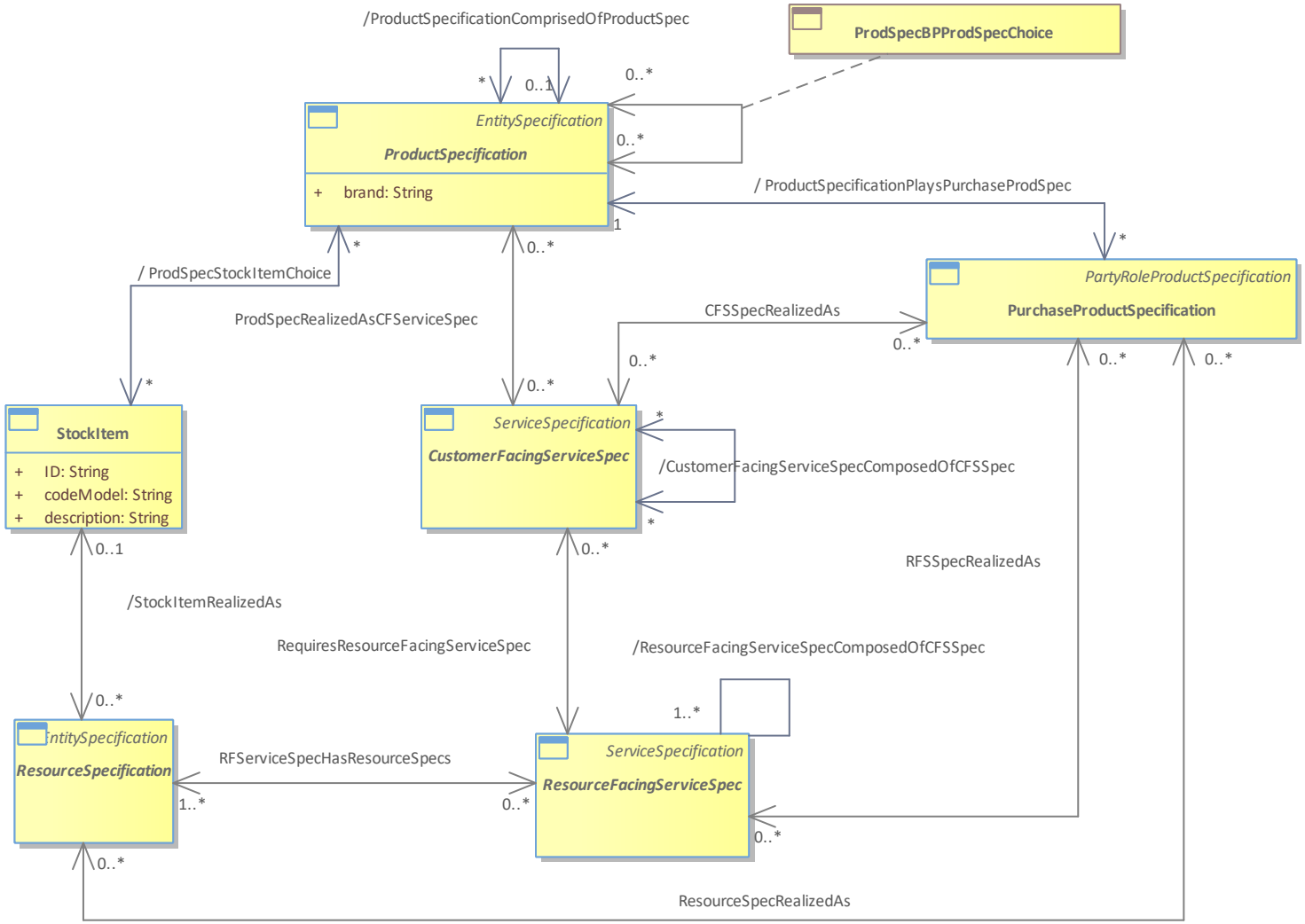

Figure PR.02c - Global catalog view

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Figure Pr.03 - Product Specification - Basic Entities

Products are things (tangible or intangible) which enterprises, such as service providers, market, sell or lease to customers to create profit. In a changing world, monopoly telcos with a custom-built portfolio have been replaced by a value network of Suppliers and Service Providers offering complex communications-based Products. The way in which those Products have to be specified is not as simple as it used to be.

A product serves as a window for customers and users into the world of the services and resources used to realize the product.

Customer/user always deals with products, unless the individual/organization they represent is playing the role of a service technician or some similar role. A user uses a Product to obtain its value promise, e.g. to make a phone call

Neither the Product User nor the Customer can see the CustomerFacingService. They only can see the Product.

A ProductSpecification may be simple (atomic) or a composition of other ProductSpecifications [Fowler-Specification]. A ProductSpecification may represent a tangible object or something intangible provided to customers, realized as a 'service' but not to be confused with a network service here. The SID modeling of a CompositeProductSpecification covers both the physical build of more than one ProductSpecification and one or more ProductSpecifications supplied together in a collection. ProductSpecifications may also exist within groupings, such as product categories or product lines or product types.

Enterprises may assemble products themselves or they may buy them in from other suppliers (through contractual agreements where the enterprise takes the role of customer - see SID Common Domain guide book); or they may do a combination of these things.

Product Specification Example - The "3G Mobile super saver pack"

This is sold via intermediaries ("Dog-n-Bones-R-Us" phone shops) as a "BigTelephone" branded product. Nokia supply the phones and provides a 2-year guarantee. The "BigTelephone" network is used for voice and they supply a phone number and a SIM card. The SMS function is provided by "Best Messaging", as "BigTelephone" doesn't have an SMS function. The customer signs the contract with "Dog-n-Bones-R-Us" for 2 years with minimum monthly calls and Dog-n-Bones-R-Us bills the customer directly. Dog-n-Bones-R-Us has a separate contract with "Best Messaging" to cover the use of their SMS function. In looking at the detail of ProductSpecification, the following concepts are modeled by the figure below:

- Relationships between different ProductSpecifications for bundling or composite specifications;

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- Different attributes for different types of ProductSpecification; for example, some Products the customer can take home with them, others they experience as a Service; for example, one delivered across a network;
    - Atomic ProductSpecifications which cannot be broken down further.
    - CompositeProductSpecifications which comprise a number of other ProductSpecifications, any of which may provide more than one instance per CompositeProductSpecification.

Note: A ProductSpecification cannot contain itself, but it can reference other ProductSpecifications.

At least one action is allowed for a ProductSpecification (AllowedProductAction) such as create, update…that might be limited to one or many SalesChannels. An AllowedProductAction has a type represented by a ProductTypeAction.

For ProductOfferingSpecification a Customer could upgrade their offering on the web, but to downgrade they need to come into a store or contact a service representative.

An AllowedProductAction might imply changes from one ProductConfigSpec to one or many ProductConfigSpec.

Thus, it is possible to define actions such as upgrade or downgrade and specify which SalesChannel enables these actions.

In all combinations of products, there must be clearly defined relationships between the products in order to ensure consistency of service to the customer. Some examples of such ProductSpecificationRelationships would be migration, exclusivity, dependency and substitution, as indicated in the Figure below.

Often it becomes essential to group product specifications. SID supports following two ways in which product specifications can be grouped

- Based on how the product specs are organized in a product portfolio
- Based on common characteristics

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These groupings are supported by ProductSpecificationType entity and its two subclasses Product Line (Based on how the product specs are organized in a product portfolio) and Product Category (Based on common characteristics). Product Line represents the coarsest form of grouping of product specifications.

“ProductSpecificationType” is a specialization of “EntitySpecificationType” and as such it inherits the recursive association “EntitySpecificationTypeRelationship”. This can be used to represent how a Product Line is associated to multiple Product Categories. For example, ProductLine “Home Security” is composed of different Product Categories “Door Alarms”, “Security Lights”, “GPS & Tracking” etc.

Same Product Category can also be used across different Product Lines. For example, same set of “Mobile phones” Product Category reused across Product Lines “Mobile Residential” and “Mobile Enterprise”.

The decision regarding when to model a ProductLine or ProductCategory is often left to the discretion of the Product/Marketing managers.

There are also costs associated with creating, developing and maintaining a ProductSpecification; the enterprise records these against the ProductSpecification. There is an implied link between the ProductSpecificationCost entity and business information, but this has not been explored in this area of the SID model.

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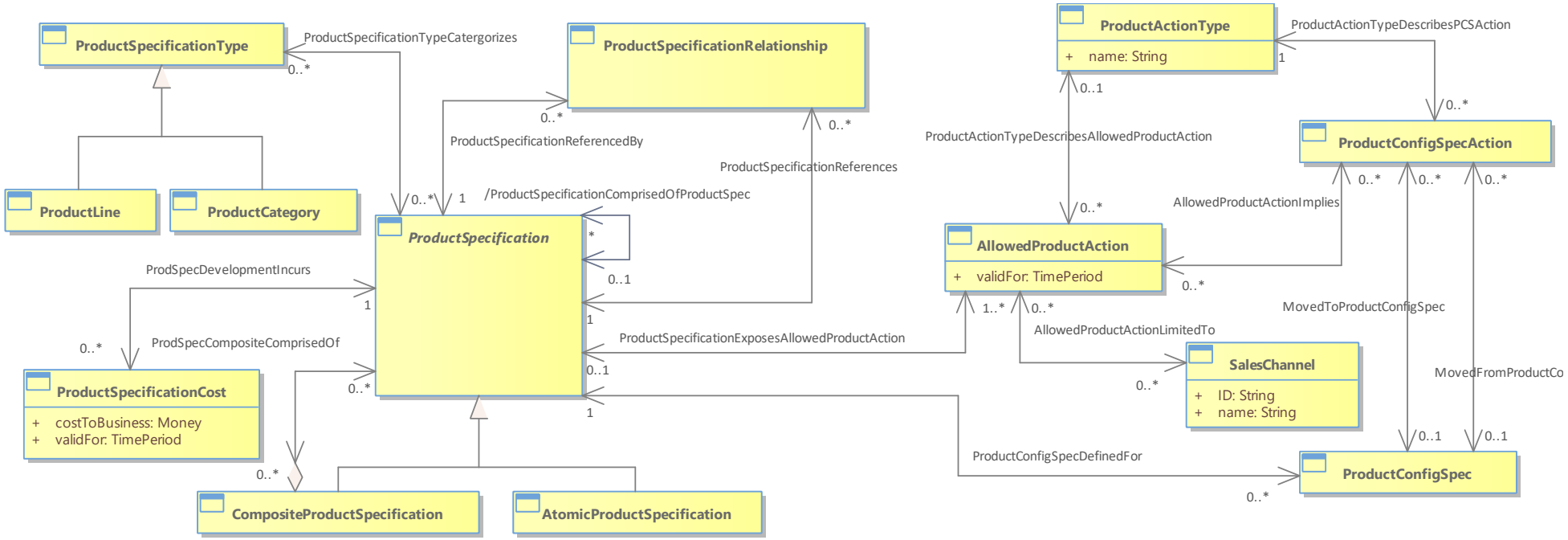

Figure Pr.03 - Product Specification - Basic Entities

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Figure Pr.03a - Product Specification - Examples

The following diagram gives examples of subclasses of AtomicProductSpecification that can be defined when it is required to specify particular relationships or attributes for specific ProductSpecification.

In SID, only the specificities of NetworkProductSpec, UsageVolumeProductSpec and LoyaltyProductSpecification have been detailed for the moment.

The NetworkProductSpec defines intangible products. It represents the common behavior and description of an installed network product that will be provisioned in the network and that enables usages, like a Mobile Line.

A UsageVolumeProductSpec defines buckets of usage, such as 2 hours national calls. A UsageVolumeProductSpec requires one or many NetworkProductSpecs from which Usages will debit the bucket.

A RatePlanProductSpec defines criteria to be used to gain special usage tariffs like the period (Ex: evenings and week-ends) or phone number (Ex: friends and family). The ProductOfferingSpecification packaging a RatePlanProductSpec defines the Usages' ProductOfferingPrice conditioned by the criteria. A RatePlanProductSpec requires a NetworkProductSpecs.

The GoodsProductSpec corresponds to the specification of material goods.

The LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked to identify the actions to apply. A dedicated guide book describes all what is related to Loyalty.

The FieldOperationProductSpec is a ProductSpecification corresponding to the intervention of an employee (from the marketing operator or one of his Suppliers) such as "Internet installation at home", "Internet training" … It requires an Appointment.

A DirectoryProductSpec is a ProductSpecification corresponding to publication of information (like associated subscriber name and address) of a telecommunication line with its phone number in a telephone directory.

A ShippingProductSpec defines the possible choices of configuration for the shipment of one or several GoodsProductSpecs like the place to ship, the delay…

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A ServiceLevelProductSpec is a type of ProductSpecification that defines KPI that have to be evaluated and the threshold that must be respected. These KPIs are described by a ServiceLevelSpec. Note: Service Level Spec ABE has to be updated according to Metric ABE.

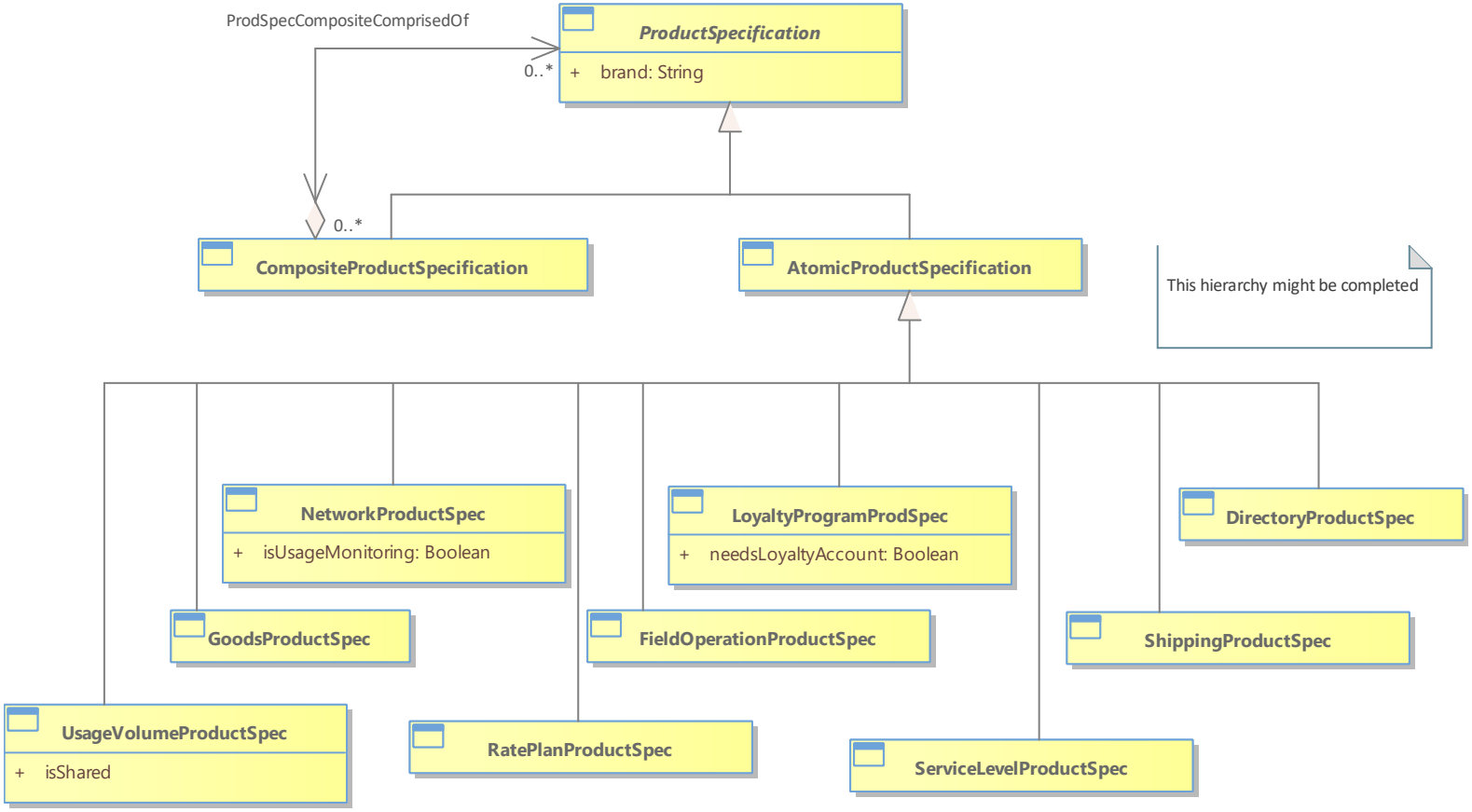

Figure Pr.03a - Product Specification - Examples

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Illustration Figure Pr.03a-I01 - Mobile Line and related ProductSpecifications

This instantiation of the model intends to illustrate the use of ProductSpecification subclasses and ProductSpecificationRelationships.

A Mobile line ProductSpecification enables ProductUsageSpecs (like Voice, SMS, MMS…), therefore this is a NetworkProductSpec.

A Mobile line needs a SIM card to function. A Service Provider SIM Card is a tangible good, therefore this is a GoodsProductSpec.

An evening & week-end product defines the usages' tariff to be applied to the usage of the Mobile line, therefore this is a RatePlanProductSpec.

A SMS package defines a predefined quantity of product usage; therefore, this is a FixedPricePackageProdSpec.

A TV on Mobile ProductSpecification enables ProductUsageSpecs (like watch channel…), therefore this is a NetworkProductSpec.

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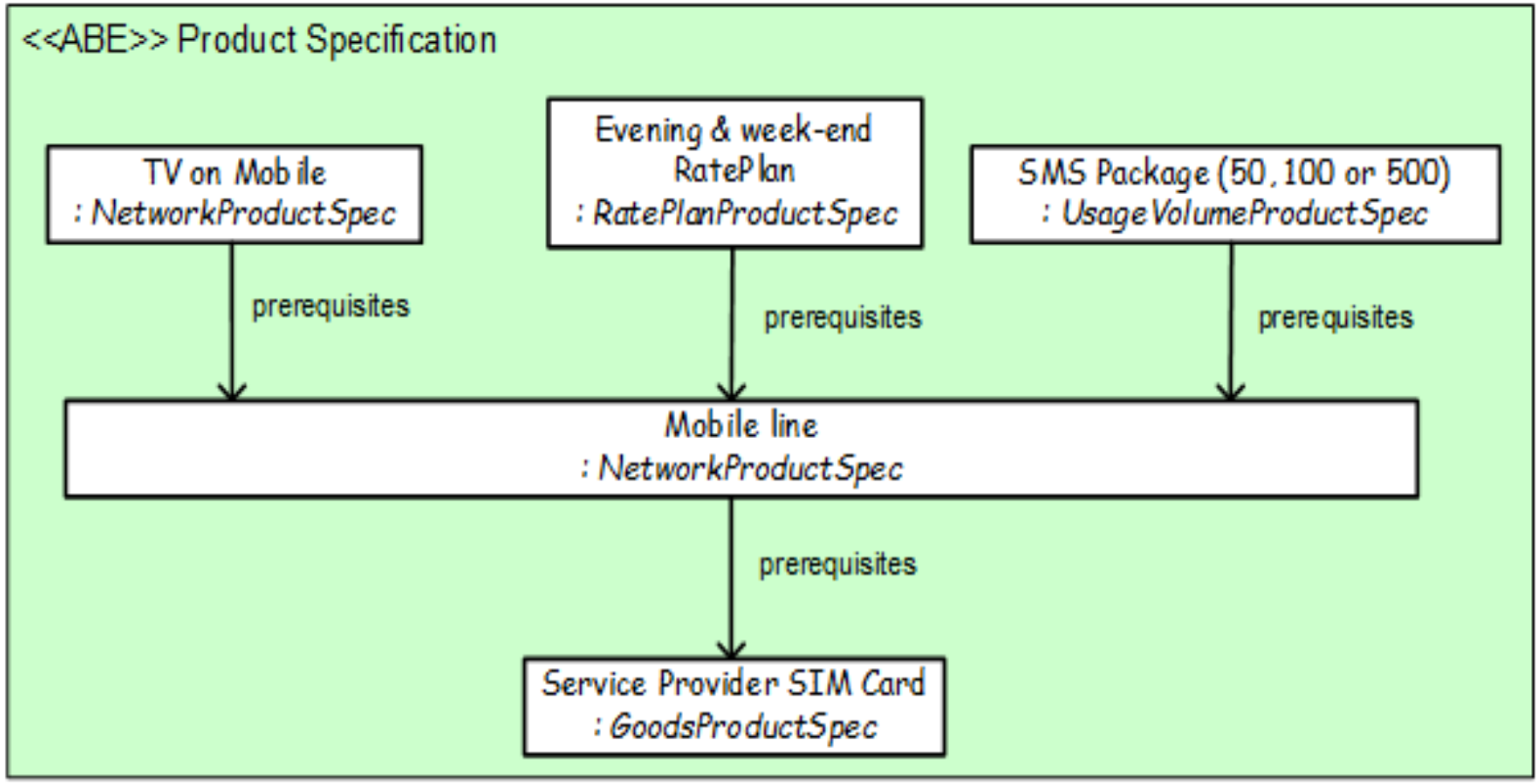

Illustration Figure Pr.03a-I01 - Mobile Line and related ProductSpecifications

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Figure Pr.04 - Goods Product Spec and Shipping Product Spec

The GoodsProductSpec corresponds to the specification of material goods.

A ShippingProductSpec defines the possible choices of configuration for the shipment of one or several GoodsProductSpecs like the place to ship (Ex: at home or pick-up store), the delivery schedule (Ex: in 1 hour, in 48h)…

A ShippingProductSpecification pre-requires at least one GoodsProductSpecs.

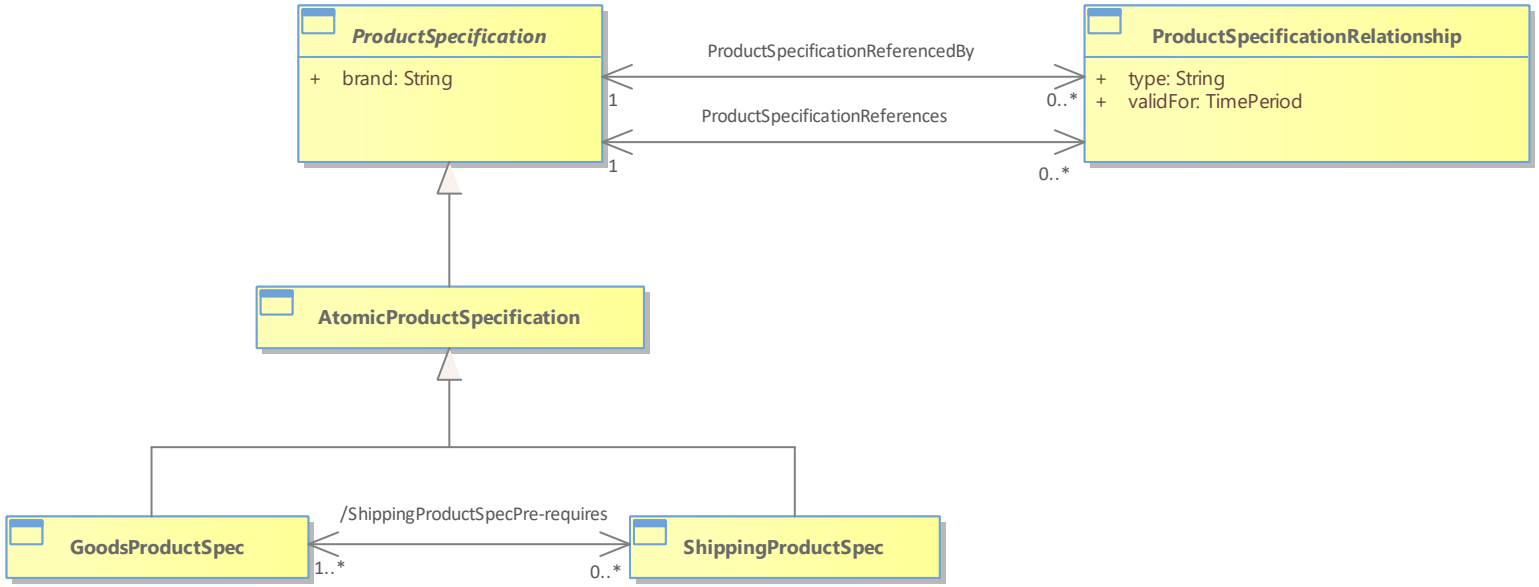

Figure Pr.04 - Goods Product Spec and Shipping Product Spec

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Illustration Figure Pr.04-I01 - Goods Product Spec and Shipping Product Spec

Let's consider a ShippingProductSpec that proposes the following choices:

- Shipment Type: At Home, Pick-up & Go location, By-Hand by technician and By-Hand in Store
- Delivery Time: Immediately, With Installation, 24h and 48h

In addition, a characteristic is needed to describe the shipping address.

But all combinations of configuration are not available:

- for At Home Shipment Type only Delivery Time 24h and 48h are available
- for Pick-Up & Go location Shipment Type only Delivery Time 24h and 48h are available
- for By-Hand by technician Shipment Type only Delivery Time With Installation is available
- for By-Hand in Store Shipment Type only Delivery Time Immediately is available

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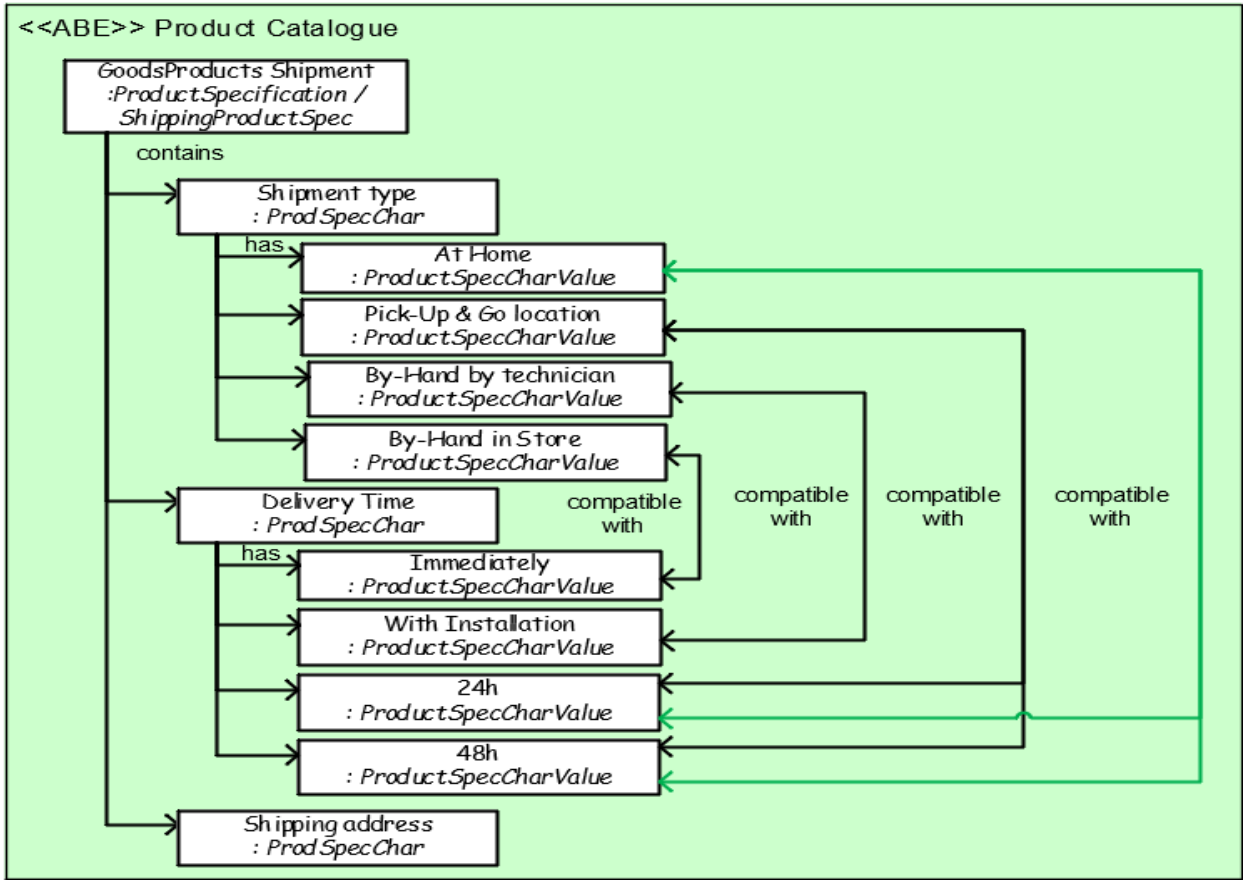

Illustration Figure Pr.04-I01 - Goods Product Spec and Shipping Product Spec

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Figure Pr.40 - Usage Volume Product Specification

A UsageVolumeProductSpec is a type of AtpmicProductSpecification.

It represents a predefined quantity of product usage (aka bucket), like a 2 hours national calls, often known as an Allowance.

It could be either a quantity or an amount in a currency (i.e. It could represent a fixed number of SMS, MMS, minutes of calls, Quantity of data as well as a specific amount in € …).

A UsageVolumeProductSpec requires one or many NetworkProductSpecs from which Usages will debit the bucket.

A UsageVolumeProductSpec also defines UsageVolumeChargingRules specifying how to decrement the bucket.

A NetworkProductSpec might be monitored or not. The wording monitored means that the user can't realise a type of usage if he hasn't any FixedVolumePackageProd for this type of usage (i.e. without an empty balance). In this case, the attribute isUsageMonitoring is TRUE.

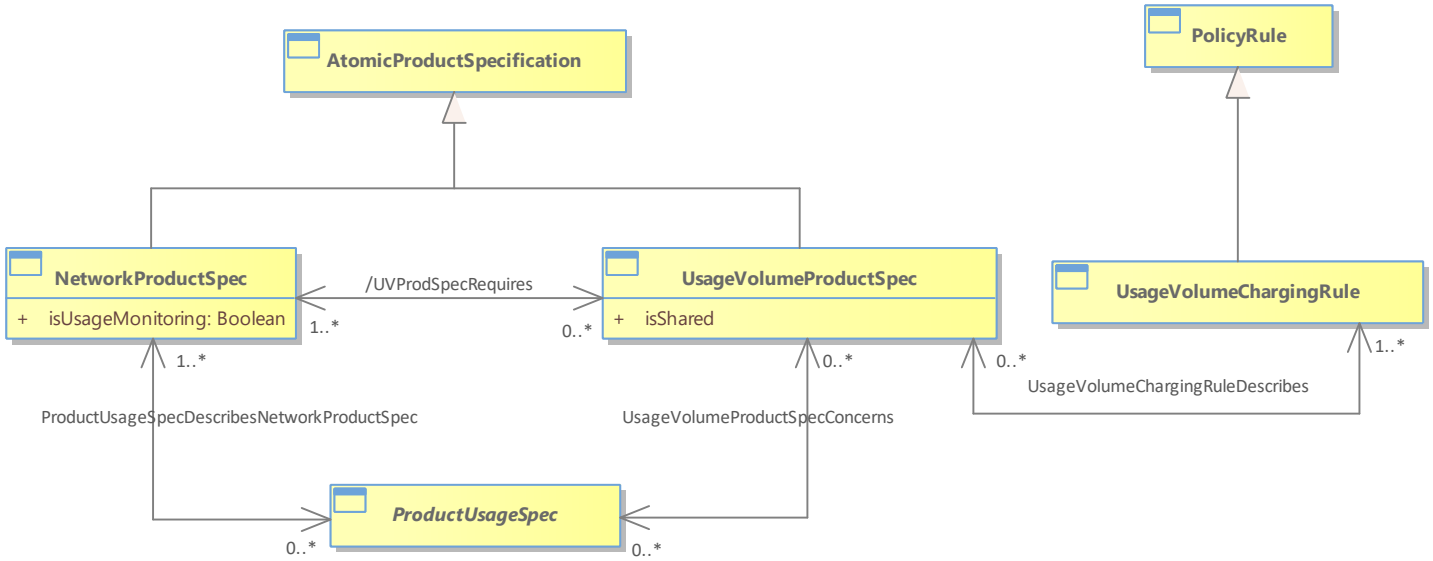

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Figure Pr.40-I01 – Illustration National Voice Usage Volume Product Specification

In this example, the ProductOfferingSpecifications proposed by the CSP corresponds to:

- A Basic Mobile Line enabling National Voice usages for a price of 0.1$ per minute
- A National Voice Usage Volume

The National Voice Usage Volume

- Has two possible volumes (2 hours or 10 hours) with respectively 10$ and 30$ prices
- The volume is credited each month
- Each ProductUsage's duration is decremented from the current balance.

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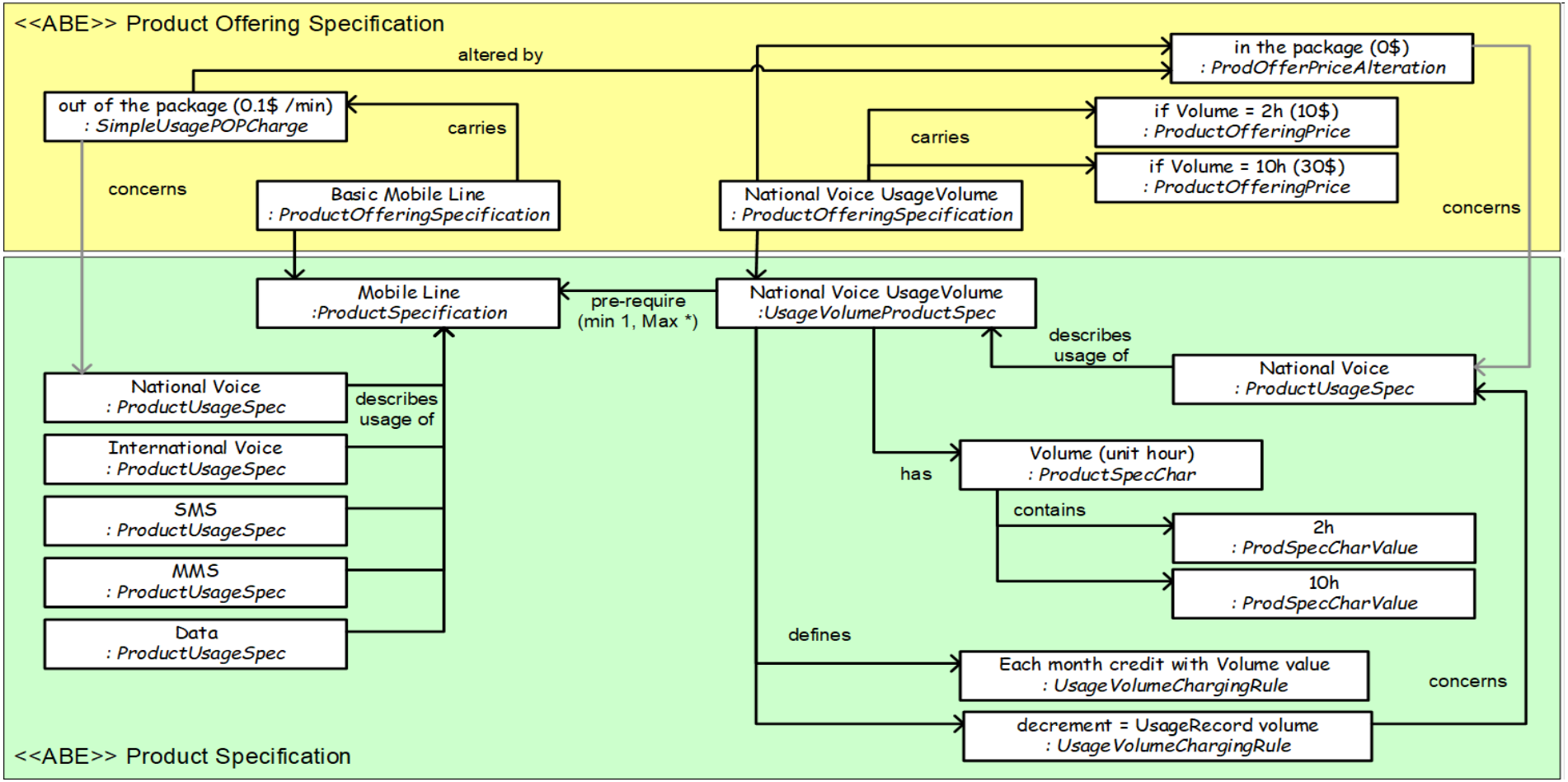

Figure Pr.40-I01 – Illustration National Voice Usage Volume Product Specification

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Figure Pr.41 - UsageVolume and Network Product

A ProductOfferingSpecification packaging a UsageVolumeProductSpec may also define ProductOfferingPriceAlterations. These ProductOfferingPriceAlterations overwrite the regular Usage charges (SimpleUsageProdOfferPriceCharge or TariffUsageProdOfferPriceCharge) carried by the ProductOfferingSpecification that packages the NetworkProductSpec.

If ProductOfferingPriceAlterations are defined out of the package (i.e. when the bucket is empty), it means that the usage is not monitored (i.e. that the user can still use the network product even when the bucket is empty). Else, it corresponds to a blocked package (aka prepaid package) and the user can't use any more its network product when the bucket is empty.

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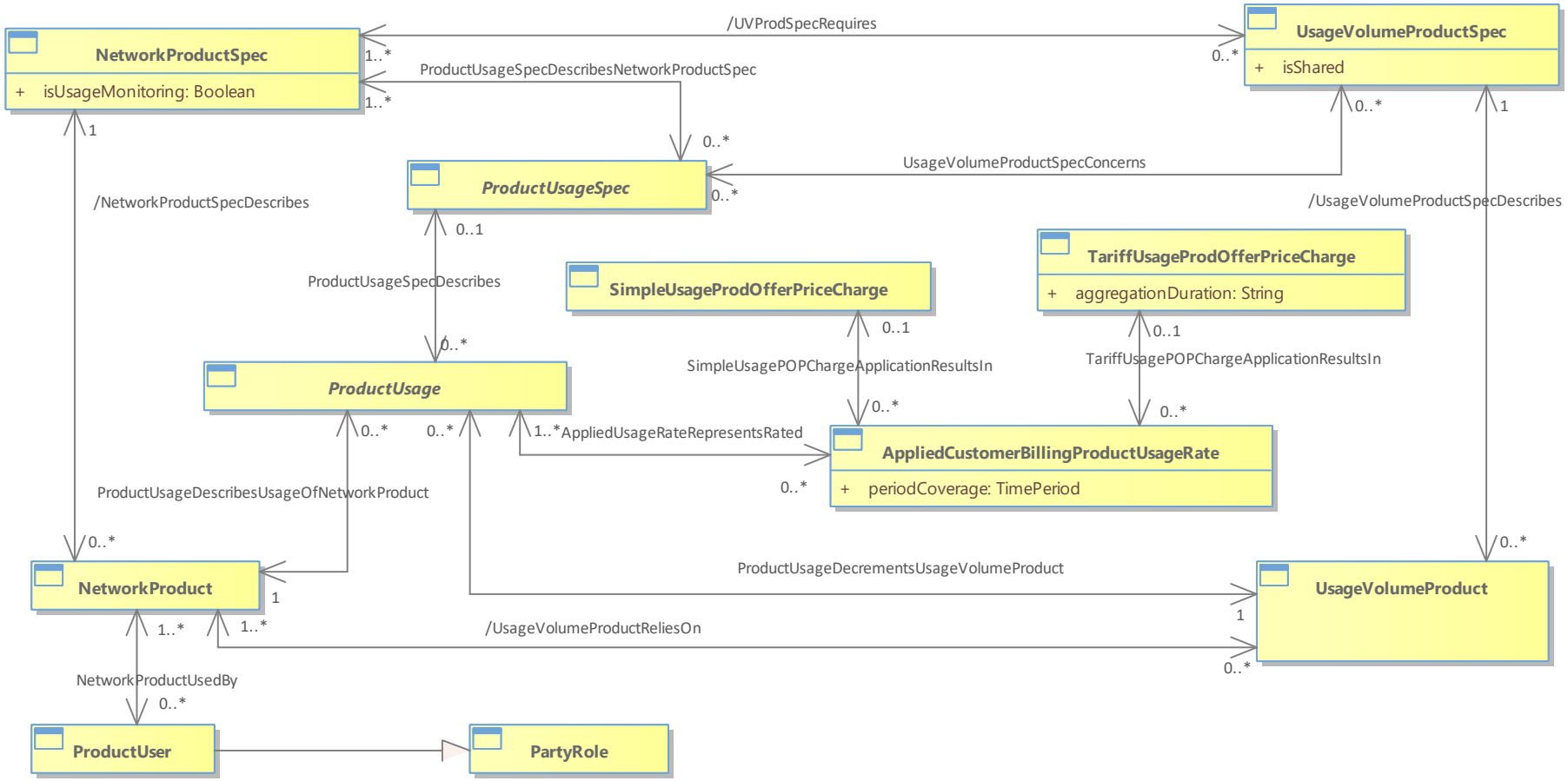

Figure Pr.41 - UsageVolume and Network Product

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Figure Pr.41-I01 – Illustration UsageVolume and Network Product

Kate has bought a Mobile Line and a 2 hours National Voice Usage Volume.

Her phone number is 0601010101.

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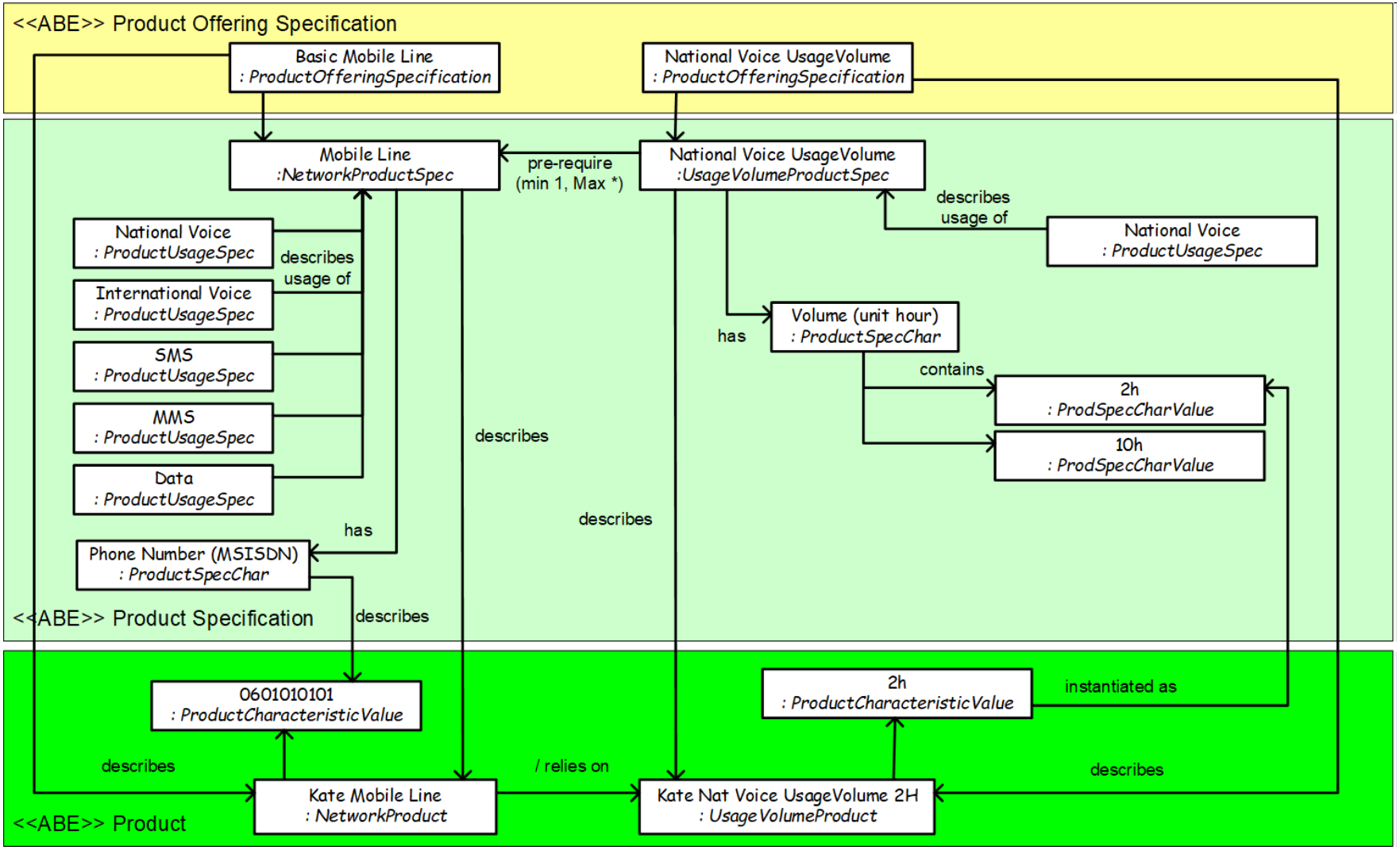

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Figure Pr.42 - UsageVolume in detail

A ProductUsage might be

- Charged by one or several UsageVolumeBalanceDebit that decrements the corresponding balance
- And might be rated by one or several ProductOfferingPrices that will be charged to an invoice.

A UsageVolumeProduct is an installed UsageVolumeProductSpec.

Each UsageVolumeProduct has at least one balance (UsageVolumeBalance) valid at a given point. A UsageVolumeBalance is initialized when a UsageVolumeChargingRule applies a credit for a new valid period.

Each balance might be decremented by one or several UsageVolumeBalanceDebits and credited by one or several UsageVolumeBalanceCredits.

A UsageVolumeBalanceDebit results from the charging of a ProductUsage.

A ProductUsageCredit results from the application of the rules specified on the UsageVolumeProductSpec through its characteristics (frequency of the credit, quantity of the credit).

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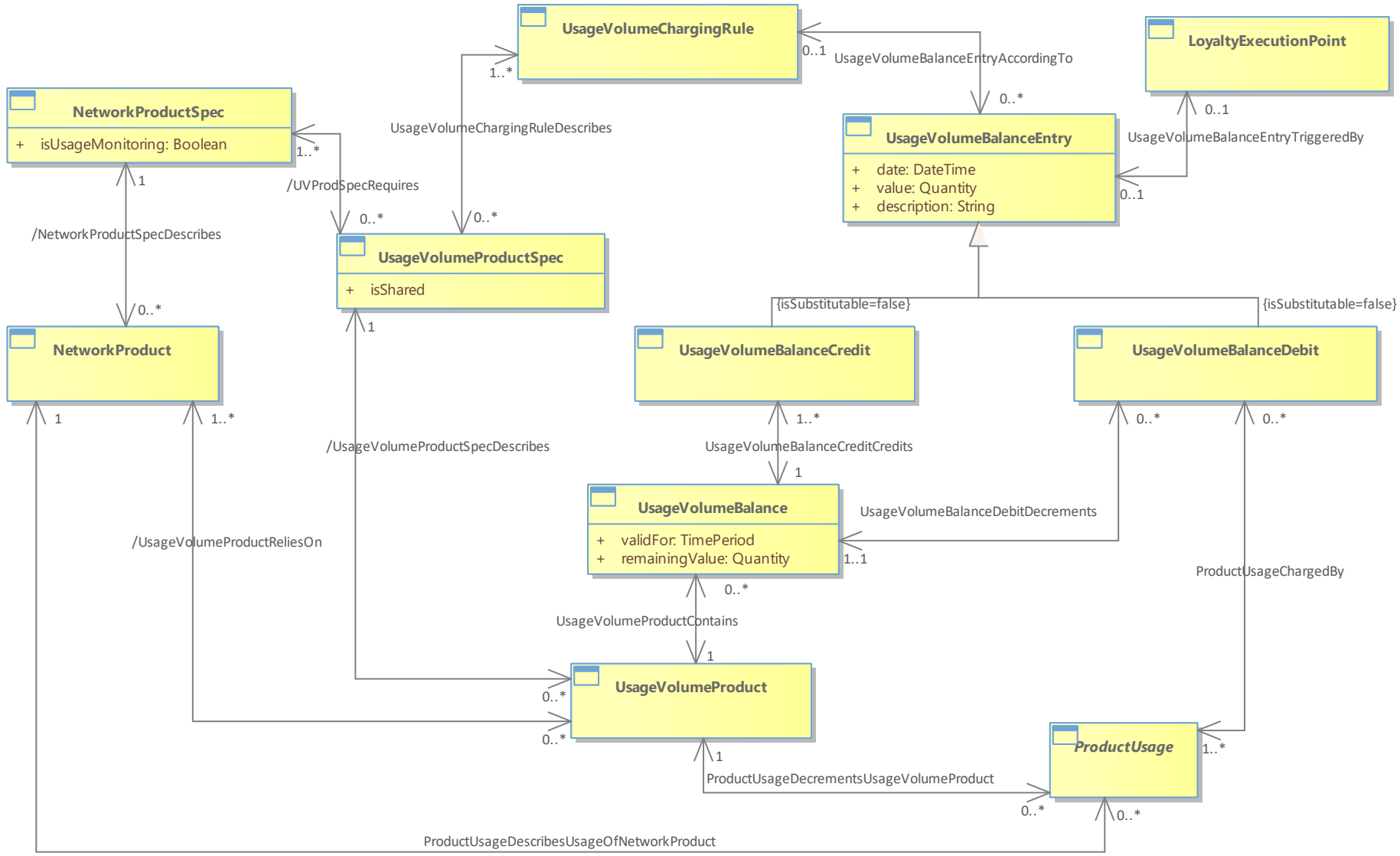

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Illustration Figure Pr.42 – I01 - National Voice UsageVolume Balance

Each Usage Balance has a two months validity period and each credit triggers the creation of a new balance.

Kate has a National Voice Usage Volume product with a 2 hours volume..

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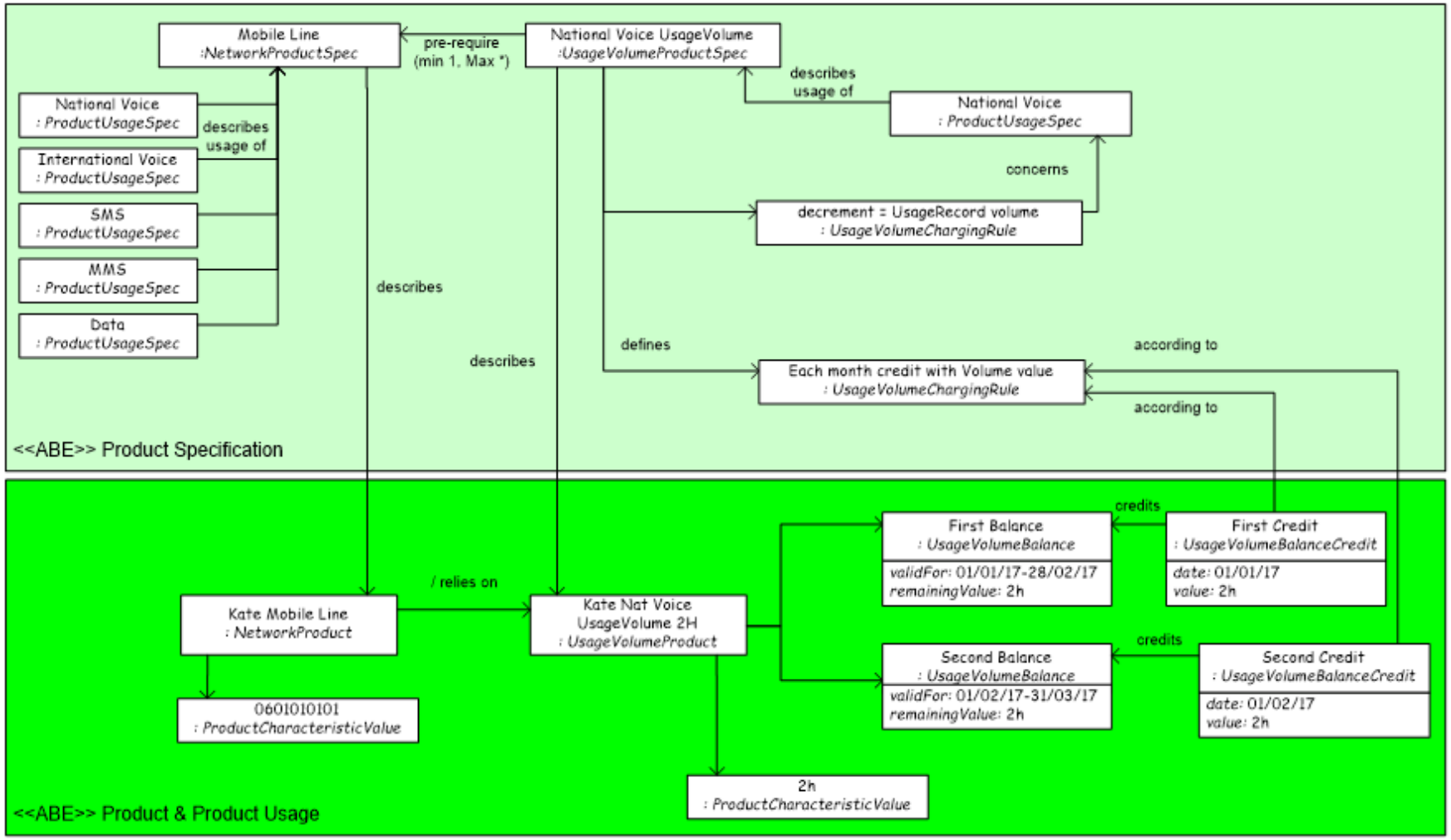

Illustration Pr.42 I01

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Illustration Figure Pr.42 – I02 - National Voice UsageVolume Debits

Kate had two calls:

- A long call of two hours, then the first balance is decremented. After applying the usage volume charging rule, the remaining amount is 0.
- A second call of 5 minutes. After applying the usage volume charging rule, the remaining amount is 1h55.

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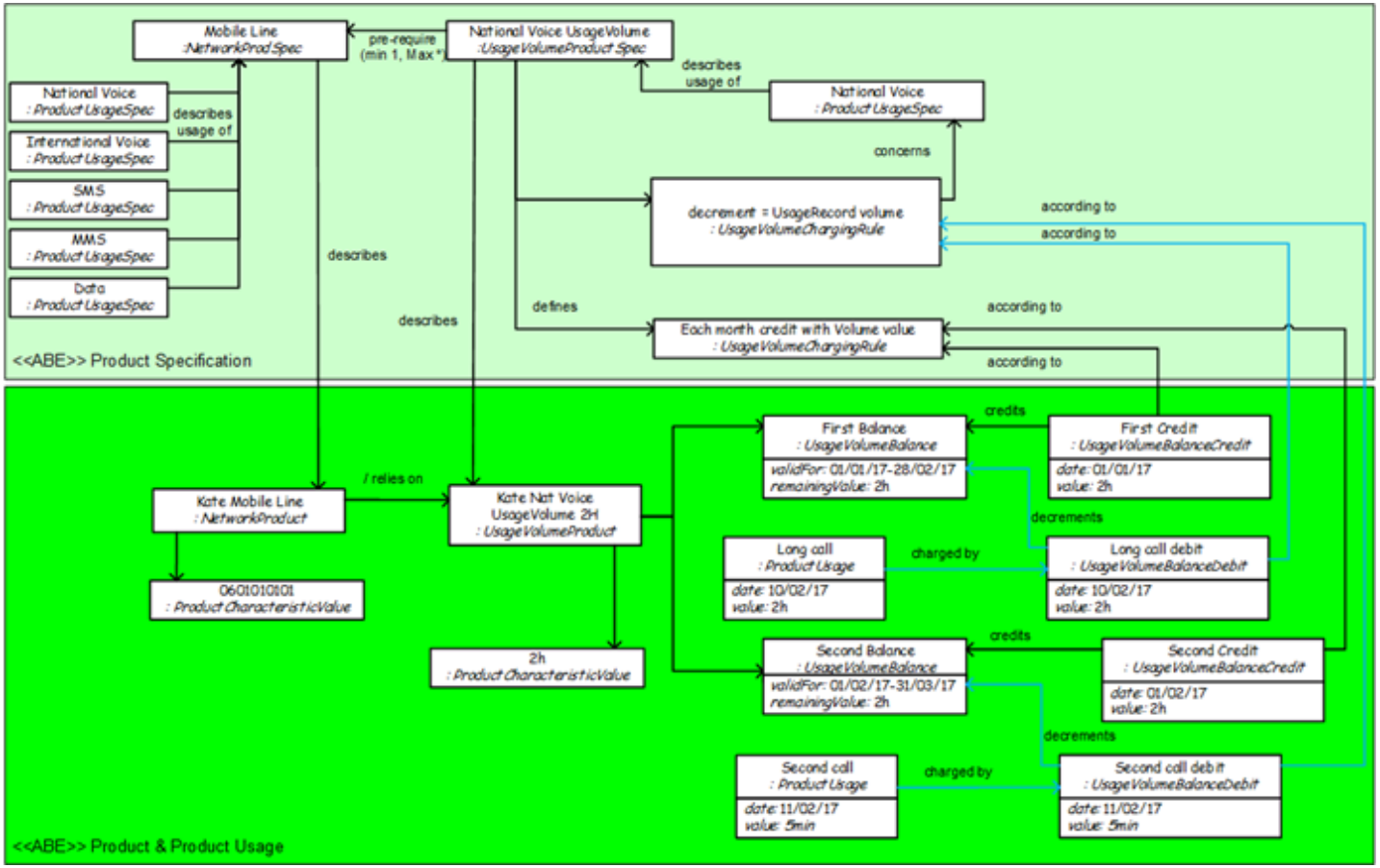

Illustration Pr.42 I02

Illustration Figure Pr.42 – I02 - National Voice UsageVolume Debits

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Figure Pr.43 - Usage Consumption Summary

A ConsumptionSummary give measures about the usage quantity consumed.

It might concern UsageVolumeBalances and / or UsageVolumeProducts and / or NetworkProducts and / or ProductUsers.

A ConsumptionSummary aggregates UsageVolumeBalanceCredits and / or UsageVolumeBalanceDebits and / or AppliedCustomerBillingProductUsageRates.

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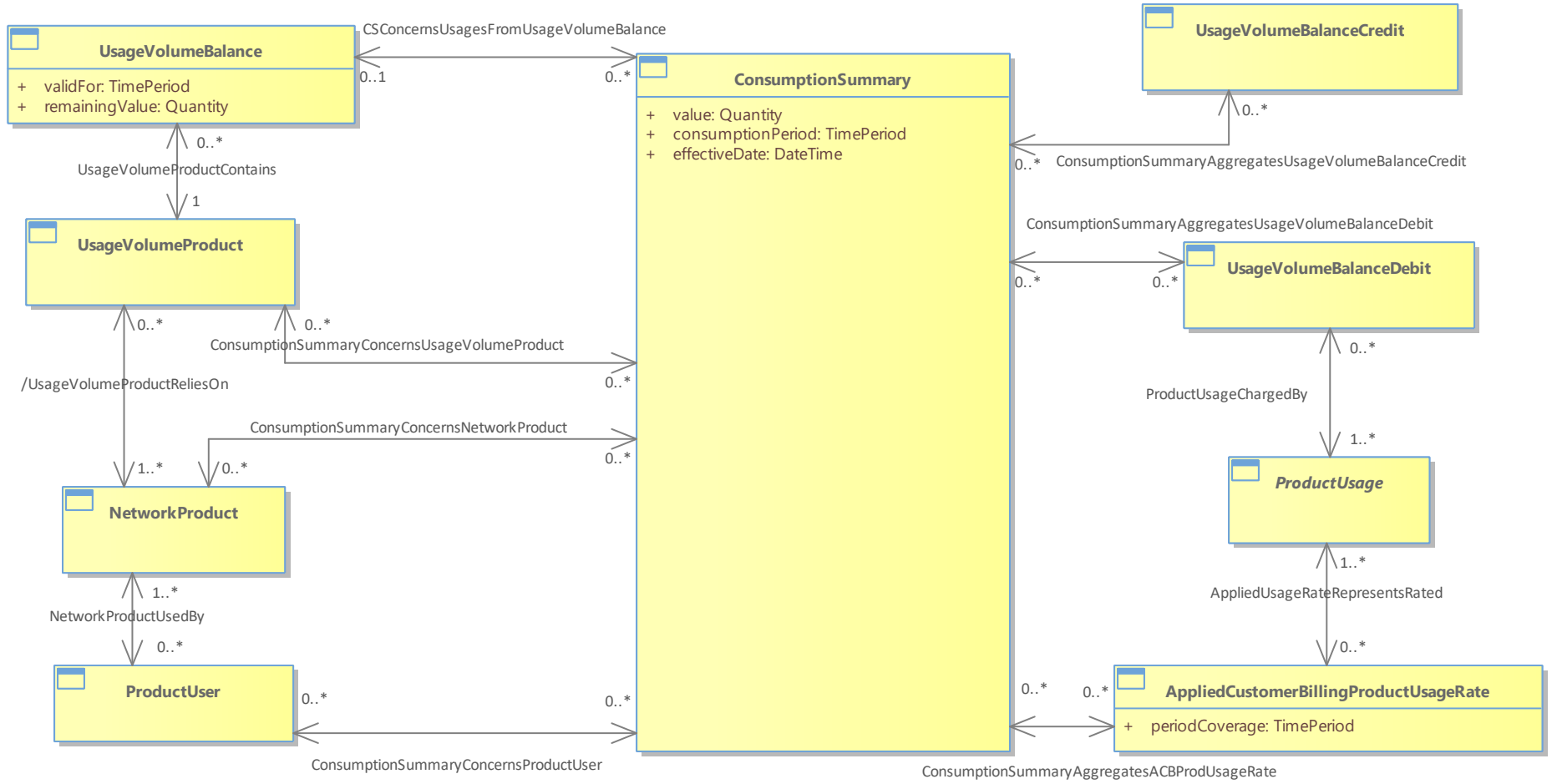

Figure Pr.43 - Usage Consumption Summary

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Figure Pr.44 - Product Specification - Characteristics and Versions

ProductSpecifications also have a variety of characteristics, such as a printer's dimensions, paper capacity, print speed, color, and so forth, and possible values associated with the characteristics. The SID uses ProductSpecification Characteristic and ProductSpecCharacteristicValue to represent the properties for a ProductSpecification [Fowler-Properties] as shown in Figure Pr.4 on the following page. Additional information regarding this pattern can be found in Common Domain guide book section Root Business Entities ABE.

Characteristics can be grouped into four different types:

- If a characteristic has a discrete value (e.g. color or business coding), it may be derived from a range of allowable values;
- If a characteristic describes the product within a parameter range (e.g. dimension), it may be defined with a 'high value' and a 'low value' [Fowler-Range] ;
- If a characteristic value is 'derived', it will use other parameters in a formula. N.B. There are business rules implied here that require more work in a future phase of the SID Product definition.
- If a characteristic has no predefined values and unconstrained values (no business rules and no ranges/allowed values)

Characteristics can also be bundled together into packages by using the ProductSpecCharRelationship. For example, a number of electrical characteristics can be grouped together using an "electrical properties" characteristic that represents a composite of the detailed properties, such as power requirements, plug requirements, and so forth. This entity can also be used for a number of other relationships between characteristics, such as mutually exclusive, inclusive, and so forth.

CharacteristicValues can reference other CharacteristicValues.

As ProductSpecifications are upgraded, and as the market and technology move on, they are subject to version control. If there is a significant change of capabilities, a new ProductSpecification is created.

When a product is created based on a ProductSpecification (see Product Offering below), the party may also have the option to choose from a set of properties values, such as color, or may be able to choose one from a number of product properties. Where a ProductSpecificationCharacteristic is interchangeable with others in this way, it is modeled as a ConfigurableProductSpecificationCharacteristic.

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Product Characteristic Example – Email ProductSpecification

ProductSpecCharacteristics and their related ProductSpecCharacteristicValues can be used to support capabilities provided by ProductSpecifications. For example, an Email may have a number of features, such as Spam protection, storage, message size, virus scanning and cleaning, mail forwarding, archiving, and so forth.

One way this type of product can be introduced into a model is by sub-classing the Product entity and including each feature as an attribute of an EmailProduct subclass. But this means that each time a new type of product is introduced the information architecture and most likely the application that supports the Product ABE would have to be changed. This is sometimes viewed as a static extension to information architecture.

A more dynamic approach is to consider these features as instances of a ProductSpecCharacteristic. In this example each feature has a value associated with it. The value associate with Spam protection is “SpamBlock”; the value associated with storage is “1GB”; the value associated with message size is “10BM”; the value associated with virus scanning and cleaning is “included”; the value associated with mail forwarding is “included”; the value associated with archiving is “included. Note that note all characteristics have to have values at specification time. For example, user ID are capabilities, but the value is assigned when an instance of a Product is created.

The example will be continued in the Product section of this addendum.

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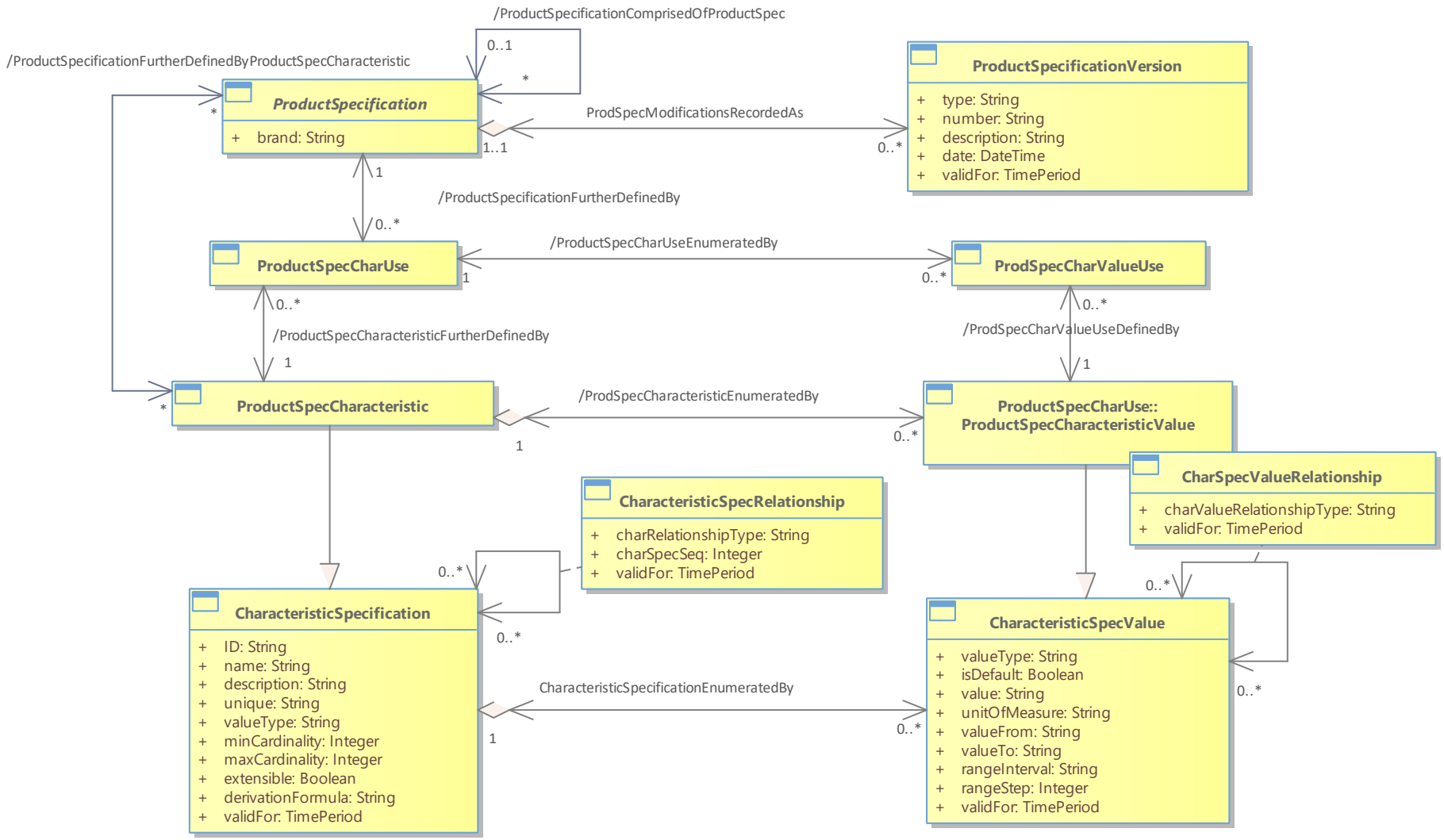

Figure Pr.44 - Product Specification - Characteristics and Versions

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Illustration Figure Pr.44-I01 - ProductSpec characteristics related to Mobile Line and other ProductSpecifications

The phone number (aka MSISDN) of a Mobile Line corresponds to a ProductSpecCharacteristic without any predefined value.

The SMS Package has a Volume with predefined valued (50, 100, 500). The value will be chosen by the customer during the order handling. The configuration chosen will impact the delivery and will be evaluated in the rules that determine the price to apply.

The Evening & Week-end Rate Plan has a Characteristic Period with two possible values “Night and week-ends” and “Week days”. The configuration chosen will be evaluated in the rules that determine the usage price to apply.

The SIM Card has a type characteristic that is used in the delivery process to provide a physical resource corresponding to this choice.

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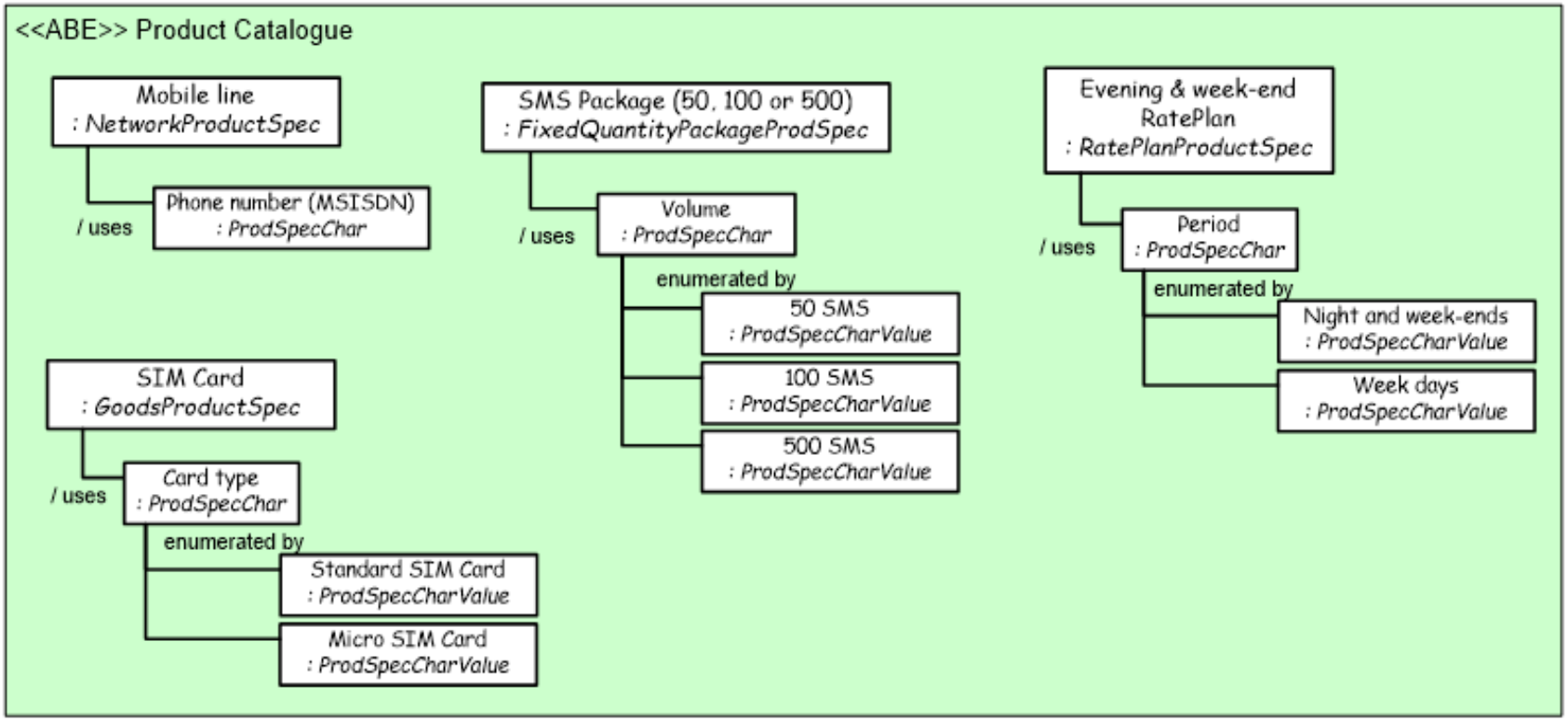

Illustration Figure Pr.44-I01 - ProductSpec characteristics related to Mobile Line and other ProductSpecifications

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Figure Pr.44a - Product and Service Characteristic Associations

There are cases when an instance of a ProductSpecCharacteristic and/or a ProductSpecCharacteristicValue must be translated to a corresponding ServiceSpecCharacteristic and/or a ServiceSpecCharacteristicValue. For example, there may be a ProductSpecCharacteristic that represents the level of service provided as defined by associated Platinum, Gold, and Sliver instances of ProductSpecCharacteristicValues. Each of these values may be translated into specific upload and download bit rates as defined by ServiceSpecCharacteristicValues.

The following Figure shows the associations between characteristics in the Product and Service domains that enable this translation. Similar translations can be made between characteristics in the Product, Service, and Resource domains.

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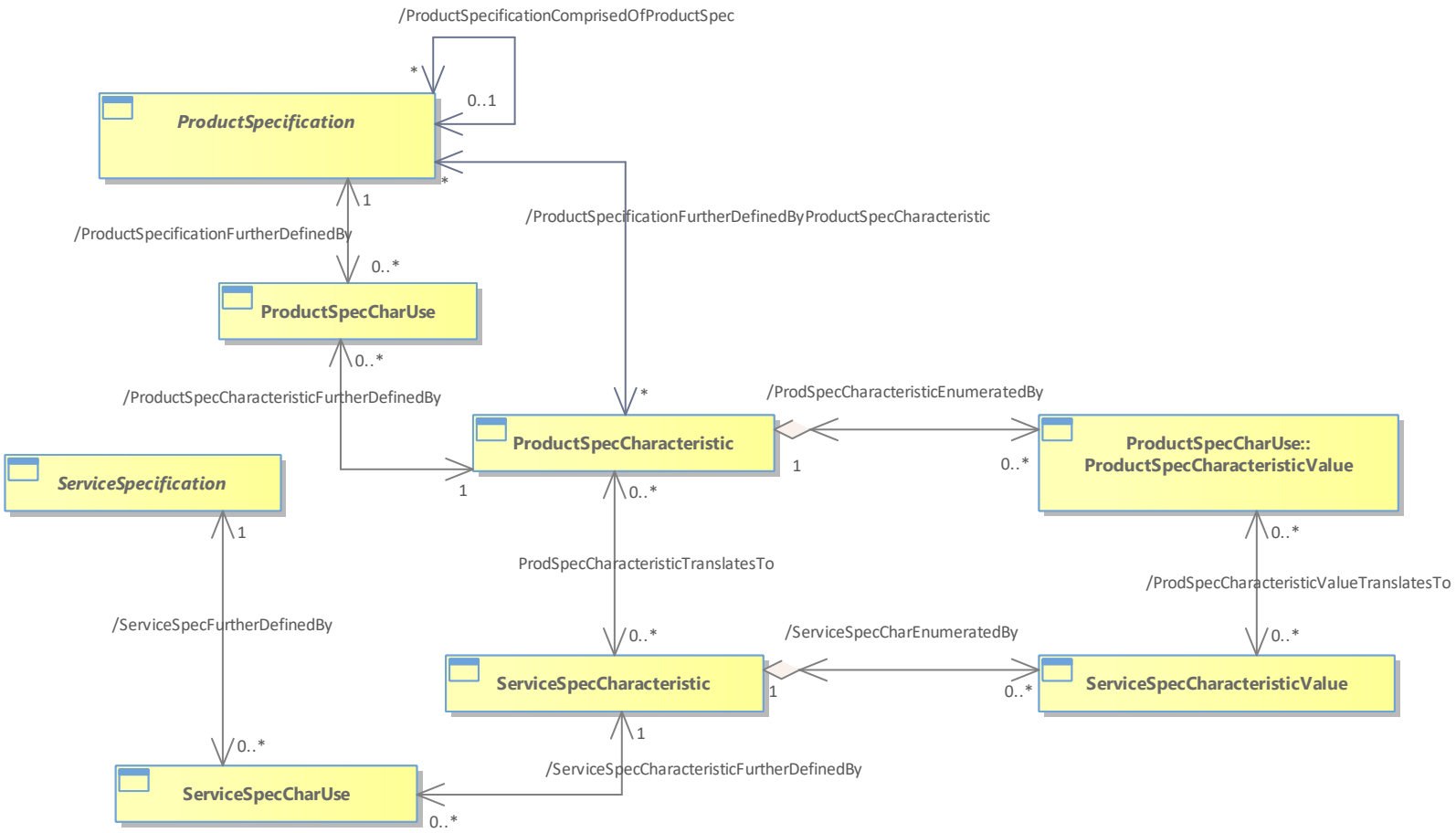

Figure Pr.44a - Product and Service Characteristic Associations

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Illustration Figure Pr.44a – I01 - ProductSpec characteristics translated to ServiceSpec characteristics

In a telecommunication operator's know-how, a Mailbox may have a size of 10Mo, 20Mo or 30 Mo.

But marketing may decide to market only Mailbox with a size of 10Mo and 20Mo and use more marketing wording (Basic and High Volume).

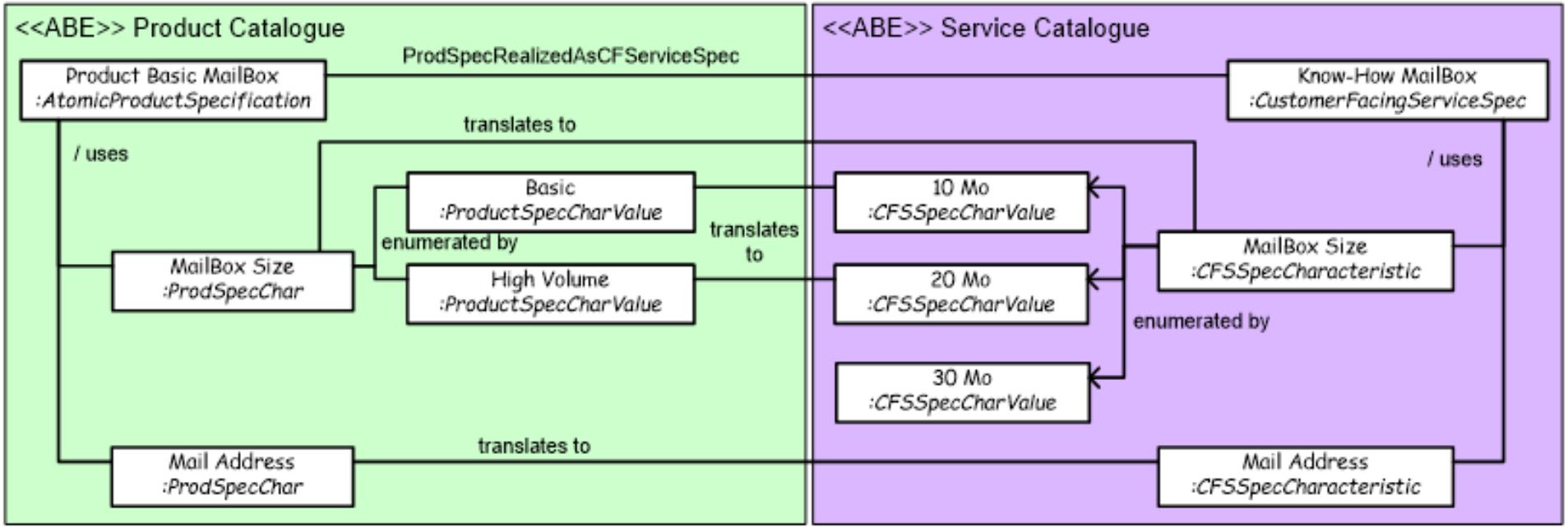

Illustration Figure Pr.44a – I01 - ProductSpec characteristics translated to ServiceSpec characteristics

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Illustration Figure Pr.44a – I02 - ProductSpec characteristics values uses a sub-set of CFSSpec characteristics values

And based on the same CFSSpec, many ProductSpecification might be built up to being able to sell them in different ProductOffering with different possible configurations.

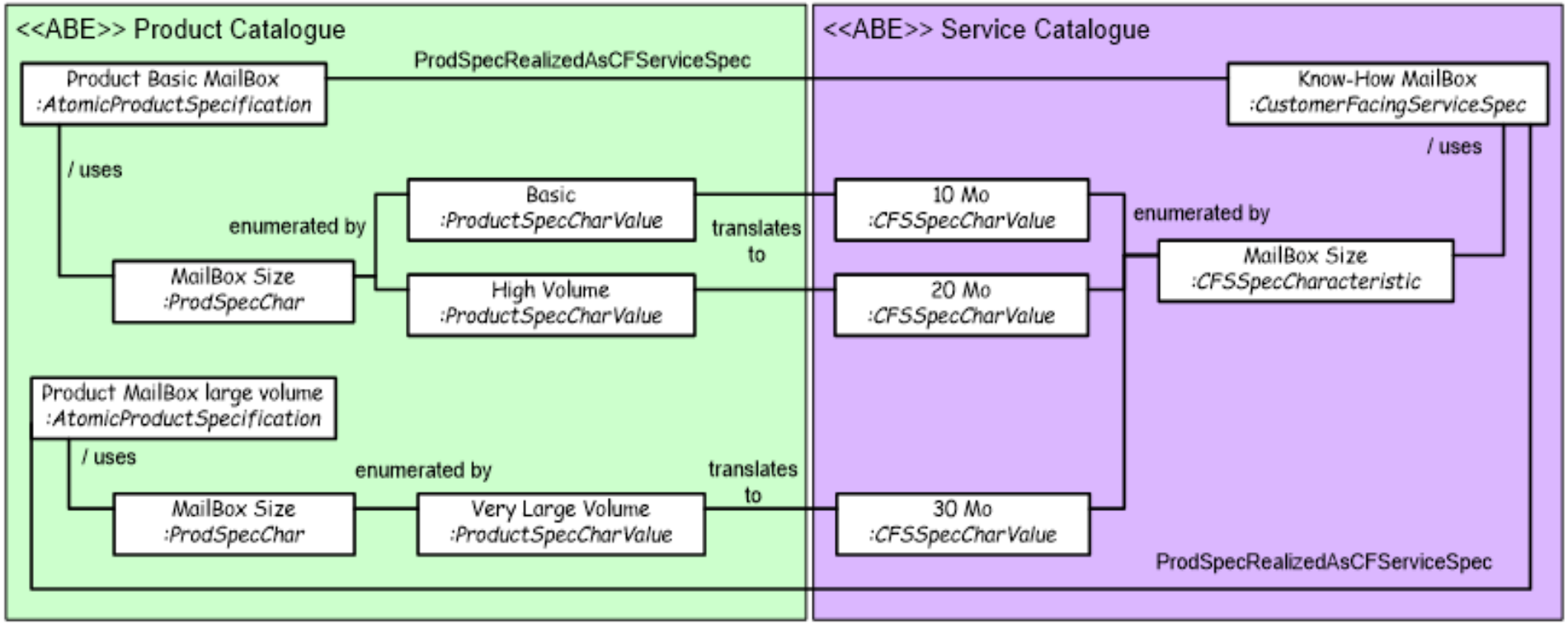

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Figure Pr.44b - Product and Resource Characteristic Associations

There are cases when an instance of a ProductSpecCharacteristic and/or a ProductSpecCharacteristicValue must be translated to a corresponding ResourceCharacteristic and/or a ResourceSpecCharacteristicValue. For example, there may be a ProductSpecCharacteristic that represents the color provided as defined by associated Brilliant and Dark instances of ProductSpecCharacteristicValues. Each of these values may be translated into real colors as defined by ResourceSpecCharacteristicValues such as Withe and Black.

The following Figure shows the associations between characteristics in the Product and Resource domains that enable this translation. Similar translations can be made between characteristics in the Service and Resource domains.

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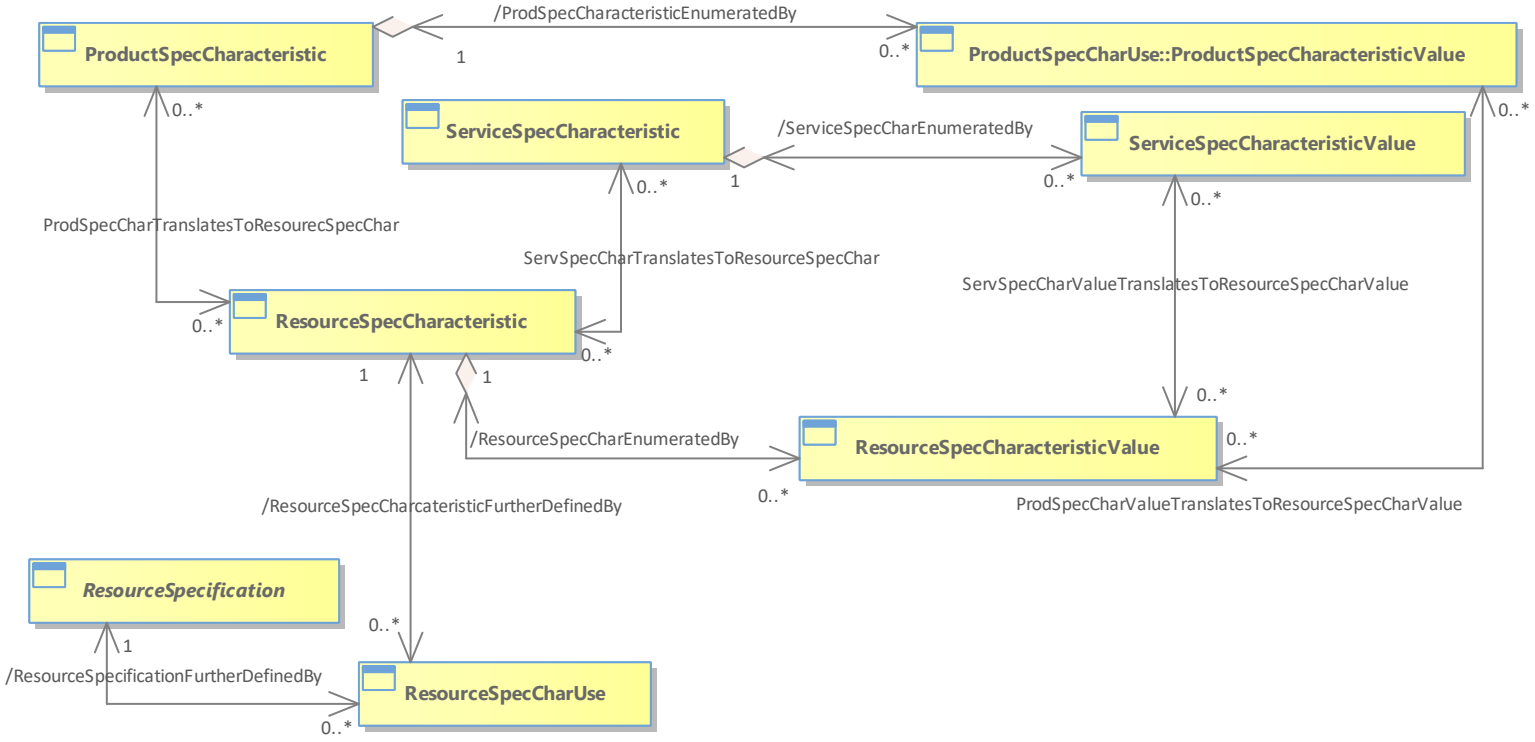

Figure Pr.44b - Product and Resource Characteristic Associations

4.2.1 Example Product Specification Entities ABE

The Example Product Specification Entities ABE contains examples of ProductSpecification subclasses that can be used.

Figure EPrS.01 - Example Product Specification Entities ABE Related Elements

Following are the business entities aggregated under the **Example Product Specification Entities** Aggregate Business Entity.

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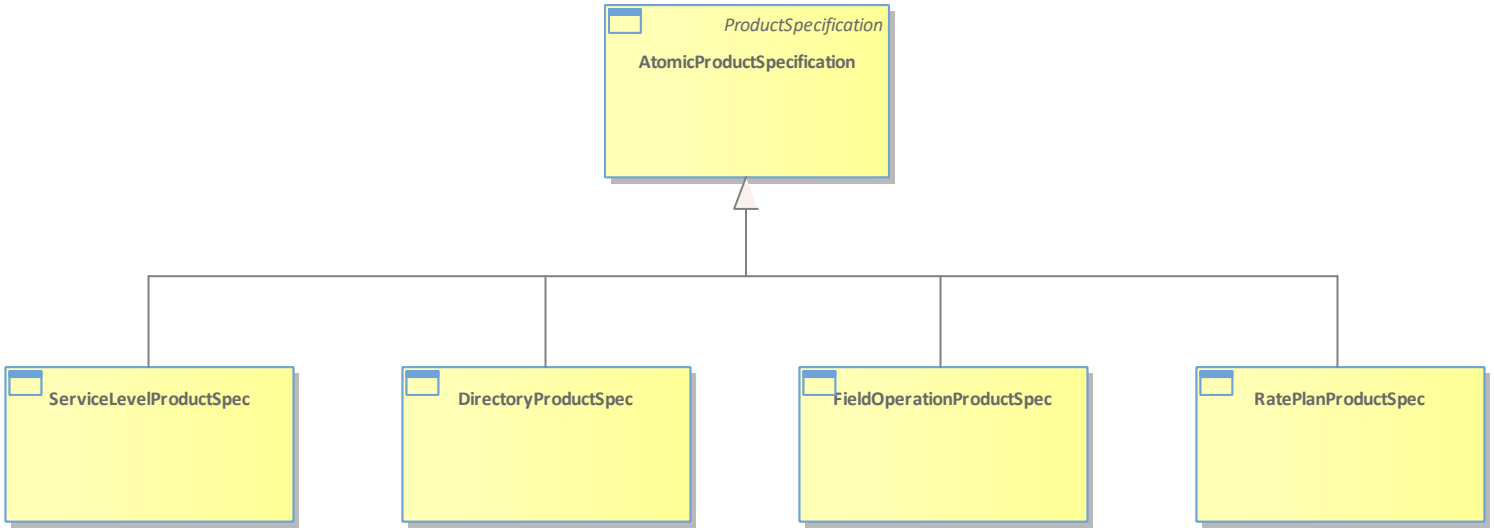

Figure EPrS.01 - Example Product Specification Entities ABE Related Elements

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4.2.2 Goods and Shipping Product Spec ABE

Goods and Shipping Product Spec ABE describe specificity of Goods and Shipping Product Spec and their relationships.

Figure GSPS.01 - Goods and Shipping Product Spec ABE Related Entities

Following are the business entities aggregated under the **Goods and Shipping Product Spec** Aggregate Business Entity.

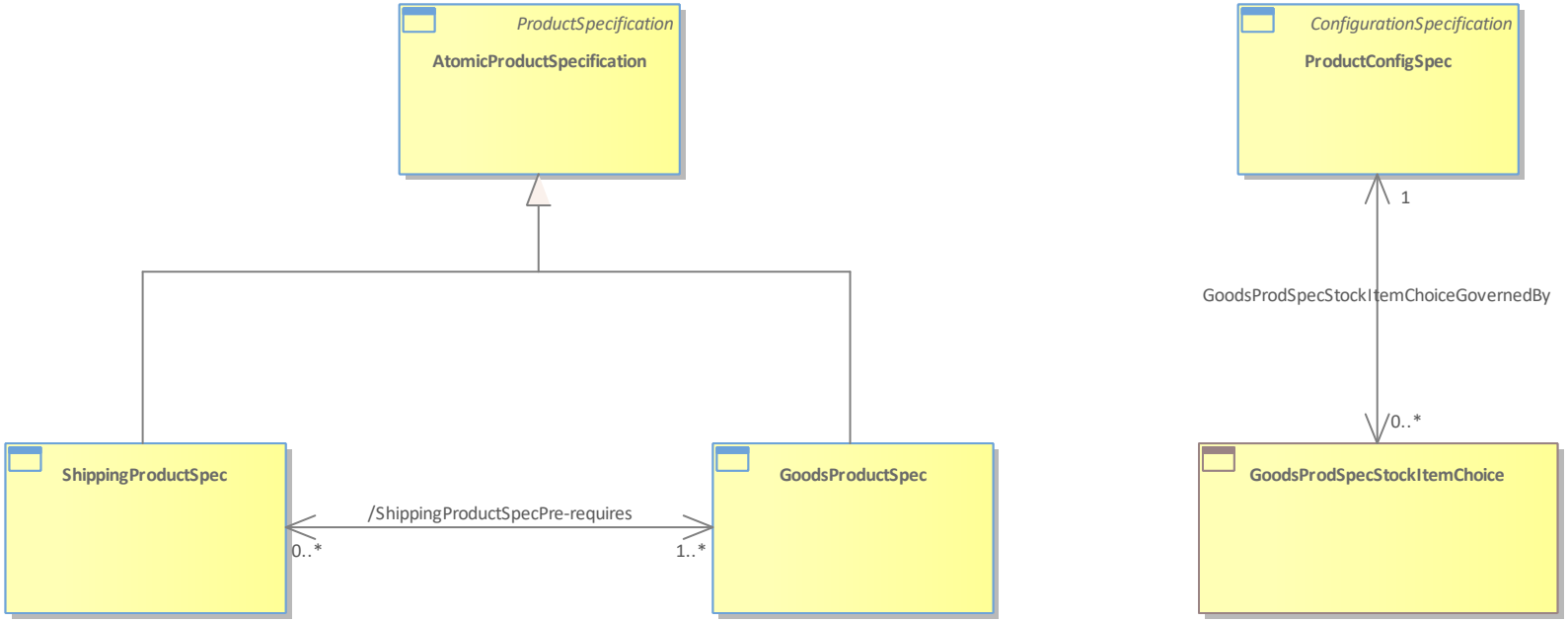

Figure GSPS.01 - Goods and Shipping Product Spec ABE Related Entities

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4.2.3 Network Product Spec ABE

The Network Product Spec ABE contains the NetworkProductSpec description and relationships.

Figure NPS.02 - Network Product Spec ABE Related Entities

Following are the business entities aggregated under the Network Product Spec Aggregate Business Entity.

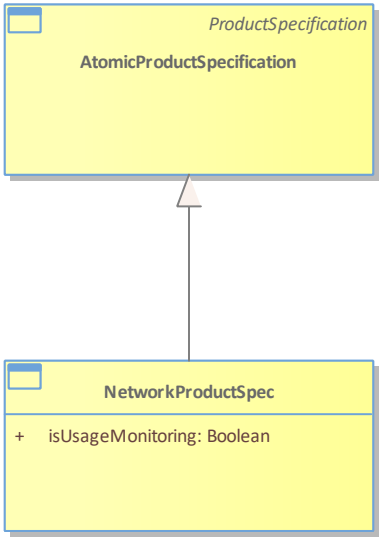

Figure NPS.02 - Network Product Spec ABE Related Entities

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4.2.4 Usage Volume Product Spec ABE

The Usage Volume Product Spec ABE contains the UsageVolumeProductSpec and all the entities specific to this type of product.

Figure UVPS.01 - Usage Volume Product Spec ABE Related Entities

Following are the business entities aggregated under the **Usage Volume Product Spec** Aggregate Business Entity.

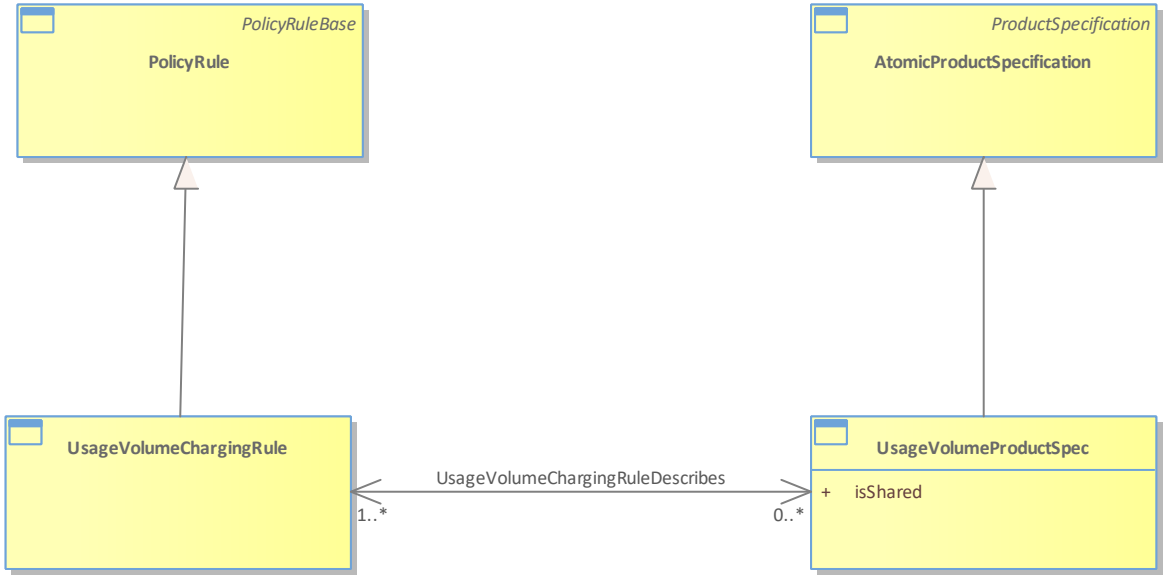

Figure UVPS.01 - Usage Volume Product Spec ABE Related Entities

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4.3 Product Offering Specification ABE

The **Product Offering Specification ABE** describes tangible and intangible goods made available for a certain price to the market in the form of product catalogs. This ABE is also responsible for targeting market segments based on the appropriate market strategy.

Figure Pr.05 - Product Offering - Basic Entities

ProductOfferingSpecifications are targeted at one or more MarketSegments based on the appropriate MarketStrategy. Enterprises create ProductOfferingSpecifications from ProductSpecifications that have additional market led details applied over a particular period of time. The ProductOfferingSpecification may also be defined by the Location(s) at which it is available (i.e. Place). Locations in the SID model are represented by the Location business entity. Customers do not need to know anything about the detailed ProductSpecification, as the enterprise tailors the ProductOfferingSpecification to their needs.

Where a business market strategy dictates, ProductOfferingSpecifications may be bundled together and re-priced as necessary. In addition, the time period for a CompositeProductOfferingSpecification will depend upon those for its component offerings.

An enterprise puts ProductOfferingSpecifications into the marketplace by developing and maintaining a catalog of ProductOfferingSpecifications across one or more (market) DistributionChannels, for example a catalog on a website and a glossy brochure have different attributes and may also contain different sets of ProductOfferingSpecifications. The ProductOfferingSpecifications set out in the ProductCatalog are ProductSpecifications with additional details that enable a contract to be struck for their sale/purchase. This includes not only the ProductCharacteristics but also other product-dependent details, such as SLA parameters, invoicing and shipping details.

There are typically a number of ProductOfferingSpecifications associated with a single ProductSpecification, for example to reflect offers to the market for different time periods. In cases where a ProductSpecification's value to the business is solely as a component part of a CompositeProductSpecification, it does not have a ProductOfferingSpecification associated with it as it is not be sold on its own. Cases where these items are supplied to the Customer separately as 'spares', e.g. under warranty, are covered under Product below.

When Customers select Products from the ProductCatalog, it is the ProductOfferingSpecification details that they are looking at and which are reflected in what they agree to contractually.

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Each ProductOfferingSpecification is marketed by a PartyRole. If need be, more processes responsibilities such as order handling or billing can be specified using Party Product Specification and Offering (see. Engaged Party guide book for more details).

Each ProductOfferingSpecification might be valued by ProductOfferingPrices that are asked for or allowed when a ProductOfferingSpecification is bought, rented, or leased.

At least one action is allowed for a ProductOfferingSpecification (AllowedProductAction) such as subscription, update…that might be limited to one or many SalesChannels. For example, a ProductOfferingSpecification might be upgraded directly by the Customer on the web, but to downgrade it they need to come into a store or contact a service representative.

An AllowedProductAction has a type represented by a ProductTypeAction.

Note: Another way of specifying the actionType would be to have a relationship with Process.

The main ProductOfferingSpecification entities are shown in the Figure below.

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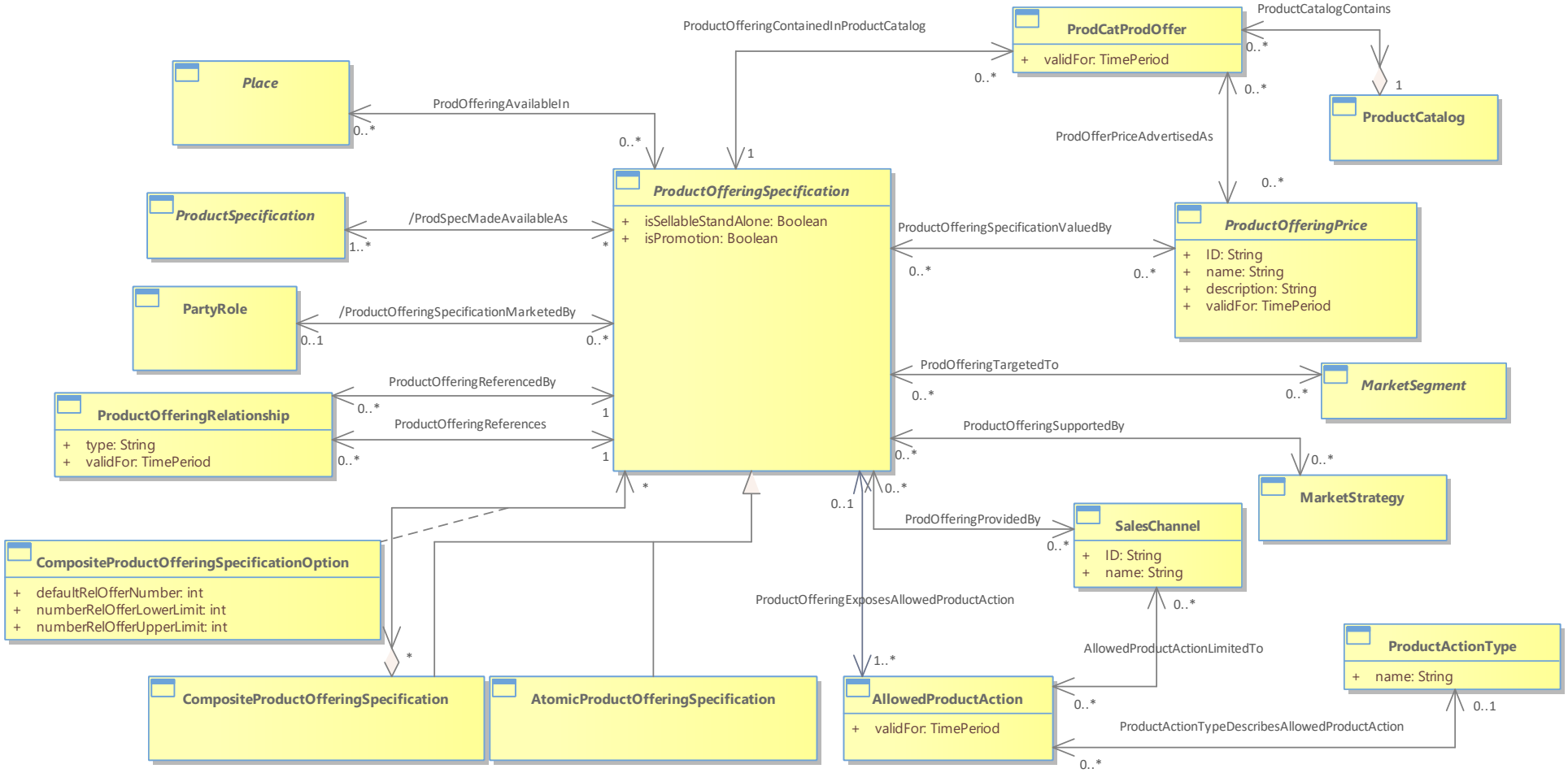

Figure Pr.05 - Product Offering - Basic Entities

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Figure Pr.05b - Product Spec - Product Offer Relationship

Smooth Telco normally markets a range of mobile phones as individual items. For a limited period around February 14th, it decides to market, a CompositeProductOfferingSpecification called the Valentine Special, one model with a special red cover, boxed in pairs and accompanied by two champagne glasses. As an added attraction, the deal includes a discount for calls made between the telephone numbers allocated to each pair for a set period. It is a huge success and likely to be repeated the following year when a new ProductOfferingSpecification with the new valid time period will be created. These ProductOfferingSpecifications exist side by side with the standard ProductOfferingSpecification for the mobile phone model when the phones are sold individually.

A question arises with this example. Should the Valentine Special or any CompositeProductOfferingSpecification have an associated ProductSpecification?

One alternative could allow for this by representing the association between ProductOfferingSpecification and ProductSpecification as 0, 1. But this association could lead to a*n* AtomicProductOfferingSpecification not having a related ProductSpecification.

Note: This way of doing isn't the one recommended by SID, so this relationship isn't in the SID model as only one solution can be in the model.

A better representation of this association would be to only relate an AtomicProductOfferingSpecification to a ProductSpecification as shown in the following Figure. This association allows a CompositeProductOfferingSpecification to not have a related ProductSpecification while requiring that a AtomicProductOfferingSpecification has one.

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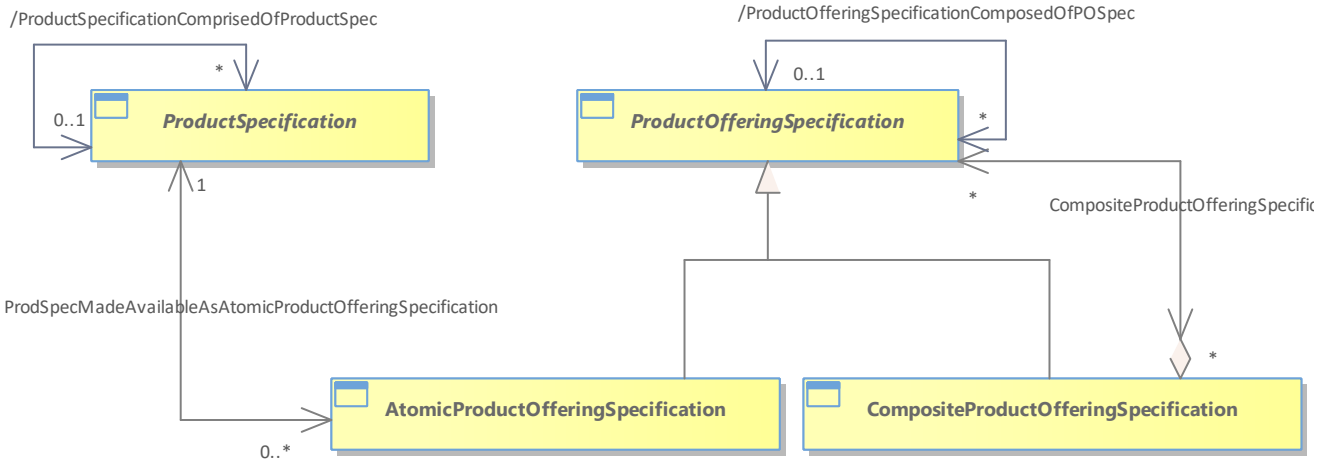

Figure Pr.05b - Product Spec - Product Offer Relationship

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Figure Pr.05c - CompositeProductOfferingSpecification Details

In some cases, there are limits to the number of ProductOfferingSpecifications that can be procured as part of a CompositeProductOfferingSpecification. For example, when subscribing to a mobile telephone service up to five phones can be procured at a special price. The BundledProdOfferOption handle these cases by allowing lower and upper limits to be set for ProductOfferingSpecifications that comprise a CompositeProductOfferingSpecification as shown in Figure Pr.5c. The figure also shows via the defaultRelOfferNumber attribute that is used to initialize the number of ProductOfferingSpecification instances when ordering the ProductOfferingSpecification that is contained within an instance of CompositeProductOfferingSpecification.

In addition, it might be useful describing that the customer must choose minimum X1 and maximum X2 ProductOfferingSpecifications between X contained in a CompositeProductOfferingSpecification. This is specified through the two attributes offerLowerLimit and offerUpperLimit carried by CompositeProductOfferingSpecification.

Some Information Framework users may want a ProductOfferingSpecification instance that is part of a CompositeProductOfferingSpecification instance to not be individually procurable. For example, a free SMS offering cannot be procured by itself. A separate AtomicProductOfferingSpecification can be created to allow for add-on SMS service. Other Information Framework users may feel that that this may result in ProductOfferingSpecification instance explosion. They want a ProductOfferingSpecification instance that is part of a CompositeProductOfferingSpecification instance to be individually procurable rather than creating another instance of ProductOfferingSpecification that is individually procurable.

Prior to release 17.5 of this guide book these two options were described as implementation options. In release 17.5 an isSellableStandAlone attribute shown in Figure Pr.5c was added satisfy the needs of both groups of users. If the isSellableStandAlone attribute is TRUE, it indicates that a ProductOfferingSpecification instance, which is part of a CompositeProductOfferingSpecification instance, can also be offered stand-alone, if FALSE means that the ProductOfferingSpecification instance can only be acquired within a CompositeProductOfferingSpecification instance. NOTE: This attribute only applies to offering instances (bundles and simple offerings) that are part of a CompositeProductOfferingSpecification instance.

There are some considerations that can be taken into account to assist in determining when this attribute should be set to FALSE or TRUE for applicable instances of ProductOfferingSpecification.

If the attribute is set to FALSE there is an increase in the number of ProductOfferingSpecification instances as previously mentioned.

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If the attribute is set to TRUE and associations to all related entity instances, such as, but not limited to, ProductOfferingPrice and ProdSpecCharValueUse, remain the same there are no other considerations.

If the attribute is set to TRUE and some associations, such as, but not limited to, ProductOfferingPrice and ProdSpecCharValueUse, are to different instances of these entities then there are other considerations. Note that this is not a complete list of considerations, but each association should be analyzed to determine if the ProductOfferingSpecification instance is related to different entity instances:

- Requires the use of the Product Offering Price Rule ABE to define a rule to determine the price for the ProductOfferingSpecification in the bundle and the price for the stand-alone ProductOfferingSpecification. This introduces approximately 20 policy entity instances to define the rules rather than creating an additional instance of a ProductOfferingSpecification.
- For a ProductOfferingSpecification instance there may be different instances of ProdSpecCharValueUse that apply to the offering when it is part of a bundle versus when it is sellable stand-alone. For example, a single device color may be made available in a bundle, but multiple colors available for the stand-alone offering. It appears that this consideration may require the creation of an association between ProdSpecCharValueUse and BundledProdOfferOption; the current association between ProductOfferingSpecification and ProdSpecCharValueUse could be used for the stand-alone offering.
- For a ProductOfferingSpecification instance there may be different instances of ProductOfferingTerm that apply to the offering when it is part of a bundle versus when it is sellable stand-alone. For example, a device in a bundle may have a different agreement length than for the stand-alone offering. It appears that this consideration may require the creation of an association between ProductOfferingTerm and BundledProdOfferOption; the current association between ProductOfferingSpecification and ProductOfferingTerm could be used for the stand-alone offering.
- This is a catalog consideration that is based on if there is a price for the stand-alone offering and different price for the offering in a bundle. If the price is the same then it could appear once in the catalog as part of the bundle with a notation that it can be individually acquired. If there are different prices, then a determination must be made on how to represent the prices. For example, both prices could be shown for the offering with a notation on the price for when the offering is acquired stand-alone.

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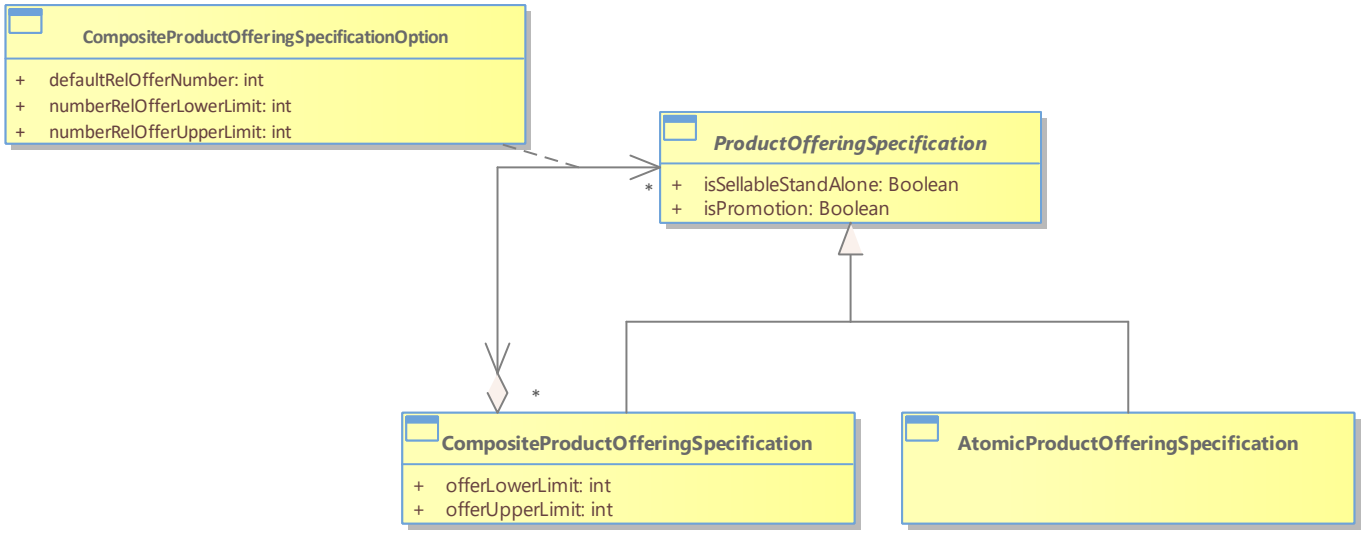

Figure Pr.05c - CompositeProductOfferingSpecification Details

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Illustration Pr.05c - I01 - CompositeProductOfferingSpecification Details

For the example of minimum 3 and maximum 5 channels that can be ordered with a specific channel ordered only once, it could be described like described in the illustration.

Like this:

- each AtomicProductOfferingSpecification is optional and can't be ordered more than one time thanks to LowerLimit=0 and UpperLimit=1 (on BundledProdOfferOption)
- among the 6 TV Channel AtomicProductOfferingSpecification, minimum of 3 must be ordered (OfferLowerLimit =3 on the CompositeProductOfferingSpecification) and maximum of 5 can be ordered (OfferUpperLimit =5 on the CompositeProductOfferingSpecification)

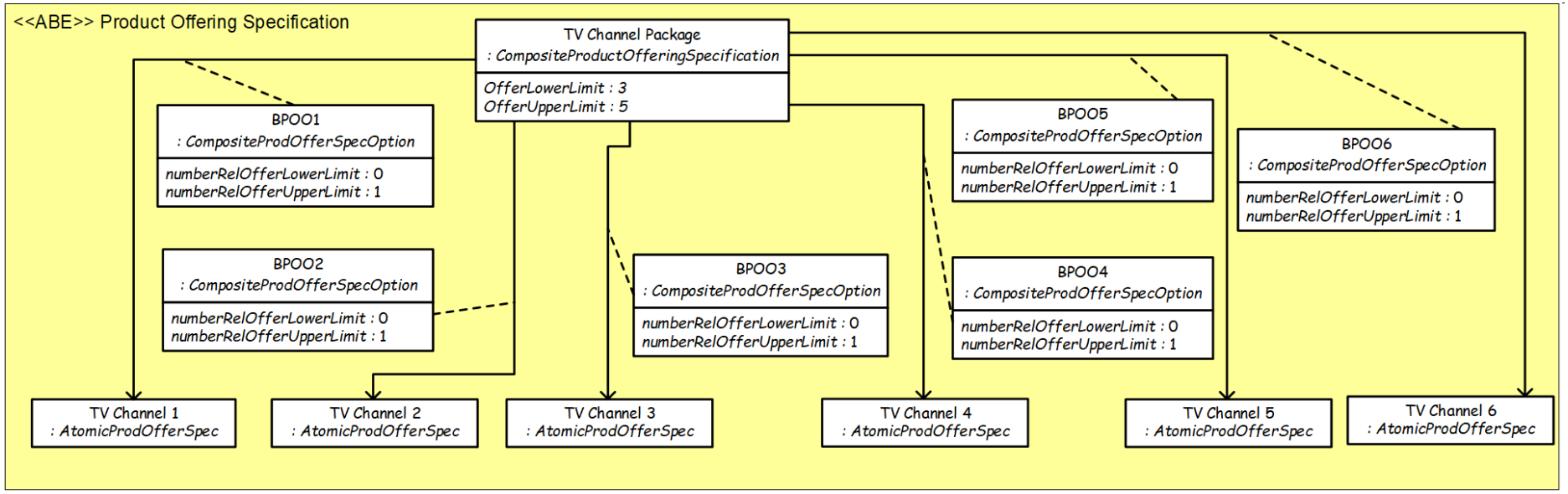

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Figure Pr.06 - Product Offering Terms

The ProductOfferingSpecification is primarily the way to position products in the marketplace to create profit; the amount of profit depends upon both the original pricing of the product and the eventual terms of the sales agreements struck for that product offering.

The ProductOfferingPrice is based on both the basic cost to develop and produce products (see Product Specification) and the enterprise's policy on revenue targets. The pricing policy mechanism is represented in this model by 'term'. Terms are the conditions under which the ProductOfferingSpecification is made available to Customers. For example, product finance term is only one of many which may come into play. In addition to a definition of the list price, financial terms may also include acceptable methods of payment for this type of product. As examples of other types of ProductOfferingTermOrCondition, the SID model could also include product service term and product shipment term.

When a price has been created, it will appear as the list price for ProductOfferingSpecification in the ProductCatalog. For a AtomicProductOfferingSpecification, this may also be the effective list price but, if ProductOfferingSpecifications are bundled together, the overall price will not necessarily be the combination of component list prices and pricing policy will be required. The Customer will refer to the list price in selecting a Product but, depending upon their relationship with the enterprise making the ProductOfferingSpecification available, this price may be further revised through discounting, such as for bulk buying, loyalty and so forth. This is modelled through the AgreementTermOrCondition class, shown in Figure Pr.6.

The price, applied for a ProductOfferingSpecification may also be influenced by the ProductOfferingTermOrCondition, the customer selected – e.g. a ProductOfferingSpecification can be offered with multiple terms, like commitment periods for the contact. A 12-month contract may be cheaper than a contract with a 24-month commitment period.

The AgreementItemGovernsProduct relationship represents the items in an Agreement that govern the Product. Examples include:

- A service level agreement clause defining maximum allowed downtime for the Product.
- A framework agreement clause defining the terms under which the Product was purchased.

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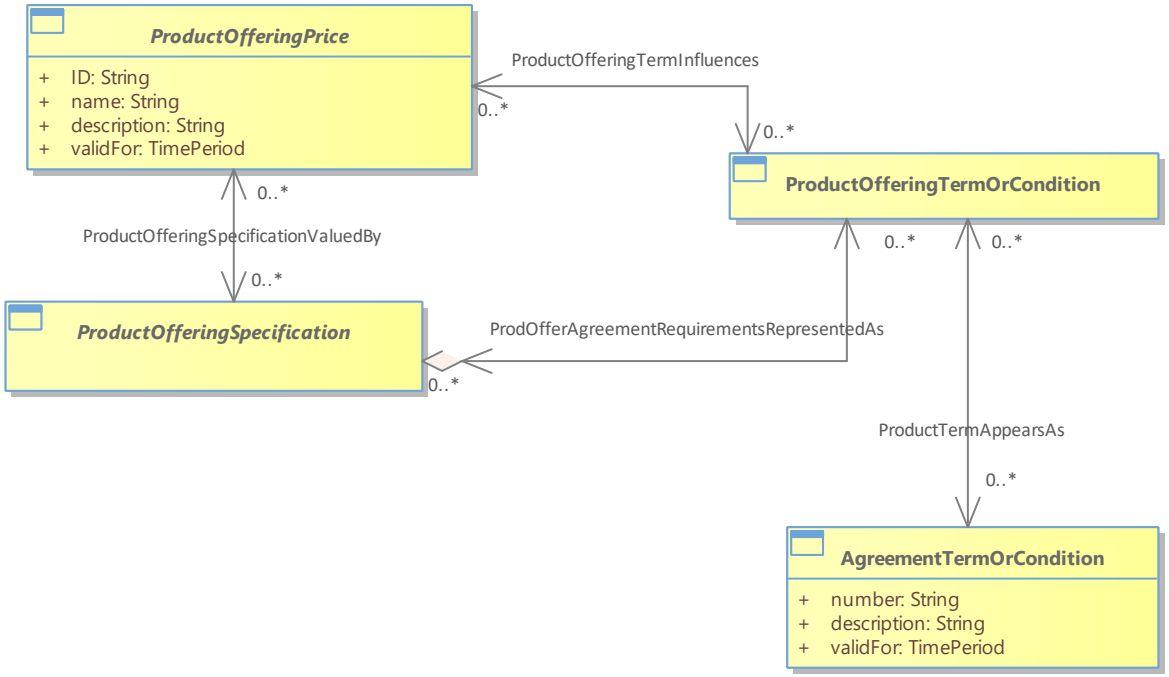

Figure Pr.06 - Product Offering Terms

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Figure Pr.08 - Product Offering Price

The price of a ProductOfferingSpecification or a bundle may be simple or quite complex. A simple price could be the charge associated with the monthly rental of a cable modem. Or, a price could encompass many different components. A typical introductory wireless offering, for example, will have an activation cost, a monthly cost, a number of free minutes, a cost for extra minutes, and some sort of promotional component. It might also include a corporate discount for business users. Each of these elements must be described in order to paint an accurate picture of the overall price charged for any ProductOfferingSpecification.

This part of the SID model presents a set of components that can be combined to offer a complete and accurate description of the price charged for an offering. A secondary consideration is to define those components in such a way that they can be readily implemented in a typical billing system.

A ProductOfferingSpecification may have any number of prices, each of which is made up of a single component or a composite of a number of components. The Product Offering Price model employs the composite/component modeling pattern used throughout the SID to represent simple and complex prices, as illustrated in the figure below.

ProductOfferingPrices can be dependent upon a ProductSpecCharValueUse as shown in the figure above. For example, a provider may offer a photo printing service for mobile phone pictures. The price may depend on the size of the photo or the type of finish of the photo; $.20 for a 3 by 5, plain finish, $.25 for a 5 by 7, plain finish, and so forth. The values associated with the size (3 by 5, 5 by 7, and so forth) and finish (plain, glossy, matte, and so forth) can be stored as ProductSpecCharacteristicValues.

ProductOfferingPrice, may have a complex structure (composition) and involve complex billing / rating capabilities.

The ProductSpecification to ProductOfferingPrice association allows only the definition of:

- either templates to create the ProductOfferingPrices associated to the ProductOfferingSpecification that will package the ProductSpecification,
- or prices under which the ProductSpecification would not be sold (example: because selling under these prices would generate loss).

Figure Pr.08 also shows an association between ProductOfferingPrice and PriceEvent. The PriceEvent business entity represents an occurrence (happening) or schedule that triggers the creation of a billing charge for a ProductOfferingSpecification. For example, a PriceEvent instance may represent the end of a month when monthly recurring billing charges are created. In some cases, the creation of a charge associated with a

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ProductOfferingPrice may be triggered by multiple events. For example, a penalty (ProductOfferingPrice) may be triggered by over- or under-usage of a service. Under- and over-usage are represented as two separate PriceEvent entity instances.

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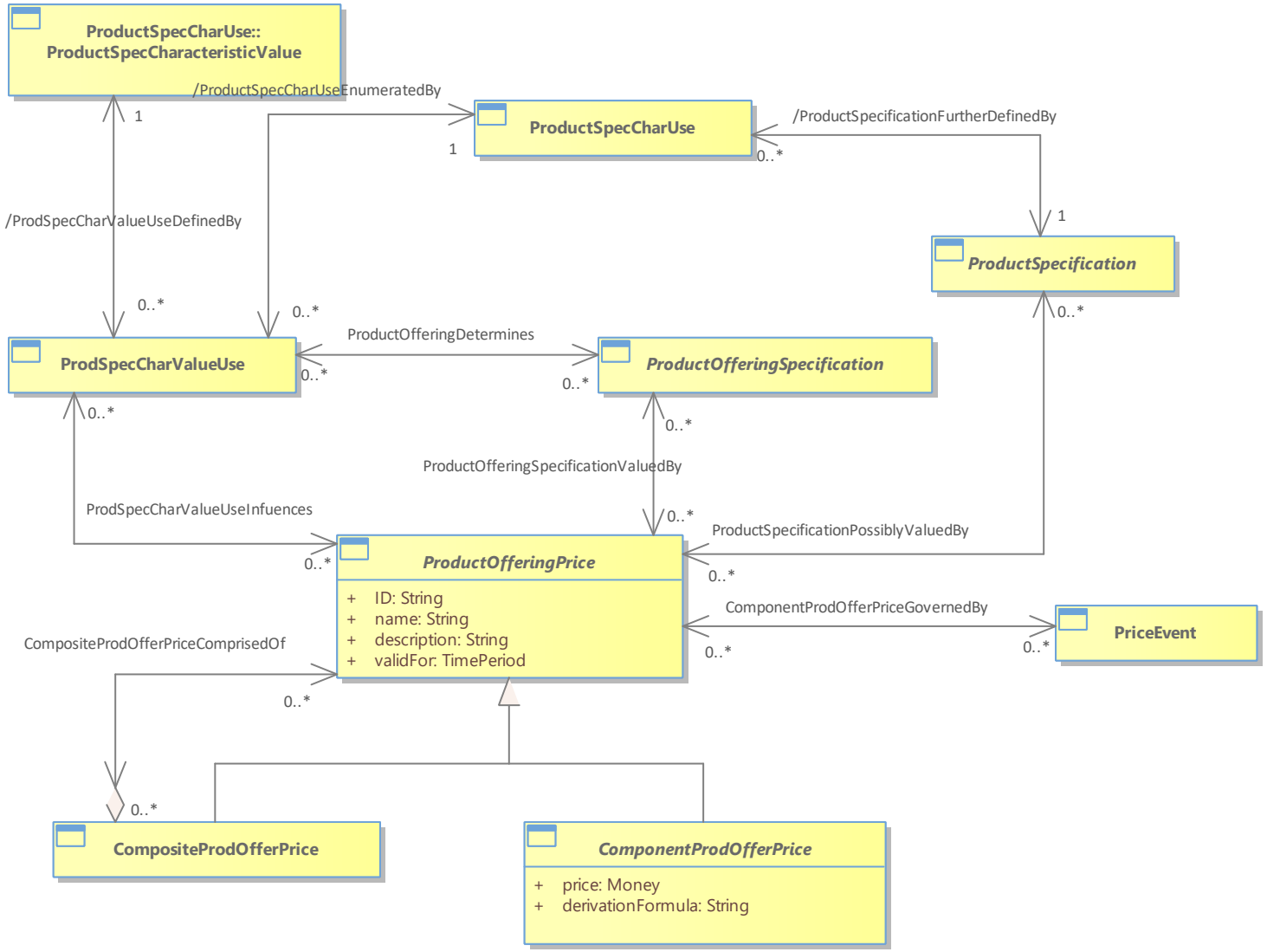

Figure Pr.08 - Product Offering Price

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Figure Pr.09 - Product Offering Price's Associations with Other Business Entities

The price paid for a ProductOfferingSpecification may appear in ProductCatalogs, may be offered via certain DistributionChannels, or in certain GeographicAreas. The figure below depicts the associations a ProductOfferingPrice has with these business entities.

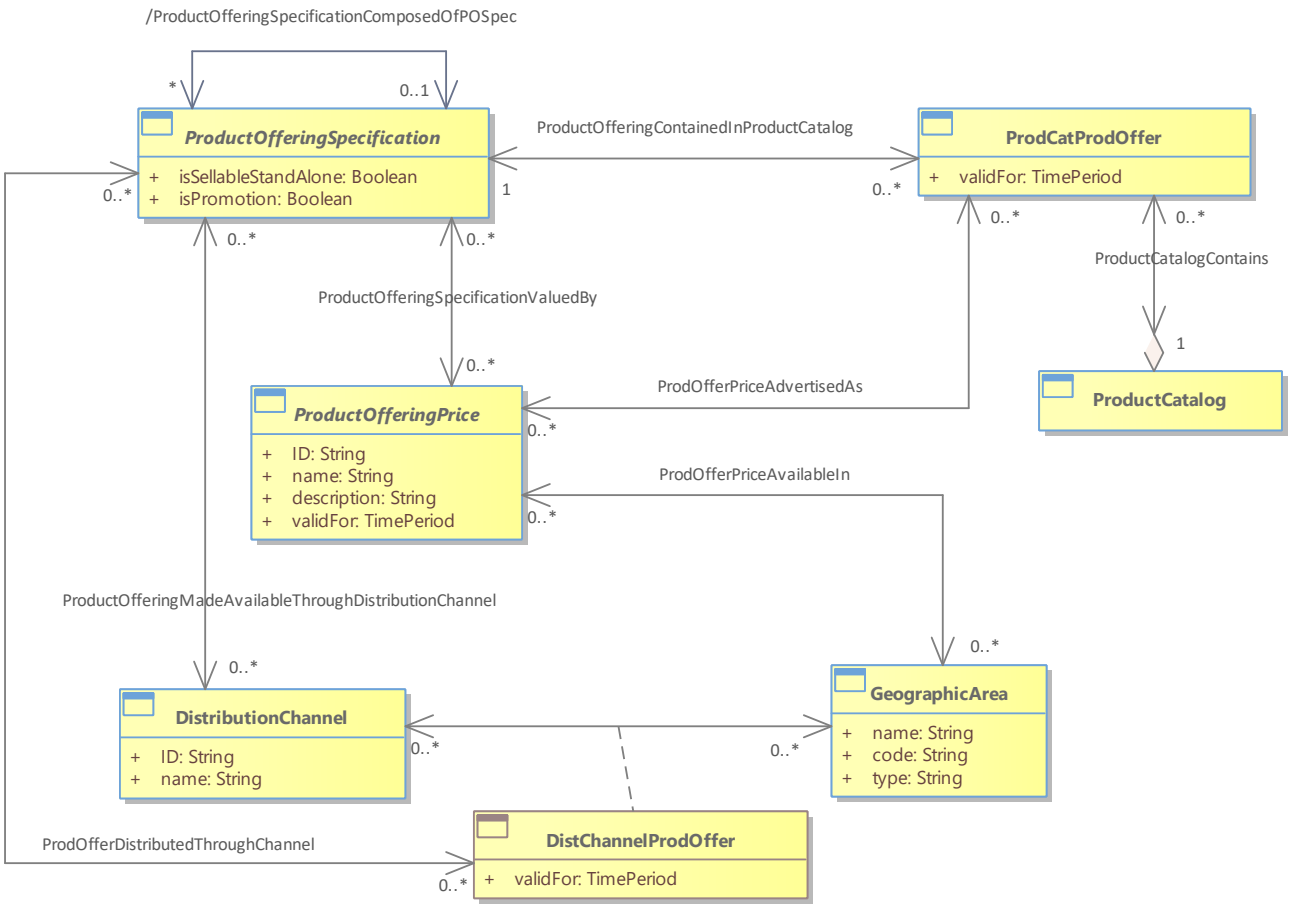

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Figure Pr.10 - Product Offering Price Components

Each component of a ProductOfferingPrice can take on one of two forms.

A component may represent the amount charged (ProdOfferPriceCharge) such as a monthly internet connection fee.

Or, a component may represent an alteration (ProdOfferPriceAlteration) that will be made to the amount charged such as an installation allowance of $49.95 to the first month's bill, which offsets an installation charge. An alteration impacts ProductOfferingPrice that can be a component as well as a composite price. For example, a tax alteration may impact the result of a charge reduced by an alteration (Ex: 100$ -10%). In this case, the Alteration impacts a composite price.

The figure below depicts the two types of ComponentProdOfferPrices.

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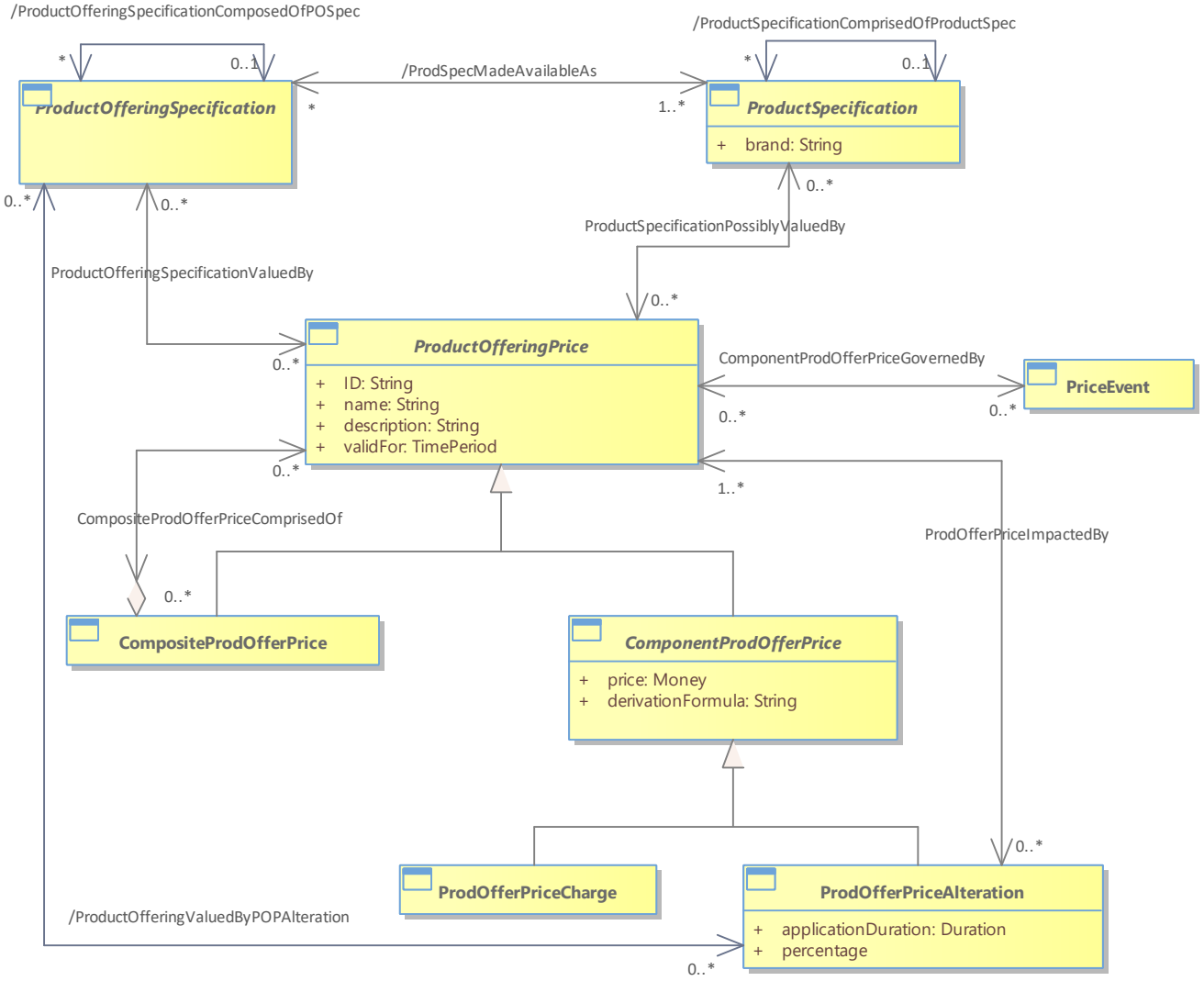

Figure Pr.10 - Product Offering Price Components

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Figure Pr.11 - Types of Product Offering Price Charges

Charges incurred for a ProductOfferingSpecification can take many different forms, such as

- Recurring Charge – A recurring charge is levied repeatedly. It has a period, such as monthly, quarterly, or yearly. An example would be a monthly charge for long distance. A recurring charge may also limit the application duration. For example, a recurring charge applied only during 6 months for an expensive device.
- One Time Charge – A one-time charge is levied once. An example is the cost of a cell phone handset or installation fee (FeeProdOfferingPrice valuing an AllowedProductAction).
- Usage – A charge levied for usage (ProductUsageSpec), based on measuring events. This model distinguishes between simple usage and usage where the charge is described by a tariff.

0 Simple Usage – Simple usage describes a rate where the charge is the same for all events. A 10 cents a minute for long distance plan is one example.

0 Tariff Usage – Tariff usage describes a rate where there are different charges for different types of events. This could include plans, which have different prices for evening minutes and anytime minutes. This usage aggregates a number of simple usages.

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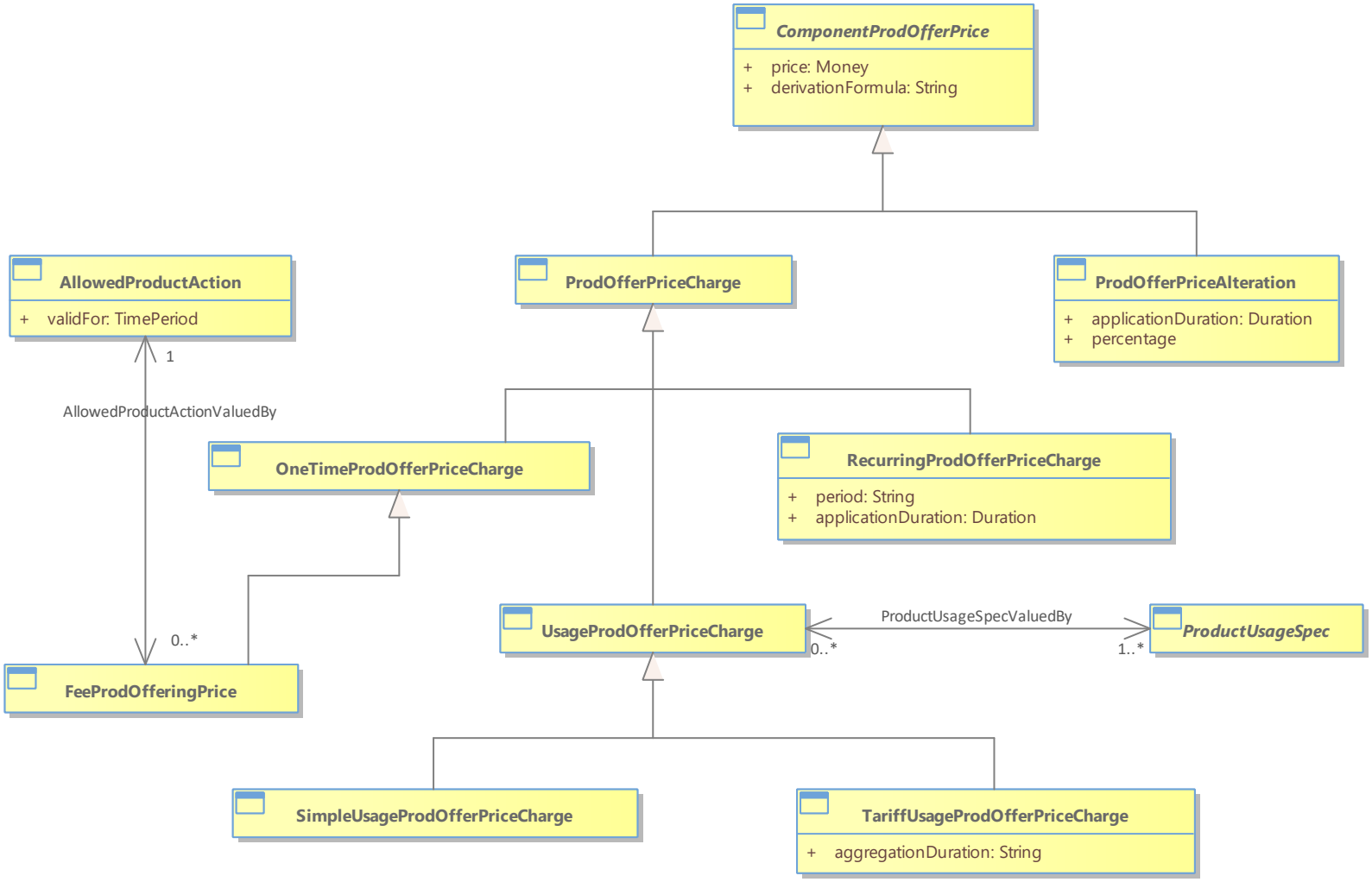

Figure Pr.11 - Types of Product Offering Price Charges

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Figure Pr.12 - Types of Product Offering Price Alteration

Similarly, alterations made to other ProductOfferingPrice(s) incurred for a ProductOfferingSpecification can take many different forms, such as

- **Discount** – A discount is a reduction in price. It could be a flat fee such as 10 dollars or a percentage, such as 50% off. A discount always applies to, and thus is associated with, one or more other ProdOfferPrices. So, a discount would be 10 dollars off the installation fee or 50% off the per-minute rate.
- **Replacement** – A price that replaces another ProductOfferingPrice (charged in lieu of another price). For example, purchasing two of a single product offering may result in a discounted price for both.
- **Taxation** – A tax amount is an increase in price due to a compulsory levy by an authorized jurisdiction. For ProductOfferingPrice this is a projected amount. The actual tax amount will be applied in the ProductPrice. See the dedicated section below on tax for fuller details.

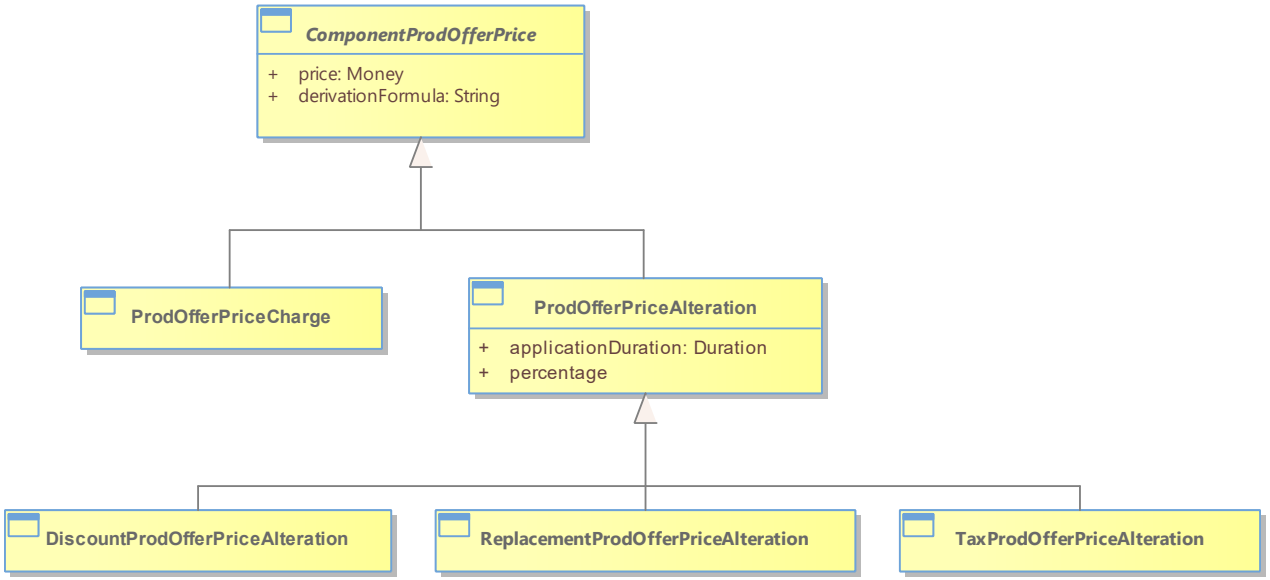

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Figure Pr.13 - Complete Product Offer Price Model

Putting charges and alteration in a single example provides a complete view of a ProductOfferingPrice. A simple wireless plan includes a 75 dollar phone, a 50 dollar activation fee, a 20 dollar monthly charge replaced by 10 dollar during 6 months, 10 cents a minute for additional minutes, and 25 cents a minute long distance. The components are

- A 75 dollar One Time Charge (OneTimeProdOfferPriceCharge)
- A 50 dollar Fee (FeeProdOfferingPrice)
- A 20 dollar Recurring Charge per month (RecurringProdOfferPriceCharge)
- A 10 dollar replacement of the recurring charge during 6 months (ReplacementProdOfferPriceAlteration)
- A Tariff Usage comprising a 10 cent per minute Simple Usage and a 25 cent per minute Simple Usage for long distance (SimpleUsageProdOfferPriceCharge)

Adding on to the example, suppose Sue is an employee of a company, that has a volume discount agreement with the service provider. The agreement is to waive the installation fee and provide a 20 percent discount on local minutes. This would manifest itself as:

- A composite price that associated the 50 dollar Discount (DiscountProdOfferPriceAlteration) with the 50 dollar Fee
- A composite price that associated the 20 percent Discount (DiscountProdOfferPriceAlteration) with the 10 cent per minute Simple Usage

In addition a Tax (TaxProdOfferPriceAlteration) might be applied on each component or composite ProductOfferingPrice.

The complete ProductOfferingPrice model is shown in the figure below.

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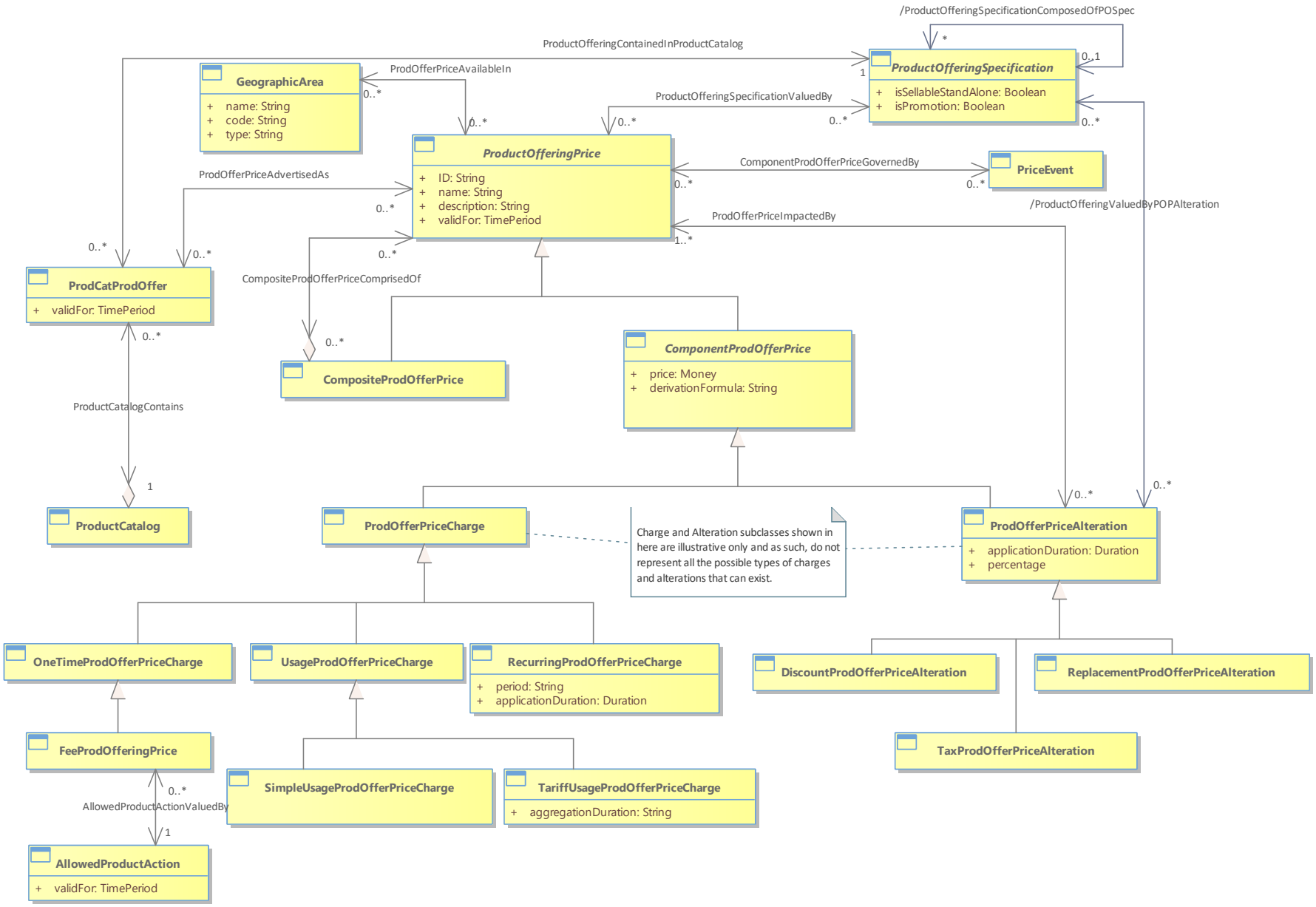

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Figure Pr.13 - Complete Product Offer Price Model

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Figure Pr.14 - Product Offering Price Governed by Policy

The price of a Product is often dependent upon the procurement of another Product for a particular price, another ProductOfferingSpecification, or another ProductOfferingSpecification of the same type. For example, a Family Plan mobile phone service includes two free phones. Within this offer, the cell phones are associated with a zero value price. In another offering, there may be a choice of some number of phone accessories that are offered at no or greatly reduced cost.

Both of these offers require the employment of a policy rule that governs the price of the accessories. In the first case, the PolicyRule is used to ensure that two of the phones are provided at zero cost, as long as the terms of a particular ProductOfferingSpecification are met. In the second case, the PolicyRule defines which accessories can be purchased at what price.

Price rules such as these require another set of business entities in addition to the more basic rules set forth in Product Offering Price and its components.

The intent of this part of the SID model is to present a model that can be used to define simple cross-product price policy rules. This will be done by extending the SID Policy model. As such, the model presented here is not intended to provide for elaborate pricing schemes. In these cases, it may be necessary to employ business entities as yet to be defined within the policy model or that may require the development of code specific to the rule. Still, it is significant that the existing SID Policy model, whose primary use case was to support the definition of networking policies, can be easily extended to meet the needs of a very different domain.

The figure below shows the relationship that a price rule has with ProductOfferingSpecifications and with Policy.

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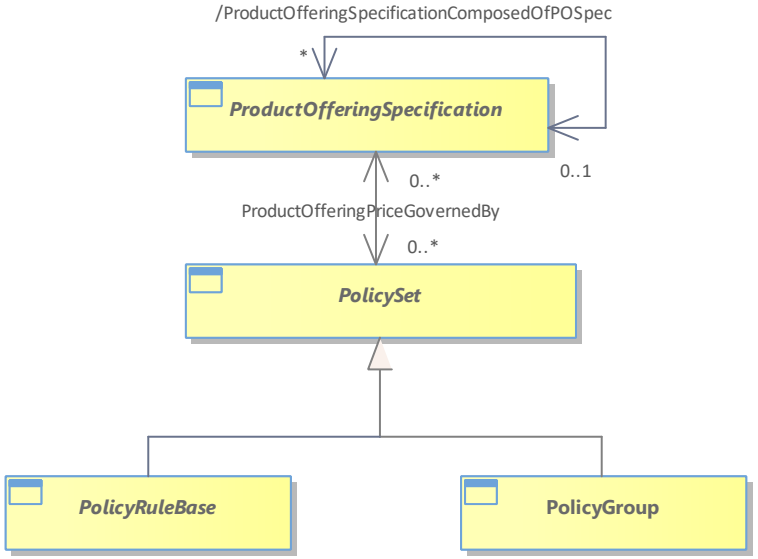

Figure Pr.14 - Product Offering Price Governed by Policy

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Figure Pr.15 - Policy Rule Components

The price of a ProductOfferingSpecification is governed by a PolicySet. The PolicySet may be a single PolicyRule or a PolicyGroup which ties together a number of PolicyRules. The use of both single rules and grouped rules will be explained via examples.

For example, suppose there is a ProductOfferingSpecification called “Subscribe for a Year, Get Two Accessories Free”. The free accessories can be selected from list of available accessories. In this example, the accessories are a cell phone re-charger, hands-free holder, cell phone wallet, or earphone and microphone. Included in the offering are the cell phone service and the accessories. There is an instance of PolicySet rule related to this offering.

All PolicyRules consist of three types of components, which define the semantics of the rule. (Please see Policy guide book for more information.) This is shown in the figure below.

The semantics of a rule that governs the price of a ProductOfferingSpecification builds upon this generic structure. The first component is a PolicyEvent, which triggers the evaluation of the PolicyRule. The second defines the PolicyCondition(s) of the PolicyRule being evaluated. The third defines the PolicyAction(s) that are taken if the PolicyCondition(s) of the rule are met. In the “Subscribe for a Year” offering:

The PolicyEvent is the time of purchase

The PolicyCondition is the particular type of cell phone service offering being purchased (since different pricing rules could be attached to different types of cell phone offerings)

The PolicyAction taken as a result of this particular condition being true is that two of the accessories are free.

Note that a PolicyRule aggregates at least one or more PolicyConditions and at least one or more PolicyActions. (Policy guide book defines a general set of semantics that specify how a PolicyRule is evaluated, including the concepts of rule priority and a decision strategy for combining different PolicyConditions.) In addition, there are other constructs, such as a Policy Group, that can aggregate multiple PolicyRules. This enables more complex pricing rules to be developed by defining appropriate subclasses of PolicyRule, PolicyCondition and PolicyAction. Note that PolicyCondition and PolicyAction both use the composite pattern; please see Policy guide book for more information.

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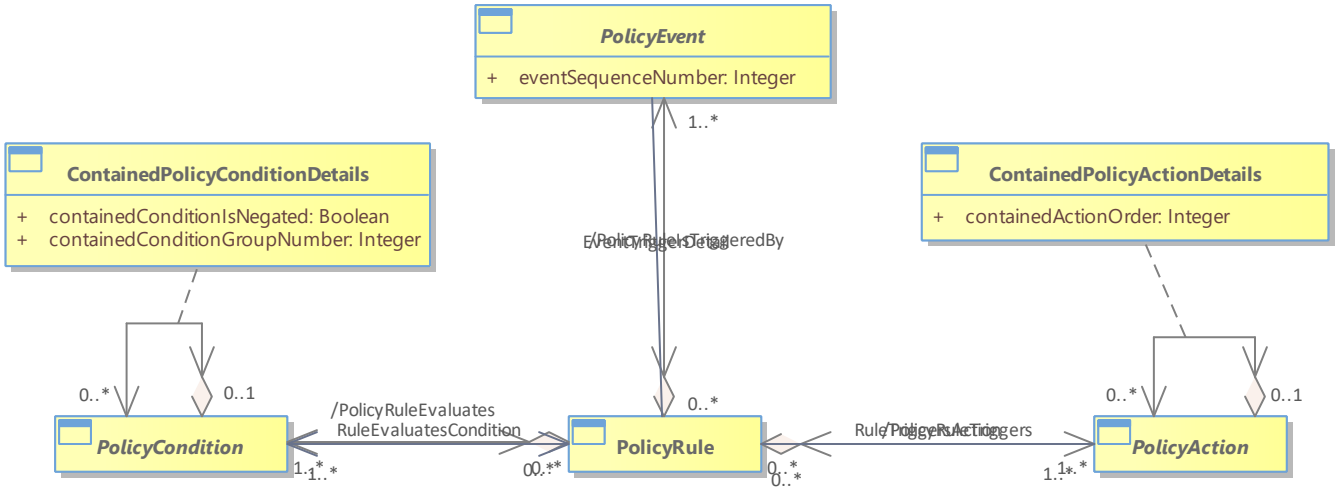

Figure Pr.15 - Policy Rule Components

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Figure Pr.16 - Product Offer Price Condition and Policy

Next, a subclass of PolicyConditionAtomic, called ProdOfferPriceCondition will be defined. Its purpose is to add semantics that determine if one or more PolicyAction(s) should be taken.

The attributes of the policy subclasses do not completely define the semantics of the condition governing the price of a ProductOfferingSpecification. To complete the definition of the condition, associations to other business entities specify what must be procured or what must have been procured. The related business entity may be a ProductOfferingSpecification, a similar type of product, or a product at a particular price. In the “Subscriber for a Year” offering example, the ProdOfferPriceCondition instance is associated to the cell phone service instance of ProductOfferingSpecification.

In the example, the condition type (which is represented by the priceRuleConditionType attribute) is “buy”. Other values for the type will be described in later examples.

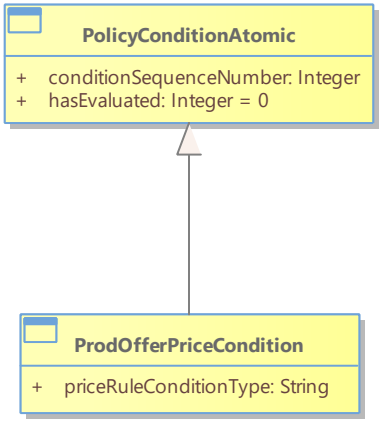

Figure Pr.16 - Product Offer Price Condition and Policy

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Figure Pr.17 - Product Offer Price Condition and Policy Statement

The condition is fully defined using the PolicyStatement entities in the Policy model, as shown in the figure below.

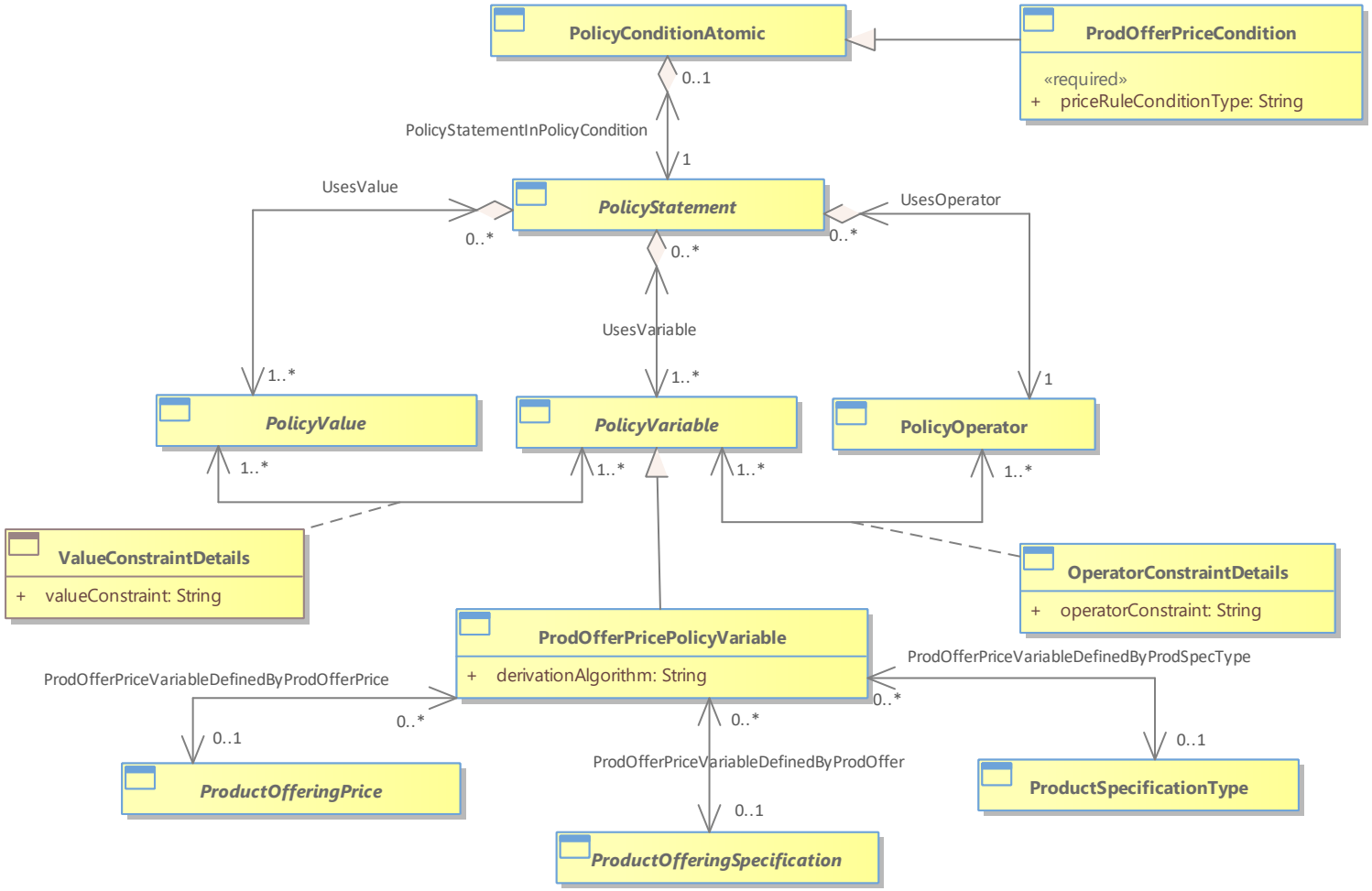

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Figure Pr.18 - Product Offer Price Action and Policy Statement

The complete PolicyStatement for this condition is: The ProdOfferPolicyVariable is the quantity ordered for the cell phone instance of ProductOfferingSpecification. The PolicyOperator is “equals”. The PolicyValue is 1. The condition is interpreted as “if one instance of the cell phone is ordered” then some action will be taken. The power of representing a PolicyStatement in this way is that each part of the PolicyCondition is object-oriented and reusable. This sets the stage for representing more complex pricing rule conditions.

Satisfying a condition may govern the entire price of an offering or one or more of the components of the offering. In this example, it governs the entire price of the offering for the accessories. That is, two can be chosen for free.

The action to take is defined by ProdOfferPriceAction, which is a subclass of PolicyAction. These entities are shown in the figure below, along with the association between PolicyAction and PolicyStatement.

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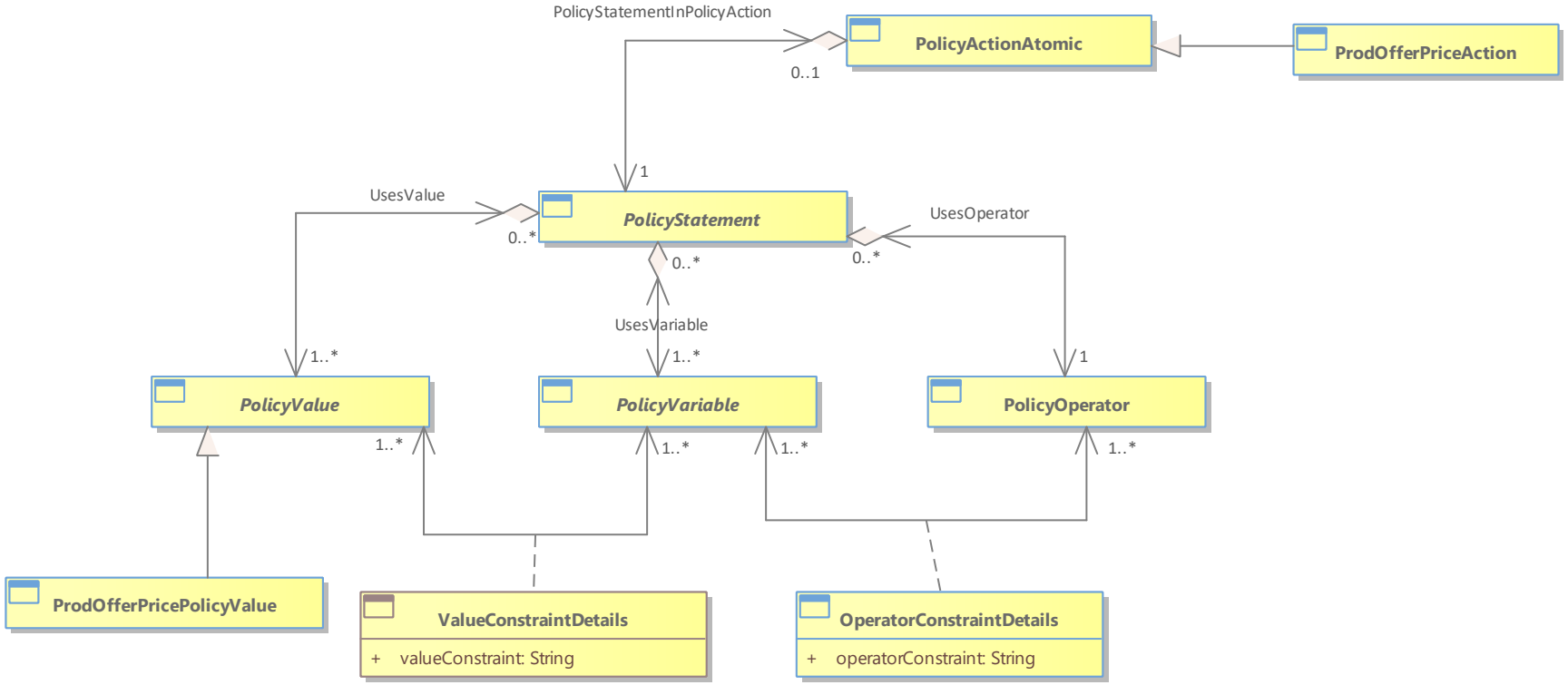

Figure Pr.18 - Product Offer Price Action and Policy Statement

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Figure Pr.19 - Product Offer Price Action and Product Offering Price

The complete PolicyStatement for this action is: The Prod Offer Policy Variable is the quantity for the accessories. The Policy Operator is “equal to or less than”. The PolicyValue is “2”. The action is interpreted as “set the quantity allowed for free accessories to no more than 2”.

There is one missing variable in this action. The missing variable is the price (0) to be paid for the accessories and what accessories are free. This variable is expressed using an association between ProdOfferPriceAction and ProductOfferingPrice. Instances of this association specify the ProductOfferingSpecification and the Price associated with the action. In this example, the instance of the action would be associated with instances of zero value prices for cell phone re-charger, hands-free holder, cell phone wallet, and earphone/microphone. The figure below shows this association.

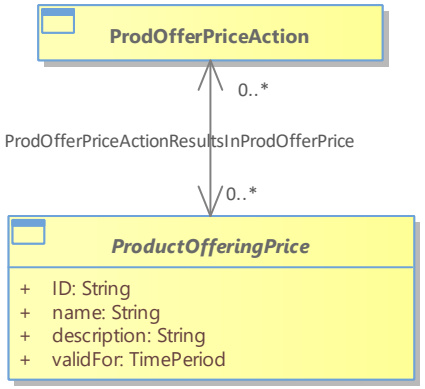

Figure Pr.19 - Product Offer Price Action and Product Offering Price

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Figure Pr.20 - Product Price Value and Product Offering Price

To support the action associated to a complex rule, another extension is added to the Policy model. The value of the discounted price is not a fixed value, but one associated with a specific Product Offering Price. This can be modelled by first creating a subclass of PolicyValues, called ProdOfferPricePolicyValue, and then creating an association between ProdOfferPricePolicyValue and ProductOfferingPrice, as shown in the figure below.

Note: In all the pricing rule examples, if a condition is not TRUE, then nothing happens.

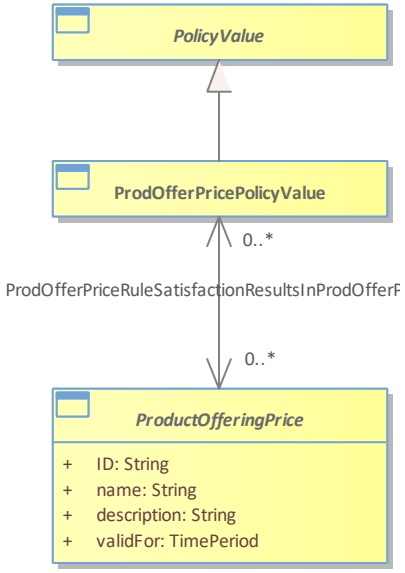

Figure Pr.20 - Product Price Value and Product Offering Price

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Figure Pr.21 - Complete Product Price Rule Model

The following diagram represents the complete Product Price Rule model.

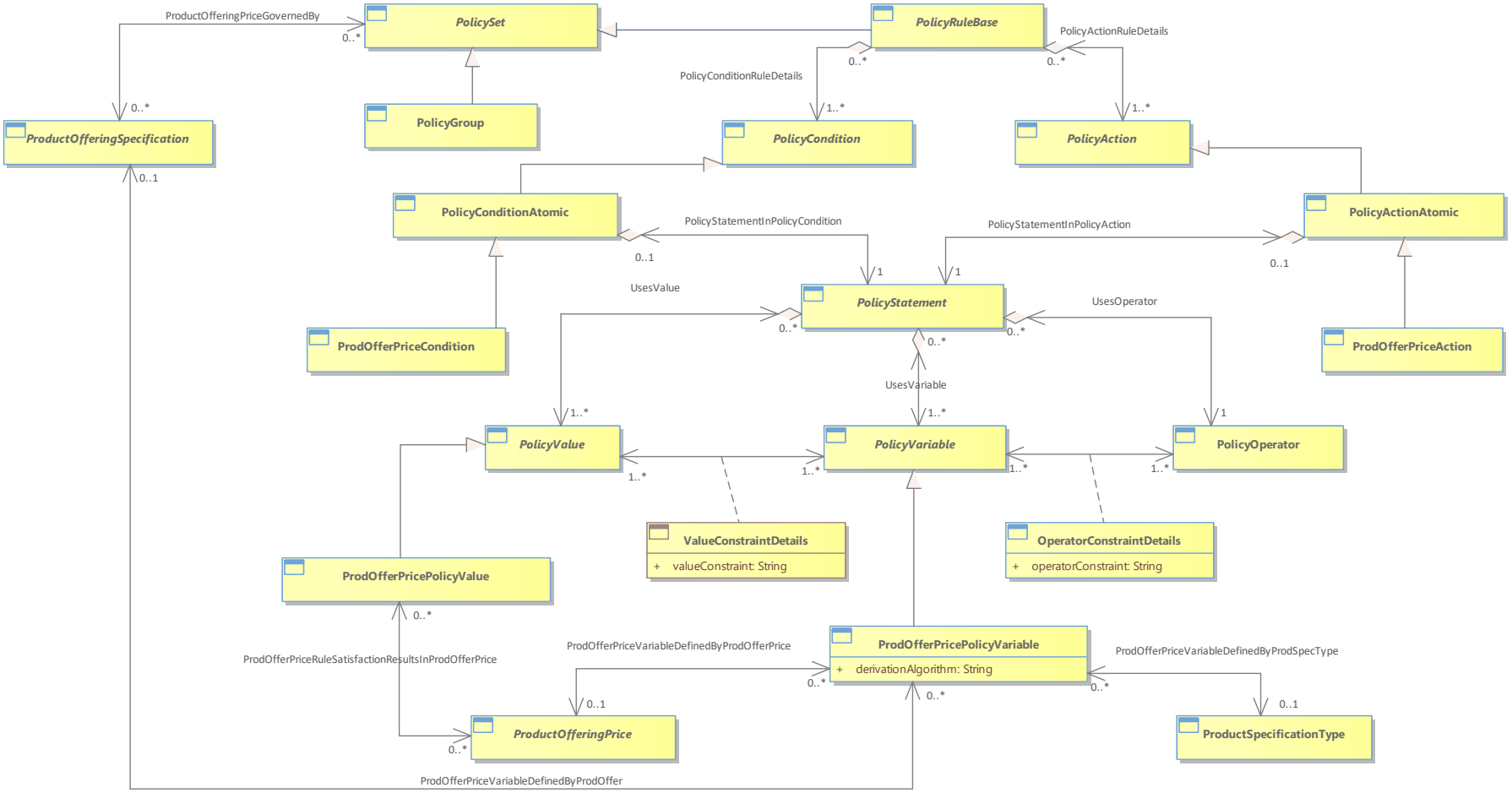

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Figure Pr.21-I01 – Subscribe for a Year with Free Accessories

The figure below shows the instances of all business entities that define product price rules within the context of the model for the Subscribe for a year with Free Accessories example.

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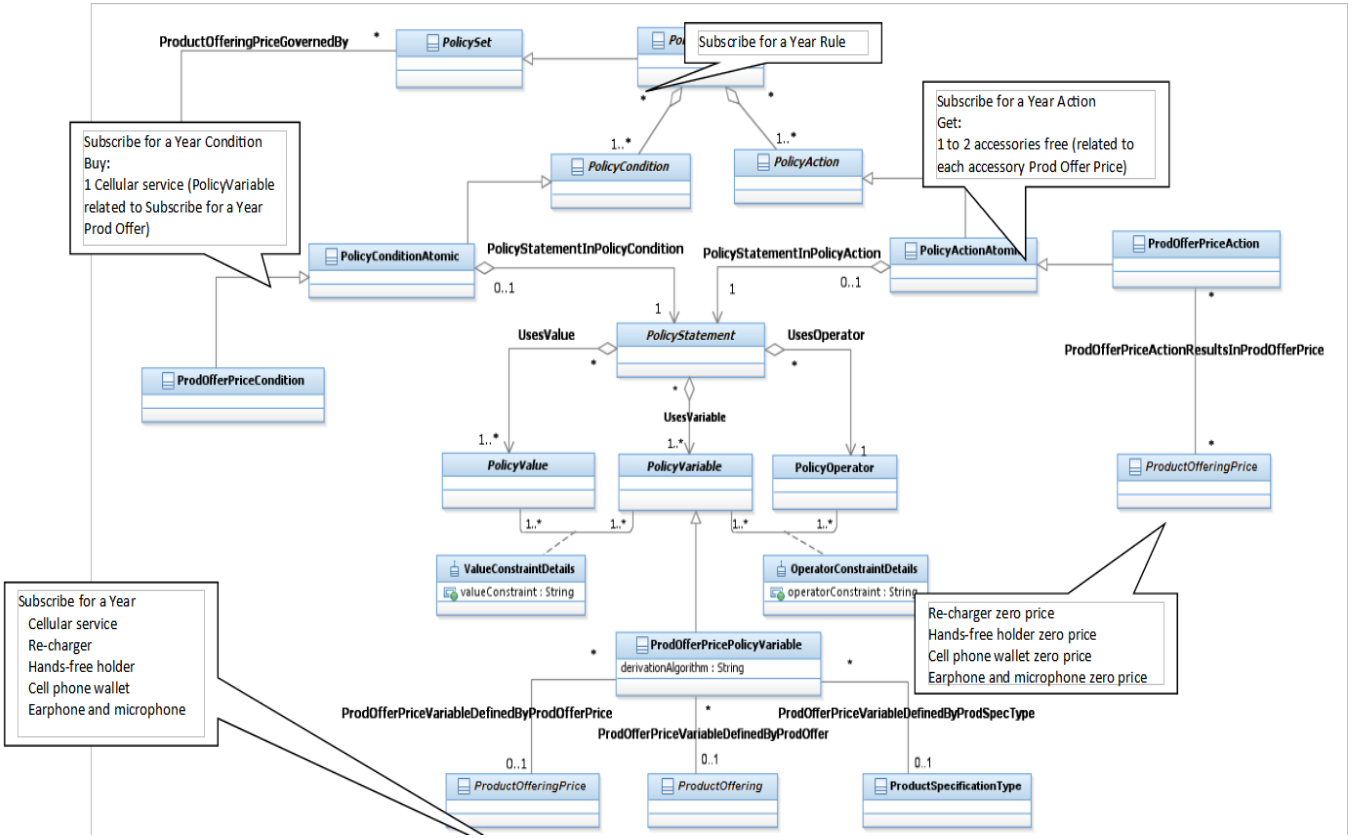

Figure Pr.21-I01 – Subscribe for a Year with Free Accessories

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Figure Pr.21-I02 – WLAN Subscription with Discounts

It is not always necessary to relate an instance of ProductOfferingSpecification to a price rule. In some cases, obtaining a product offering at a specific price may enable the procurement of other product(s) for a particular price. This price may be represented by an instance of an entire ProductOfferingPrice or may be represented by an instance of a ProductOfferingPriceComponent.

For example, suppose that if a customer subscribes to wireless Local Area Network (WLAN) access at the regular price, the customer is entitled to a discount on their DSL internet access or a discount on their cellular phone service.

Handling this rule requires the use of a PolicyGroup. The PolicyGroup subclass of PolicySet brings together multiple PolicyRules and applies them as an atomic set of rules. This enables different PolicyRules to be defined to serve stand-alone needs, yet be reused in other applications. The first rule checks to see if the price of the ordered WLAN access is equal to or greater than the regular price. If this condition is true, then an event is generated that triggers an option that allows the user to select one of the two items offered at a discount. The second rule, also triggered by this event, determines which choice of options has been made and sets the price to the discounted price.

Expressed as a PolicySet, the rules' conditions and actions are:

User chooses WLAN access in GUI screen #1. This selection triggers an event, which fires the following rule:

```
PolicyRule 1: WLAN_Access_Sale
  ON EVENT customer_choice DO
    IF (order >= min_price) THEN
      SET Event_WLAN_Access_Options to TRUE
    ENDIF
  ENDEVENT
```

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Setting the Event to TRUE initiates the display of GUI screen #2. The user chooses an option, which is recorded in a choice variable, which initiates the following nested PolicyRule (remember, a PolicyRule can contain a PolicyRule):

```
PolicyRule 2: Determine_Option_Choice
  ON EVENT WLAN_Access_Options DO
    IF (choice == DSL) THEN
      SET Business_Interaction_Item_Price = Discounted_DSL_Price
    ENDIF
    IF (choice == Cell) THEN
      SET Business_Interaction_Item_Price = Discounted_Cell_Price
    ENDIF
  ENDEVENT
```

The figure below shows the instances of these business entities for the WLAN example within the context of the model.

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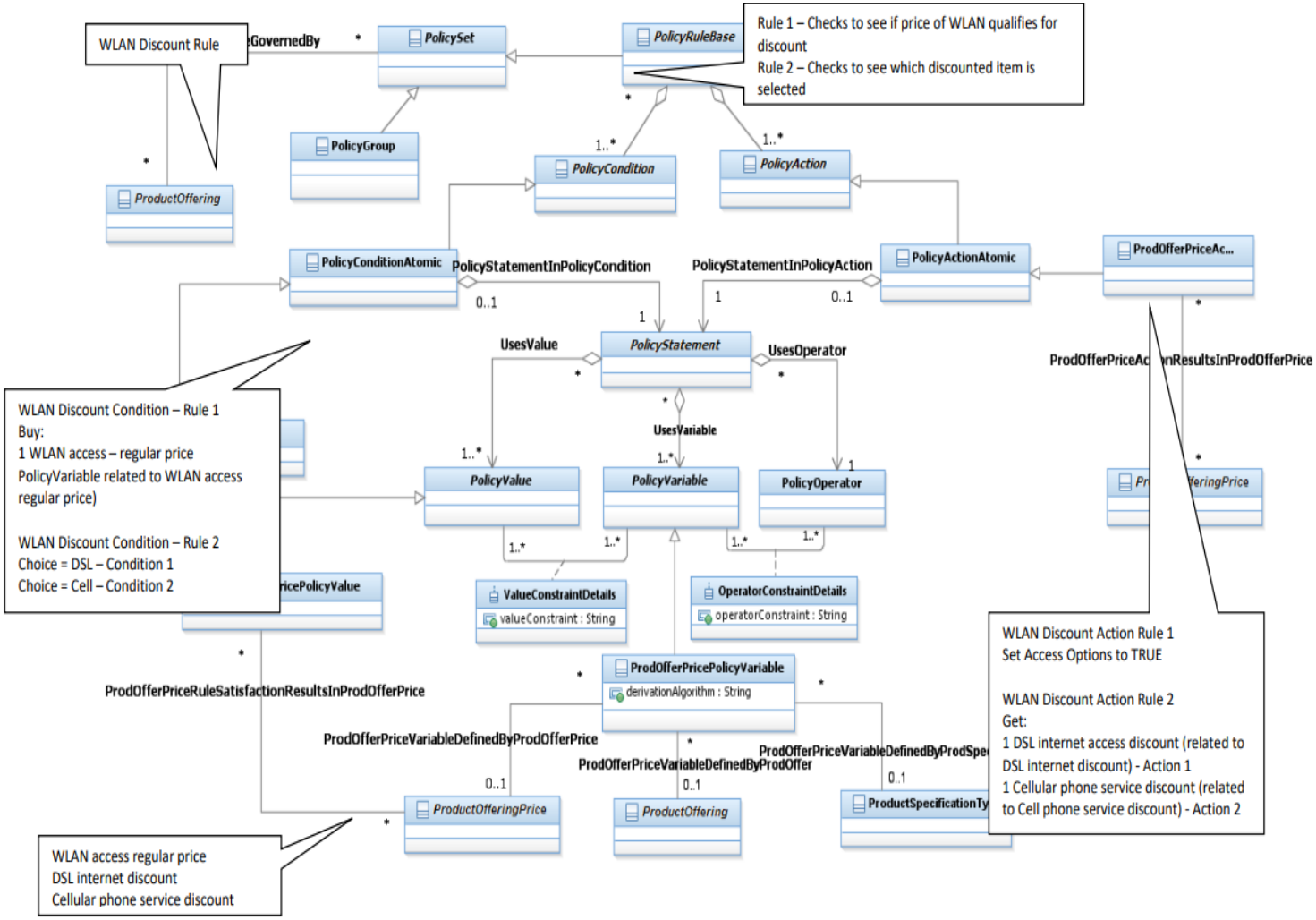

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Figure Pr.21-I03 - Product Price Rule Model – DSL Service and Router Rebate

The next example is a ProductOfferingSpecification that combines ordering DSL service with one of several types of routers that offer varying functionality. If both the DSL service and either of the routers are purchased, a subscriber (customer) gets a fifty-dollar rebate on the router of choice (we'll keep it simple and assume that there are only two types of routers to choose from in this example). In this case the ProductOfferingSpecification instance, "Buy DSL Service and Router, Get $50 Back!", has as its product offering components, the DSL service and one of two different routers.

Since the price rule involves purchasing two products, a composite PolicyCondition is employed. The model for representing a composite PolicyCondition follows the Composite/Component pattern used throughout the SID model and in the Product domain, so the PolicyConditionComposite class is not shown here.

The price rule has six condition instances, three atomic condition instances and three composite condition instances (the composite instances aggregate two or more atomic instances). The first atomic condition's ProdOfferPolicyVariable is related to the quantity for the DSL service. The PolicyOperator is "equals". The PolicyValue is "1". The second atomic condition's ProdOfferPolicyVariable is related to the quantity for one type of router. The PolicyOperator is "equals". The PolicyValue is "1". The third atomic condition's ProdOfferPolicyVariable is related to the quantity for the other type of router. The Policy Operator is "equals". The Policy Value is "1".

In order to represent the completed composite PolicyCondition, two of the composite conditions join the DSL service with the routers using an "and" Boolean operator. The third composite joins these two composites using an "or" Boolean operator. This entire condition is: (DSL service "and" router #1) or (DSL service "and" router #2).

The ProdOfferPriceAction is related to the $50 rebate for the DSL service. The action is: "Set allowed quantity for the DSL service rebate equal to 1". The description of the price indicates it is a rebate on the purchased router. The figure below shows the instances of these business entities within the context of the model.

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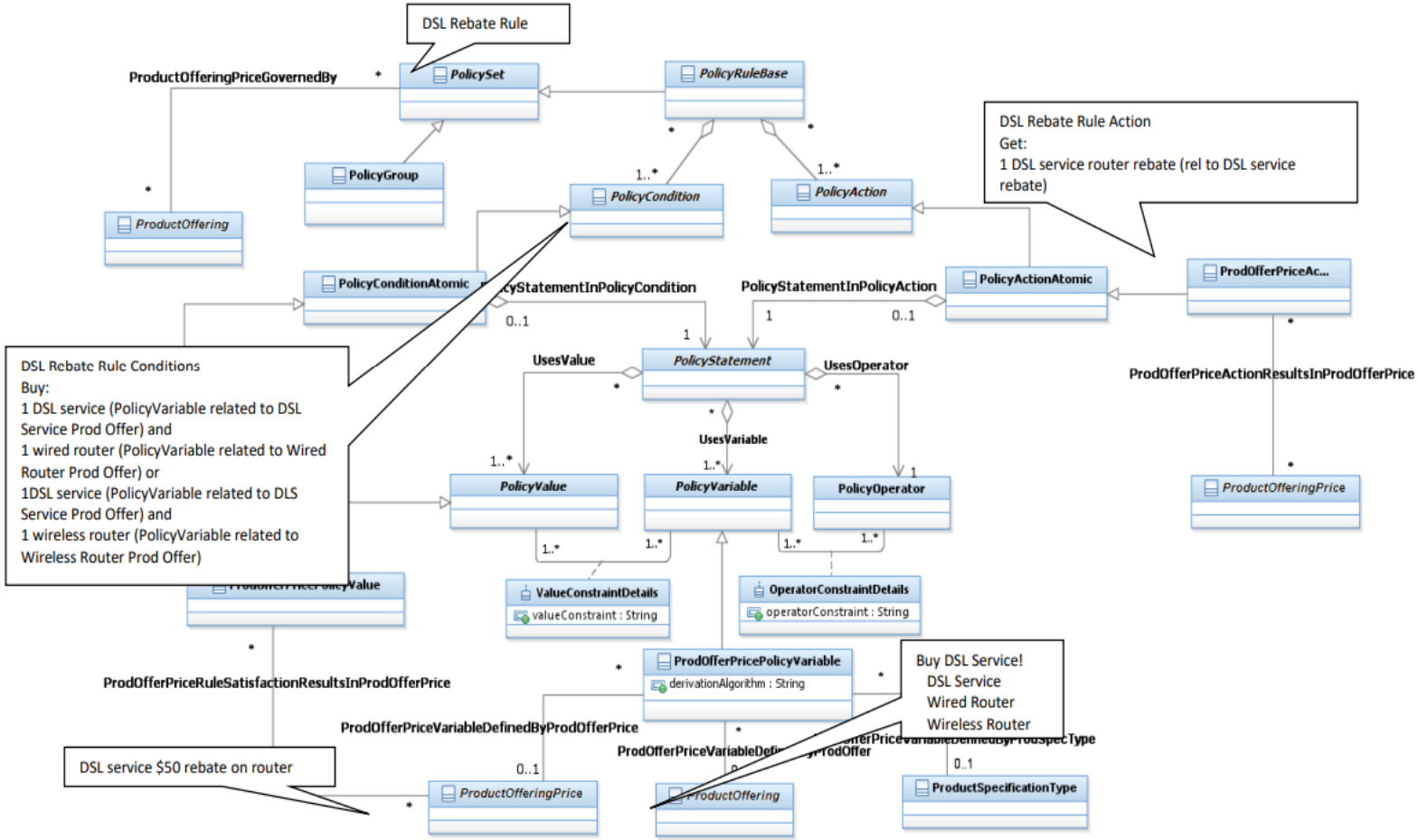

Figure Pr.21-I03 - Product Price Rule Model – DSL Service and Router Rebate

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Figure Pr.21-I04 - Product Price Rule Model – Price Rules for Like Products

The next example shows a price rule that allows a discount on like products. Suppose that for any mobile phone accessory purchased at the regular price, one whose price is equal to or less than the accessory purchased at the regular price may be purchased at a 50% discount. In this case the ProductOfferingPrices represent regular and discounted prices. In this example, rule condition types of “buy” and “price” are used. The “buy” condition specifies that two accessories must be obtained. The “price” condition specifies the relationship between the prices of the two accessories; the price of the discounted accessory must be less than or equal to the price of the accessory obtained at the regular price.

Of course, the policy model cannot represent all the behavior implied in this example; application code would have to be written to interpret a rule such as this. For example, when a customer purchases a cell phone wallet (an accessory), the application would have to check to see if there are price rules associated with compatible accessories. It would find the “buy one get one at 50%” rule. The customer could be queried to see if a purchase of a second accessory is of interest. The figure below shows the instances of these business entities within the context of the model.

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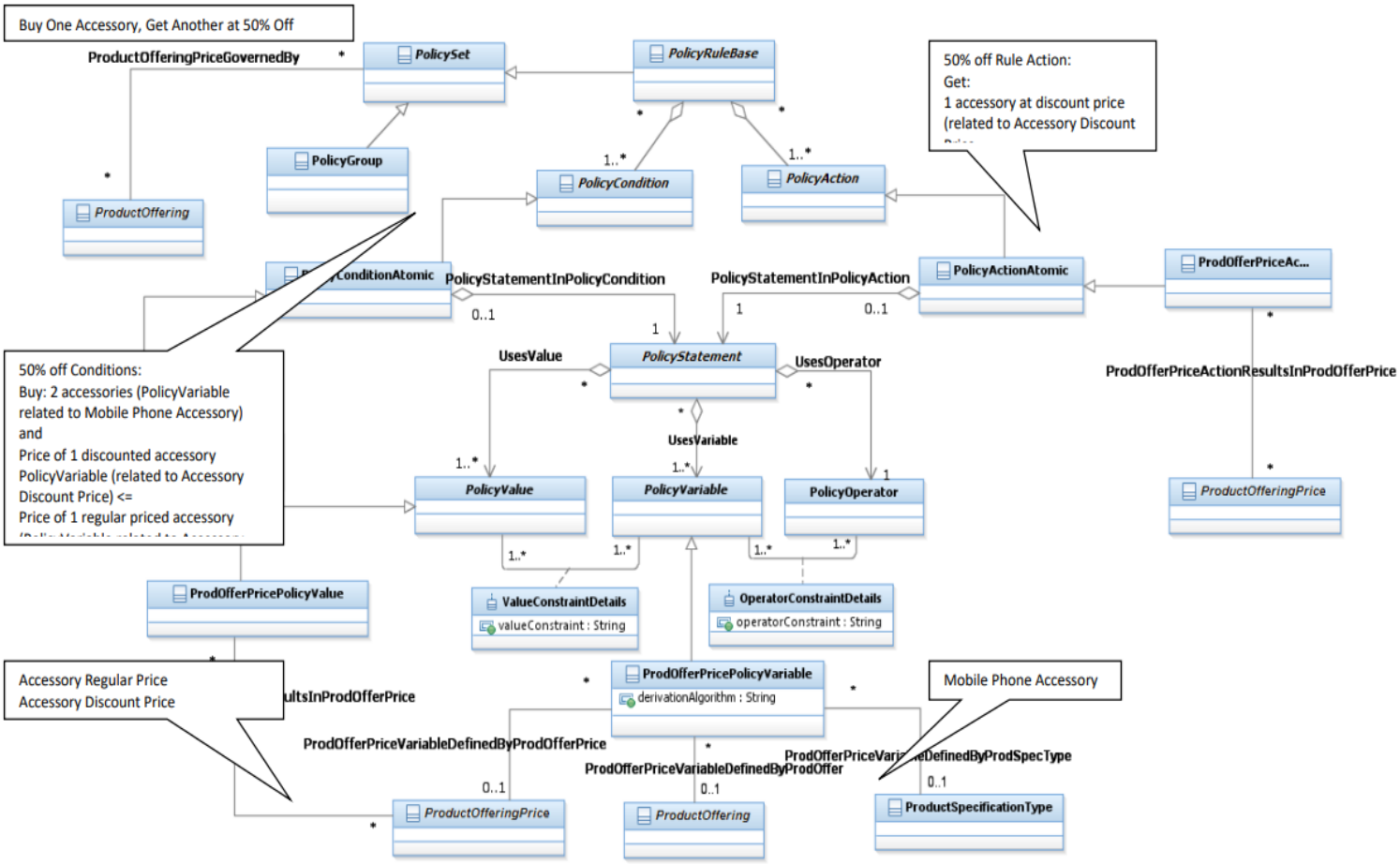

Figure Pr.21-I04 - Product Price Rule Model – Price Rules for Like Products

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Figure Pr.25 - PolicySet Association with Business Interaction

The ProductOfferingSpecification price rules described here can also be used to represent many other types of pricing rules. Some additional examples include:

- Defining conditions based on the prior procurement of a product offering possibly within a certain time period
- Defining quantity based discounts
- Defining quantity-tier based discounts

As can be seen by the examples provided here, this simple model provides a variety of pricing options that can be used by a provider.

Prices are often associated with ProductOfferingSpecifications involved in BusinessInteractions, such as CustomerOrders, Agreements, RequestsForQuotes, and so forth. Additionally, these prices are sometimes influenced by rules that govern ProductOfferingPrices. Figure below, shows these associations.

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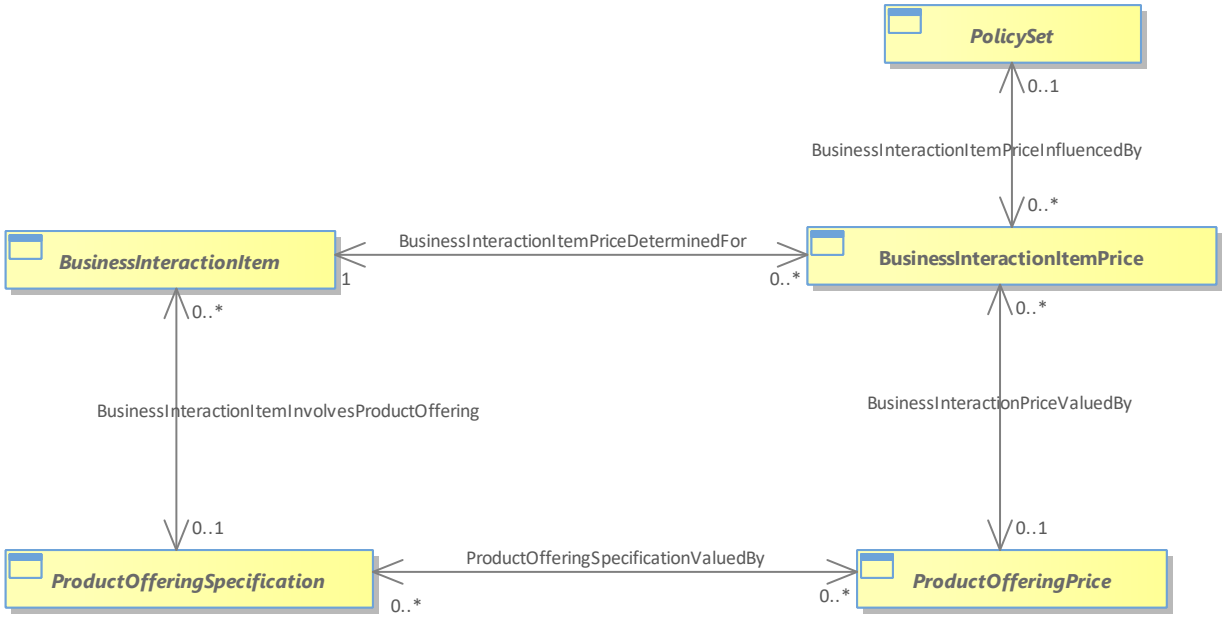

Figure Pr.25 - PolicySet Association with Business Interaction

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Illustration Pr.25 – I01 - Pricing Logic Algorithm

The PricingLogicAlgorithm (PLA) entity is designed to support the link to a “black box” that is used for rating events. This is the common case for usage rating but this scenario exists also for rating recurring and one-time charges. In such cases the price is not kept in the catalog nor the rules used to determine the price, only the interface to the black box algorithm is modelled and prices are determined by sending the relevant information using the interface, and getting the rate for the event from this black box algorithm. The PricingLogicAlgorithm and its related entities (especially PricingLogicAlgorithmSpec) are designed to model flexible interface definition in order to provide maximal reuse for existing rating algorithms.

Use Case 1: Linear Call Rating

This use case describes one of the simplest possible rating algorithms for call usage events – a linear rating according to call duration. Such an algorithm may require two parameters in order to be instantiated (just an example):

Rate for a minute

Interval for calculating the rate

This algorithm is charging the call based on call duration in intervals according to the second parameter, so if the interval is 5 seconds, a call between 1 and 5 seconds would be charged for 5 seconds, between 6 and 10 seconds would be charged for 10 seconds and so on. In any case the charge will be for a whole number multiplication of 5 seconds.

The following table illustrates such an algorithm – in all cases the rate is $0.30/minute

Using such an algorithm, instantiation of $.12/minute and interval of 5 seconds will always yield a whole number (in cents) and so $.10/minute and 6 seconds interval. On the other hand, a rate of $.07/minute and 1 second interval will almost always yield charges with fractions of cents.

Note that in order to calculate the rate you need to set the rate and interval (which is something that is communicated to the customer and typically done during the customer order), and also pass the Usage event details (Usage class in SID) to the rating algorithm (in order to get event relates data, in our use case only the call duration is important).

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Call Duration (secs)	Interval (secs)	Call charged for (secs)	Call cost ($)
20	5	20	0.10
20	6	24	0.12
32	6	36	0.18
32	10	40	0.20
32	16	32	0.16
32	20	40	0.20

Illustration Pr.25 – I01 - Pricing Logic Algorithm

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Illustration Pr.25 – I02 - Time and Customer Information Sensitive Algorithm

This use case describes a more complex algorithm in which there are more parameters which are also more complex. Let's assume we have an algorithm which charges differently for peak and off-peak calls (and can also split the charge of a call that starts peak and ends off-peak or vice versa). Let's also assume that the rate per minute is different according to customer type (student, normal, VIP)

In order to instantiate such an algorithm, we need to set the following parameters.

- Define which time is peak and which is off-peak (we can have one instantiation in which peak is 10-15 Mon-Thu and another one which is 8-18 Mon-Thu)
- Define the rate per combination of customer type and peak/off-peak. Such a matrix could look like

Note – this matrix is not necessarily two dimensional in some cases it has more dimensions (the rate depends on time (peak/off-peal), customer type and local/long distance rates)

	Peak	***Off-Peak***
Student	$.15/min	$.10/min
Normal	$.20/min	$.12/min
VIP	$.10/min	$0/min (free)

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Figure Pr.26 - Pricing Logic Algorithm Base Entities

What we try to model in the SID is the interface to these algorithms. The algorithms can be implemented in many different ways by different systems but we need to model the interface to them with all the parameters in order to be able to use these algorithms in our systems

PricingLogicAlgorithm is a classic case of the EntitySpec/Entity pattern as described in the above use cases.

This diagram uses the generic EntitySpec/Entity pattern so that PricingLogicAlgorithmSpec automatically gets the characteristics that are associated with EntitySpecification. The PricingLogicAlgorithm is an instantiation of PricingLogicAlgorithmSpec after the parameters were set, so in the case of the simple use case above it represents linear usage rating, $0.10/minute with 6 seconds interval. The parameters ($0.10 and 6) are characteristic values.

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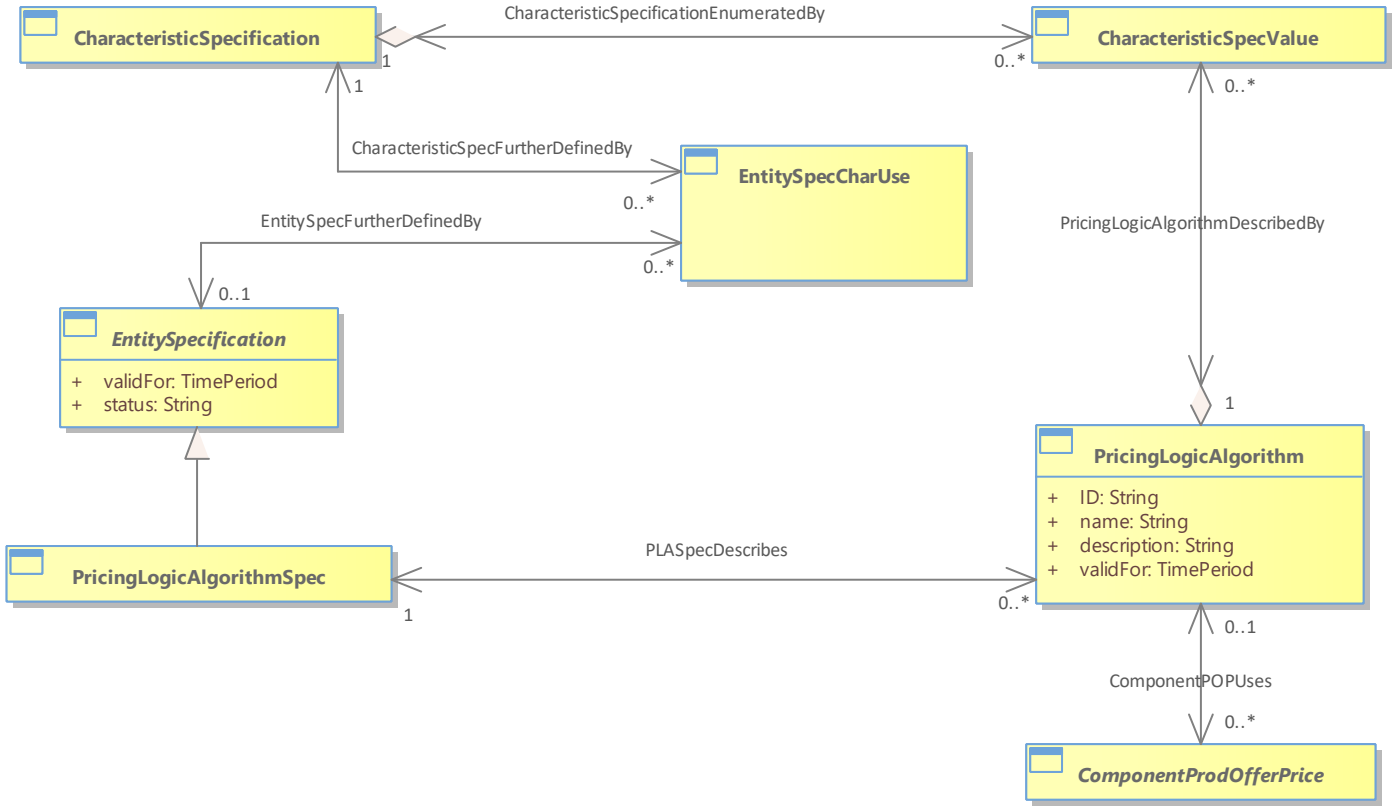

Figure Pr.26 - Pricing Logic Algorithm Base Entities

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Figure Pr.27 - Types of PLA Spec and associations to different Spec Characteristics

This diagram shows the association of the PricingLogicAlgorithmSpec to the other sources of parameters to the algorithm (not only PLA characteristics, but also ProductSpecCharacteritics and UsageSpecCharacteristics).

It completes the definition of PricingLogicAlgorithmSpec and together with Figure Pr.26 it shows all the sources of parameters which are needed to be defined and associated with the PLA spec in order to create an instance (PricingLogicAlgorithm) which has enough information (with Product and Usage information) in order to rate an event.

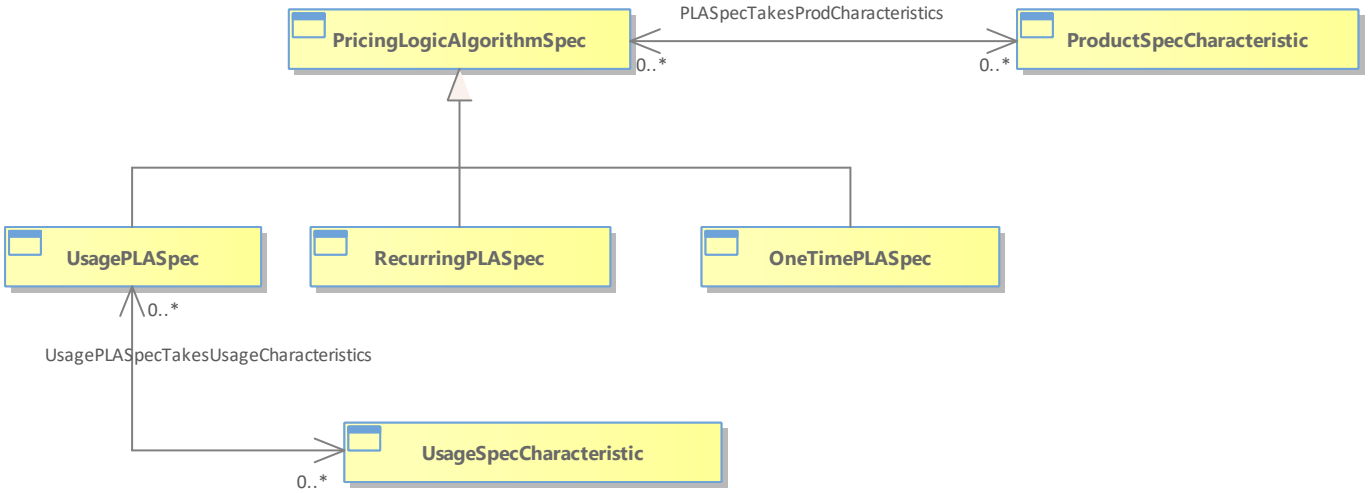

Figure Pr.27 - Types of PLA Spec and associations to different Spec Characteristics

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Figure Pr.28 - Types of Pricing Logic Algorithms and associations to different characteristic values

This diagram shows how the PricingLogicAlgorithm uses the ProductCharacteristicValue and the Usage record information and associated UsageCharacteristicValue in order to generate a charge out of an event:

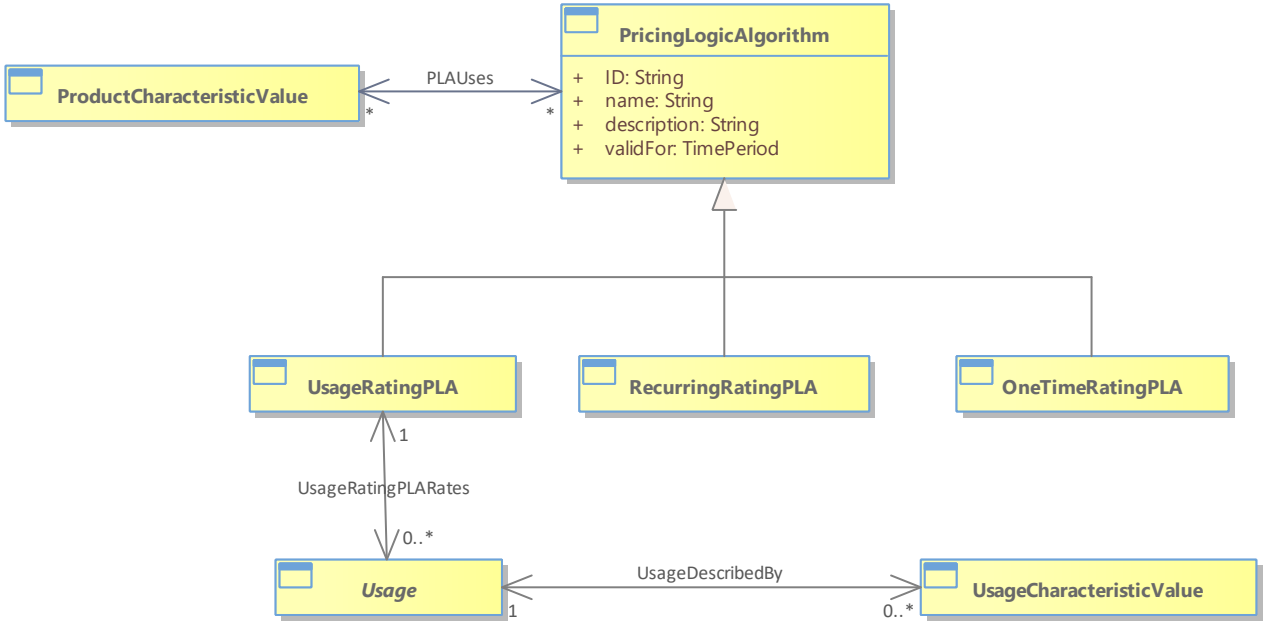

Figure Pr.28 - Types of Pricing Logic Algorithms and associations to different characteristic values

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Figure Pr.29 - Pricing Logic Algorithm full picture

The following diagram shows the entire picture – how PLA Spec is defined, how PLA is instantiated, and the association to the possible event related characteristic values.

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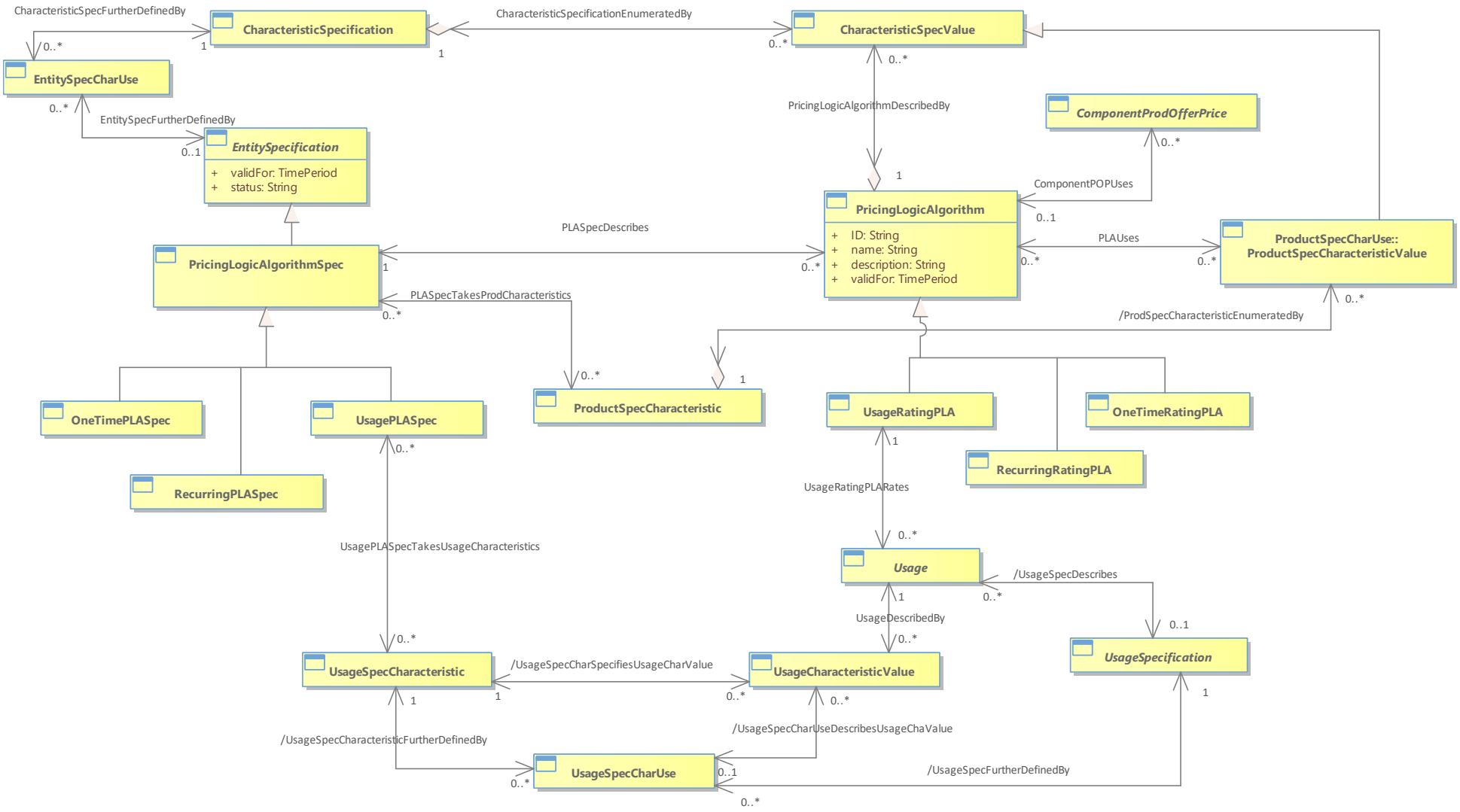

Figure Pr.29 - Pricing Logic Algorithm full picture

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Figure Pr.30 - Modeling a matrix with dimensions based on PLASpec and ProductSpec Characteristics

As described in the use cases, a typical implementation of PricingLogicAlgorithm requires parameters which are multi-dimensional matrices, in which each dimension is defined by the valid values for a specific characteristic. For example, if we have PLA which rates events based on customer type, we can have one instantiation in which customer type is either residential or commercial (and in which case the size of the dimension of the matrix is 2) and in other instantiation the customer type can be student, normal and senior citizen (in this case the size of the dimension is 3).

Modeling the matrix.

As described above, the matrices are defined by the possible valid values of characteristics (from ProductSpec or PLASpec), so we can define the matrix itself using the self-association CharacteristicSpecificationReferences. This was correct if CharacteristicValue and ProductCharacteristicValue were the same entity. Since currently these entities do not share a common base class there is a need to create some more complex modeling in order to create a matrix.

Note: MatrixSpecDimention is associated to a single entity either ProductSpecCharacteristic or CharacteristicSpecification – but it is always associated to one (association to none or both entities is not allowed, so this class (MatrixSpecDimention) is a specification for a single dimension of the matrix.

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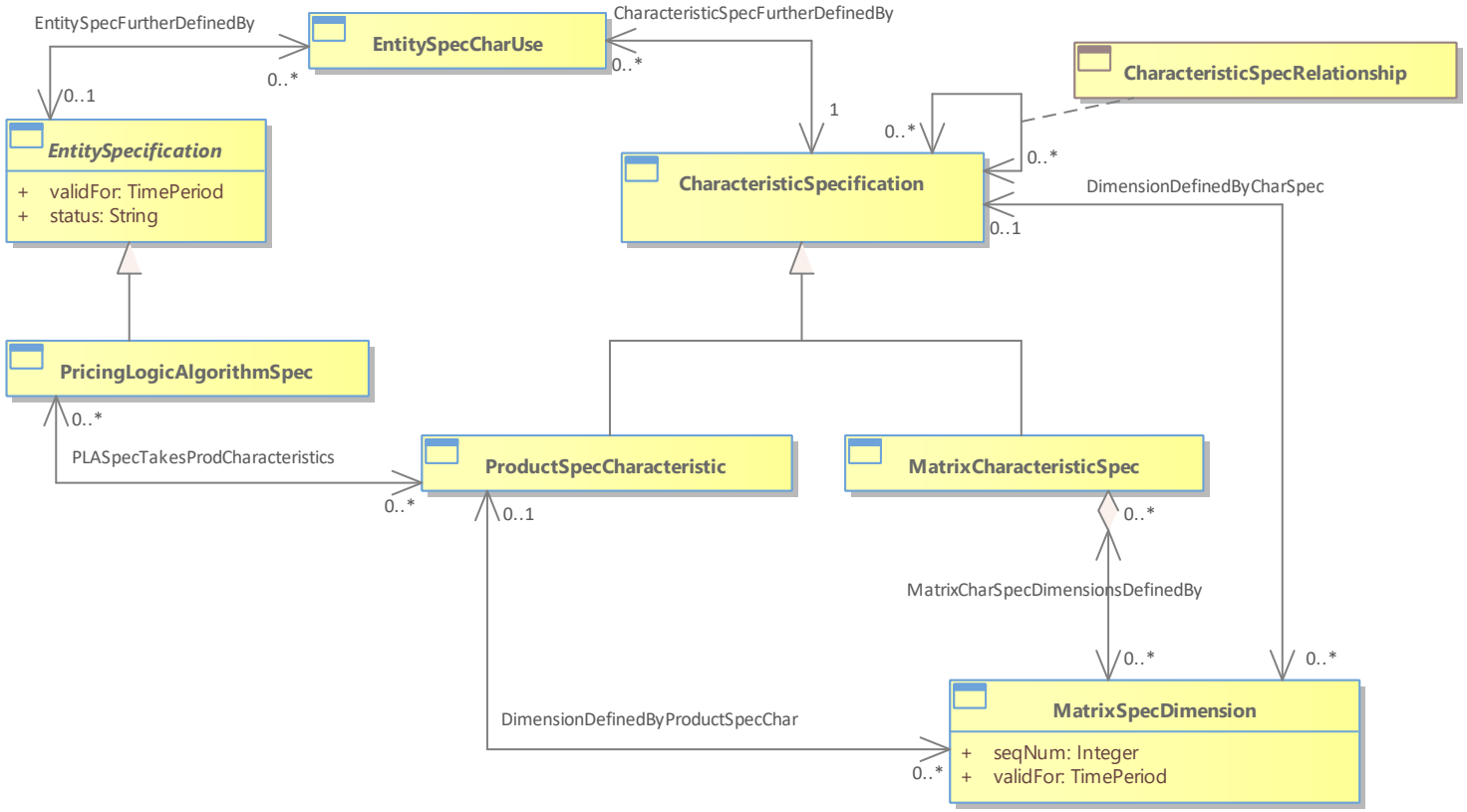

Figure Pr.30 - Modeling a matrix with dimensions based on PLASpec and ProductSpec Characteristics

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Figure Pr.31 - Modeling matrix values

We can look at each value (cell) of n dimensional matrix as n+1-tupple of Characteristic Values, one describes the cell value and the other n are the indices of the dimensions (not numeric indices, but the values are themselves indices). Since each index can be either CharacteristicValue or ProductCharacteristicValue we cannot use the self-association for the CharacteristcValue class. Instead we define the class MatrixCharValueIndex which associate either CharacteristcValue or ProductCharacteristicValue. In addition we define a class which is derived from CharacteristicValue which represent a matrix value and includes associations to the matrix dimensions (defined by either CharacteristicValue or ProductCharacteristicValue) and aggregates the matrix cell values which are derived from CharValue (to include the value) and associated with the MatrixCharValueIndex for the indices of this cell value.

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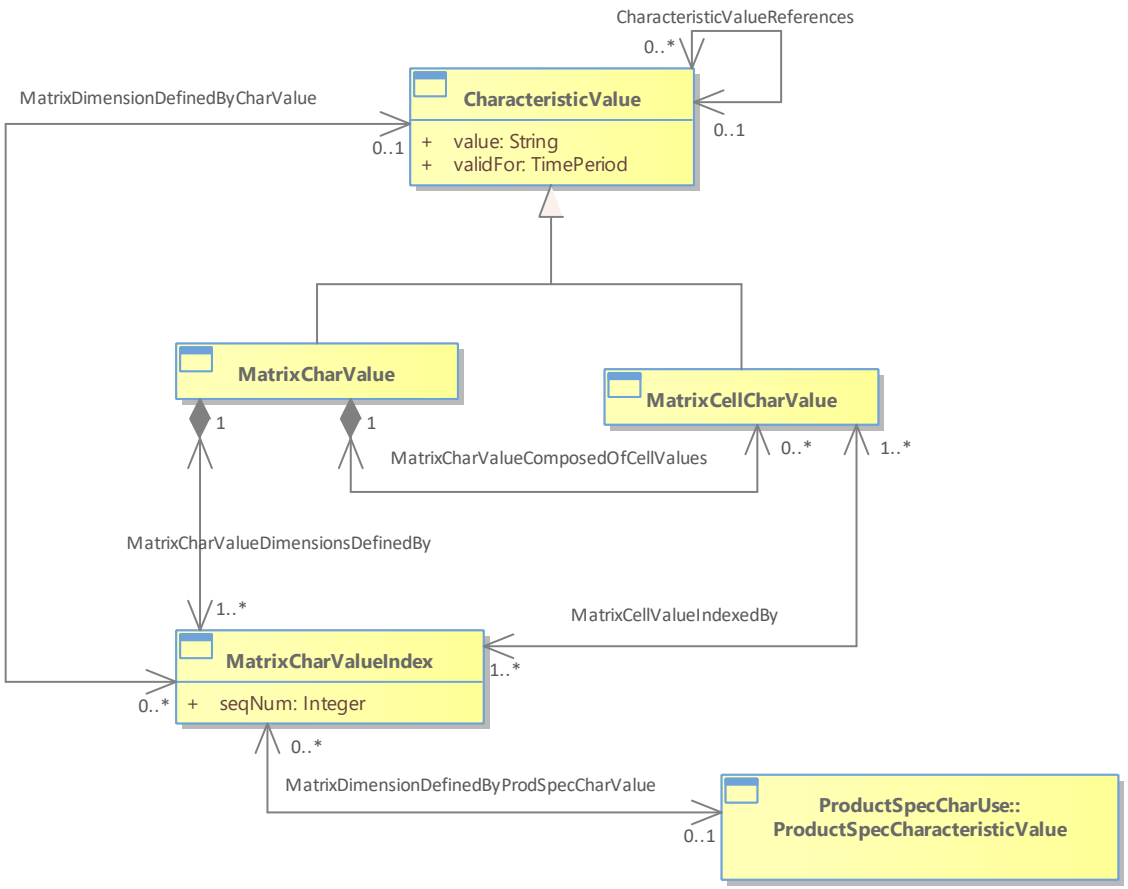

Figure Pr.31 - Modeling matrix values

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Figure Pr.44c - Product Spec Char Values and Product Offer

The Figure shows the Characteristic Value Use association between characteristic and ProductOfferingSpecification, that allows the offering to restrict the possible values from the complete list at the Product Specification level. This association can be applied, if the ProductOfferingSpecification determines the applicable characteristic values – e.g. the ProductOfferingSpecification “silver plan” only allows bandwidths 2000 Mbit and 5000 Mbit out of the values 1000Mbit, 2000Mbit, 5000Mbit and 8000Mbit that are possible for the ProductSpecCharValueUse “bandwidth” in general.

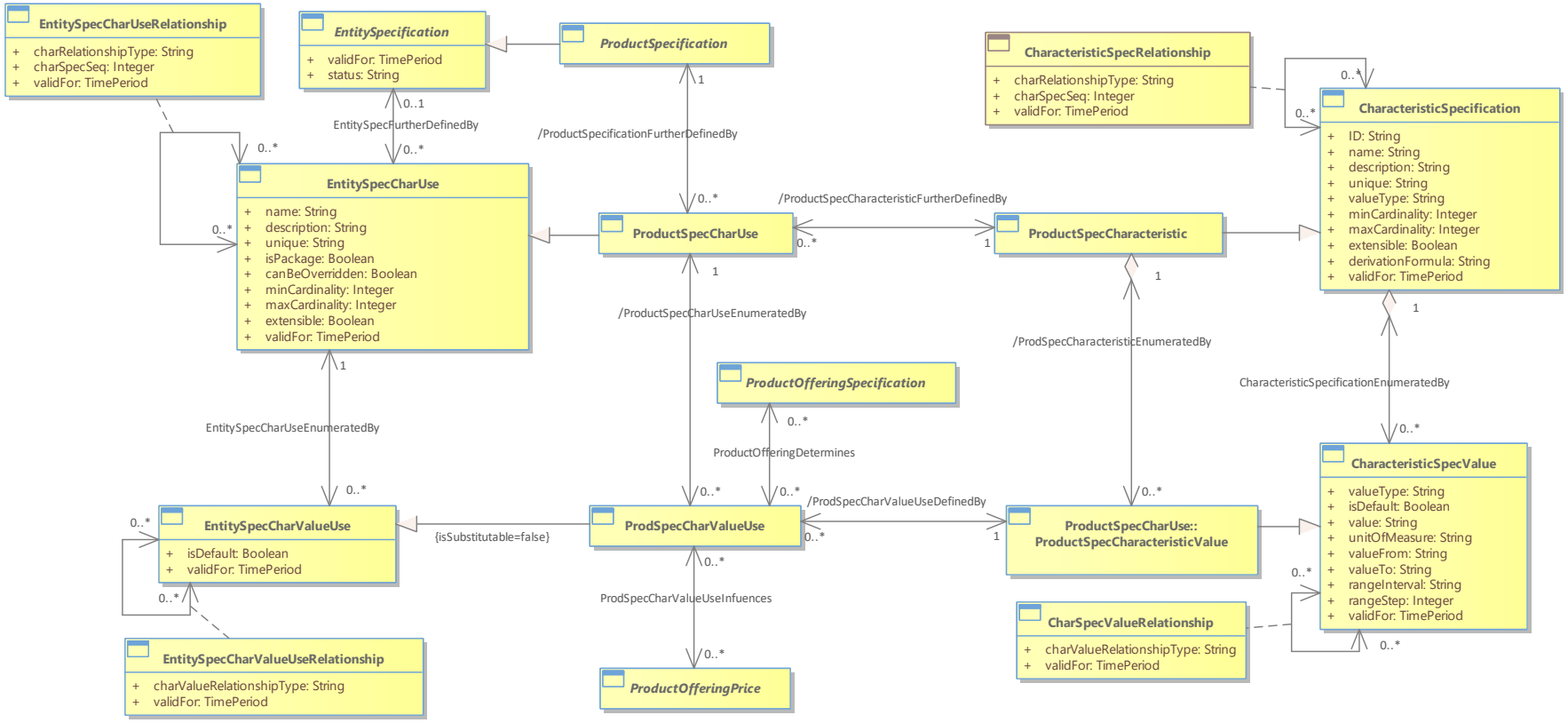

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Figure Tax.01 - Tax Definition

A tax is levied by an authorized TaxJurisdiction. A jurisdiction is a specific PartyRole, that includes the administrative level (e.g. country, state) that represents the extent of the governmental authority.

There are many different types of tax (including income tax, municipal property tax, etc.), but for the purposes of this document we consider only taxes that apply on amounts paid by Parties for Products (whether one-time purchase fees or ongoing charges). A single price may have multiple taxes applied, by the same or different jurisdiction, e.g.:

- Federal Tax – levied by the US government
- State Tax – levied by the State of California
- County Tax – levied by the County of Santa Monica

A category specifies a level of tax for a specific type of TaxDefinition type such as Standard, Alimentary, Luxury…

If definesExemption attribute is TRUE, the TaxDefinition corresponds to a tax exemption that replaces a standard tax if a corresponding TaxExemptionCertificate grants it.

For example, a voluntary organization may be granted a complete exemption from value-added tax, to pay 0% of the standard tax.

The actual amount of tax to be paid may be:

- A fixed percentage of the price – for example Value Added Tax of 17%
- A fixed amount – for example an Eco-tax of 0.5 euros per handset device, irrespective of the device's price
- A complex calculation based on the price and other factors – for example a progressive percentage (10% for a low price, 20% for a high price); this would be represented as a policy rule.

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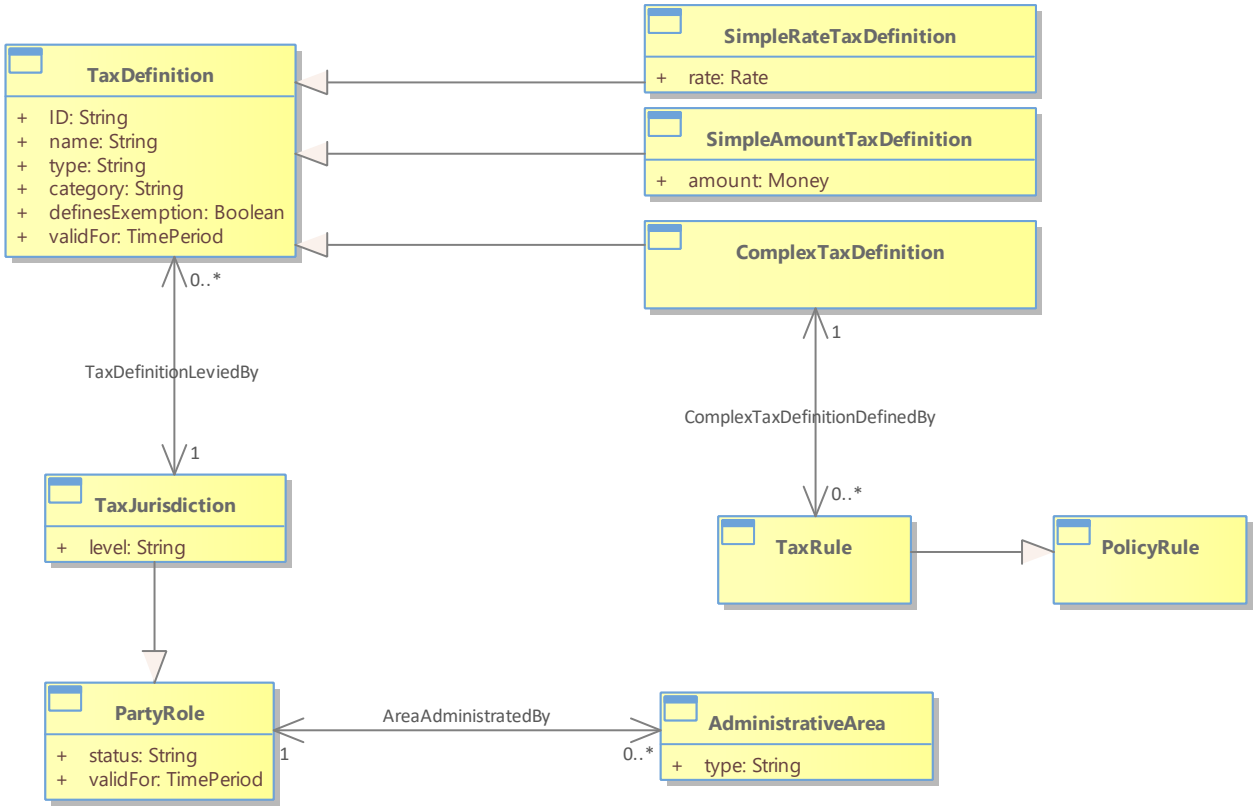

Figure Tax.01 - Tax Definition

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Figure Tax.02 - Tax ProductOfferingPriceAlteration and ProductPriceAlteration

As part of defining a ProductOfferingPrice in the product catalog, the implementor will need to define the taxes that will potentially be levied for the price. This will also allow the tax amounts to be presented as part of sales and ordering processes. Taxes are defined as alterations on the price, specifically TaxProdOfferPriceAlteration. The alteration points to the definition of the tax.

When a Product is instantiated, the actual taxes to be levied are represented as alterations on the ProductPrice, specifically TaxProdPriceAlteration. Again, the alteration points to the definition of the tax.

Finally, the tax amounts are invoiced and collected as part of the billing process. This is represented by the entities AppliedCustomerBillingTaxRate (customer-centric model) and AppliedPartyBillingTaxRate (engaged party model).

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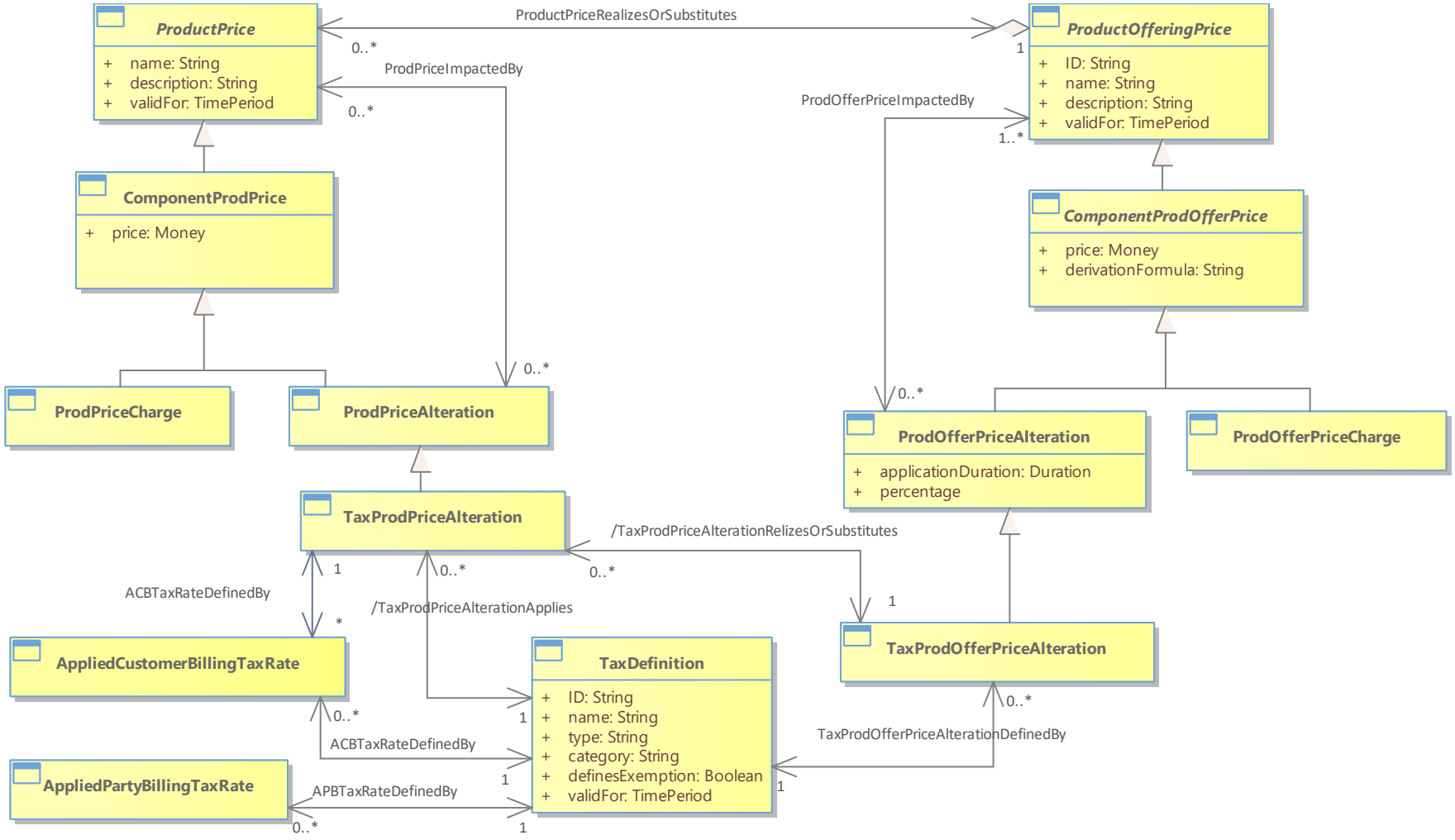

Figure Tax.02 - Tax ProductOfferingPriceAlteration and ProductPriceAlteration

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Figure Tax.01 & Tax.02 – I01 – Tax Example

The following example shows how various taxes are modelled and implemented.

The example shows two tax definitions (Sales Tax and US Federal Tax) defined by two different tax jurisdictions (USA and California).

One of the TaxDefinition is a simple tax rate (15% sales tax) levied by a California State, and the other one is a complex tax rate (10% + a fixed amount according to the amount taxed) levied by the US government.

Two alterations applying different TaxDefinitions, impact the same one-time charge and similarly two alterations applying different TaxDefinitions, impact the same recurring charge.

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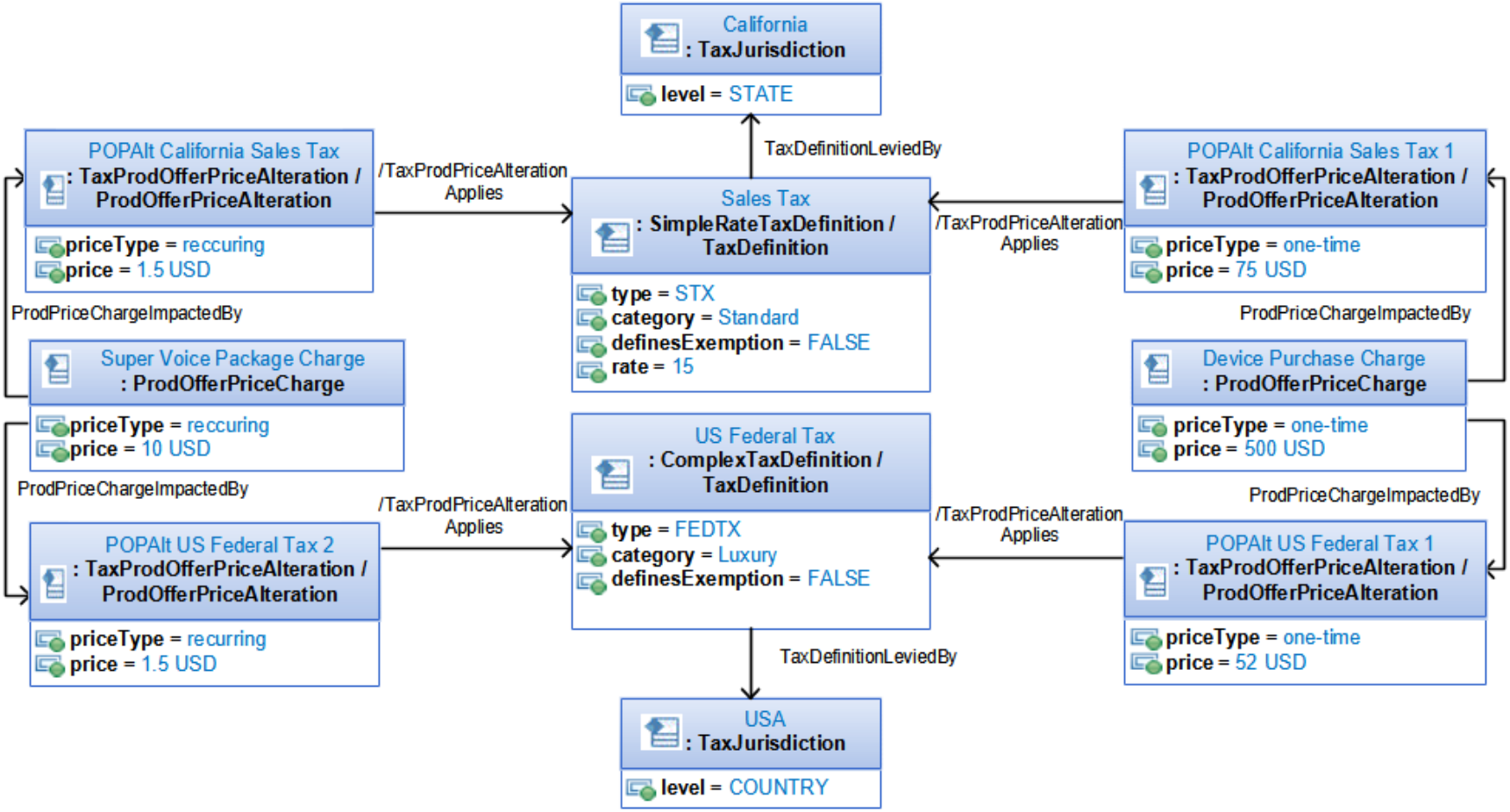

Figure Tax.01 & Tax.02 – I01 – Tax Example

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Figure Tax.03 - Tax Exemption

The actual tax to be paid may be modified by TaxExemptions, that can apply to:

- A Party – for example a war veteran may be granted a partial exemption from purchase tax, to pay 7% instead of 15%, or a voluntary organization may be granted a complete exemption from value-added tax, to pay 0%.
- A Location – for example a disadvantaged neighborhood may be granted a partial exemption from purchase tax, or a tourist zone may be granted a complete exemption from value-added tax.

The TaxExemption will be issued by the same TaxJurisdiction that levies the tax, and will typically be represented by some certificate of exemption that has an ID and perhaps a printed or electronic representation to attest the granting of the exemption.

The amount of the exemption is determined in the same way as the tax amount itself, by simple or complex calculation.

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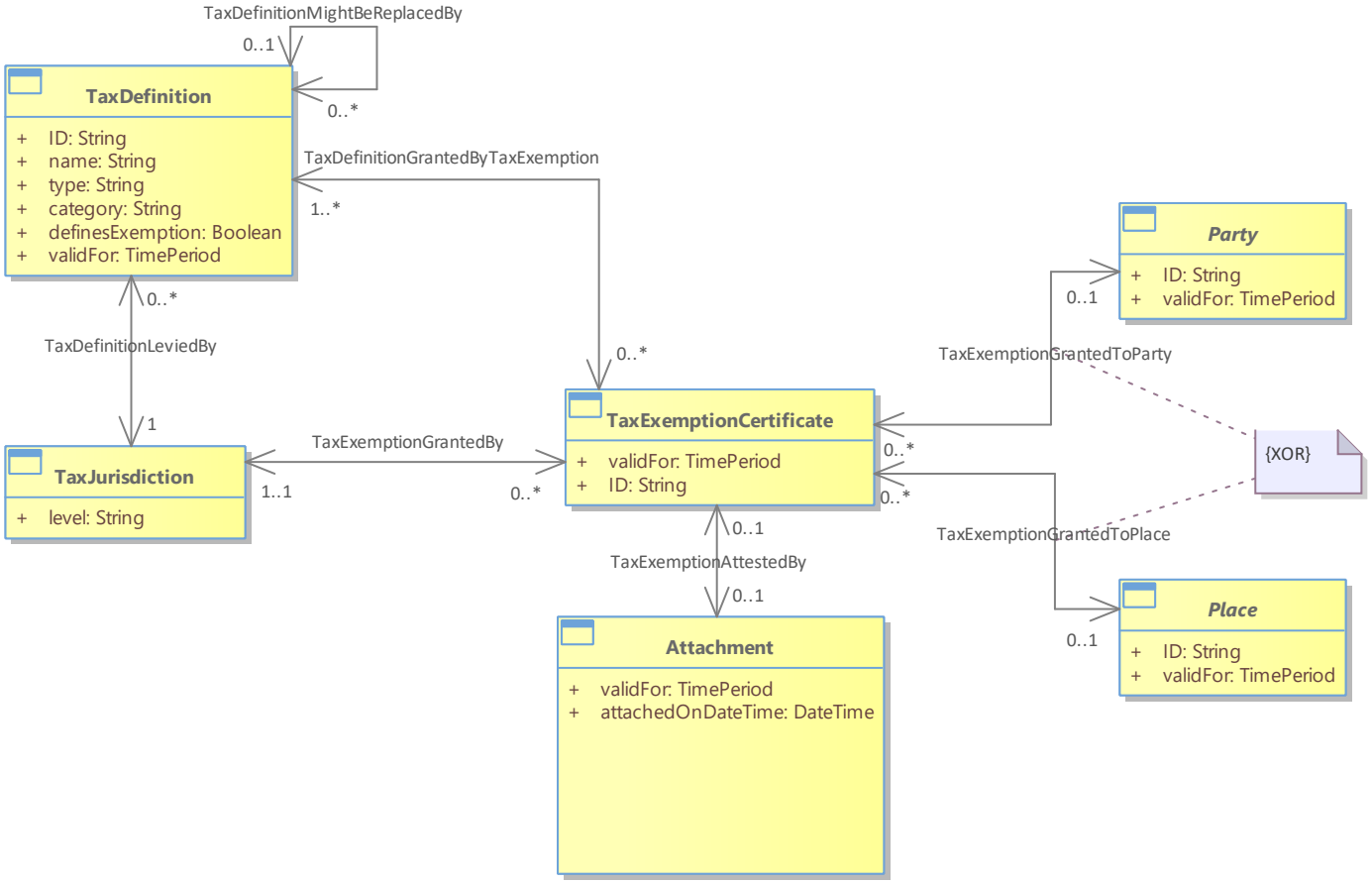

Figure Tax.03 - Tax Exemption

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Figure Tax.03 – I01 – Tax Exemption Example

The following example explains how to use the tax exemption model.

The example shows a TaxJurisdiction that defines that defines two TaxDefinitions for the same type and category: a basic and an exemption tax definition for Standard Sales Tax.

The basic tax is 15% and the reduced one is 10% if the Party has specific characteristics such as “younger than 26 years old”.

Both tax definitions impact the same charge (ProdOfferPriceCharge) but the exemption tax is only applied if the Party has a tax exemption certificate granting the exemption tax definition.

John Smith is a Party (Individual) that is granted by this exemption certificate.

So the pricing process will apply for John Smith the exemption tax.

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4.3.1 Pricing Logic Algorithm ABE

The Pricing Logic Algorithm ABE contains entities representing an instantiation of an interface specification to external rating function (without a modeled bahavior in SID).

Figure PLA.01 - Pricing Logic Algorithm ABE Related Entities

Following are the business entities aggregated under the Pricing Logic Algorithm Aggregate Business Entity.

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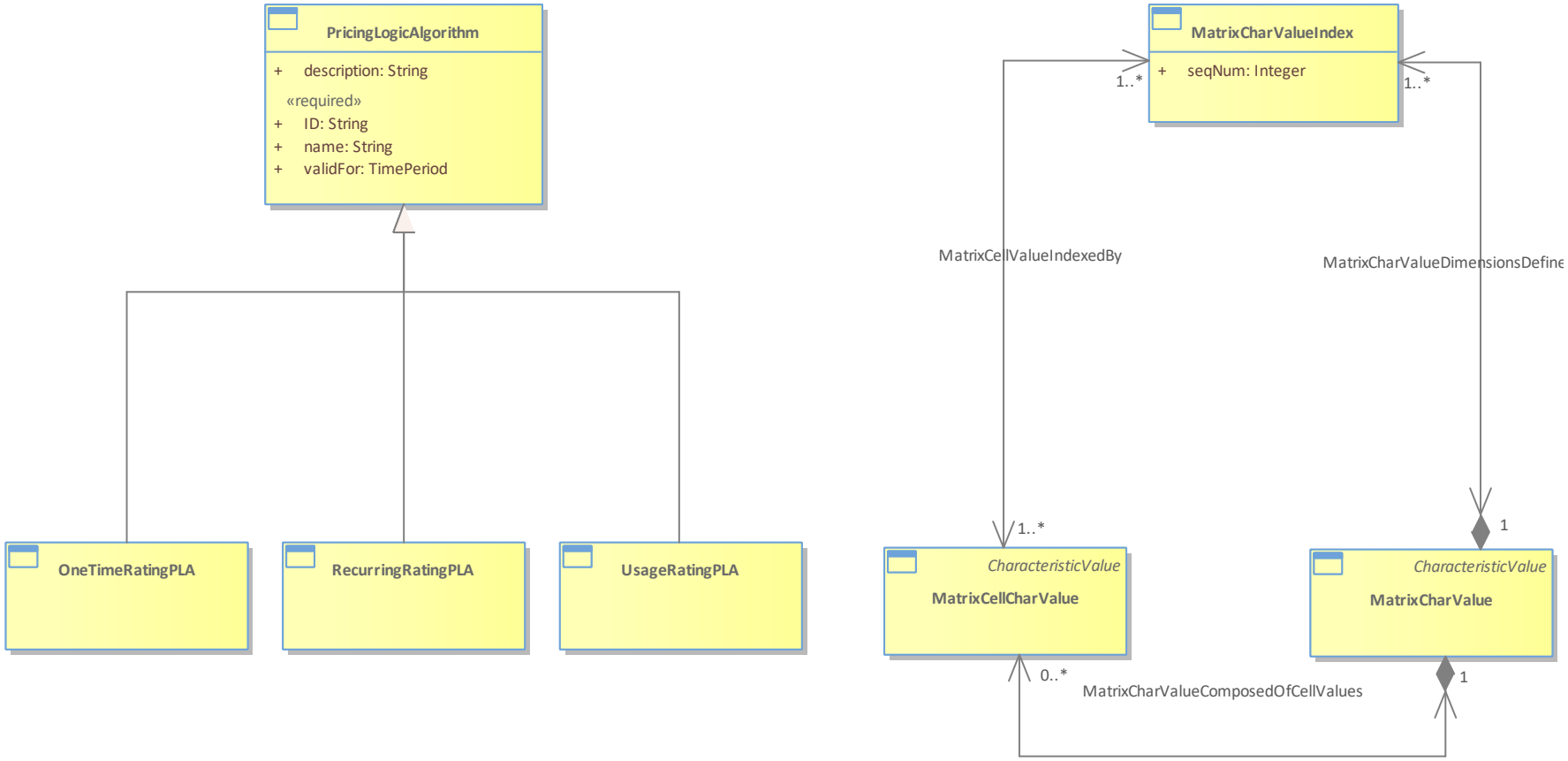

Figure PLA.01 - Pricing Logic Algorithm ABE Related Entities

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4.3.1.1 PLA Spec ABE

The PLA Spec ABE specifies Pricing Logic Algorithm.

Figure PLAS.01 - PLA Spec ABE Related Elements

Following are the business entities aggregated under the **PLA Spec** Aggregate Business Entity.

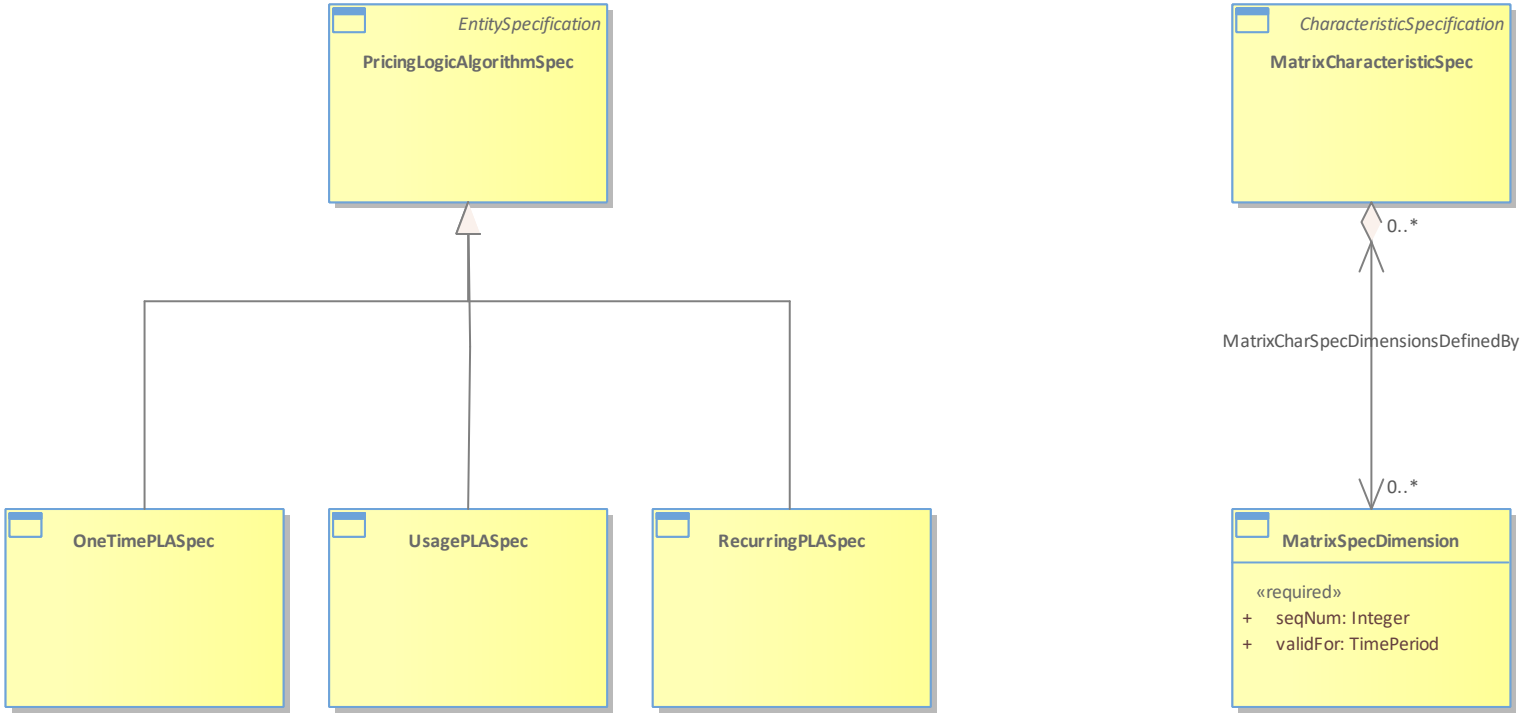

Figure PLAS.01 - PLA Spec ABE Related Elements

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4.3.2 Product Catalog ABE

Product Catalog ABE is a collection of entities that represent the specification of ProductCatalogs and catalogs described by them that contain ProductOfferingSpecifications and their associated ProductOfferingPrices.

Figure PC.01 - Product Catalog ABE Related Elements

Following are the business entities aggregated under the **Product Catalog** Aggregate Business Entity.

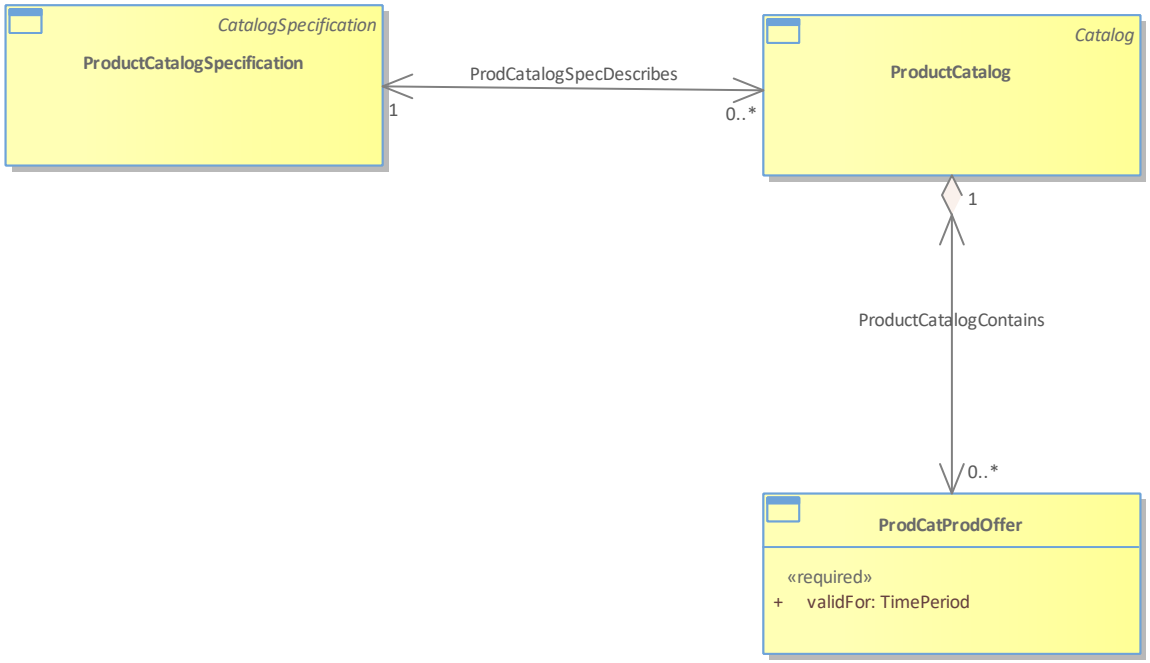

Figure PC.01 - Product Catalog ABE Related Elements

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4.3.3 Product Offering Price ABE

The Product Offering Price ABE describes all the prices related to ProductOfferingSpecifications.

Figure POP.01 - Product Offering Price ABE Related Entities

Following are the business entities aggregated under the **Product Offering Price** Aggregate Business Entity.

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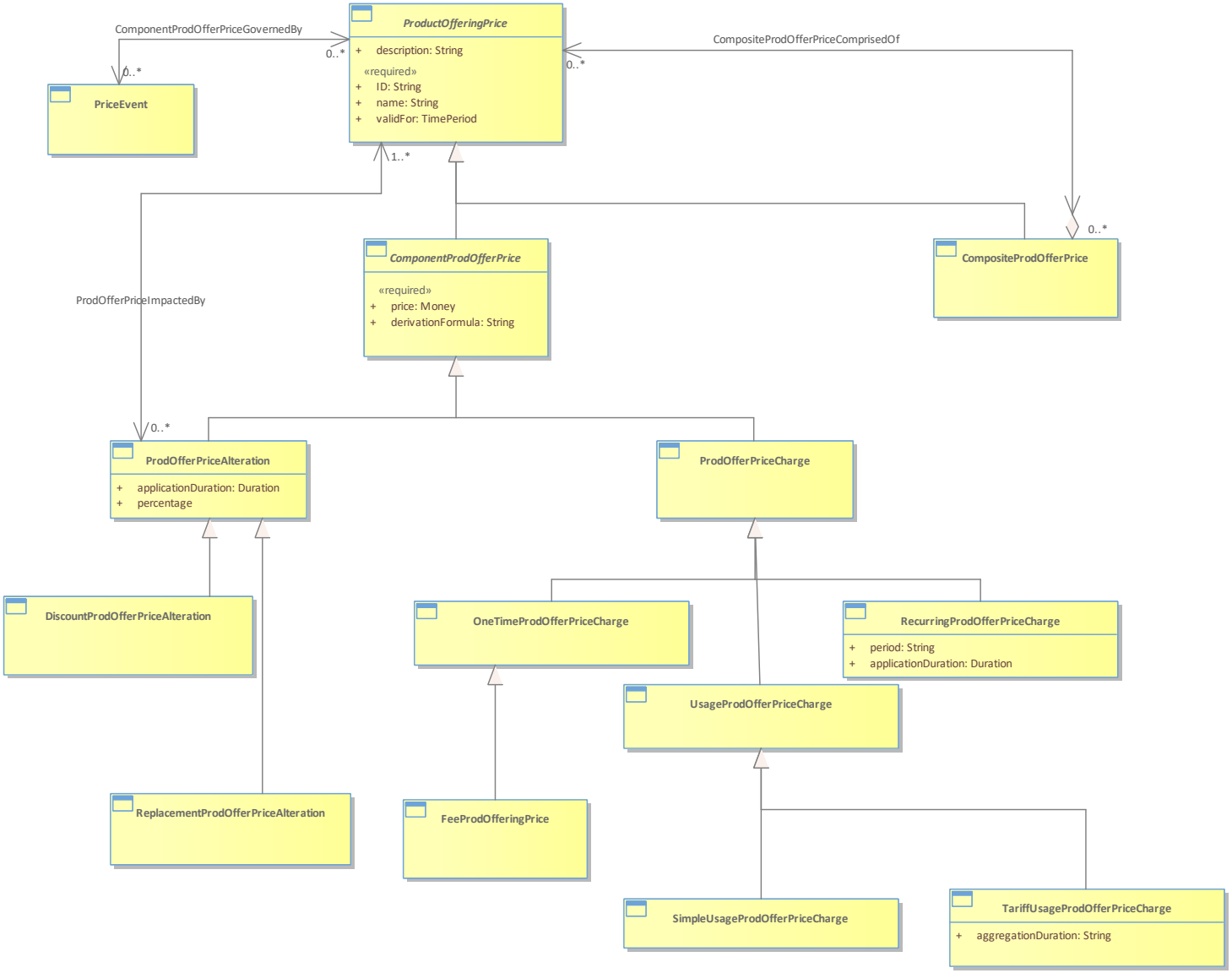

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4.3.4 Product Offering Price Rule ABE

The Product Offering Price Rule ABE describes all rules related to prices concerning ProductOfferingSpecifications.

Figure POPR.01 - Product Offering Price Rule ABE Related Elements

Following are the business entities aggregated under the Product Offering Price Rule Aggregate Business Entity.

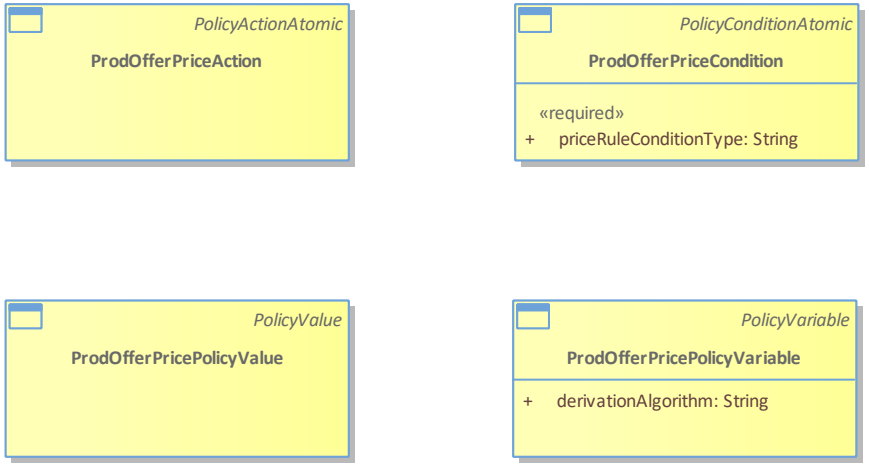

Figure POPR.01 - Product Offering Price Rule ABE Related Elements

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4.3.5 Product Offering Price Tax ABE

The Product Offering Price Tax ABE describes how taxation is modelled in the context of ProductOfferingPrice and ProductPrice. The model covers the following aspects:

- TaxDefinition and TaxExemption - these specify what the tax is and how it is calculated, including the influence of specific circumstances on the calculation
- TaxProdOfferPriceAlteration - this is an alteration to a ProductOfferingPrice that projects how much tax will be added for a ProductOfferingSpecification
- TaxProdPriceAlteration - this is an alteration to a ProductPrice that represents how much tax will be added for a Product that was actually purchased and delivered to a Party. The tax amounts are ultimately represented on the invoice as AppliedCustomerBillingTaxRate (or AppliedPartyBillingTaxRate as per the Engaged Party model).

Figure POPT.01 - Product Offering Price Tax ABE Related Entities

Following are the business entities aggregated under the **Product Offering Price Tax** Aggregate Business Entity.

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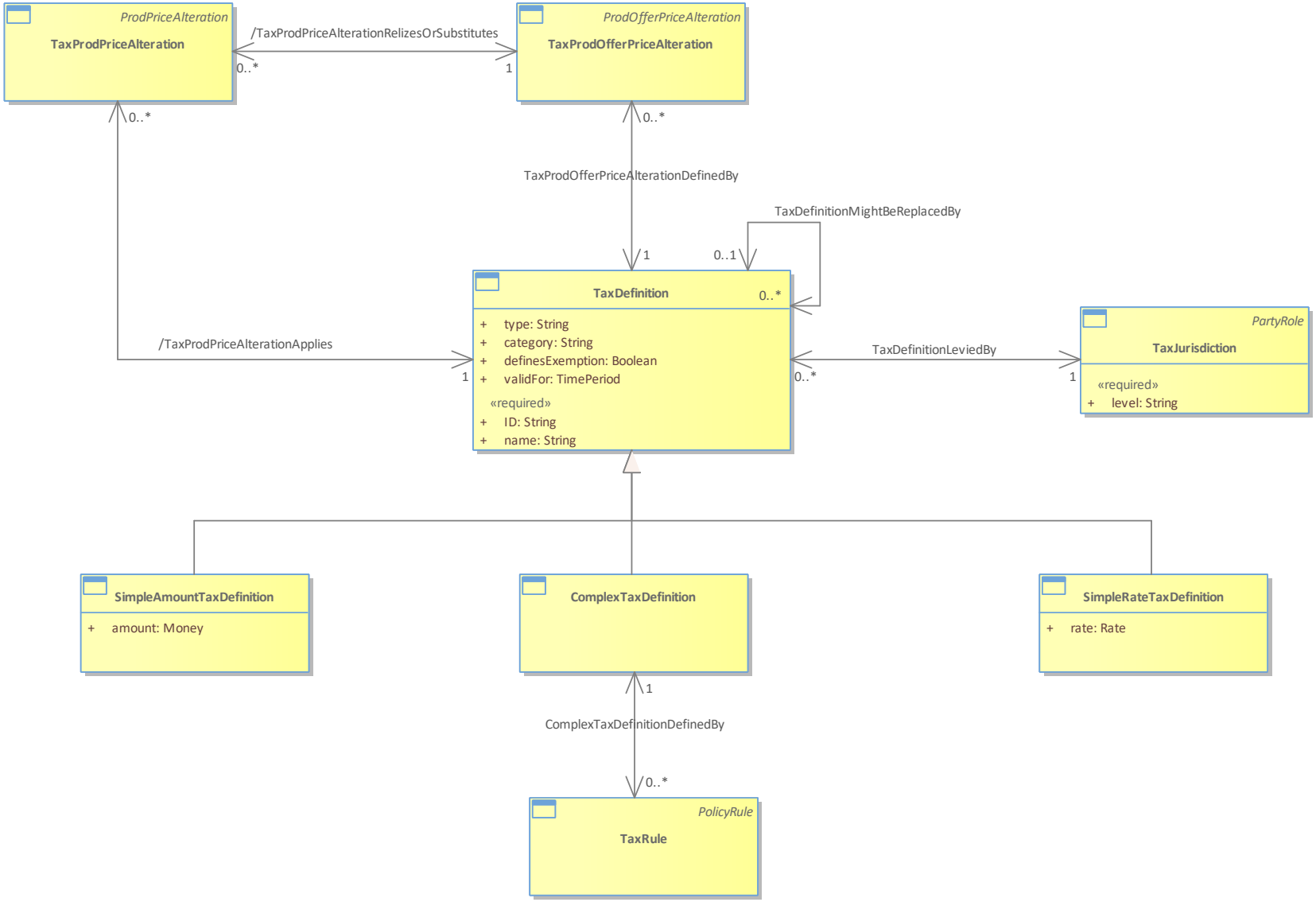

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4.4 Product and Offering Instance ABE

Product and ProductOfferingInstance represent, in the product inventory, respectively an instance of a ProductSpecification and ProductOfferindSpecification subscribed or not yet subscribed to by a Customer from the CSP or by the CSP from a BusinessPartner. The Product carries the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The **Product and Offering Instance ABE** also tracks the links to services and/or resources through which the Product is realized as well as the applied tariff and commitment duration related to the ProductOfferingInstance.

Figure Pr.07a1 - Simple Product & Offering Spec and instances

Each Product corresponds to a unique ProductSpecification provided to a party. A Product might be either contained in another Product (CompositeProduct) or in a unique ProductOfferingInstance at one time.

Each ProductOfferingInstance corresponds to a unique ProductOfferingSpecification provided to a party. It might be either be composite (CompositeProductOfferingInstance) and then contain ProductOfferingInstance or be atomic (AtomicProductOfferingInstance) and contain a unique Product.

A ProductOfferingSpecification is marketed exclusively by the CSP playing a PartyRole of MarketingOperator or by a PartyRole Supplier played by another company.

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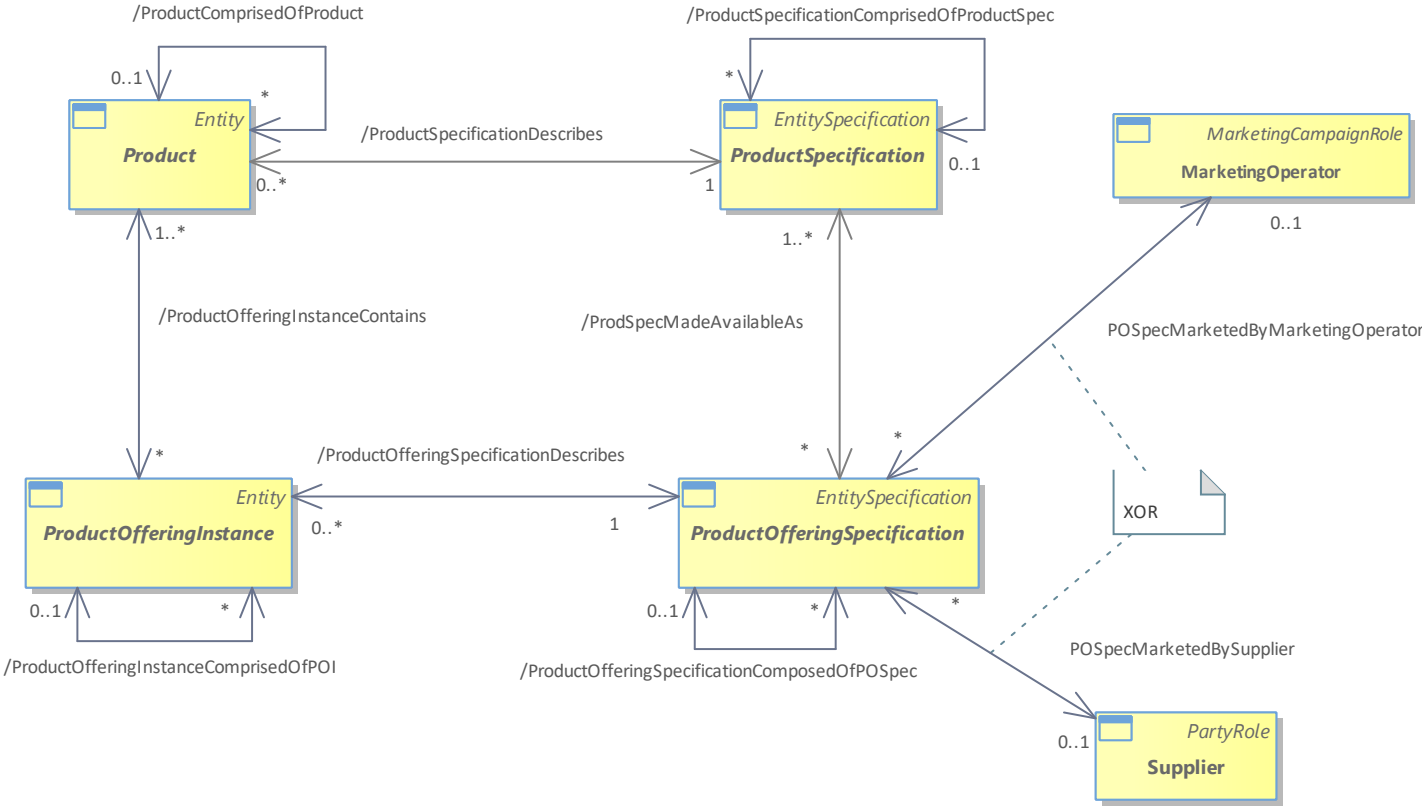

Figure Pr.07a1 - Simple Product & Offering Spec and instances

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Figure Pr.07a2 - CSP Product & Offering Spec and instances

When the ProductOfferingSpecification is marketed by the MarketingOperator i.e. the CSP, when a Customer buys it, he will hold the corresponding ProductOfferingInstance and by navigation the contained ProductOfferingInstances and the related Products.

In this case, the ProductOfferingInstance is charged to one or many CustomerBillingAccounts.

Note: The ProductOfferingInstance might be charged to many CustomerBillingAccounts but no split billing rules are specified for the moment in SID.

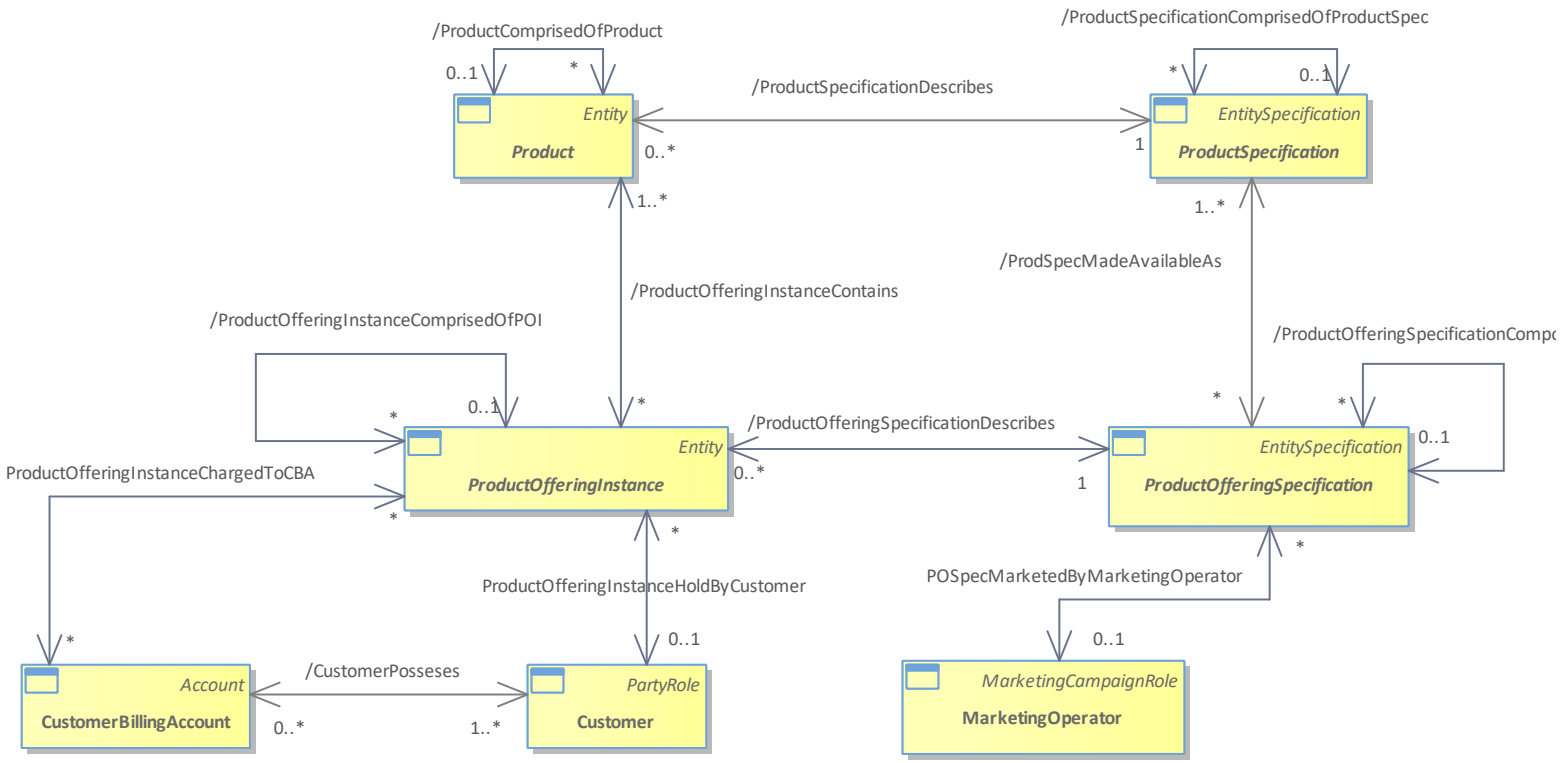

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Figure Pr.07a3 - Supplier Product & Offering Spec and instances

When the ProductOfferingSpecification is marketed by Supplier, when the ServiceProvider buys it, he will hold the corresponding ProductOfferingInstance and by navigation the contained ProductOfferingInstances and the related Products.

In this case, the ProductOfferingInstance is charged to one or many BusinessPartnerAccounts.

Note: The ProductOfferingInstance might be charged to many BusinessPartnerAccounts but no split billing rules are specified for the moment in SID.

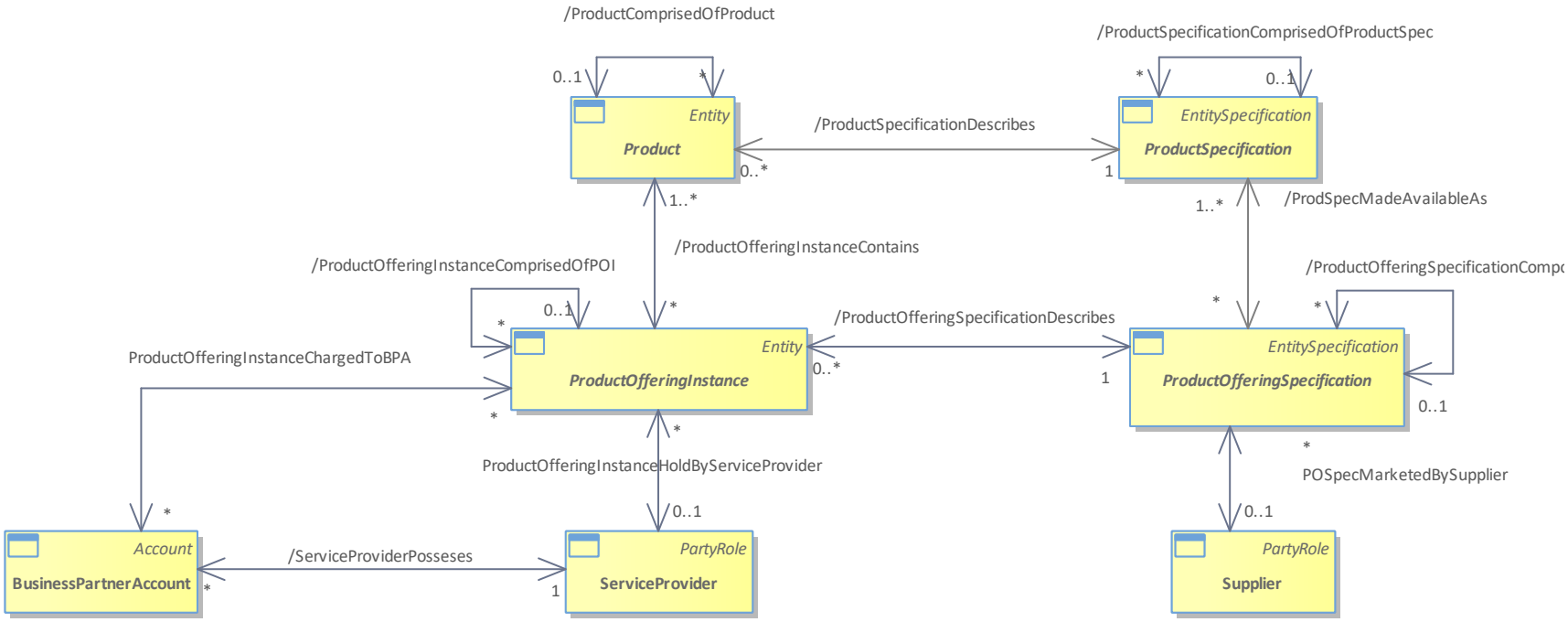

Figure Pr.07a3 - Supplier Product & Offering Spec and instances

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Figure Pr.07a4 - Complete Product & Offering Spec and instances

The following diagram provides a complete view of the Product & Offering Spec and Instances.

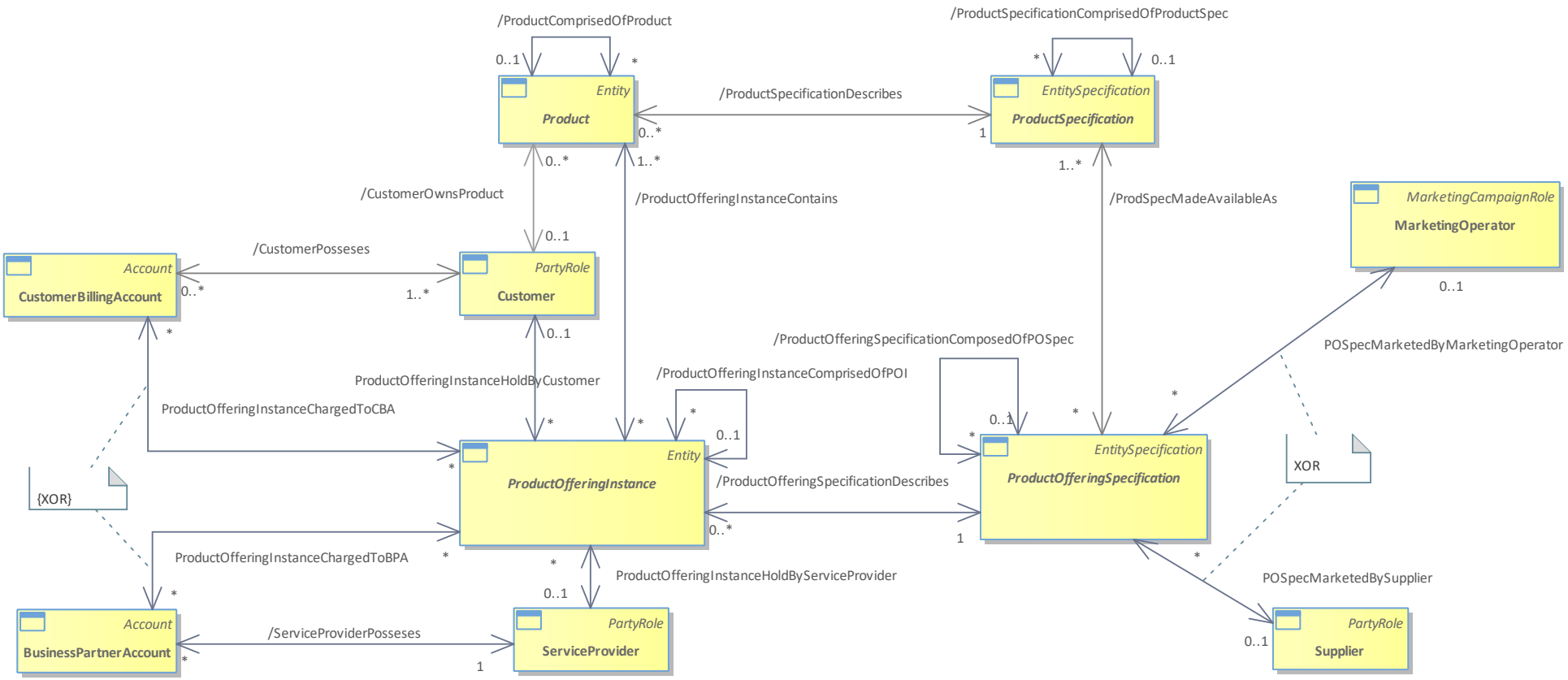

Figure Pr.07a4 - Complete Product & Offering Spec and instances

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Figure Pr.07b - Product Entities

A Product represents the realisation of a configured ProductSpecification for a Customer. Product follows the same Composite / Atomic pattern as ProductSpecification.

A Product is the functional view of what is used by the ProductUser and is always provided through an AtomicProductOfferingInstance that carries the corresponding commercial part such as the price.

When A Customer makes a selection, they may also have a number of options to select from. These are associated to a ProductSpecification and modeled in ProductCharacteristicValue (see Figure below), which defines the Product delivered to the Customer. The ProductCharacteristicValue comprises the detail of the ProductSpecCharacteristics, such as color, selected by the Customer.

Examples of characteristics might be the Voice package duration or Internet speed.

When an enterprise agrees to deliver a Product to a Customer, there is a significant amount of information that needs to be recorded to implement that Product on the provider's infrastructure or Resources and to activate the Service via with the Product is realized. This is modeled in Product and associated entities below.

The users/recipients that are entitled to use the Product are specified through PartyRoles ProductUsers related to the Product.

The ProductUser may be the signatory to an agreement, particularly if the Customer is an Individual, but they may also be an agent or Employee of the Customer where that Customer is a large Organization.Note that the ProductUser might be played by a Party different from the one who plays the role of Customer. For example, the mother might be the customer who holds a mobile line ProductOfferingInstance and the son might play the role of ProductUser of the mobile line Product.

In the context of a Company, the company holds the mobile fleet and each company's employee plays the role of ProductUser of one Product of the mobile line fleet.

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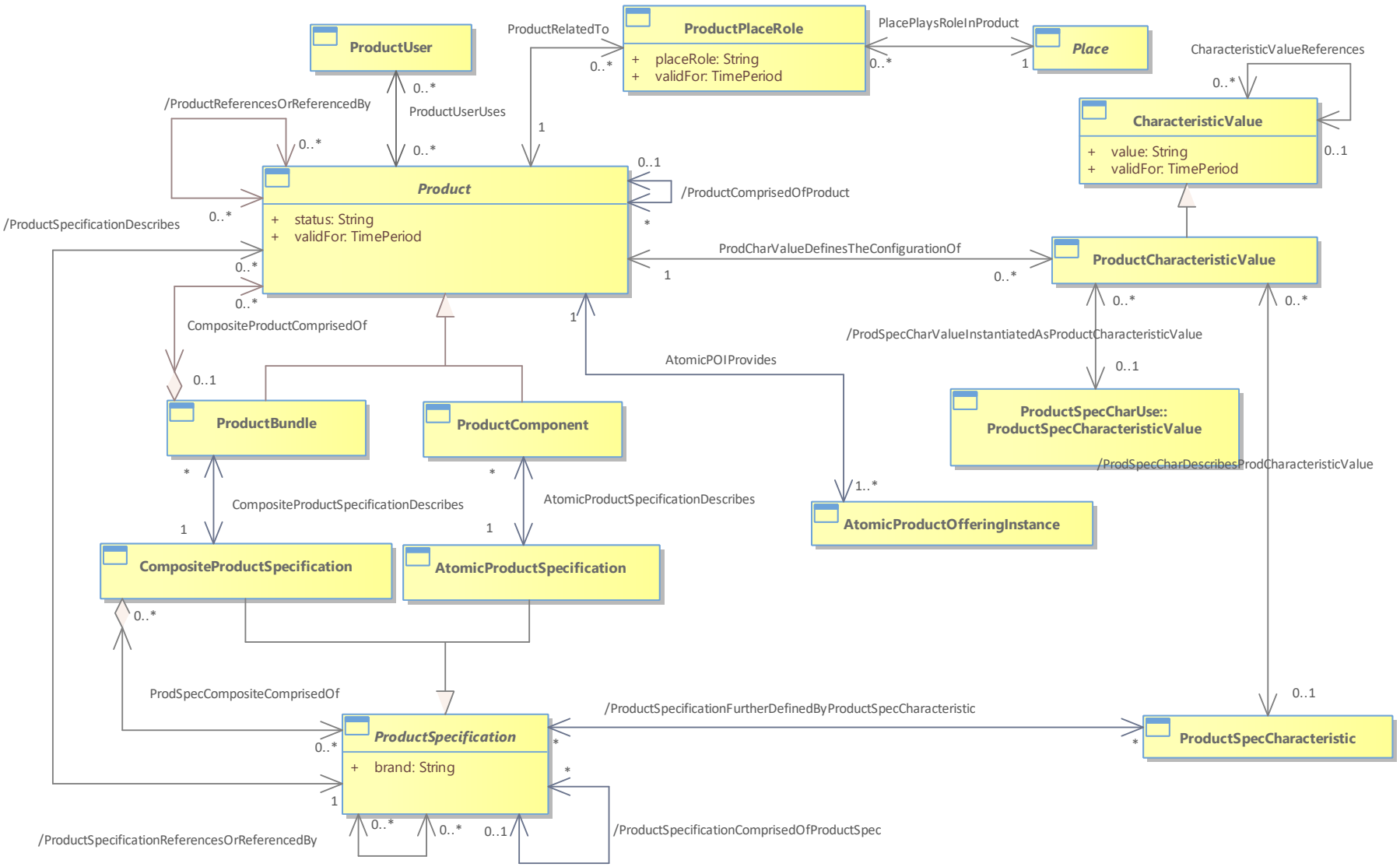

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Figure Pr.07b – I01 - 1.4.1.1.ProductCharacteristicValue Example – Email Product

The features of a Product to which a Customer subscribes are represented by ProductCharacteristicValues. In the email example, the features Spam protection, storage, message size, virus scanning and cleaning, mail forwarding, and archiving all have fixed values. These values are sometimes referred to as invariant characteristics. User name and password are variant characteristics in that the values are specified for each Product. The values may come from a list of permitted values specified in ProductSpecCharacteristicValues or may be simply be entered Product by Product. User name and password are examples of the latter. In the example, user name may take on a value of “Gsmith” and password may take on a value of “mom001”.

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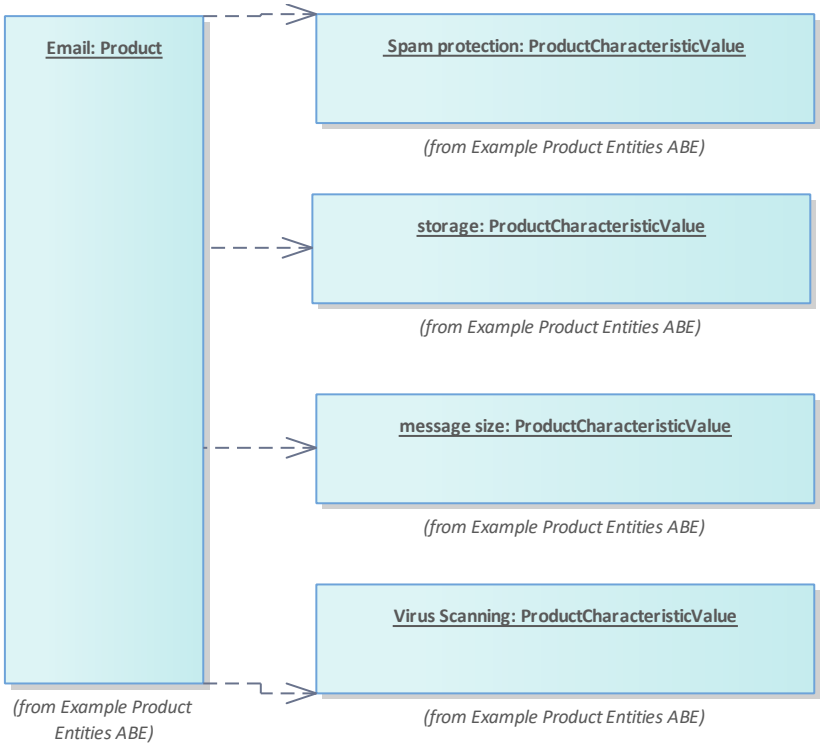

Figure Pr.07b – I01 - 1.4.1.1.ProductCharacteristicValue Example – Email Product

Figure Pr.07c - Product Offering Instance Entities

When A Customer makes a selection, they may also have a number of options to select from. These are associated with the ProductOffering via its association to a ProductSpecification and modeled in ProductCharacteristicValue (see Figure below), which defines the Product delivered to the Customer. The ProductCharacteristicValue comprises the detail of the ProductSpecCharacteristics, such as color, selected by the Customer.

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When an enterprise agrees to deliver a Product to a Customer, there is a significant amount of information that needs to be recorded to implement that Product on the provider's infrastructure or Resources and to activate the Service via with the Product is realized. This is modeled in Product and associated entities below.

A Product bundle in an instantiation of the BundledProductOffering that was used when the customer purchased a product or products from the provider. The ProductBundle holds the information (usually business information) that relates to the specific product offering (such as special prices or commitment terms) in case the customer changes its products and is no more eligible to the special business terms that are attached to the ProductOffering.

If the customer, purchased, as example a triple play bundle (VoIP, High speed internet and Cable TV) and as a result is eligible for a 10$ monthly discount on the Cable TV charges, then this information should be kept so when the customer cancels the VoIP product the TV discount is removed. In the same way the bundle can also have changes on commitment terms and other business terms. The information which BundledProductOffering was used for the purchase is kept in the ProductBundle entity and when a product is cancelled, possibly the ProductBundle may change (to reflect the fact that now the product business terms are derived form a different ProductOffering

A Product component in an instantiation of the SimpleProductOffering that was used when the customer purchased a product or products from the provider. The ProductComponent holds the information (usually business information) that relates to the specific product offering (such as special prices or commitment terms) in case the customer changes its products and is no more eligible to the special business terms that are attached to the ProductOffering.

If the customer, purchased, as example a VoIP line and as a result is eligible for a low recurring charge because of a special promotion in order to penetrate a new city, then this information should be kept so when the customer changes her address the special rate is replaced with a normal monthly rate. In the same way the product component can also have changes on commitment terms and other business terms. The information which SimpleProductOffering was used for the purchase is kept in the ProductComponent entity and when a product is cancelled, possibly the ProductComponent may change (to reflect the fact that now the product business terms are derived from a different ProductOffering.

ProductOfferings may be obtained by the Customer in a number of ways. Two different examples are Products are equipment obtained directly by the Customer, compared to Products that are realized by Services provided across the provider's Resources and which the Customer accesses and uses, rather than owns. In some cases, a provider may give the Customer access to a Product resident on a third party's resources. Where a third-party Product is involved, it may be branded/labeled as coming from another provider or it may be re-branded/labeled as the first provider's Product.

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Whichever of these is applicable, once the customer is using the Product, we are talking about a realization of the Product Offering that is based on a standard Product Specification.

The price someone pays, the rate charged for usage of a Product, discounts and allowances related to a Product are represented by the ProductPrice entity. ProductPrice is a separate SID ABE that is further detailed in this addendum.

We are also talking about the user/recipient that is entitled to use the Product. This is represented by a PartyRole, which is in turn related to the Product using the Product Involvement Role business entity and its involvementRole derived attribute.

The Product that the Customer has obtained is not truly realized until it is in use. The user may be the signatory to an agreement, particularly if the Customer is an Individual, but they may also be an agent or Employee of the Customer where that Customer is a large Organization.

Finally, a CustomerAccount may also be involved with a Product. For example, the role could be the CustomerAccount that pays for a Product; the CustomerAccount to which a Product was charged; the CustomerAccount associated with ordering a Product. This is represented by the CustomerAccountProductInvolvement business entity.

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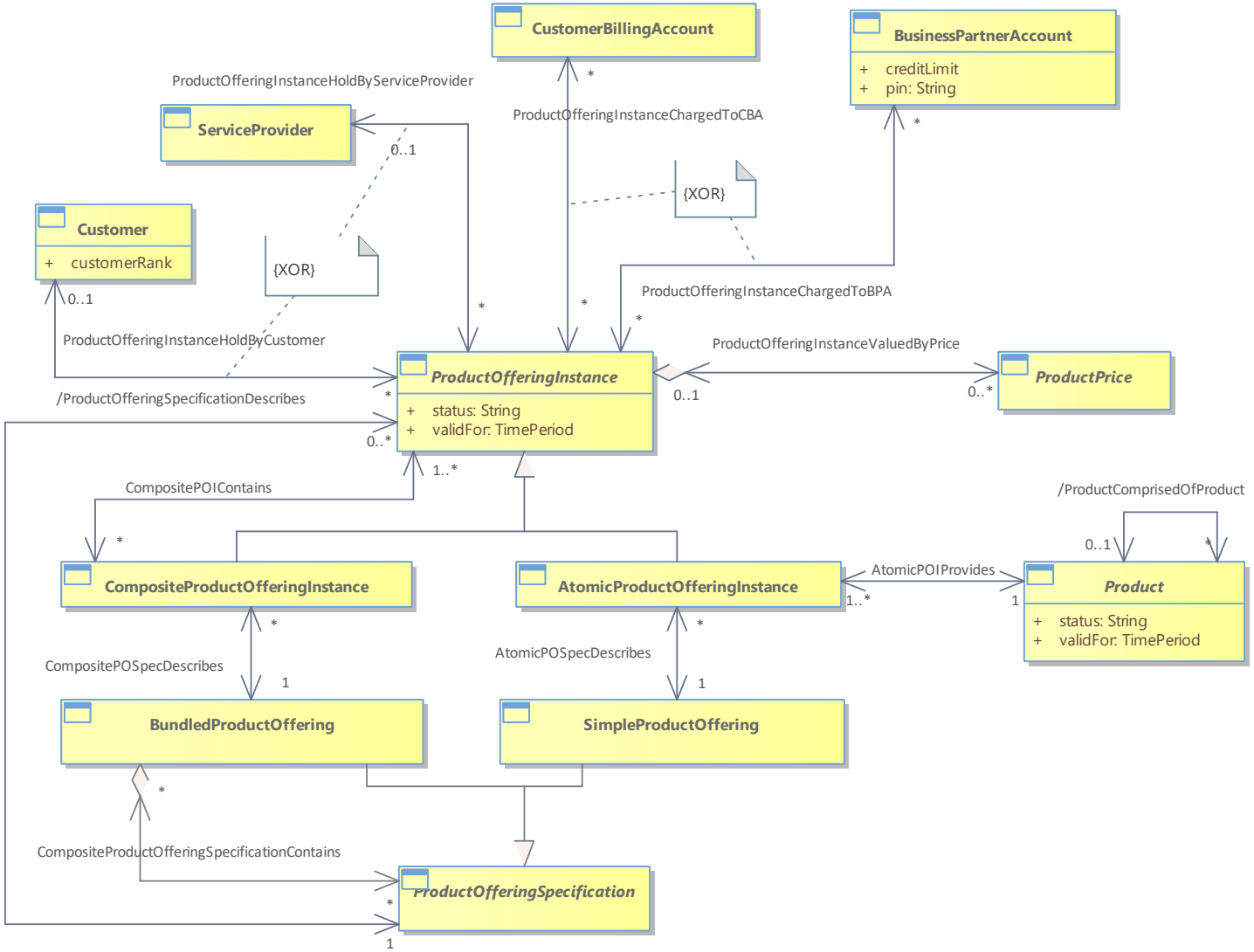

Figure Pr.07c - Product Offering Instance Entities

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Figure Pr.32 - Product Price

When an enterprise agrees to deliver a product to a customer, one or many ProductOfferingPrice might be identified as applicable for this Product and therefore one or many prices (ProductPrice) are associated with the Product.

The figure below presents the set of entities that are used to hold this information. The ProductPrice represents an amount, usually of money, that represents the actual price paid by a customer for a purchase, a rent or a lease of a Product.

The ProductPrice is associated with the Product entity and the ProductOfferingPrice entity. The structure of ProductPrice subclasses resembles the structure of ProductOfferingPrice subclasses.

The ProductPrice validFor attribute represents the time period during which it is applicable. For example, an alteration might be applicable only 6 months on a recurring charge.

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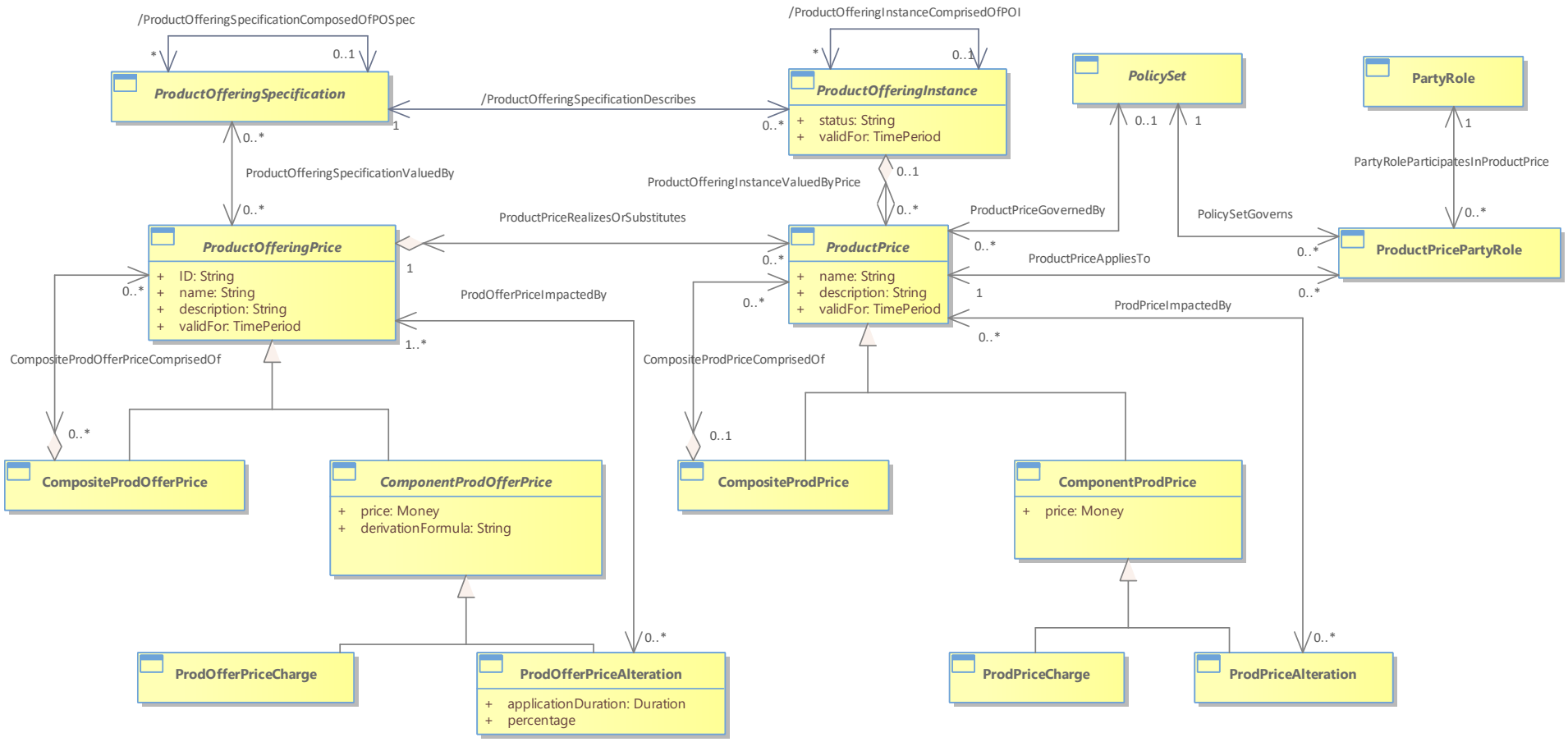

Figure Pr.32 - Product Price

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Figure Pr.32a - Product Price Charge hierarchy

The following diagram shows the ProdPriceCharge hierarchy.

- A **UsageProdPriceCharge** represents a charge applicable to ProductCharges for a Product.
- A **OneTimeProdPriceCharge** represents a charge to apply for a Product action that has been done such as creation fee.
- A **RecurringProdPriceCharge** represents a charge that must be applied on a recurring basis for a Product.

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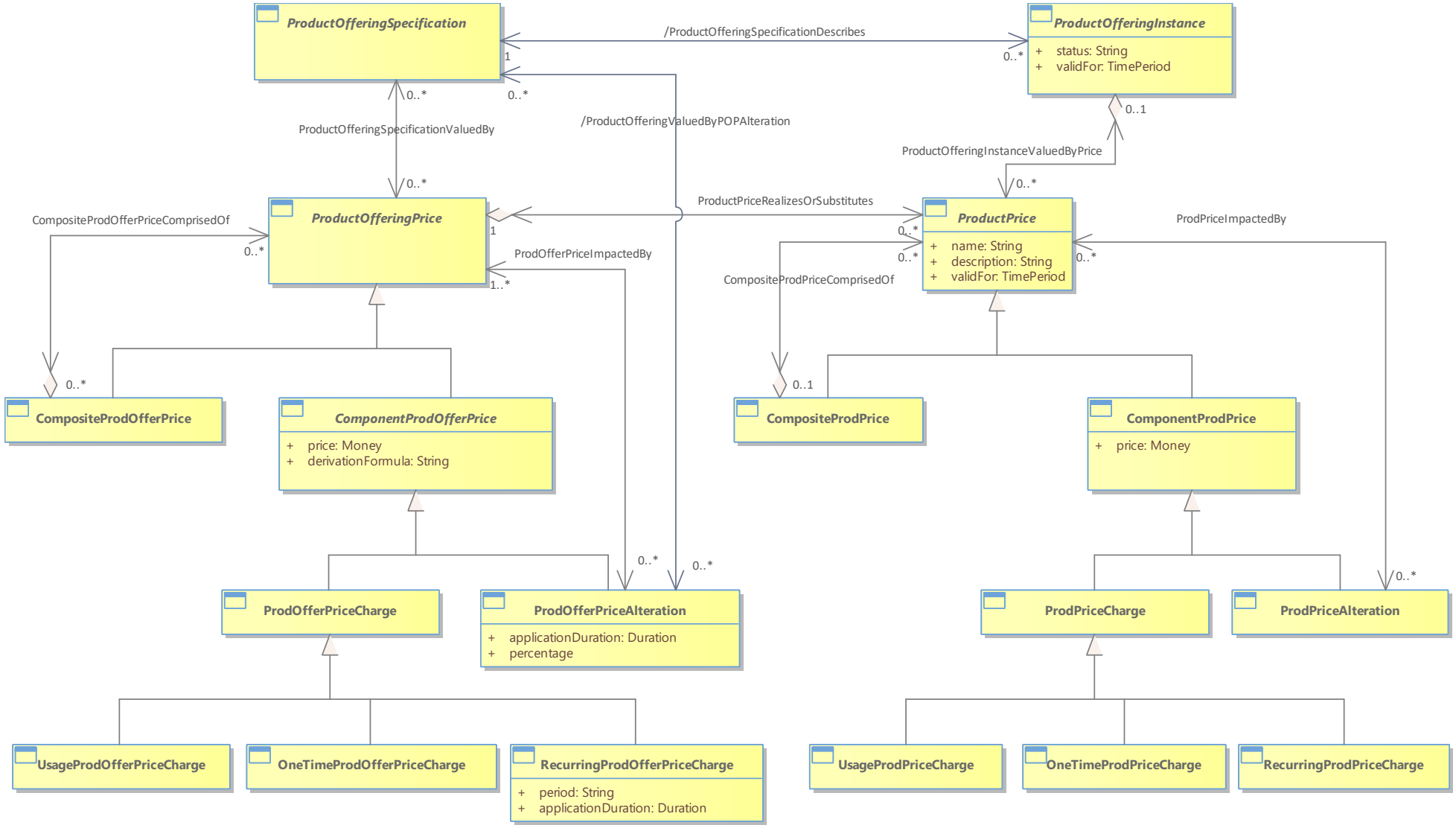

Figure Pr.32a - Product Price Charge hierarchy

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Figure Pr.32b - Product Price Alteration hierarchy

The following diagram shows the ProdPriceAlteration hierarchy.

- A ReplacementProdPriceAlteration represents a new price that replaces a charge for a Product.
- A DiscountProdPriceAlteration represents a discount (a reduction of price) for a Product.
- A TaxProdPriceAlteration is a type of ProdPriceAlteration that applies a TaxDefinition.

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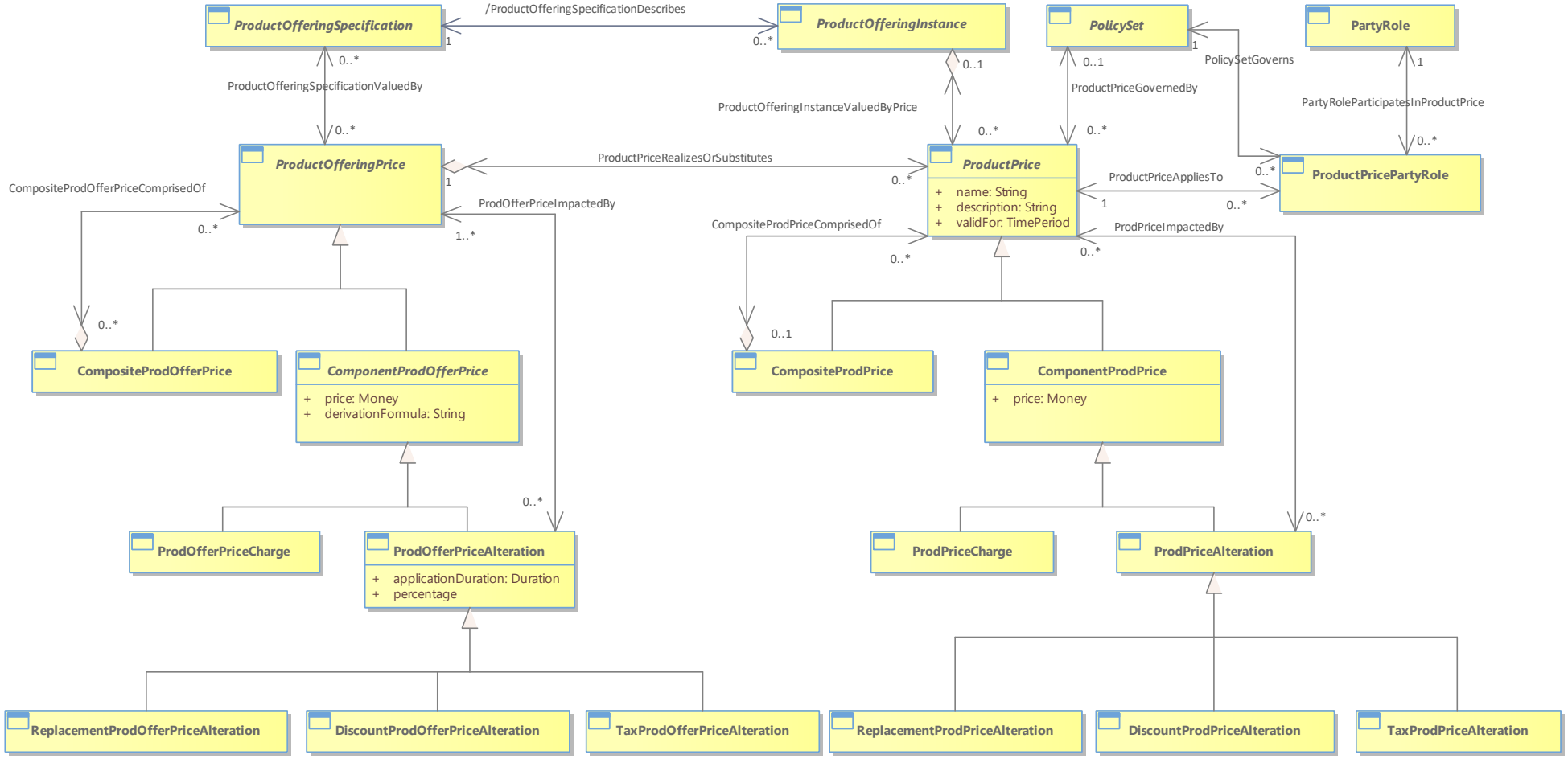

Figure Pr.32b - Product Price Alteration hierarchy

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4.4.1 Product ABE

Represents an instance in the product inventory of a ProductSpecification subscribed or not yet subscribed by a Customer to the CSP or by the CSP to a BusinessParnter, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses.

It may come from a CustomerProductOrderItem or a BusinessPartnerOrderItem. At one time, it might be contained in a unique ProductOfferingInstance.

The Product ABE also tracks the links to services and/or resources through which the product is realized.

The Product ABE also contains detailed concerning Network Products and Usage Volume Products as well as other examples of Products.

Figure Prod.01 - Product ABE Related Entities

Following are the business entities aggregated under the Product Aggregate Business Entity.

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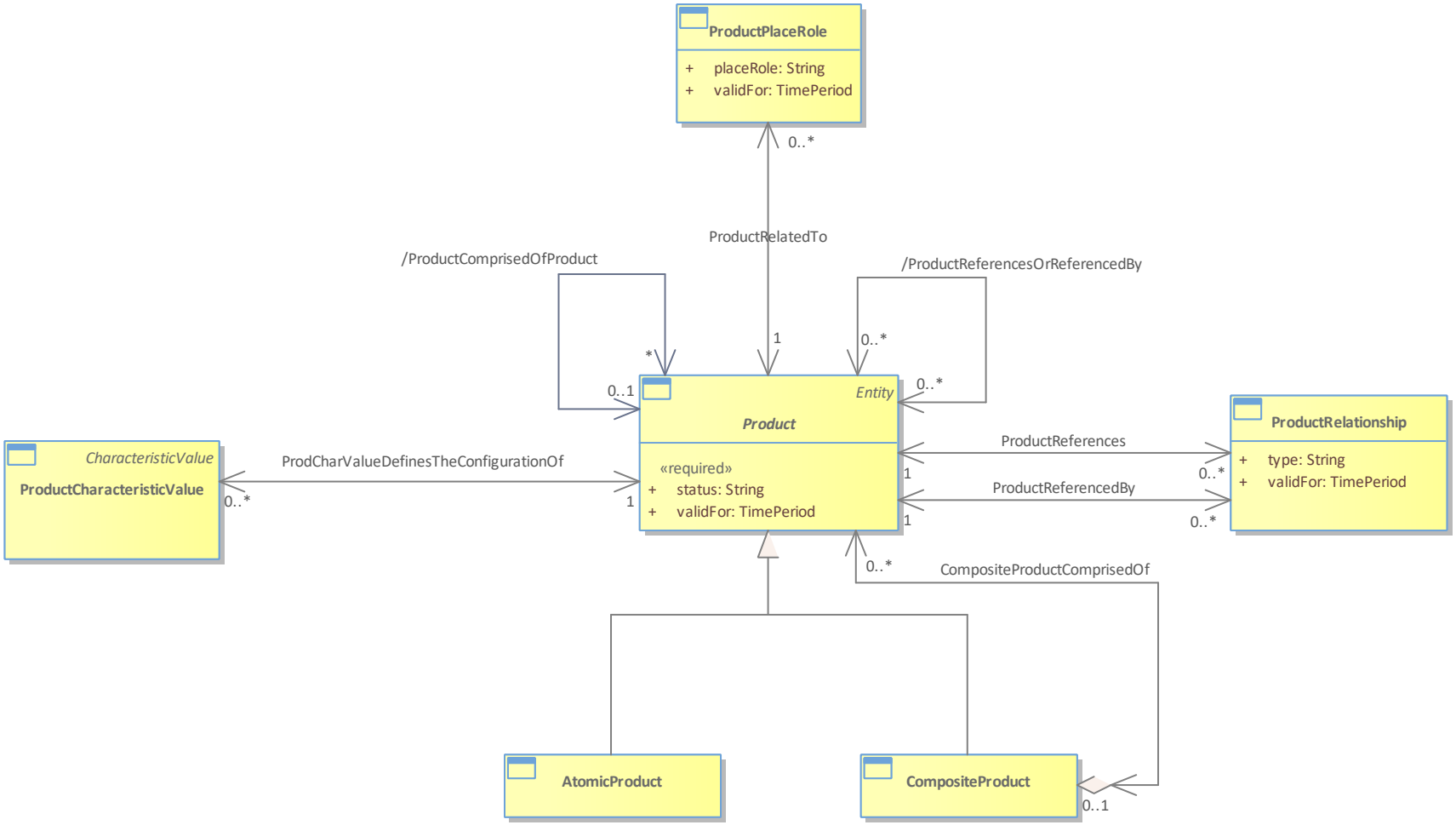

Figure Prod.01 - Product ABE Related Entities

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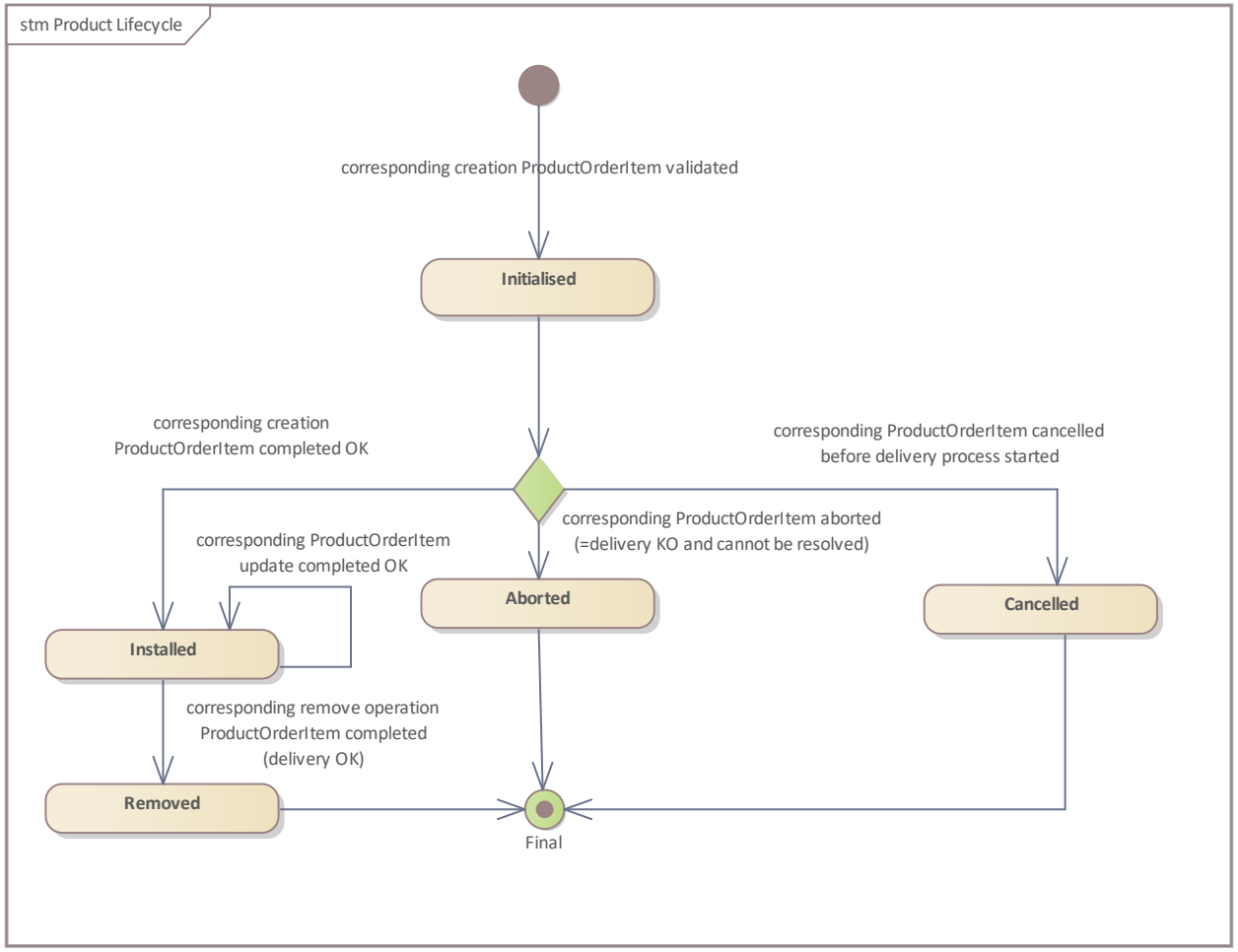

Product Lifecycle

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4.4.1.1 Example Product Entities ABE

The Example Product Entities ABE contains examples of Products subclasses that can be used.

4.4.1.2 Network Product ABE

The Network Product ABE contains the NetworkProduct description and relationships.

Figure NP.01 - Network Product ABE Related Entities

Following are the business entities aggregated under the **Network Product** Aggregate Business Entity.

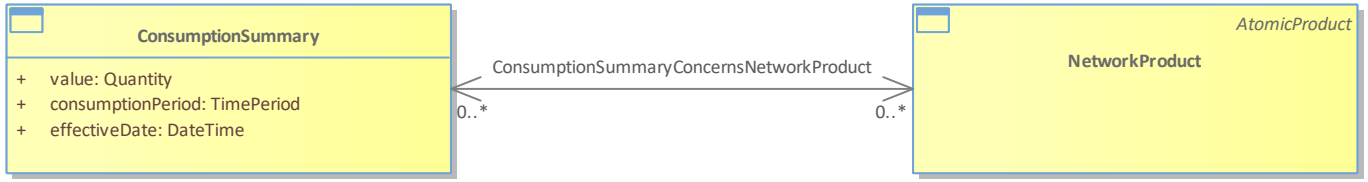

Figure NP.01 - Network Product ABE Related Entities

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4.4.1.3 Usage Volume Product ABE

The Usage Volume Product ABE contains the UsageVolumeProduct and all the entities specific to this type of product.

Figure UVP.01 - Usage Volume Product ABE Related Entities

Following are the business entities aggregated under the **Usage Volume Product** Aggregate Business Entity.

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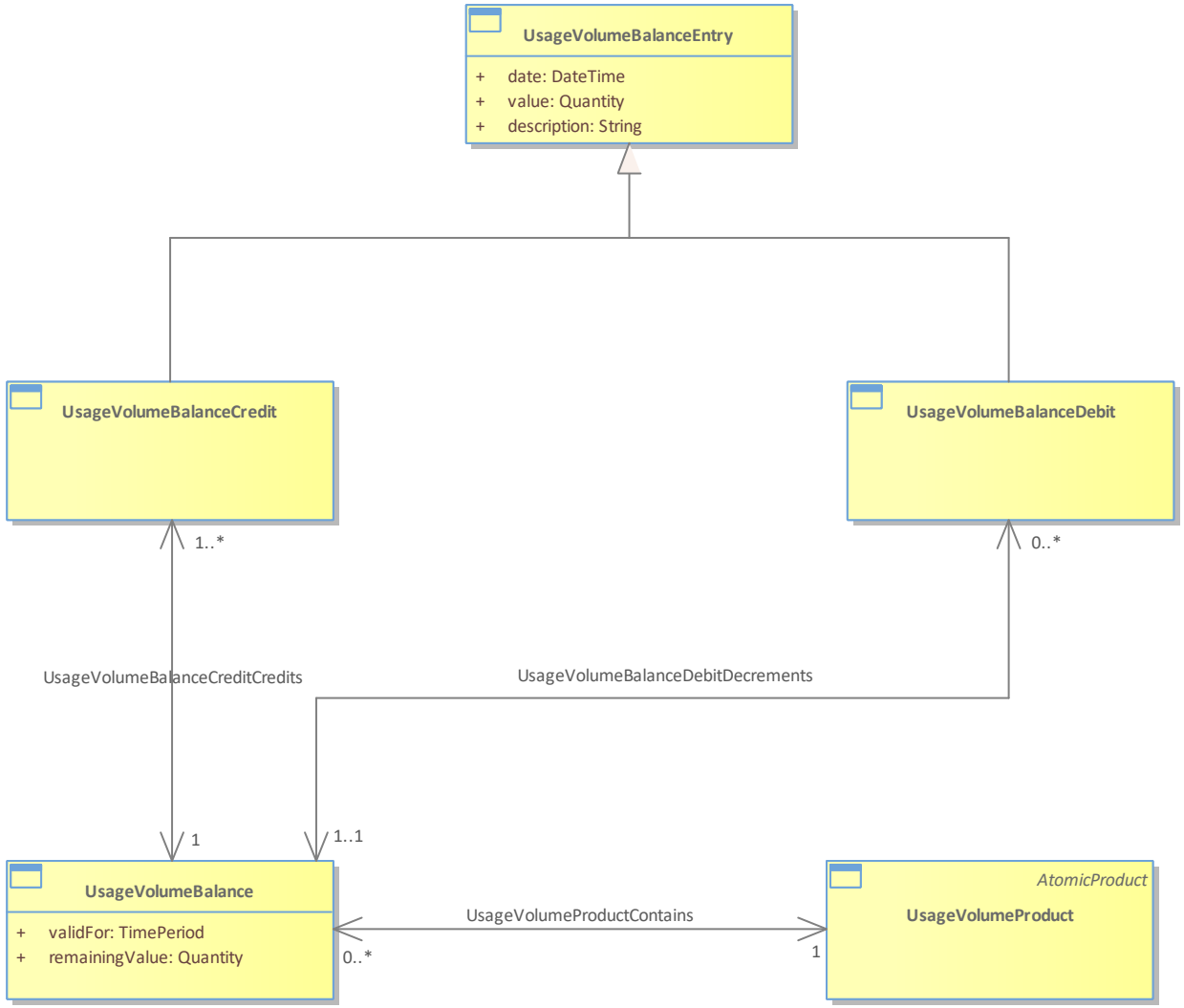

Figure UVP.01 - Usage Volume Product ABE Related Entities

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4.4.2 Product Offering Instance ABE

A ProductOfferingInstance represents the instantiation of a ProductOfferingSpecification for a specific Customer. It may carry ProductPrice and commitment choice.

Figure POI.01 - Product Offering Instance ABE Related Entities

Following are the business entities aggregated under the Product Offering Instance Aggregate Business Entity.

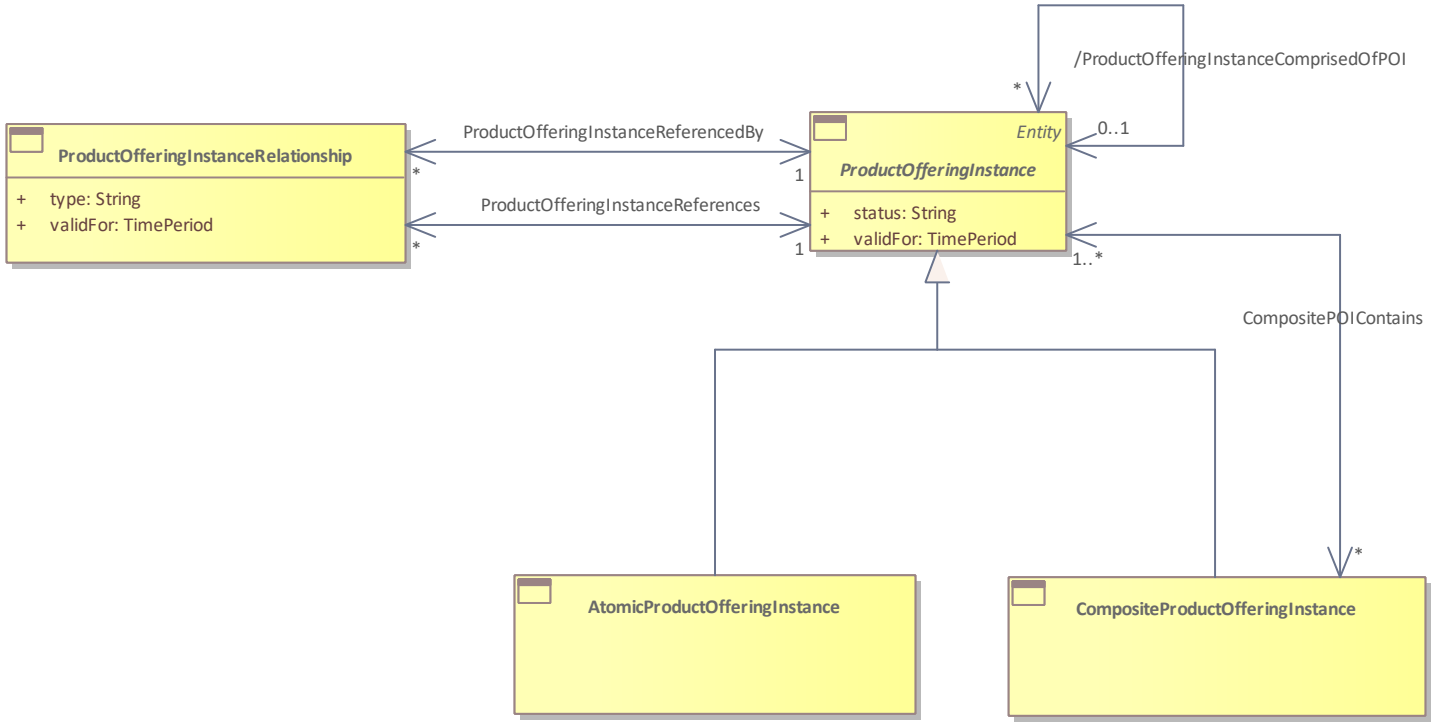

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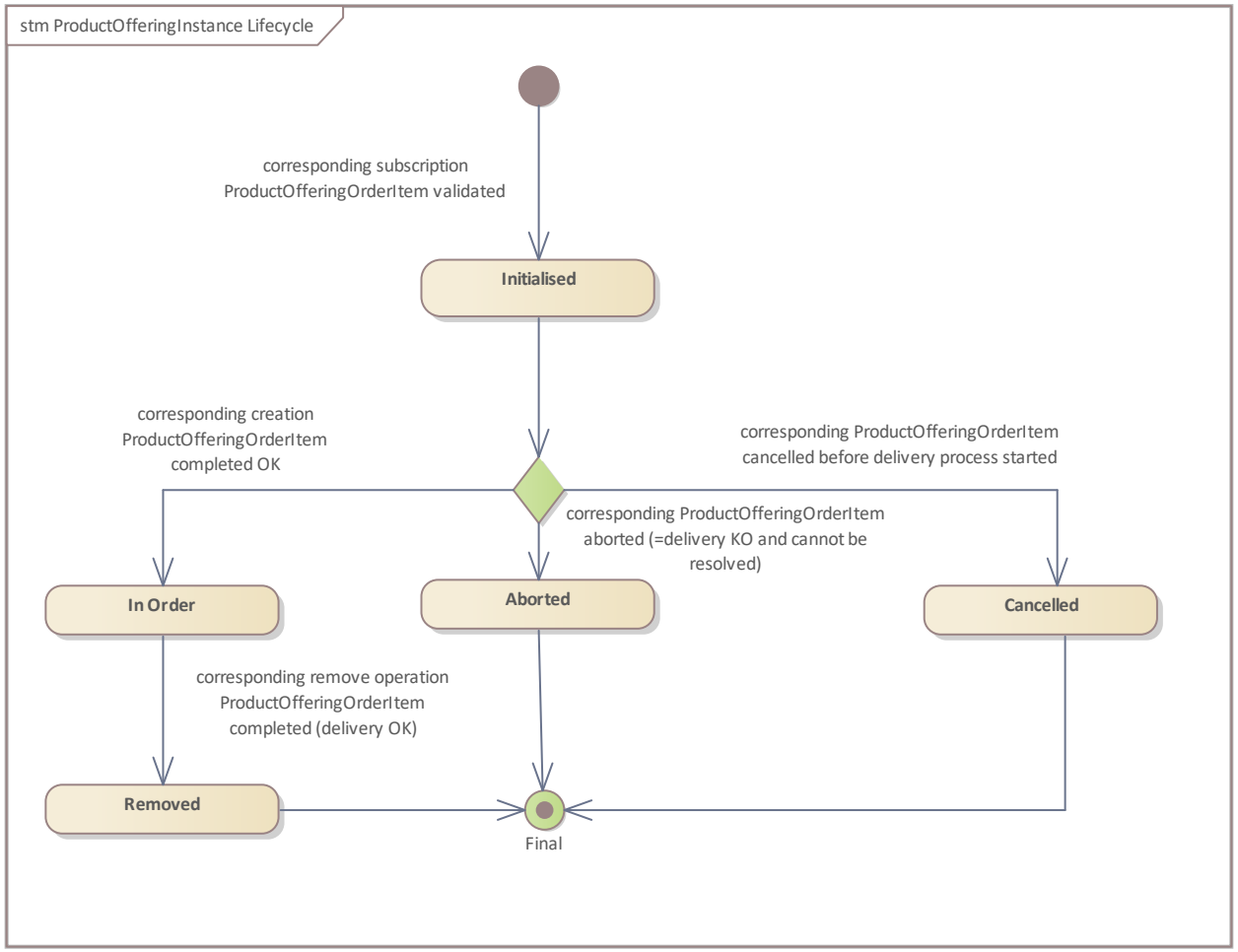
Figure POI.01 - Product Offering Instance ABE Related Entities



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4.4.2.1 Product Price ABE

The Product Price ABE describes all the prices applied to Products.

Figure PP.01 - Product Price ABE Related Entities

Following are the business entities aggregated under the **Product Price** Aggregate Business Entity.

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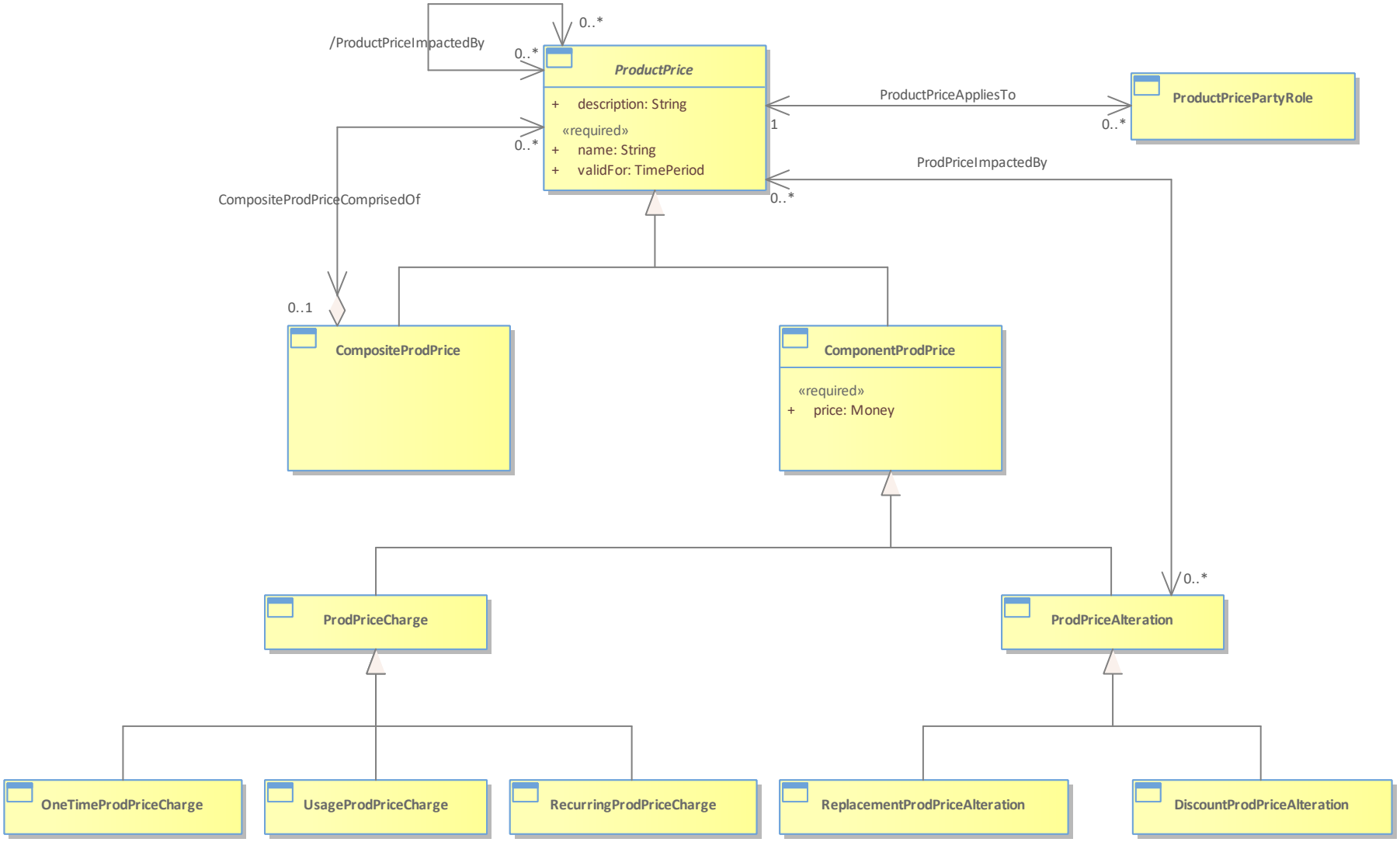

Figure PP.01 - Product Price ABE Related Entities

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4.5 Product Order ABE

The **Product Order ABE** contains all entities required to specify a communication used to procure or update one or many Products and ProductOfferingInstances in the context of ProductSpecifications and ProductOfferingSpecifications. This pattern is used to define:

- CustomerProductOrder corresponding to ProductOfferingInstances ordered by a Customer from the Service Provider
- and BusinessPartnerProductOrder corresponding to ProductOfferingInstances ordered by the Service Provider from a Business Partner

Note: Some but not all enterprises consider an Order to be a type of an Agreement. From a SID model perspective, an Order can be formalized by an Agreement and the Agreements relationship between the Agreement and the ProductOrder. This philosophy has the advantage of clearly separating two different concepts: the order from the legal formalisms associated with the order. For example, several orders may refer to the same Agreement.

Figure ProdO.01 ProductOrder overview

A ProductOrder represents a request used to procure, update or remove one or many Products and ProductOfferingInstances in the context of a ProductSpecifications and ProductOfferingSpecifications through all its ProductOfferingOrderItems and ProductOrderItems.

Note that ProductSpecifications / Products cover 2 cases: 1) ProductSpecifications / Products bought by a Customer from the Service Provider, 2) ProductSpecifications / Products bought by the Service Provider from Business Partners.

ProductOrder is a sub-class of RootEntity and therefore inherits from all its attributes and relationships.

At least one CommunicationInteraction is the source of a ProductOrder. Then, additional interactions might happen such as the Customer asking why his order isn't yet delivered or the Service Provider informing the Customer his order is completed.

A ProductOrder might refer to an Agreement specifying all what has been approved by the PartyRoles involved in the ProductOrder.

A ProductOrder, ProductOfferingOrderItems and ProductOrderItems might be further described by Attachments.

A ProductOfferingOrderItem may need other ProductOfferingOrderItems to be completed; this dependency is represented through the ProductOfferingOrderItemRelationship.

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A ProductOrderItem may need other ProductOrderItems to be completed; this dependency is represented through the ProductOrderItemRelationship.

ProductOrder / ProductOfferingOrderItem / ProductOrderItem are sub-classed as

- BusinessPartnerProductOrder used to procure, update or remove ProductOfferingInstances for the Service Provider
- CustomerProductOrder used to procure, update or remove ProductOfferingInstances for Customers

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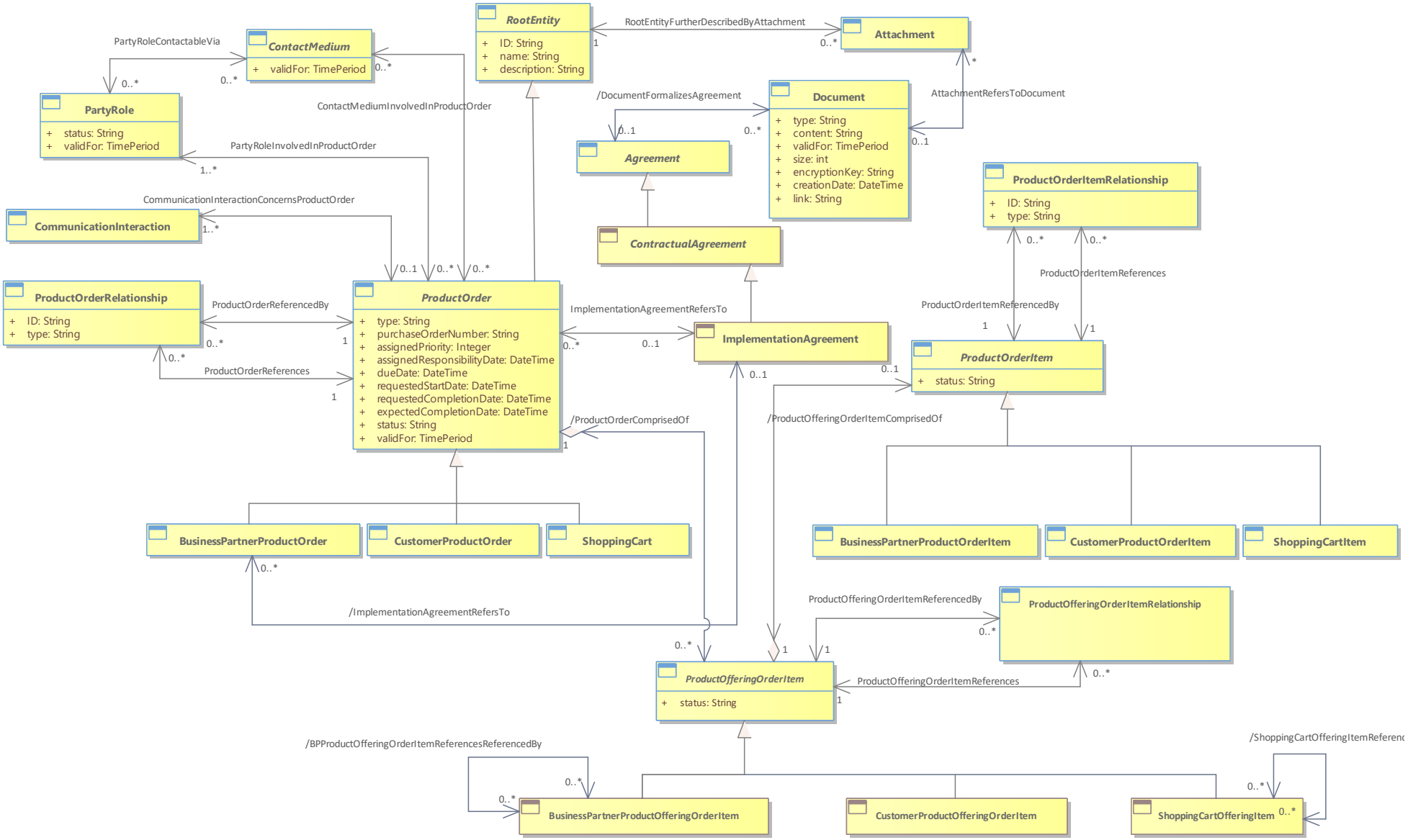

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Figure ProdO.02 ProductOrderItem detailed view

One or many PartyRoles might be involved in a ProductOrder / ProductOfferingOrderItem / ProductOrderItem such as Distributor, Holder, Buyer or Product User.

A ProductOrder, ProductOfferingOrderItem and ProductOrderItems might be further described by Attachments related to Documents.

Each ProductOrderItem requires an action (AllowedProductAction) on a ProductSpecification.

An AllowedProductAction is either an action on ProductSpecification or exclusively on a ProductOfferingSpecification.

The configuration required is described by the related Product even if it isn't delivered yet.

A ProductOrderItem may contain ProductOrderItems according to ProductSpecification (Composite / Atomic ProductSpecification) structure.

To deliver the corresponding Product, Place(s) to deliver and Appointment(s) might be needed. An Appointment is an arrangement to do something or meet someone at a particular time and location.

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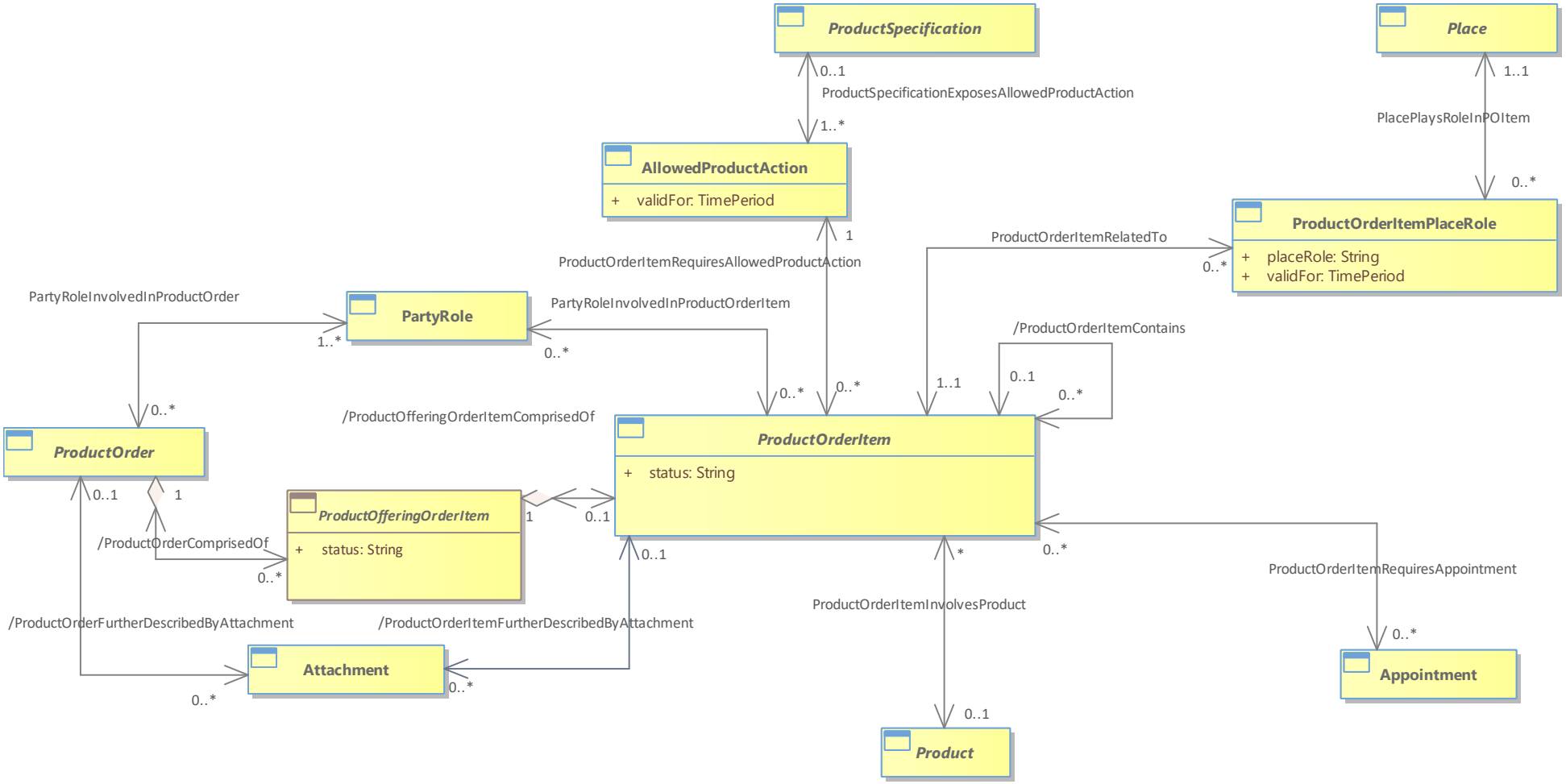

Figure ProdO.02 ProductOrderItem detailed view

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Figure ProdO.02b ProductOfferingOrderItem detailed view

One or many PartyRoles might be involved in a ProductOrder / ProductOfferingOrderItem such as Distributor, Holder, Buyer or Product User.

A ProductOrder and ProductOfferingOrderItems might be further described by Attachments related to Documents.

Each ProductOfferingOrderItem requires an action (AllowedProductAction) on a ProductOfferingSpecification.

An AllowedProductAction is, in the context of this diagram, an action on a ProductOfferingSpecification.

The prices applicable are related to the ProductOfferingInstance or to ProductOfferingOrderItem / ProductOrder when it corresponds to prices not specific to a ProductOfferingInstance such as a global of 5% on ProductOrder one-time fees.

A ProductOfferingOrderItem may contain ProductOfferingOrderItems according to ProductOfferingSpecification (CompositeProductOfferingSpecification / AtomicProductOfferingSpecification) structure.

A ProductOfferingOrderItem may also specify the commitments chosen for a ProductOfferingSpecification (i.e. CommitmentTermOrCondition).

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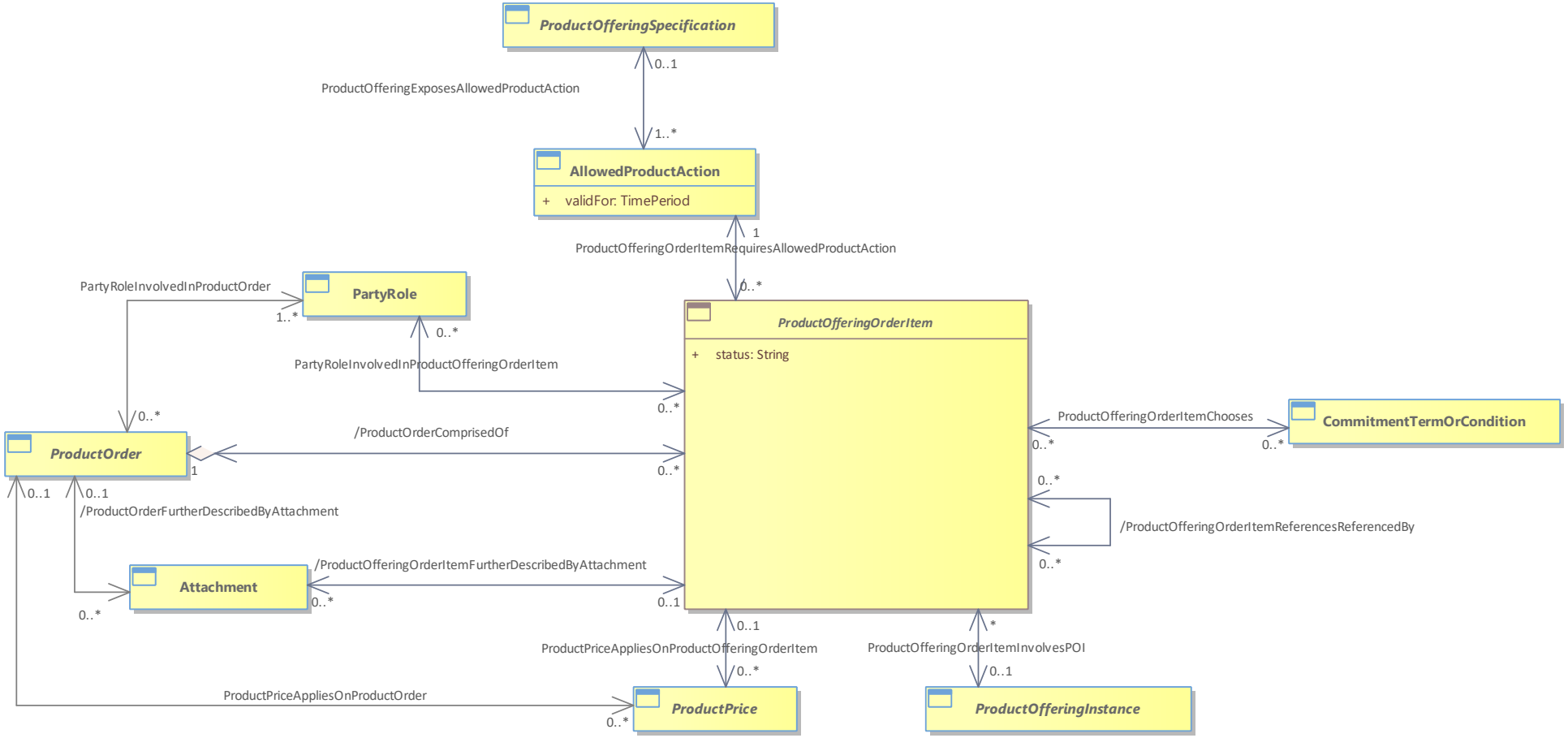

Figure ProdO.02b ProductOfferingOrderItem detailed view

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4.6 Product Usage ABE

Product Usage ABE represents specification of usages enabled by Products and usages of products tracked.

Figure Pr.33 - Usage Specification

Put simply, usage is how much service is used, by whom is it used, where and when is it used and circumstances under which it is used. Normally, when a usage event occurs, it is stored in a network element, for instance in a switch, router, gateway or in an application system. Resources (applications, network and computing capabilities) usually store usage data in proprietary formats that are not understood by the billing system. Depending on the polling strategy, a mediation engine connects to resources, collects usage data and formats them into a format that is understood by the billing system. Output of a mediation engine are Usage Detail Records (UDRs). Examples of UDRs are Call Detail Records (CDRs – used to describe usage details of voice call services) and Internet Protocol Detail Records (IPDRs – used to describe usage details of Internet Protocol based services). In this document we will use the Usage abstract business entity to describe any resource-, service- or product-based usage that the billing system can read, update and process.

The following two Figures show business entities that are used to model usage. The model shown below is generic; particular usage models should define their own usage business entities by extending the Usage entity.

The UsageSpecification class describes a type of usage. In order to achieve a flexible structure of the UsageSpecification all its attributes are stored as characteristics. Alternatively, different types of usage could be modelled as sub-classes of the Usage entities (Usage, ProductUsage, ServiceUsage, ResourceUsage).

The UsageSpecification is comprised of UsageSpecCharacteristics that define all characteristics (attributes) known by a particular type of usage. Each characteristic is described by its name, category, presence and a set of allowed values.

Name defines a unique name that identifies the characteristic.

Category defines one of the high-level aspects of the usage information described by the characteristic. These categories are commonly referred to as: Who, When, What, Where, and Why.

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Presence defines whether the characteristic is required or optional.

The set of allowed values is modelled with UsageSpecCharacteristicValue. This class describes all allowed values or value ranges.

Additional information regarding the CharacteristicSpecification/CharactersticValue pattern can be found in GB922– Root Business Entities.

The UsageSpecification entity has the following subclasses:

- ServiceUsageSpec (that represents a service usage);
- ResourceUsageSpec (that represents a resource usage);
- ProductUsageSpec (that represents a product usage).

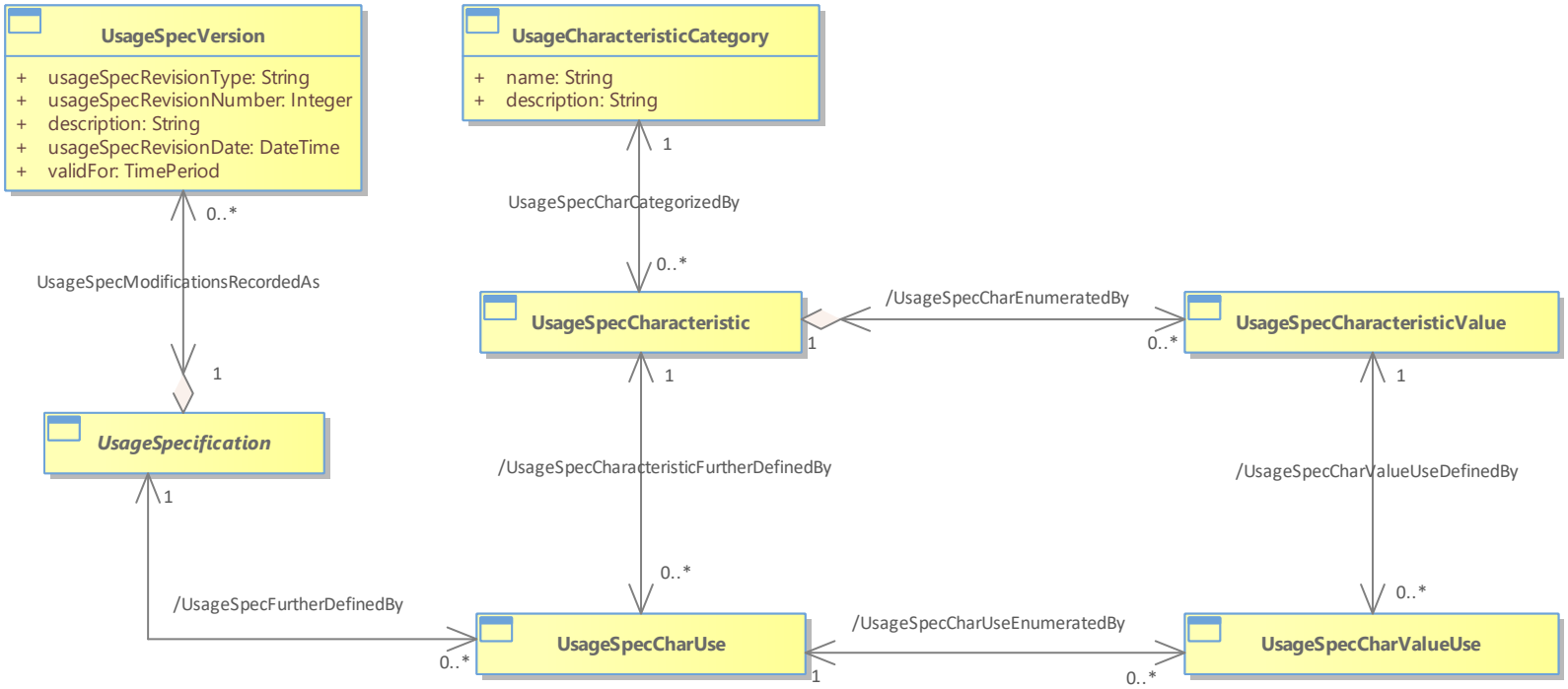

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Figure Pr.33 - Usage Specification

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Figure Pr.34 - Usage

The Usage class represents a usage event. It is described by the UsageSpecification and comprised of UsageCharacteristicValues that hold data attributes of a usage specified by UsageSpecCharacteristics. The UsageCharacteristic can either store its value in the usageCharValue property or reference predefined UsageSpecCharacteristicValue.

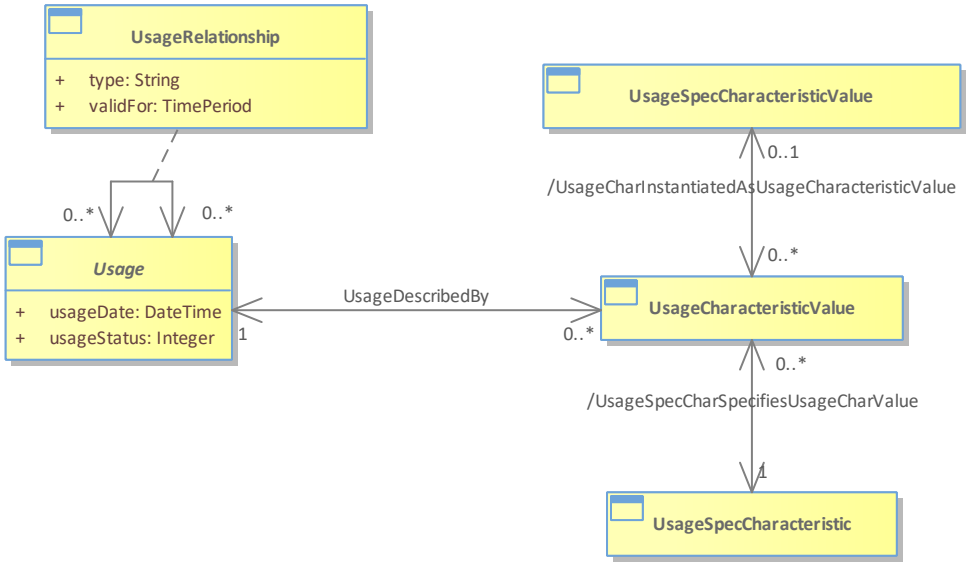

Figure Pr.34 - Usage

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Figure Pr.35 - Product Usage, Part 1

A Product is realized as one or more Service(s) and/or Resource(s). A Product is created by instantiating a ProductOfferingSpecification and adding it to a specific customer account. In order to charge a service (or a resource) usage, the service (or resource) has to be associated to a product.

The following Figure shows relationships among the ProductSpecification, the Product, the ProductUsageSpec and the ProductUsage ABEs. The ProductUsageSpec is defined for the certain ProductSpecification. As the ProductSpecification has 2 subclasses (the AtomicProductSpecification and the CompositeProductSpecification), the ProductUsageSpec needs 2 subclasses as well: the AtomicProductUsageSpec and the CompositeProductUsageSpec (associated with the AtomicProductSpecification and the CompositeProductSpecification entities respectively).

The ProductUsage describes the usage of an instantiated Product. The Product ABE is extended with the ProductComponent and the ProductBundle business entities. Consequently, the ProductUsage is extended with ProductComponentUsage and the ProductBundleUsage business entities (that are associated with the ProductComponent and the ProductBundle respectively).

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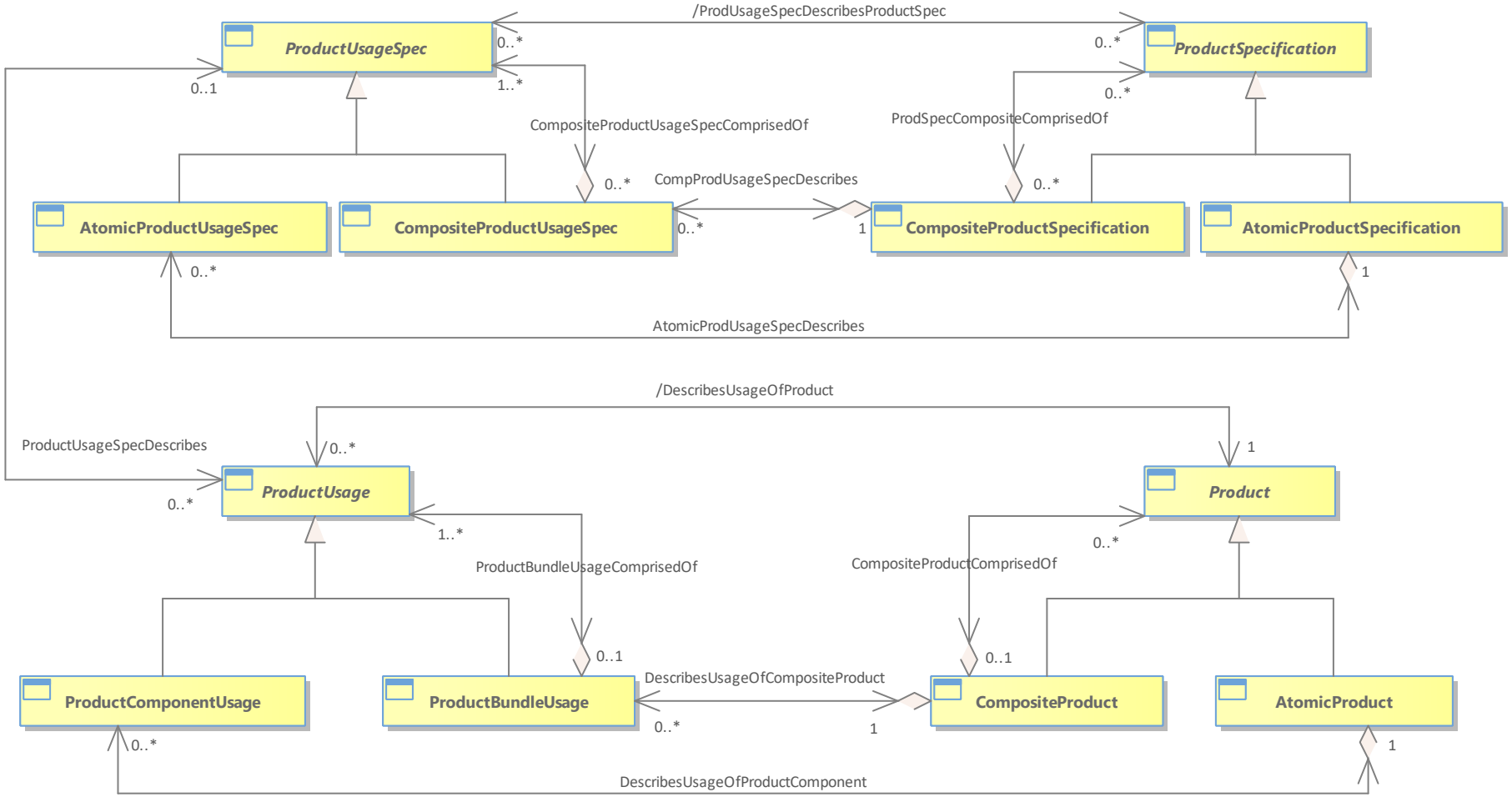

Figure Pr.35 - Product Usage, Part 1

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Figure Pr.36 - Product Usage, Part 2

As mentioned above, a product is realized as one or more services and/or resources. Consequently, the ProductUsageSpec business entity has to be associated with the ResourceUsageSpec and the ServiceUsageSpec business entities. The ProductUsage, the ServiceUsage and the ResourceUsage business entities have to reflect these associations as well. The ProductComponentUsage can be associated with either ServiceUsage or ResourceUsage and included in a ProductBundleUsage. These entities and associations are shown on the following Figure.

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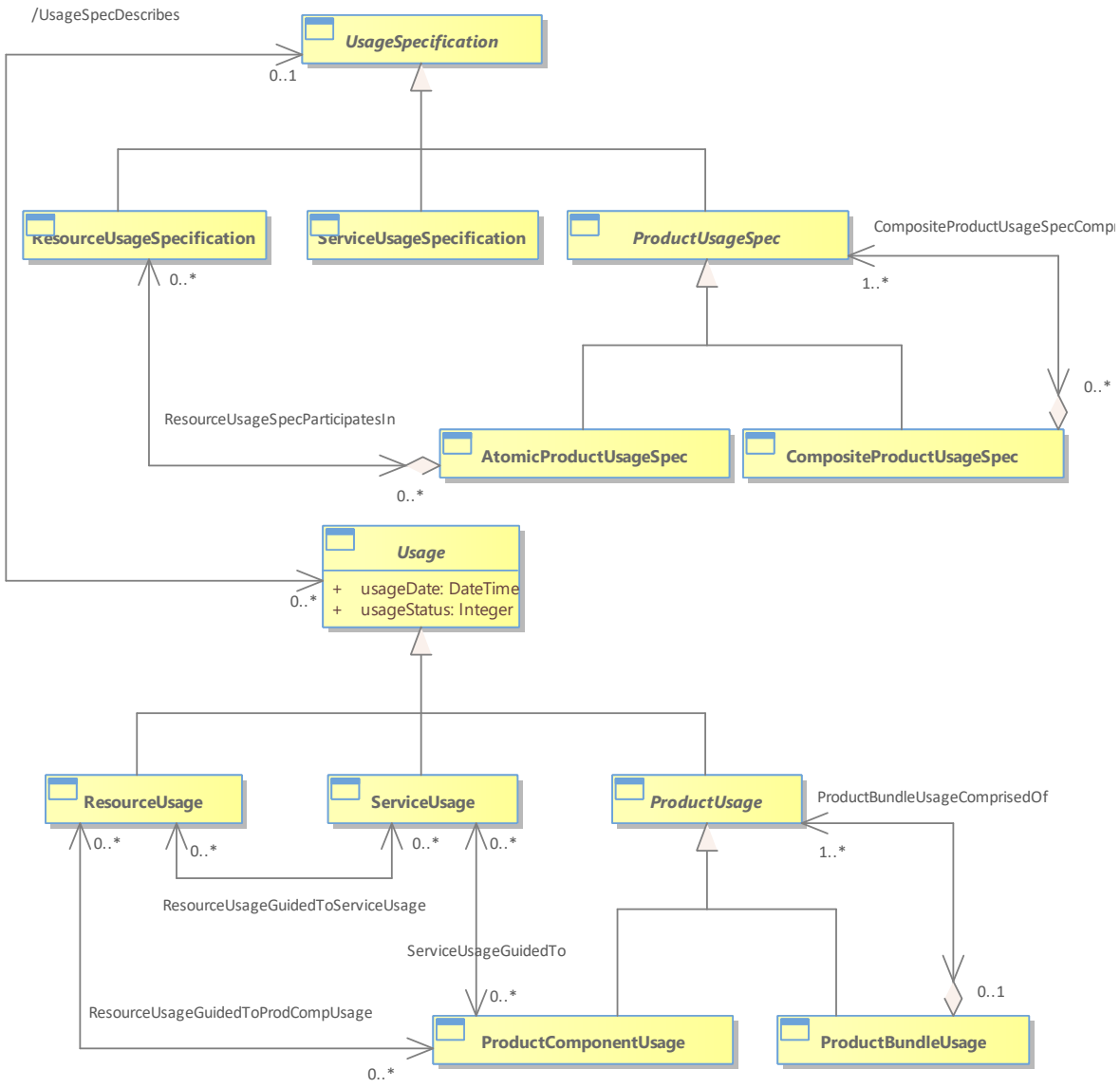

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4.6.1 Product Usage Entity ABE

The Product Usage Entity ABE collects product consumption information, and generates ProductUsage according to ProductUsageSpecification for use by other business entities.

Figure PrE.01 - Product Usage Entity ABE Related Entities

Following are the business entities aggregated under the **Product Usage Entity** Aggregate Business Entity.

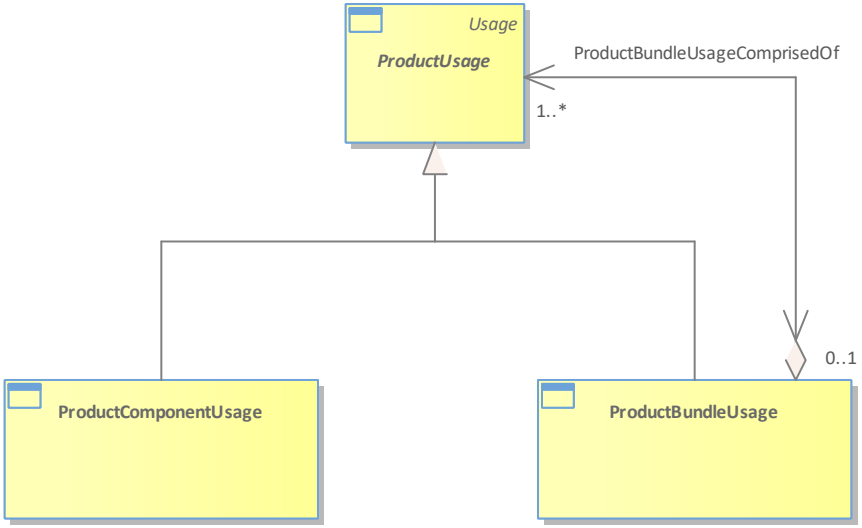

Figure PrE.01 - Product Usage Entity ABE Related Entities

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4.6.2 Product Usage Specification ABE

The Product Usage Specification ABE specifies types and characteristics of usage of ProductSpecifications.

Figure PUS.01 - Product Usage Spec ABE Related Entities

Following are the business entities aggregated under the **Product Usage Specification** Aggregate Business Entity.

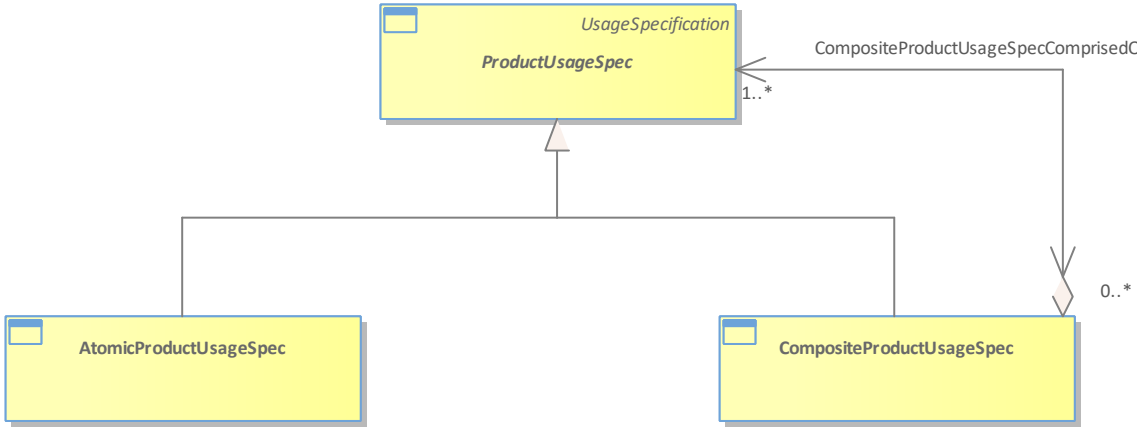

Figure PUS.01 - Product Usage Spec ABE Related Entities

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4.7 Loyalty ABE

A loyalty program is one of the tools used by the loyalty process to retain customers.

The **Loyalty ABE** contains all entities useful to specify and instantiate loyalty programs.

A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply.

Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not.

A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.

Figure Pr.37 - Loyalty Program main view

A loyalty program is one of the tools used by the loyalty process to retain customers. For example, another tool is the ProductOfferingSpecification Commitment.

An example of loyalty program could be « Change Your Mobile Offer »:

- available only for prepaid contracts, therefore incompatible with postpaid contracts,
- month after month the customer earns points and depending on the number of points he has, he can buy a new handset at a lower price using his points.

Another example could be a “more time for calls” program that gives you more time for calls according to your seniority (i.e.: 5% after six months of seniority, 10% after two years of seniority, etc.).

And these two loyalty programs could have a cost and be incompatible.

The aim of marketing is to find levers in the form of rewards to enhance customer loyalty to its operator.

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Most of the time, the conditions to reward a customer are based on

- seniority in customer relationship,
- the importance of revenue from the customer,
- or the risk of churn.

The more the customer remains with the operator or consumes, the more he will be rewarded.

Different types of rewards can be considered:

- physical rewards: accessories, terminals, points usable to buy accessories / terminals ...
- virtual rewards: free usage, vouchers ...

The rewards that can be bought thanks to loyalty earns correspond to ProductSpecification & ProductOfferingSpecification with a price allowing a specific currency (loyalty points).

A loyalty program is a part of a contract and may be either mandatory or not (packaging rule, pre-requisite).

As a loyalty program represents a cost for the operator, the latter avoids that the customer may have several loyalty programs.

To do so, Loyalty Programs might be incompatible with each other and in this case the customer has to choose only one LoyaltyProgram.

Therefore, a loyalty program is a type of AtomicProductSpecification, is packaged through ProductOfferingSpecification and is instantiated in the same manner in the installed base (LoyaltyProgramProduct that is a type of ProductComponent).

A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. A LoyaltyRule is a type of PolicyRule.

Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. This is defined by the needsLoyaltyAccount attribute on the LoyaltyProgramProdSpec entity.

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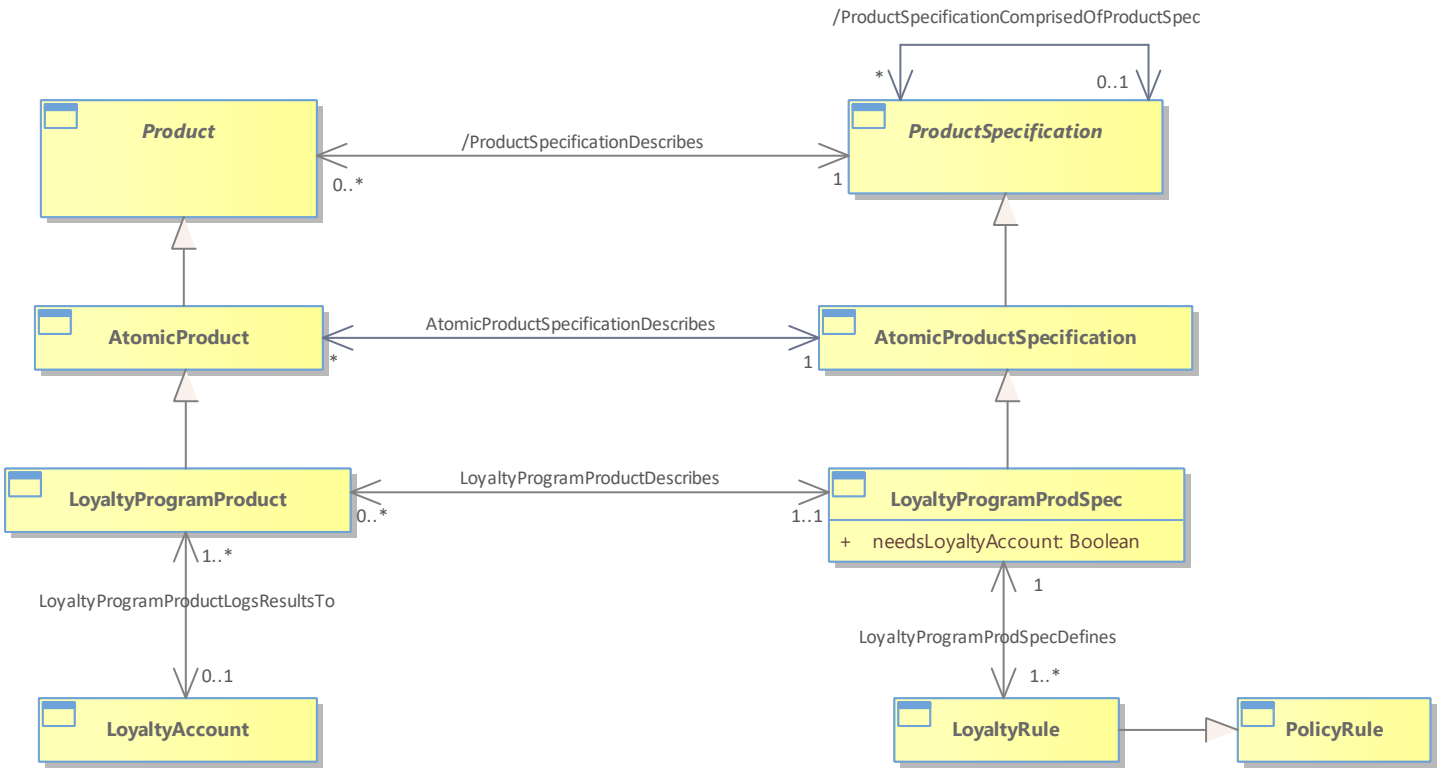

Figure Pr.37 - Loyalty Program main view

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Figure Pr.37-I01 - Loyalty Program main view

Here after, an illustration of the Figure Pr.37 - Loyalty Program main view via a loyalty program to change handset.

The Mobile contract contains:

- a ProductOfferingSpecification for the Mobile Line,
- and a loyalty ProductOfferingSpecification “Change Your Mobile”.

The loyalty ProductOfferingSpecification “Change Your Mobile” packages a LoyaltyProgramProdSpec that enables to earn points that can be burnt to buy a new handset.

Therefore, the LoyaltyProgramProdSpec defines two rules

- one to reward consumption,
- and a second one to interest young people.

As John Smith has ordered a Mobile contract he has the corresponding installed base elements including the “JS’s CYM by loyalty” LoyaltyProduct.

As John Smith has no LoyaltyAccount, “JS LoyaltyAccount” is created to collect the earns gained thanks to his “JS’s CYM by Loyalty” LoyaltyProgramProduct.

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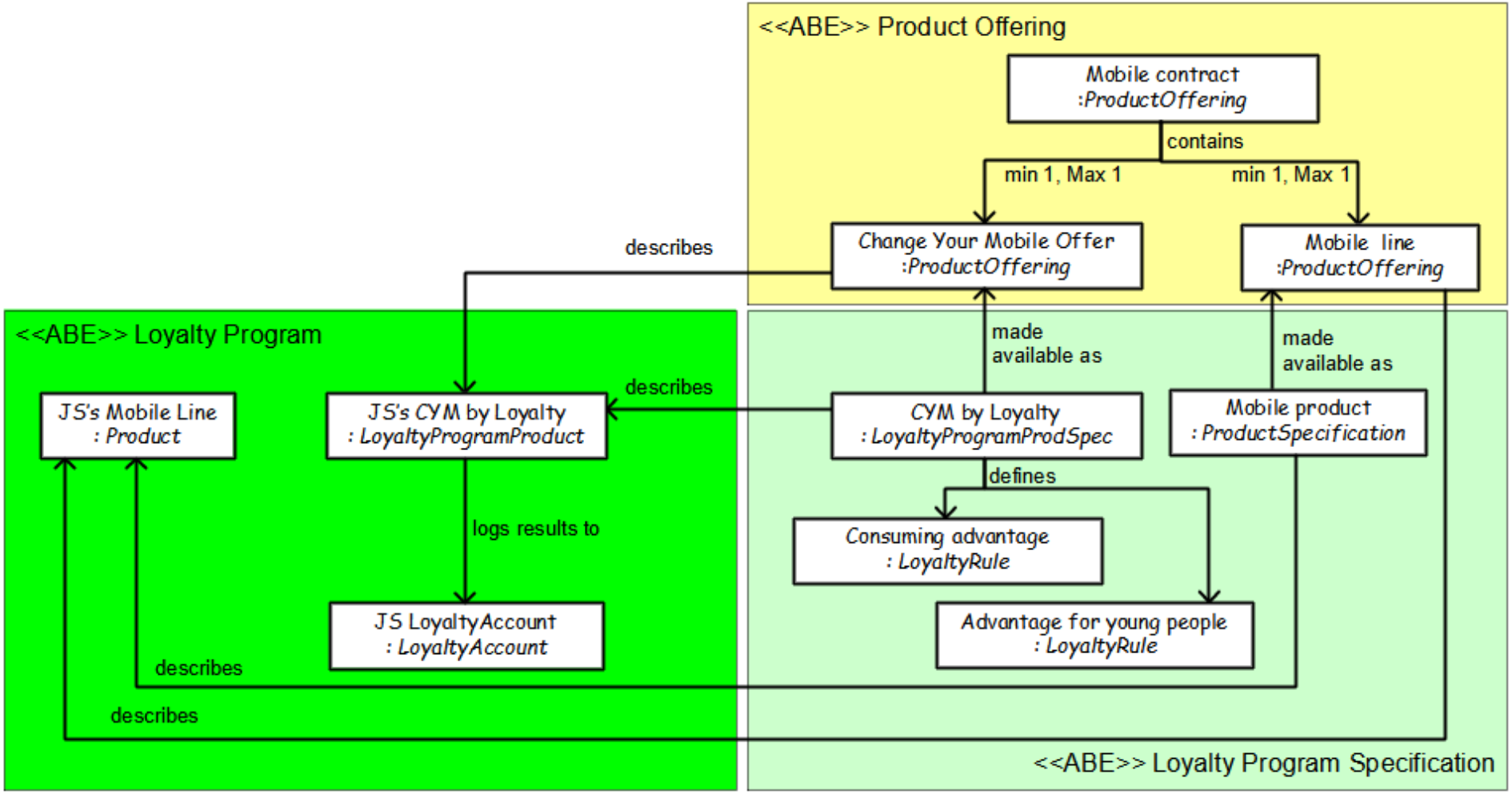

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Figure Pr.38 - Loyalty Rule

A LoyaltyRule specifies:

- Events triggering the evaluation of the rule (LoyaltyEvent)
    - Ex: Every first of a month, a CustomerOrder of Top-Up
- Conditions that have to be checked (LoyaltyCondition)
    - Ex: If the invoice amount is above 100 €, If more than 2 hours of international call
- Actions that should be performed if the condition clause is evaluated to TRUE (LoyaltyAction)
    - Ex: Give 50 SMS free, Credit 10 points on the LoyaltyAccount

Loyalty Rules may result in

- either rewards directly used / usable by the customer
    - Ex: 5% more time on the communications package after 6 months of age, and 10% after one year ...
- or loyalty earn that are gathered on an account to be used to pay later a ProductOfferingSpecification
    - Ex: 400 points to welcome a subscription then 10 points for every Euro charged. The points collected can be used to buy rewards like a USB stick, 100 SMS…

The different types of LoyaltyEvent may be event that occurs on BillingAccount (Ex: amount of Invoice, recurring charge), on PartyRole (Ex: CustomerOrder amount), on Products (Ex: monthly international usage volume).

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A LoyaltyAction may correspond either to a CustomerOrder (Ex: 100 SMS free), or to an Interaction (Ex: a SMS notifying the 100 SMS free) or to a LoyaltyEarn (Ex: 100 points on the LoyaltyAccount)

Each time a LoyaltyEvent happens, LoyaltyConditions from LoyaltyRules triggered by this type of event are evaluated.

If the result is true, a LoyaltyExecutionPoint is noted tracking the application of the rule and triggering BusinessInteraction, CustomerOrder or LoyaltyEarn. A LoyaltyExecutionPoint is a type of PolicyExecutionPoint.

Depending on event's type used in LoyaltyRule, the LoyaltyProgramProduct will specify entities from which it computes events (BillingAccount, Product…). The way to specify these boundaries from which loyalty events are collected to calculate the gains is not yet represented in this diagram and needs to be further studied.

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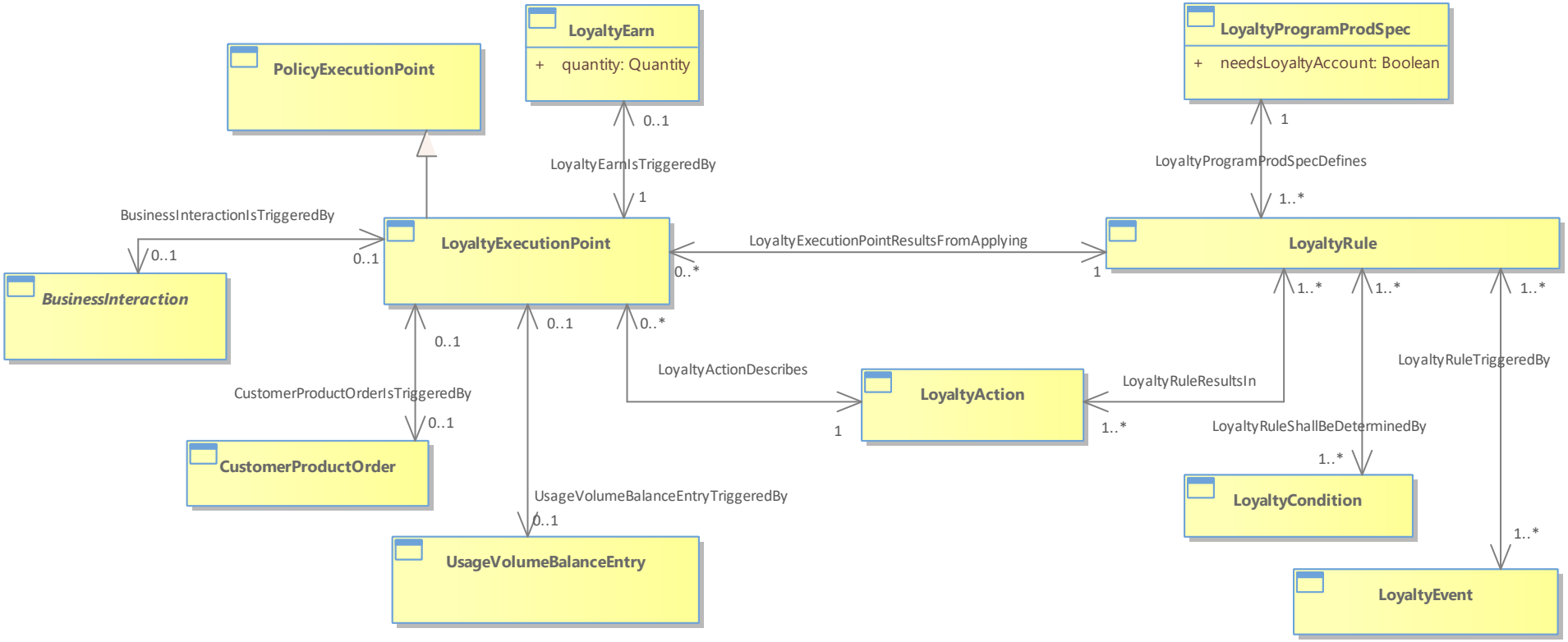

Figure Pr.38 - Loyalty Rule

Figure Pr.38-I01 - Loyalty Rule

Here after, an illustration of the Figure Pr.38 - Loyalty Rule via a focus on the loyalty program to change handset.

The “Consuming advantage” LoyaltyRule consists in:

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- evaluation for each Invoice
- only if the amount is higher than 10€
- apply a LoyaltyEarn based on the total amount of the Invoice

The « Advantage for young people » LoyaltyRule consists in:

- evaluation each 1st January
- only if the customer is younger than 26 years old
- apply a LoyaltyEarn as a result

As John Smith's Mobile contract has been invoiced, an event is collected.

As the amount is higher than 10€, the LoyaltyAction is triggered and corresponds to a LoyaltyEarn of 70 points (50 + 20).

The 1st of January, the « Advantage for young people » rule is evaluated, but unfortunately John Smith is older than 26 years, so he gets no Loyalty Earn.

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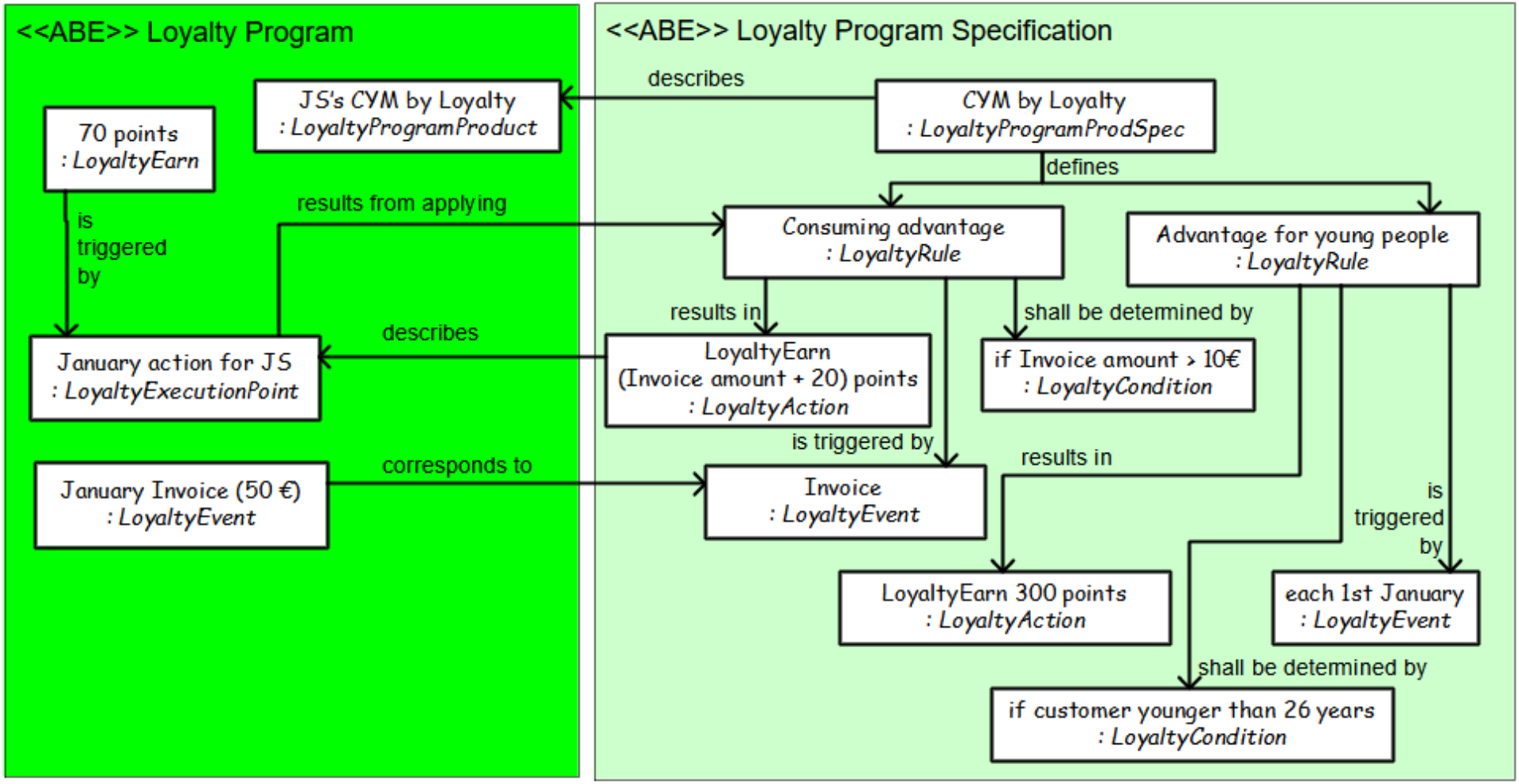

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Figure Pr.39 - Loyalty Account

A LoyaltyAccount corresponds to a set of balances to carry loyalty movement according to different valid period and unit.

A LoyaltyAccount may be required to realize a LoyaltyProgramProduct according to the rules carried by the corresponding LoyaltyProgramProdSpec.

A LoyaltyAccount may collect results from one or more LoyaltyProgramProduct.

A LoyaltyProgramMember is defined on each LoyaltyAccount. A LoyaltyProgramMember is a type of PartyRole and is granted with the rights on the LoyaltyAccount such as to execute LoyaltyBurn used to realize a CustomerPayment.

Each LoyaltyBalance gathers LoyaltyEarns and LoyaltyBurns.

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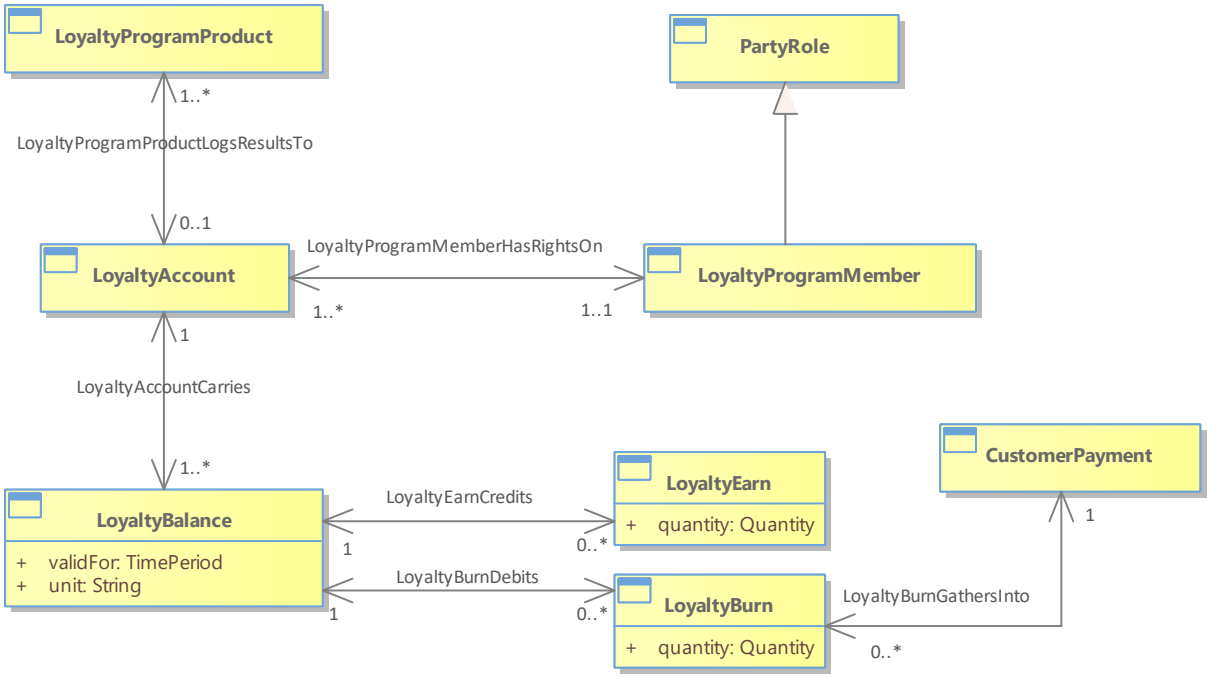

Figure Pr.39 - Loyalty Account

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Figure Pr.39-I01 - Loyalty Account

Here after, an illustration of the Figure Pr.39 - Loyalty Account via a focus on the application of the loyalty program to change handset.

The LoyaltyEarn is gathered into the LoyaltyBalance of JS LoyaltyAccount.

Mary Smith is declared as the LoyaltyProgramMember of the LoyaltyAccount.

Therefore, she can use the gains collected on it (LoyaltyEarns).

She buys a new handset with part of the payment done with 50 loyalty points corresponding to 5€. So she executes a LoyaltyBurn to pay it.

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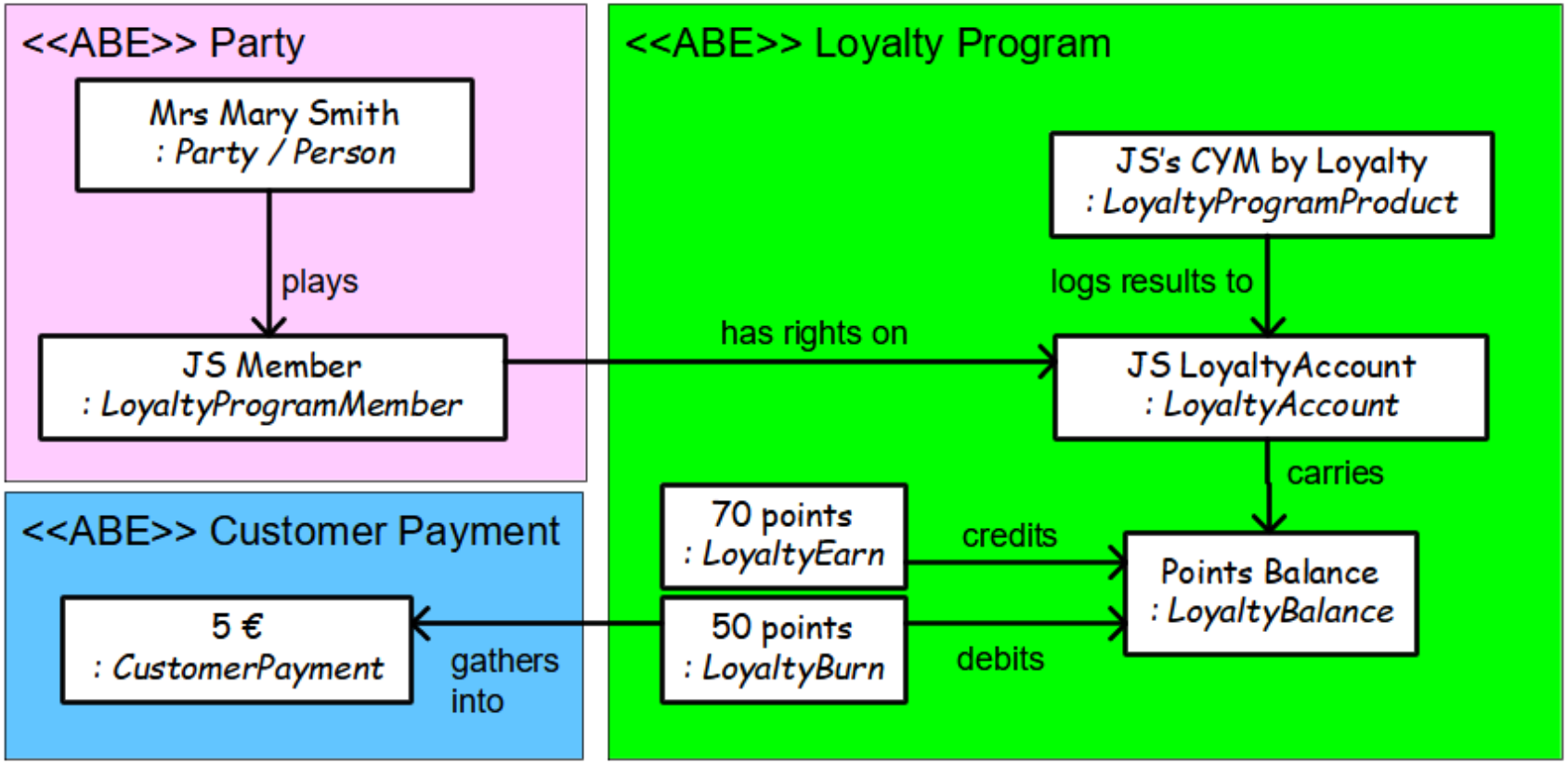

Figure Pr.39-I01 - Loyalty Account

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Figure Pr.40a - Payment Method and Plan

The means of payment is classified by PaymentMethod.

For example, payment by cash, check, bank card (credit card or debit card), third party (like bank transfer, payment tool), account balance and loyalty burn.

The LoyaltyBurnPM is a type of PaymentMethod that uses a LoyaltyAccount.

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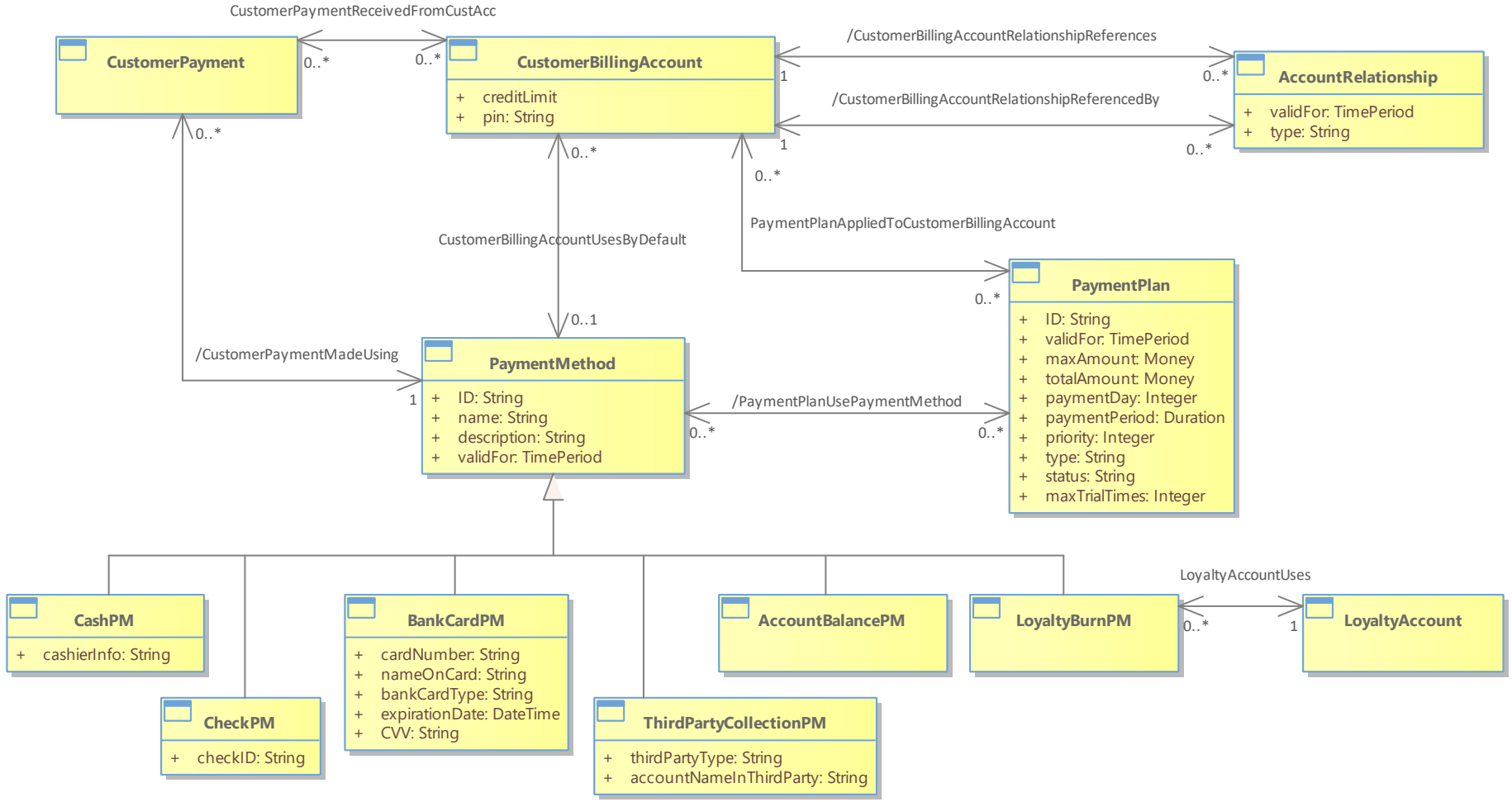

Figure Pr.40a - Payment Method and Plan

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4.7.1 Loyalty Program ABE

The Loyalty Program ABE contains all entities needed to instantiate a loyalty program for a specific customer such as LoyaltyProgram and LoyaltyAccount.

Figure LP.01 - Loyalty Program ABE Related Entities

Following are the business entities aggregated under the **Loyalty Program** Aggregate Business Entity.

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Figure LP.01 - Loyalty Program ABE Related Entities

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4.7.2 Loyalty Program Specification ABE

The Loyalty Program Specification ABE contains all entities specifying a loyalty program such as LoyaltyProgramProdSpec and LoyaltyRule.

Figure LPS.01 - Loyalty Program Specification ABE Related Entities

Following are the business entities aggregated under the **Loyalty Program Specification** Aggregate Business Entity.

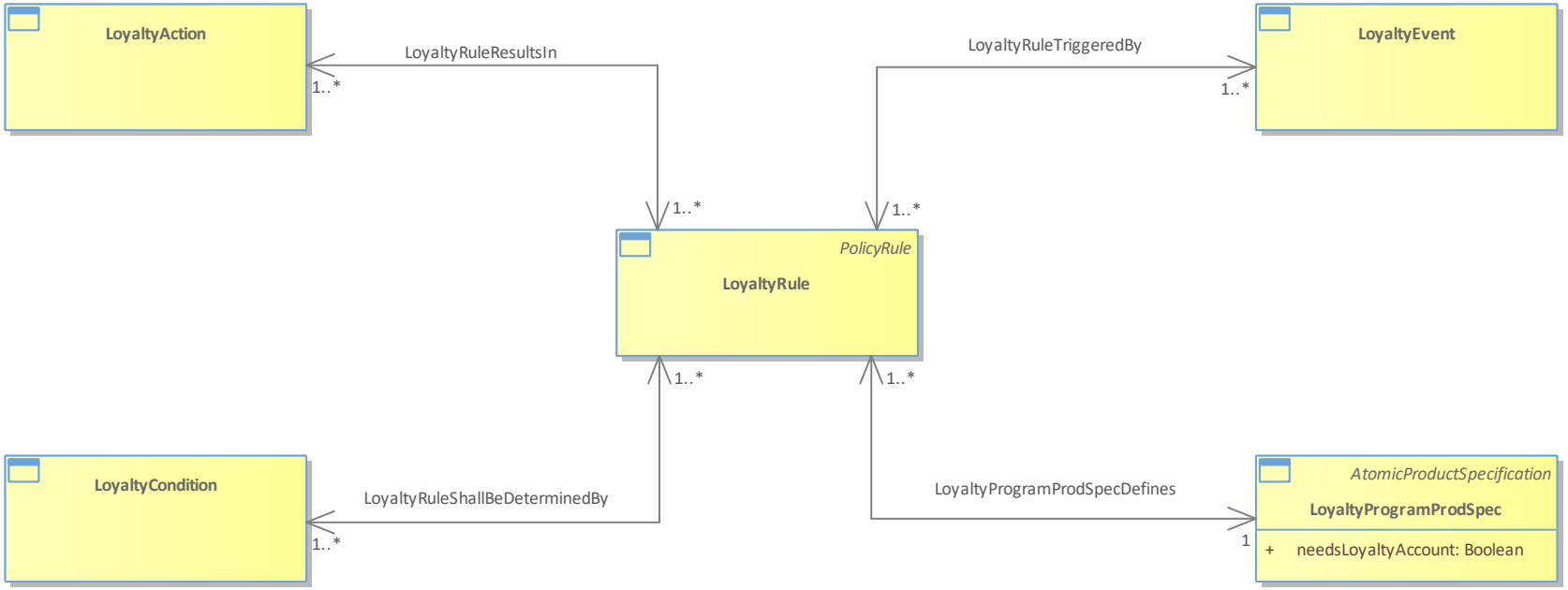

Figure LPS.01 - Loyalty Program Specification ABE Related Entities

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4.8 Product Configuration ABE

The **Product Configuration ABE** represents the specification of the different possible Product configurations as well as the Product Configurations instantiated for a Product.

Figure CP.01 - Configuration Spec and Configuration (copy from Common)

A Configuration (also referred to as a Profile) defines how a Resource, Service, or Product operates or functions. A Configuration may contain one or more parts (which is realized by using the Atomic/Composite pattern, but it is represented as a single entity - ConfigurationRelationship), and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g., using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns).

Starting from the right-side of the figure:

- The Resource, Service and Product classes are as defined in the SID model.
- The associated configuration objects are used to define additional properties (attributes) of the associated entity. For example, consider an Equipment instance (a Resource). The intrinsic properties of the Equipment instance (perhaps but not necessarily common to all Equipment instances) are defined in the Resource itself. Additional properties (which may not be common to all Equipment instances) are provided in one or one Resource Configuration instances.
- It is possible to relate one instance of Configuration to several others via the ConfigRelationship. This is useful when configuring complex (aggregate) entities. For example, to configure a circuit pack, one may also need to configure the various ports supported by the circuit pack and in turn, the connection termination points supported by the ports.

On the left-side of the figure, there are various Configuration Specifications that correspond to the various Configuration classes on the right-side of the figure. This is the typical EntitySpecification/Entity pattern already present in the SID model but now applied to a particular kind of Entity, i.e., a Configuration. The EntitySpecification represents the definition of an entity. The idea is that the various ResourceConfigSpec instances are reusable across many instances of Resource.

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Refer to Common Domain guide book section Configuration and Profile ABE for more details

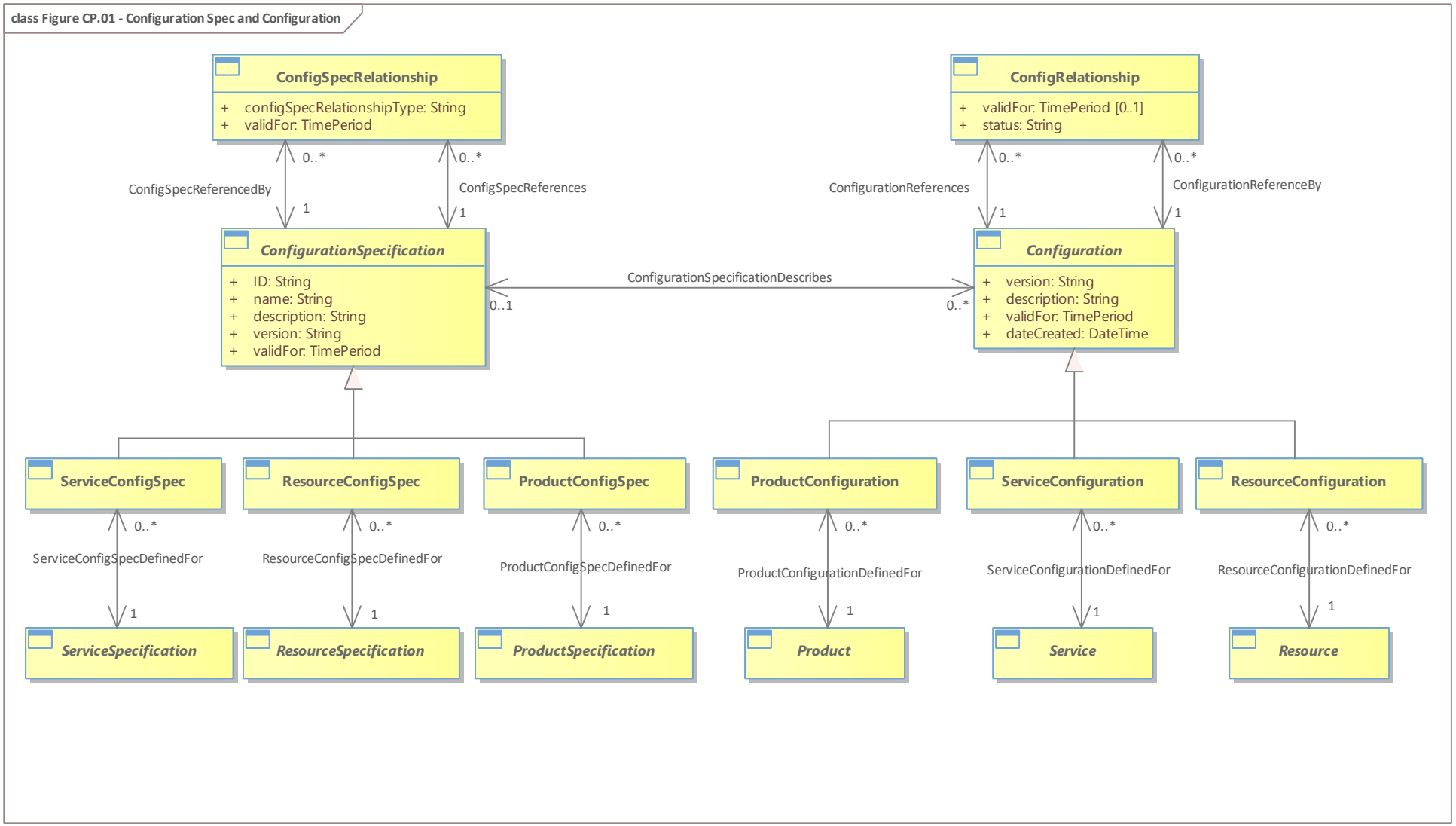

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Figure CP.01 - Configuration Spec and Configuration (copy from Common)

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4.8.1 Product Action ABE

The Product Action ABE contains Business Entities describing possible actions for ProductSpecifications, ProductOfferingSpecifications and ProductConfigSpecs.

Figure PA.01 - Product Action ABE Related Entities

Following are the business entities aggregated under the **Product Action** Aggregate Business Entity.

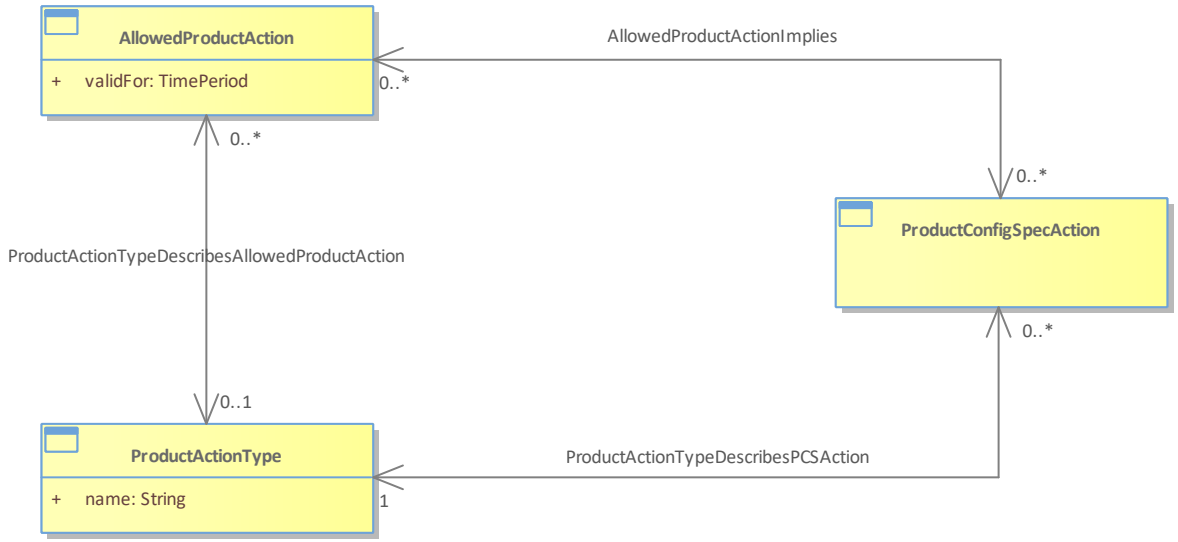

Figure PA.01 - Product Action ABE Related Entities

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Figure PA.02 - ProductOfferingSpecificationMigration and ProductOfferingSpecificationMerge

This diagram shows two AllowedProductAction specializations. ProductOfferingSpecificationMigration is a type of AllowedProductAction used to specify which migration action (or move) from one source ProductOfferingSpecification to one or several destination ProductOfferingSpecification can be performed.

ProductOfferingSpecificationMerge is a type of AllowedProductAction used to specify which merge action from a minimum of two source ProductOfferingSpecification to a unique destination ProductOfferingSpecification can be performed.

Note: This diagram does not represent all possible subclasses of AllowedProductAction. Others could be added.

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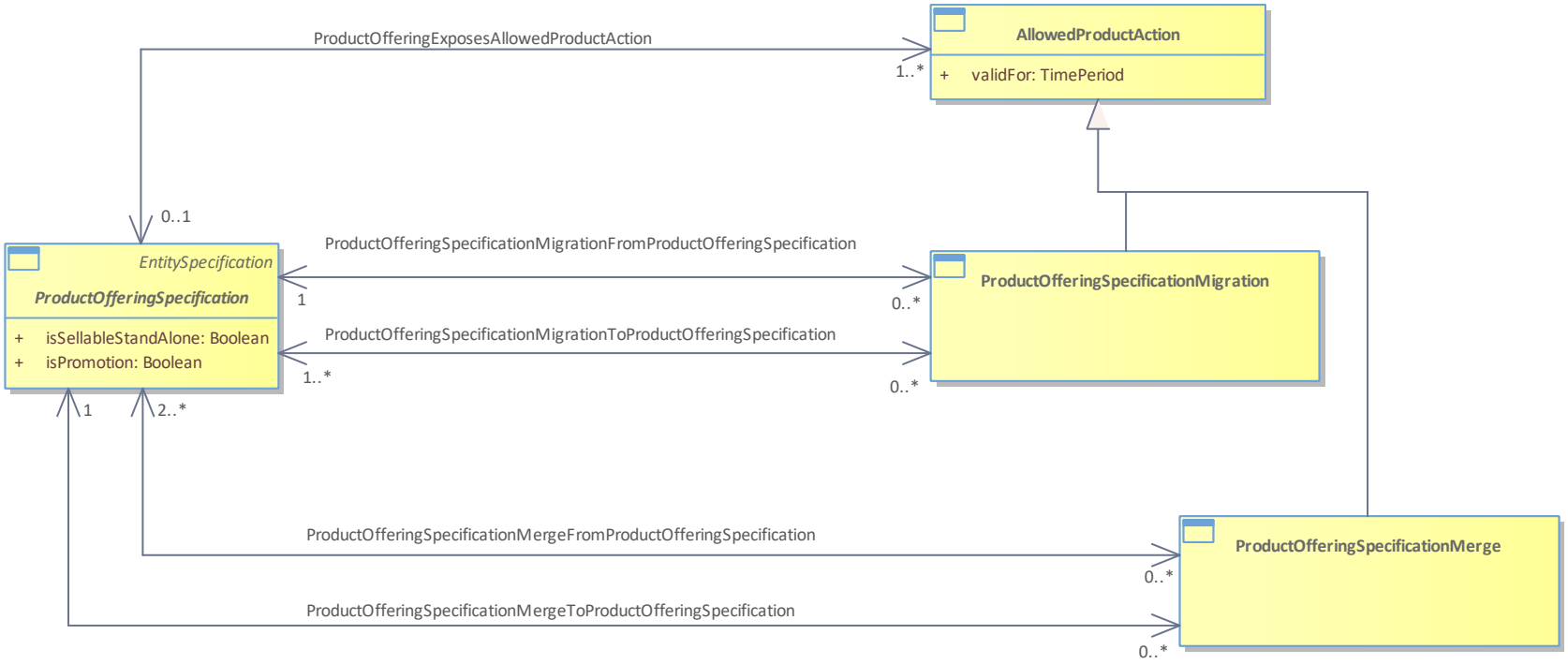

Figure PA.02 - ProductOfferingSpecificationMigration and ProductOfferingSpecificationMerge

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Figure PA.02.I01 - ProductOfferingSpecificationMigration example

This illustration provides examples of instantiation of ProductOfferingSpecificationMigration BE.

A "Low Cost Offer Migration" instance of ProductOfferingSpecificationMigration defines that it is possible to migrate:

- From "Low Cost Mobile Offer",

- To "Standard Mobile Offer" or "Premium Mobile Offer".

A "Standard Offer Migration" instance of ProductOfferingSpecificationMigration defines that it is possible to migrate:

- From "Standard Mobile Offer",

- To "Premium Mobile Offer".

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Figure PA.02.I01 - ProductOfferingSpecificationMigration example

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Figure PA.02.I02 ProductOfferingSpecificationMerge example

This illustration provides an example of instantiation of ProductOfferingSpecificationMerge BE.

A "Quad Play Offer Merge" instance of ProductOfferingSpecificationMerge defines that it is possible to merge:

- a "Standard Mobile Offer" and a "Triple Play (Internet, Fixed line, TV) Offer",

- To a "Quadruple Play (Mobile, Internet, Fixed line, TV) Offer".

A "Standard Offer Migration" instance of ProductOfferingSpecificationMigration defines that it is possible to migrate:

- From "Standard Mobile Offer",

- To "Premium Mobile Offer".

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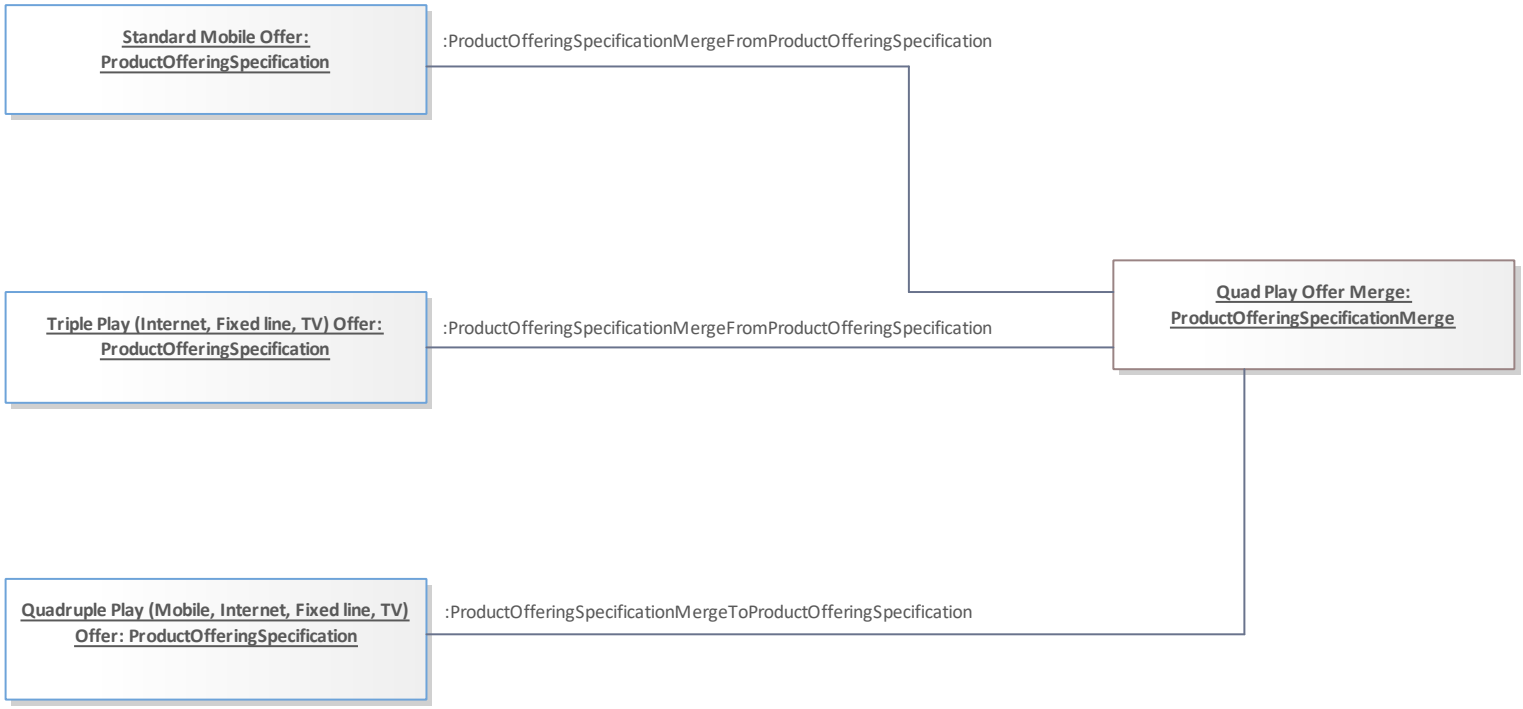

Figure PA.02.I02 ProductOfferingSpecificationMerge example

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4.9 Product Test ABE

Product Test ABE represents all information to specify tests that will be performed on Products as well as the results in measures being produced that reflect the functioning of the product under test.

Figure T.04 - Product, Service and Resource Test Entities

The following Figure shows one way to support tests for Product, Services, and Resources by applying the Entity/EntityRole pattern. This allows the same “Test Specification” to be used for many “Product TestSpecifications”, “ServiceTestSpecifications”, and “Resource TestSpecifications”.

Product, Service, and Resource Tests are defined to be generalizations of the abstract “Test” class. In this way, the “Test” model may be used for the specification of all three types of tests. Each test, such as “ServiceTest” is associated to its relative entity under test, which in this case is “Service”. The relationships therefore between products, services, and resources, are maintained to the associated tests. As an example, a service test measure such as “Frame Loss Ratio” may translate to resource test measures “Number of frames sent” and “Number of frames received”. The metric “Number of Frames Lost” is derived from the resource test measures, and the “Frame Loss Ratio” is further calculated by dividing the number of frames lost by the number of frames sent.

Refer to Common Domain guidebook for more details.

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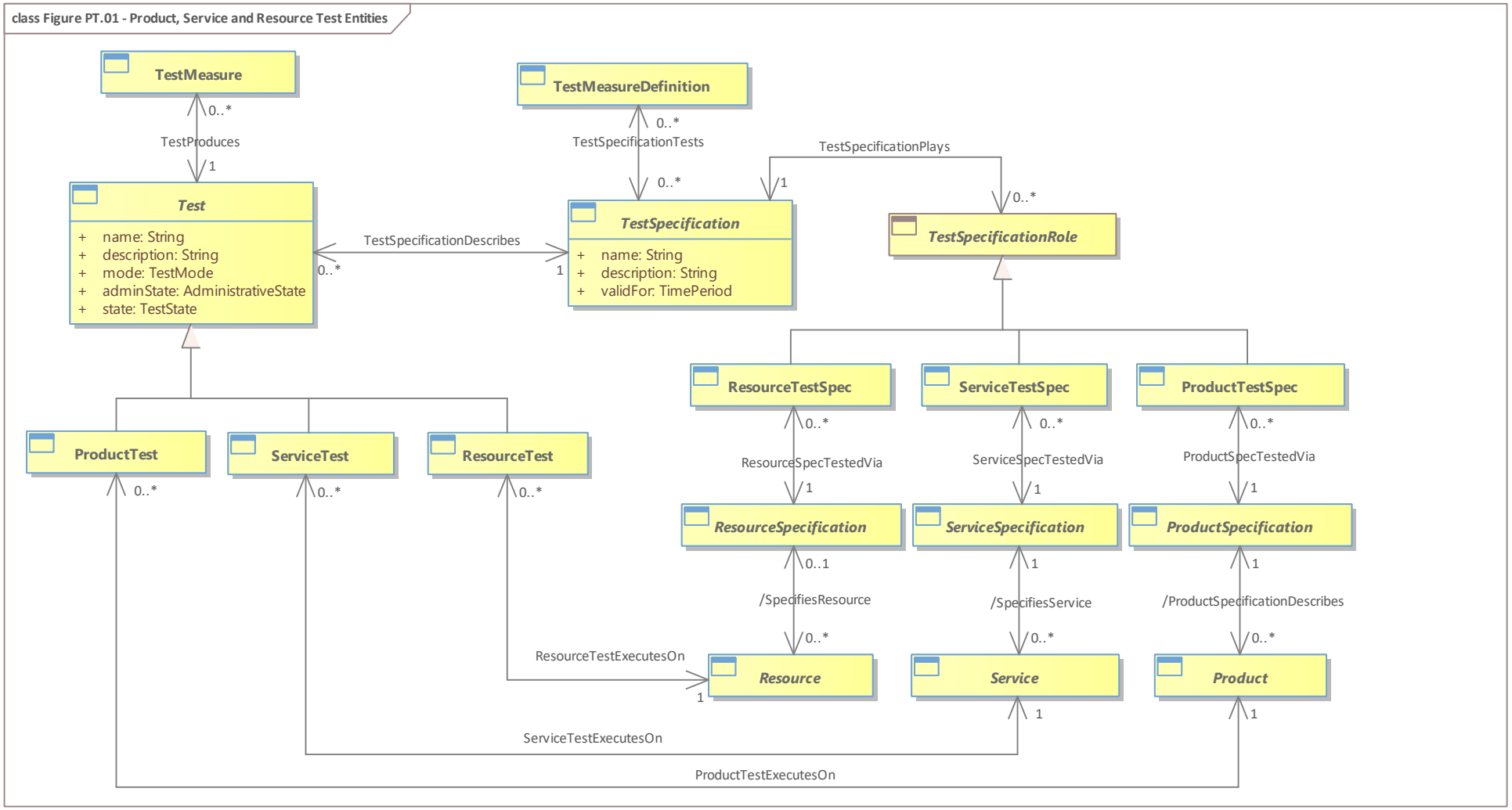

Figure T.04 - Product, Service and Resource Test Entities

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4.10 Product Capacity ABE

Product Capacity ABE represents specific ability of Products measured in quantity and units of quantity over an extended period.

Figure PC.01 - Basic Capacity Entities (Based on CM.02 from Capacity ABE in Common Domain)

A generalized capacity and capacity demand model is shown in the following Figure.

The need to satisfy the capacity requirement during the first week in June is expressed using the fromToDateTime attribute, which is a composite of from and to dates/times, and the rangeInterval attribute of ApplicableTimePeriod. Refinements to the schedule can include days of the week as shown by the daysOfWeek attribute in the ApplicableTimePeriod entity. The rangeInterval attribute is used to handle some combination of >, <, >=, <=. It is assumed this is a planned Capacity, which would be indicated using the plannedOrActutalCapacity attribute. The CapacityRelationship entity enables a planned Capacity to be associated with an actual Capacity or for Capacities to be aggregated.

The requirement for this capacity to be available in the Northeast geographic area is expressed using the relationship to GeographicPlace. Other entities that further define/refine capacity requirements could be added as needed by extending the SID.

The ability to deliver 10000 units of Product described in the previous example is expressed via the CapacityAmount, whose datatype is Quantity. Quantity is a composite of an amount attribute and a units (unit of measure) attribute. For this instance of Capacity, the capacityAmount attribute contains the amount “10000”, and the unit of measure “units”. Other examples of units of measure could be transaction volumes, storage requirements, traffic volumes, port availabilities, and so forth. The capacityAmountFrom, capacityAmountTo, and rangeInterval enable ranges of capacity to be specified.

Demands are placed on the capacity as defined by the CapacityDemand entity. The related ApplicableTimePeriod is used here to express the timeframe over which the demand exists. It may or may not be the same instance of the ApplicableTimePeriod associated with the Capacity. Therefore, it may be a demand placed on more than one CapacityAmount. Similarly, the CapacityAmount may have many demands placed against it.

The attributes are similar to those in CapacityAmount with the addition of a priority attribute that defines the relative importance of this demand.

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The AppliedCapacityDemand entity represents the amount of the CapacityDemand that has been applied against a CapacityAmount.

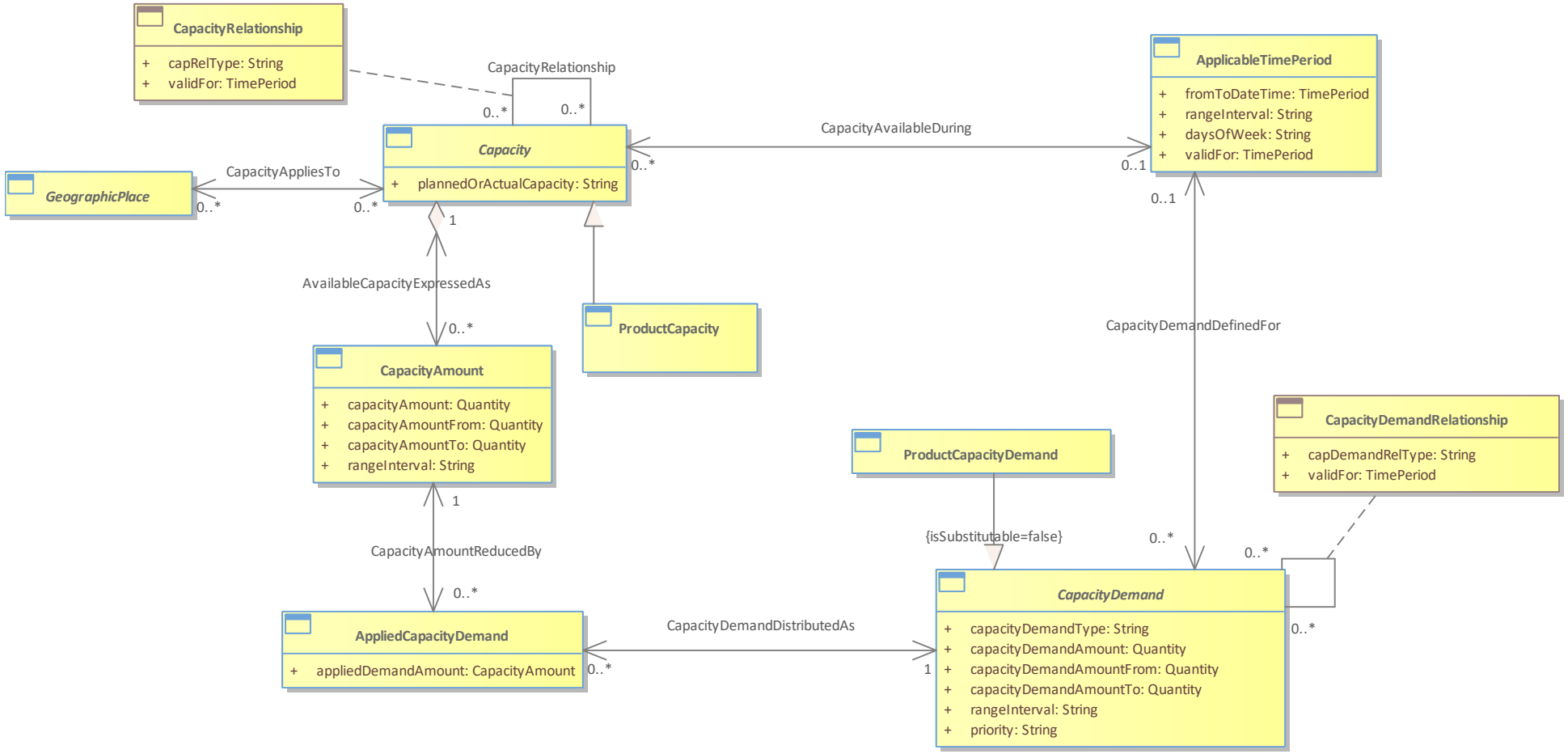

Figure PC.01 - Basic Capacity Entities (Based on CM.02 from Capacity ABE in Common Domain)

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Figure PC.02 - Basic Product Capacity Entities (CM.05 Copy from Capacity ABE in Common Domain)

The following Figure shows ProductCapacity specific relationships.

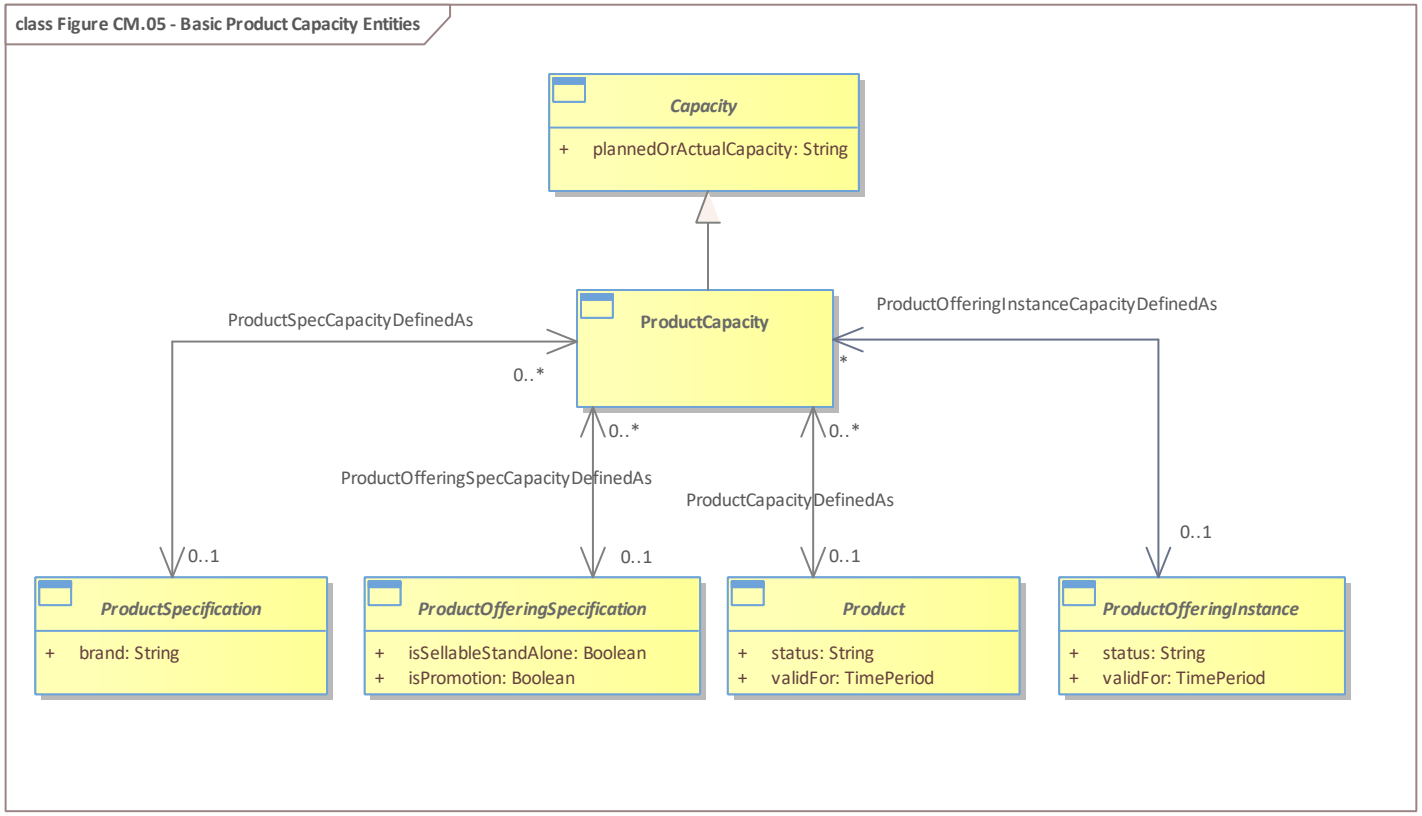

Figure PC.02 - Basic Product Capacity Entities (CM.05 Copy from Capacity ABE in Common Domain)

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Figure PC.03 - Basic Product Demand Entities (CM.08 Copy from Capacity ABE in Common Domain)

The following Figure shows ProductCapacityDemand and its specific relationships.

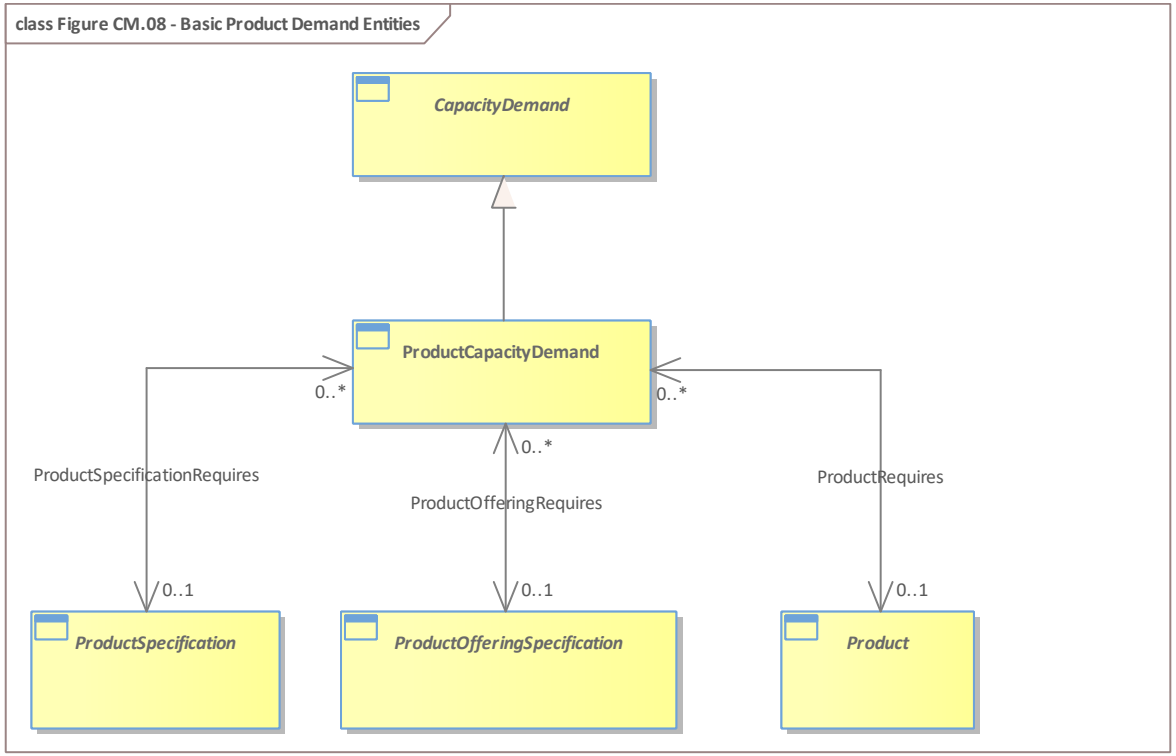

Figure PC.03 - Basic Product Demand Entities (CM.08 Copy from Capacity ABE in Common Domain)

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4.11 Product Performance ABE «notFullyDeveloped»

The **Product Performance ABE** handles product performance goals, the results of end-to-end product performance assessments, and the comparison of assessments against goals. The results may include the identification of potential capacity issues.

This ABE is not fully developed

4.12 Strategic Product Portfolio Plan ABE «notFullyDeveloped»

A strategy employed to promote a ProductOfferingSpecification. **Strategic Product Portfolio Plan ABE** is concerned with the plans of the product portfolio, which product offerings to make available to each market segment and the plans to development and deploy product offerings, as well as retirement of products. *This ABE is not fully developed*

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5 Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

5.1 About this document

This is a TM Forum Guidebook. The guidebook format is used when:

The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.

Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.

Information about the marketplace is provided, as in the report on the size of the OSS market.

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5.2 Document History

Version History

Version Number	Date	Modified by:	Purpose
0.1	Jan 2002	John Reilly	GB915 Addendum Template created from TMF407 v1.1 Guidebook template
	Feb 2002	John Reilly	TAW – updated diagrams & minor changes
0.5	May 2002	John Reilly	Change to Product Offering and Product (Instance) agreed on team calls. General changes resulting from Telstra comments including Version.
1.0	June 2002	John Reilly	Formatted for Member Evaluation release
1.1	Sept 2002	John Reilly	Phase II updates.
1.2	Sept 2002	John Reilly	Updates based on team review.
3.0	May 2003	John Reilly	Added Product Pricing, updated association and attribute names.
5.0	May 2004	John Reilly	Added Product Pricing rules, added data types to figures, added subclasses to ProductSpecificationType.
5.1	August 2004	John Reilly	Synchronized with other Addenda.
6.0	July 2005	John Reilly	Update based on review comments
6.5	November 2006	John Reilly	Included more product offering and characteristic examples; added Product Catalog ABE within Product Offering ABE; minor edits.
6.6	January, 2007	John Reilly	Minor & cosmetic Updates.
7.0	March 2007	John Reilly	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
7.5	January 2008	John Reilly	Updated for ITU-T submission
7.6	May 2008	Tina O'Sullivan	Converted to correct template and other corrections prior to posting.
7.7	July 2008	John Reilly	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
7.9	May 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
9.0	Jan 2010	Josh Salomon	Updates to reflect change requests from V9.0

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Version Number	Date	Modified by:	Purpose
9.0	Feb 2010	Josh Salomon	Added PricingLogicAlgorithm contribution.
9.1	Apr 2010	Alicja Kawecki	Minor cosmetic corrections for web posting and ME
9.2	Jun 2010	Alicja Kawecki	Updated Notice
9.3	Oct 2010	Alicja Kawecki	Updated to reflect TM Forum Approved status
9.4	Jan 2011	Dirk Rejahl	Incorporation of CRs: artf1364, artf2012, artf2079, artf2087
9.5	Mar 2011	Alicja Kawecki	Minor formatting corrections prior to web posting and ME
9.6	Sep 2011	John Reilly	Updated multiplicity on Figure Pr.04c.
9.7	Jan 2012	Josh Salomon	Updates Figure Pr.07 and InvolvementRole modeling
9.8	Mar 2012	Alicja Kawecki	Minor cosmetic corrections prior to web posting and ME
9.9	Oct 2012	Alicja Kawecki	Updated to reflect TM Forum Approved status of R12.0
9.10	Oct 2012	Raj Ravi	Removed the ProductCatalog definitions and respective documentation as it is moved to the Catalog guidebook which is introduced as a part of SID12.5 release.
9.11	Oct 2012	Alicja Kawecki	Minor style/cosmetic corrections prior to web posting and Member Evaluation
9.12	June 2013	Cécile Ludwichowski	Incorporation of CR: Artf2357 (Fig Pr30) Artf2381 (Fig Pr2,)
9.13	July 2013	Alicja Kawecki	Updated cover & header, corrected notice & footer prior to posting
9.14	Sep 2013	Alicja Kawecki	Added IPR Mode to cover, updated notice to reflect TM Forum Approved status
9.14.1	Nov 2013	Alicja Kawecki	Updated cover, header, footer & Notice
14.0.0	May 2014	Cécile Ludwichowski	Introduce Loyalty model
14.5.0	Oct 2014	John Reilly	Updated relationships shown in Figures Pr.4c and 8.
15.0.0	Apr 2015	Cécile Ludwichowski	Update Figure Pr.05
15.0.1	May 2015	Alicja Kawecki	Updated cover, Notice
15.0.2	Nov 2015	Alicja Kawecki	Updated cover and Notice to reflect TM Forum Approved status
16.0.0	May 2016	Cécile Ludwichowski	Adds

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Version Number	Date	Modified by:	Purpose
			AtomicProductSpec sub-classes illustrations Update descriptions and diagrams Remove Entities and Attributes descriptions as they are embedded in RSA.
16.0.1	25 May 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16
17.0.0	May 2017	John Reilly	Update Figure Pr.01 to show that ProductSpecification inherits lifecycleStatus and validFor from EntitySpecification.
17.0.1	June 2017	Cécile Ludwichowski	Update figures removing ProductOffering attributes according to the new inheritance from EntitySpecification.
17.0.2	26 June 2017	Alicja Kawecki	Applied rebranding and minor cosmetic edits prior to publication for Fx17
17.0.3	16 November 2017	Adrienne Walcott	Updated cover, notice and minor cosmetic changes prior to publishing
17.5.0	November 2017	Cécile Ludwichowski	Add isSellableStandAlone attribute to ProductOffering (AP-864) Add defaultRelOfferNumber to BundledProdOfferOption (AP-857) Enable many BusinessInteractionItemPrices for one BusinessInteractionItem (FP-699). Fixes typo Introduces UsageVolumeProductSpec (FP-722)
		Varun Pandhi	Added clarification about difference between ProductLine and ProductCategory. Proposed use case to suggest relationship between “ProductUsageSpec” and “ProductOfferingPrice”.
		Michel Besson	Modified text at the beginning of section 1.1.2
17.5.1	18 Dec 2017	Adrienne Walcott	Formatting/style edits prior to publishing
18.0.1	March 2018	John Reilly and Jonathan Goldberg	Added considerations to assist SID users on the use the new isSellableStandAlone attribute. Changes agreed upon by Jonathan Goldberg and John Reilly. To be reviewed by the group on an upcoming SID call.
18.0.2	June 2018	Cécile Ludwichowski	Updates according to FP-735 Miscellaneous Editorial Fixes Replaces the relation between ProductSpecification and ProductOffering by a derived relationship that can be seen as an overview Remove ConfigurableProductSpecCharacteristic entity Characteristic model clean-up for Product and ProductSpecification
18.0.3	11-Jul-2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing
18.0.4	10-Oct-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

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Version Number	Date	Modified by:	Purpose
18.5.0	03-Dec-2018	Cécile Ludwichowski	Updated according to FP-745 SID-Characteristics clean-up for Service, Resource and Usage.
18.5.1	05-Mar-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0.0	03-Jun-2019	Cécile Ludwichowski	Update ProductOfferingPrice and ProductPrice hierarchy
19.5.0	28-July-2019	Cécile Ludwichowski	Guide book generated from RSA with BIRT report.
21.0.0	02-June-2021	Cécile Ludwichowski	This release includes changes coming from the following Change Requests SID - Product Order change Split Capacity ABE in each horizontal Domain concerned
21.5.0	23-Nov-2021	Cécile Ludwichowski	Remove InvolvementRole and its sub-classes according to the Change Requests https://projects.tmforum.org/jira/browse/ISA-171 and https://projects.tmforum.org/jira/browse/ISA-272 Move TaxeExemptionCertificate to Party ABE according to the Change Request https://projects.tmforum.org/jira/browse/ISA-164 Improve ProductOrder and CustomerProductOrder according to the Change Request https://projects.tmforum.org/jira/browse/FX-1211 Improve Matrix model according to the Change Request https://projects.tmforum.org/jira/browse/ISA-134 Improve ProdOfferPriceAlteration according to the Change Request https://projects.tmforum.org/jira/browse/ISA-174 Remove redundant relationship according to the Change Request https://projects.tmforum.org/jira/browse/ISA-180
22.5.0	09-Dec-2022	Cécile Ludwichowski & Kevin Scaggs	This release includes changes per the following:

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Version Number	Date	Modified by:	Purpose
			- AppliedCustomerBillingAllowance / CustomerAllowance Entity Cleanup, including Product Placement and Product Promotion Level 2 ABE deletion per ISA-356 - Move from RSA to Sparx Enterprise Architect modeling tool - Added a diagram for ABEs containing business entities and relationships, but no diagram.
23.0.0	16-June-2023	Cécile Ludwichowski & Kevin Scaggs	Addition of a diagram showing the relationship between ProductSpec and StockItem per ISA-766.
23.5	22-Dec-2023	Kevin Scaggs	Rename ProductOfferingTerm ProductOfferingTermOrCondition per ISA-850.
24.0.0	05-July-2024	Kevin Scaggs	This release includes changes to the Product Guidebook per the following: • An example of a Product requiring another product per ISA-842 • Major updates to ProductOffering per ISA-961
24.5.0	10-Jan-2025	Kevin Scaggs	This release includes changes to the Product Guidebook per the following: • Separate business entities related to product usage into separate ABEs, namely Product Usage Specification ABE and Product Usage Entity ABE per ISA-969 • Refine ComponentProdOfferingPrice and ComponentProdPrice attributes per ISA-1000 • Delete ProductSpecActivitySpecAllocation per ISA-1112 • Enhance SID ABE Level 1 descriptions per ISA-1113

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Version Number	Date	Modified by:	Purpose
			• Provide means to handle offer migration per ISA-1128
25.0.0	18-July-2025	Kevin Scaggs	This release includes updates to the Product Guidebook given the Split of ProductOrderItem per ISA-1216.

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25.5.0	23-Jan-2026	Kevin Scaggs	This release includes updates to the Product Guidebook as follows: • Refinements related to major Agreement ABE updates per ISA-757, • Updates to Diagram Pr.07b related to ProductOfferingSpecification per ISA-1147, • Update Figures Pr.02 and Pr.02c providing an improved view on the concept of catalogue per ISA-1252, • Rename several entities using the composite/atomic pattern in Figure Pr.05 per ISA-1370, and • Update the label on a relationship between CompositeProductOfferingSpecification and Product Specification in figure Pr.05b per ISA-1396.

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Release History

Release Number	Date Modified	Modified by:	Description of changes
7.0	January 2007	SID Team	
8.0	March 2008	SID Team	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
9.0	January 2010	SID team	Various updates to diagrams and fix discrepancies in existing model. Added Pricing Logic Algorithm ABE description into this addendum
9.5	January 2011	SID team	Incorporation of CRs: artf1364, artf2012, artf2079, artf2087
9.5	September 2011	SID team	Corrected multiplicity on Figure Pr.04c
12	January 2012	SID Team	Corrected Figure PR.07 based on changes in InvolvementRole modeling as described in addendum 1 Users and Roles
12.5	October 2012	Catalog Team	Removed the ProductCatalog definitions and respective text as it is moved to the Catalog guidebook which is as a part of SID12.5 release.
13.0	June 2013	SID Team	Incorporation of CR: Artf2357 (Fig Pr30) Artf2381 (Fig Pr2,)
14.0.0	May 2014	Cécile Ludwichowski	Introduce Loyalty model
14.5.	October 2014	John Reilly	Updated relationships shown in Figures Pr.4c and 8.
15.0.	April 2015	Cécile Ludwichowski	Update Figure Pr.05
16.0	May 2016	Cécile Ludwichowski	Added AtomicProductSpec sub-classes Illustrations Update descriptions and diagrams Remove Entities and Attributes descriptions as they are embedded in RSA.

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Release Number	Date Modified	Modified by:	Description of changes
17.0	June 2017	John Reilly Cécile Ludwichowski	Update Figure Pr.01 to show that ProductSpecification inherits lifecycleStatus and validFor from EntitySpecification. Update figures removing ProductOffering attributes according to the new inheritance from EntitySpecification.
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		Cécile Ludwichowski	Updates according to FP-735 Miscellaneous Editorial Fixes Replaces the relation between ProductSpecification and ProductOffering by a derived relationship that can be seen as an overview Remove ConfigurableProductSpecCharacteristic entity Characteristic model clean-up for Product and ProductSpecification Updates according to FP 752 and FP 762, enhanced modeling of tax and tax exemption

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Release Number	Date Modified	Modified by:	Description of changes
18.0.1	10-Oct-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
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18.5.1	05-Mar-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
19.0.0	03-Jun-2019	Cécile Ludwichowski	Update ProductOfferingPrice and ProductPrice hierarchy
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23.5.0	22-Dec-2023	Kevin Scaggs	Rename ProductOfferingTerm to ProductOfferingTermorCondition per ISA-850.

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5.3 Acknowledgments

This document was prepared by the members of the TM Forum Information Framework (SID) team.

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Key individuals that reviewed, provided input, managed, and determined how to utilize inputs coming from all over the world, and really made this document happen were:

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